

MAGNETOM

Troubleshooting Guide System

Magnet System

Avanto
Espree

Document Version

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1.1 General

This chapter describes the test strategy for the **Magnet, Refrigerator system, Alarm Box** and **Magnet Supervisory Unit (MSUP)**.

The Alarm Box and Magnet Supervisory Unit are **Field Replaceable Units**. If an internal error develops, the complete unit must be replaced.

Prerequisites

n.a.

Tools

Handheld MSUP-Display Unit

Additionally required documents

n.a.

1.2 Magnet

Symptom	Area of Trouble (component or function)	Probable Cause	Service action
01. Liquid helium loss observed	Refrigerator system	Interrupted cold head operation	Maintain
		Compressor pressure too low	Leak check the helium compressor. Top off compressor pressure.
		Reduced efficiency of the cold head	Check the cold head temperature.
			Replace the cold head.
		High shield temperature	Check 50K shield link and bore temperature.
	Pressure heater	Pressure heater is permanently on	Check the status of the pressure heater. LED on the MSUP. Hand-held MSUP-Display Unit Check magnet system pressure.
		Pressure heater power > 0 W	Leak check of the venting system including quench valve. Check magnet system pressure. Ensure EIS is off.
	Venting system	16.0 psi (A) valve of the venting system is leaking	Check the magnet pressure. Perform a leak test.
		Bypass valve is open	Close the bypass valve.
	Helium fill port	Leakage on helium baffle	Check the baffle. Replace the gasket.

Symptom	Area of Trouble (component or function)	Probable Cause	Service action
02. High 50K shield temperature High cold head temp (CHT)	Water cooling	Problems with the water cooling circuit	Check the SEP. Check the Cooler.
	Cold head	Low efficiency of the cold head	Replace the cold head. Contact Siemens Magnet HSC.
	The compressor unit	Low gas pressure of the compressor unit	Refill the compressor unit.
03. Quench	ERDU	ERDU button pressed. Persistent quench	Check the ERDU buttons, release the button. Check the tamper-proof seal. Check the event log file and quench log in the MSUP. Refer to: Post Quench Procedure.
04. Magnet pressure < 15.3 psi	Pressure heater	Pressure heater is defective or off.	Check the status of the pressure heater with the Hand-held MSUP-Display Unit Ensure pressure heater LED is on. Check fuse F7 on the MSUP. Check MSUP connections.
	Venting system	Leakage in the venting system.	Leak check of the venting system including quench valve.
05. Average heater power < 0W System pressure > 15.3 psi (A)	EIS heater	EIS heater is on	Ensure that EIS heater is off, or times out.
	Helium syphon	Helium syphon is fitted	Remove syphon.
	OVC vacuum	Suspected OVC vacuum	Contact Siemens Magnet HSC.
	Refrigerator system	Long-term increase in shield temperature	Check history of shield temperature by SeSo. Contact Siemens Magnet HSC.

1.3 Refrigerator

Symptoms of the Refrigerator systems are listed in the following table.

1.3.1 Troubleshooting

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action
01. System shutdown LCD reads Helium Temp - ERR.	Compressor	Low oil level or blocked oil circulation.	Try five times to restart the compressor. If it does not start, contact MR Magnet Technology HSC .
		Faulty gas thermistor or PCB.	Contact MR Magnet Technology HSC .
	Water Supply	Water flow rate is inadequate.	Check the water flow rate.
		Water flow is reversed.	Check the water lines.
		Supply water temperature is too high	Scroll LCD display. If Water Temp-ERR is indicated, supply water temperature is too high. Provide 5° C to 25° C cooling water
02. System shutdown LCD reads Motor Temp - ERR.	Compressor	Compressor motor winding high temperature switch opens. Compressor motor windings have overheated.	Check the water flow rate. Allow 30 minutes for the windings to cool enough for the switch to reset. Contact MR Magnet Technology HSC if the problem persists.
		Reversed phase or loss of phase.	Check mains power. Fuse(s) opened or poor power quality.
		Fuse(s) opened or poor power quality.	Check fuses
03. System shutdown LCD reads Phase Seq-ERR.	Compressor	Fuse(s) opened or poor power quality.	Check fuses.
04. System shutdown LCD Return Press-ERR.	Compressor	Compressor has lost helium charge.	Refer to Helium Gas Charging F70 If no pressure, replace compressor.
05. System shutdown LCD reads FAIL instead of ERR.	Compressor	Sensor is disconnected.	Contact MR Magnet Technology HSC .

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action
06. Compressor and cold head motor do not start when the start switch is pressed.	Electrical Supply	No electrical power.	Check that the power source is on and connected.
		Wrong voltage.	Check mains voltage.
		Main power phase sequence is wrong.	Interchange any two incoming power leads (except ground).
	Compressor	Defective component in the power circuit.	Replace compressor.
		Tripped motor circuit protector on the front panel.	Re-set the protector - turn knob OFF and ON. Contact MR Magnet Technology HSC if problem persists.
	Magnet	Helium vessel pressure is very low.	Check for gross leaks on the magnet venting.
		Pressure switch faulty.	Check pressure manometer.
		Sensor cable broken or disconnected.	Check cable and connections at compressor and manometer.
07. Compressor starts but shuts down later.	Compressor	Insufficient coolant for the compressor.	Check the coolant flow and temperature. Refer to specifications.
08. System starts but gas pressure abnormally high or low.	Compressor	Wrong equalization pressure.	Refer to Service -Troubleshooting Guide - Gas Charging procedure.
		Gas line couplings are not fully engaged.	Check the gas line connections.
		Gas lines are wrongly connected.	Check the gas line connections. Re-connect.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action
09. Cold head motor does not start when the compressor starts.	Cold head	Cold head cable is not connected.	Stop the compressor. Connect the cable. Check connections at the cold head and at the compressor.
		Open circuit in the cold head cable.	Disconnect the cable. Check each conductor for continuity. Replace the cable if necessary.
		FU-4, FU-5 or FU-6 fuse is blown.	Replace the fuse.
10. Cold head motor hums but does not start.	Cold Head	Open circuit in the cold head cable.	Disconnect the cable. Check each conductor for continuity. Replace the cable if necessary.
		Valve disc stalled on the valve stem.	Check the system equalization pressure. Contact MR Magnet Technology HSC .
		FU-4, FU-5 or FU-6 fuse is blown.	Replace the fuse.
11. Cold head motor runs but there is no cooldown.	Cold head	Gas line couplings are not fully engaged.	Check gas line connections.
		Gas lines are wrongly connected.	Check gas line connections.
12. Intermittent operation.	Compressor	Compressor is cycling on and off.	Check input power, coolant flow and temperature.
13. Loss of refrigeration.	Compressor	Compressor malfunction	Check input power, coolant flow and temperature, and equalization pressure.
	Cold head	Worn displacer.	Check cold head run hours.
		Contaminated helium gas.	Contact MR Magnet Technology HSC .
14. Compressor shuts down, LCD display is blank.	Compressor	FU-1, FU-2 or FU-3 fuse is blown.	Replace the fuse.
	Compressor	Malfunctioning control boards.	Contact MR Magnet Technology HSC .

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action
15.Compressor is unresponsive to remote (DB-25 diagnostic interface) control.	Compressor	Configuration selector switch is in the wrong position.	Turn off power and move the switch to the correct position (down for 4K systems, up for 10K systems).
16.Remote status signals (DB-25 diagnostic interface) do not match actual status.	Compressor	Configuration selector switch is in the wrong position, or was in wrong position when power was supplied to the controller.	Turn off power and move the switch to the correct position (down for 4K systems, up for 10K systems).
17.LCD display shows configuration error	Compressor	Configuration selector switch is in the wrong position or was in the wrong position when power was supplied to the controller.	Turn off power and move the switch to the correct position.
18.Abnormal noise from the cold head.	Cold head	Contaminated helium gas.	Contact MR Magnet Technology HSC .
Remarks: Check the Event log of the MR system for errors and warnings generated by the refrigerator system or the water cooling system.			

1.3.2 Compressor F70

1.3.2.1 Compressor Restart time

Timeout time of the compressor:

- 20 min after power failure.
- 60 min after static pressure test.



The compressor timeout time can be 10% - 15% longer, depending on the load the MSUP controller sees.

1.3.2.2 Fuse Location F70



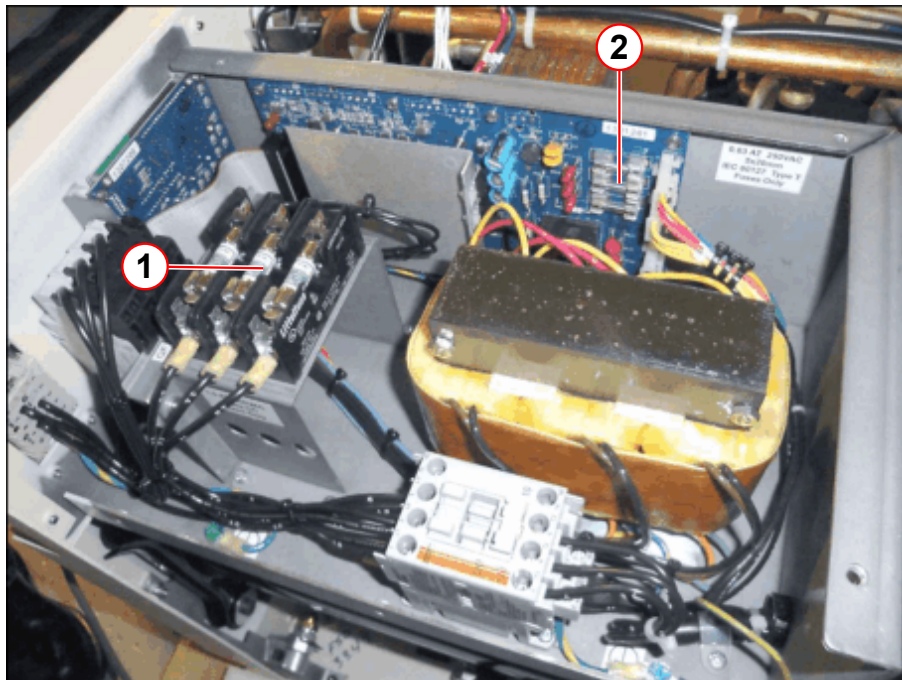
WARNING

AVOID ELECTRIC SHOCK

Failure to observe this warning can result in serious injury.

- » Disconnect the power to the compressor before troubleshooting the electrical components.
- » Permit only qualified electrical technicians to open electrical enclosures.

Fig. 1: F70 - Electrical Chassis



- (1) Fuses, FU-1, FU-2, FU-3
(2) PCB 1 with Fuses, FU-4, FU-5, FU-6

1.4 Magnet Supervision Unit (MSUP)

Error Messages

- Refer to (→ Healthcare Services Knowledge Base (SKB), <https://skb.healthcare.siemens.com/default.aspx>).

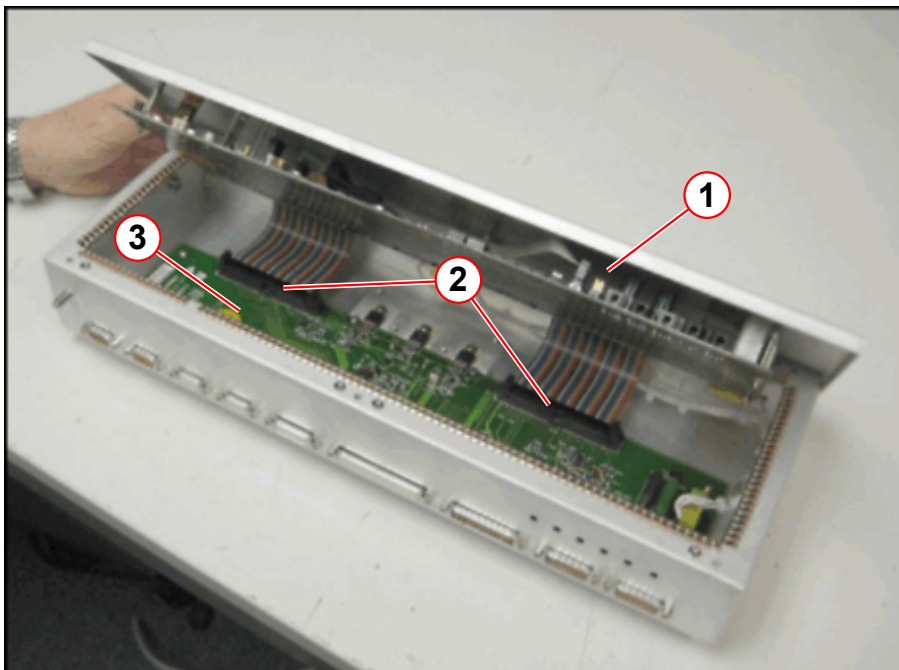
1.4.1 MSUP 2306

1.4.1.1 General

The **MSUP 2306** is a **Field Replaceable Unit**. It contains 2 assemblies:

- The **MSUP 2306 controller assembly** is attached to the underside of the lid and is a **replaceable** spare part. (→ 1/ Fig. 2 Page 14)
- The **ERDU Test Load assembly** is mounted in the base of the MSUP and is **not separately replaceable**. If faulty, the **complete MSUP 2306 unit** must be replaced. (→ 3/ Fig. 2 Page 14)

Fig. 2: MSUP 2306 complete unit



- (1) MSUP controller assembly (replaceable part)
- (2) Ribbon cables
- (3) ERDU assembly (not replaceable)

1.4.2 MSUP 2657

This chapter deals with troubleshooting the Magnet Supervision Unit (MSUP).

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
No response from MSUP	MSUP	36 V power supply incorrect.	Check fuse F2 on MSUP board. Check circuit breaker F2 in the ACC. Check fuse F3 in the ACC.	##
MSUP not serviceable with the display unit	MSUP	Defective MSU	Perform Com-test with Test Tool Replace MSUP	
Display unit suspected	Display unit	+5V is missing	Check +5 V at output of MSUP X9	
MSUP reports short circuit (S/C) after measuring helium level	Helium level probe	Incorrect level probe parameters	Check cabling Check fuse F3 on MSUP board. Alternately change pulse length and pulse height. » Change pulse length at 5 ms intervals, up to maximum 200 ms. » Change pulse height at 50 mA intervals, up to maximum 800 mA.	Speed info MR_45/04
Remarks: Check the Event log of the MR system for errors and warnings generated by the Magnet Supervision Unit.				

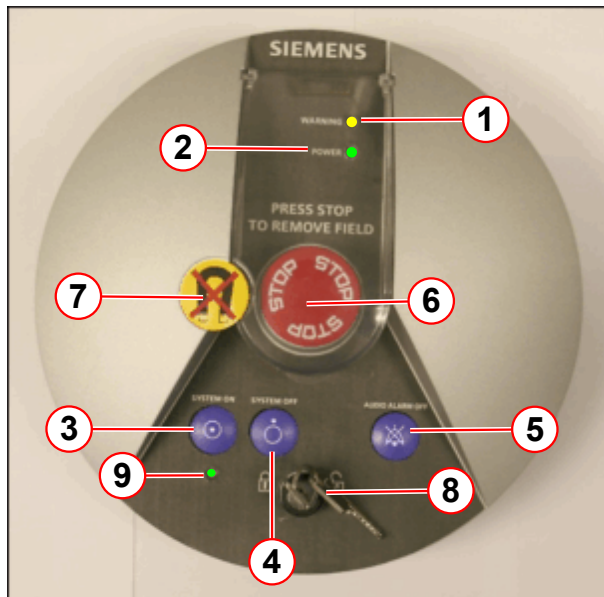
1.5 Alarm Box

The Alarm Box is an FRU. If an internal fault develops, the complete unit must be replaced.

1.5.1 Alarmbox (Mat. number 10523563) with MSUP 2306

1.5.1.1 Alarm box (Mat. number 10523563)

Fig. 3: Alarm box - LEDs and buttons



The Alarm Box is a Field Replaceable Unit (FRU). If an internal fault develops, the complete unit must be replaced.

- (1) LED WARNING = yellow
- (2) LED POWER = green
- (3) Button - SYSTEM ON
- (4) Button - SYSTEM OFF
- (5) Button - AUDIO ALARM OFF
- (6) Button - MAGNET STOP (ERDU)
- (7) Tamper-proof seal
- (8) Power switch locking key
- (9) SYSTEM ON LED = green

The alarm box provides two indicators concerning the health of the magnet system. When part of the system fails, the indicator will be illuminated to warn of the malfunction.

To silence the audible buzzer, press the AUDIO-ALARM OFF button.

Indicator - Title	Color	Function (when lit)	Service Action
System ON	Green	Indicates power has been applied to system.	n.a.
Warning	Yellow	Warns of fault in Magnet Supervisory system	Check at console for identification of problem.

1.5.1.2 LEDs

Tab. 1 Alarm box LEDs

LED	Description
LINE POWER	Power is supplied to the alarm box
WARNING	Helium level alarm or warning condition, or helium alarm Quench / switch heater error or ERDU button Magnet pressure too high / low Compressor stopped (pressure switch operated) ERDU battery voltage alarm or warning condition EIS drive error

1.5.2 Alarmbox (Mat. number 10096765) with MSUP 2657

The alarm box provides nine indicators concerning the health of the magnet system. When part of the system fails, the indicator will be illuminated to show the malfunction.

To silence the audible sounder, press the Acknowledge button.

Indicator - Title	Color	Function (when lit)	Service Action
System ON	Green	Indicates power has been applied to system.	n.a.
Line power	Green	Indicates power has been applied to alarm box	n.a.
Helium level	Red	Shows helium level in magnet is low.	Check the helium level. Check the refrigerator system.
Mag Stop	Red	ERDU button has been pressed Indicates an ERDU-related fault.	Check heater resistances via the hand held unit on the 2657 MSUP. Check the cable connections of the ERDU. Check fuses F3, F5, F6 and F7 on MSUP board Perform safety-related test. Download MSUP log. Email Magnet HSC
Pressure	Red	Shows when system pressure is too high/low.	Check the magnet pressure with hand-held unit. Ensure pressure heater circuit works correctly. Check the Event log for errors and warnings about the water cooling system. Switch on the compressor again.
Compressor	Green/Red	Red indicates that the compressor has stopped running.	Check the Event log for errors and warnings about the water cooling system. If no audible alarm sounds when the compressor stops, wait 30mins to see if the compressor restarts automatically.

Indicator - Title	Color	Function (when lit)	Service Action
Battery low	Red	Shows a fault in the emergency rundown back-up battery system.	Check fuse F4 on MSUP board. Check cable connection between battery and MSUP. Check hand held unit on the 2657 MSUP for the battery voltage. Replace the battery.
EIS	Yellow	Indicates a fault in external influence shield operation.	Check fuse F1 on the MSUP board Check hand-held unit on the 2657 MSUP for EIS condition.
Comms. Fault	Red	Indicates a break in the communications between the Magnet Supervision Unit and the alarm box.	Check cabling between Magnet Supervision Unit and alarm box. Check condition of 2657 MSUP.

1.6 Superconducting Shim (Espreo only)

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
Term 2.0 is out of specification	Superconducting Shim (Z-2 Shim)	Z-2 shim current	- Applying Z-2 shim current without using an Array Shim Device. - Perform a shim plot for calculating the Z-2 shim current	Online Help "Shim / Superconducting Shim"
Term 2.0 is out of specification	Superconducting Shim (Z-2 Shim)	Magnet was ramped	Applying the Z-2 shim current without using an Array Shim Device.	Online Help "Shim / Superconducting Shim"
Remarks: Check the Event log of the MR system for errors and warnings generated by the Magnet Supervision Unit.				

1.7 Shim power supply

Error Messages

- Refer to (→ Healthcare Services Knowledge Base (SKB), <https://skb.healthcare.siemens.com/default.aspx>).

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
SPSU suspect	Input power supply	Missing input power	Is power LED lit? Is fan running? Check input power at connector on front of SPSU	n.a.
Remarks: Check the Event log of the MR system for errors and warnings generated by the Magnet Supervision Unit.				

1.8 Magnet Power Supply

Error Messages

- Refer to (→ Healthcare Services Knowledge Base (SKB), <https://skb.healthcare.siemens.com/default.aspx>).

Symptom	Probable Cause	Service Action	Reference
Fault reported by HOST	n.a.	Check if fault is an MPS or magnet problem	n.a.
Temp error	Air vents Air temperature in the surrounding area	Clean air vents Check temperature of air in the surrounding area	n.a.
Power error	400 V is missing	Check 400 V on the power supply cable	n.a.
Overcurrent error	Defect on the MPS	Replace MPS	n.a.
MPS fails to obey HOST	Connection between HOST and MPS	Check RS232 connection between MPS and HOST	n.a.

1.9 Ramp Cable Shorting Links

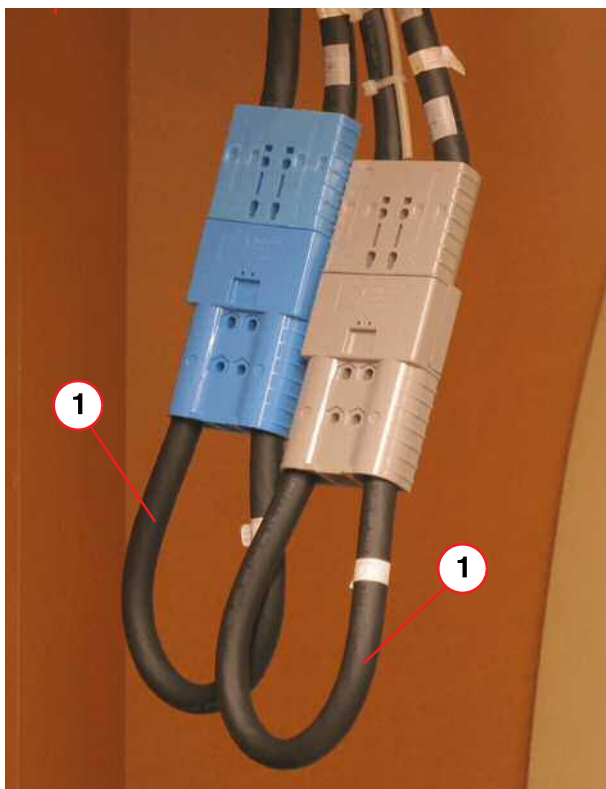
It is no longer a requirement to fit the 2 shorting links when the magnet is at field and operational.

- Magnets after 2008 are no longer fitted with shorting links.



DO NOT remove the 2 shorting links at the same time.

Fig. 4: Magnet Shorting links.



(1) Shorting links

- Remove one shorting link.
- Connect the magnet ramp lead to the extension lead.
- Connect the extension lead to the 3600 MPSU.
- Repeat the process for the other shorting link and ramp lead.



If shorting links are fitted, it is not possible for the 'Anderson' connectors to be bolted behind the filter plate. The leads will be stored in the lead tray.

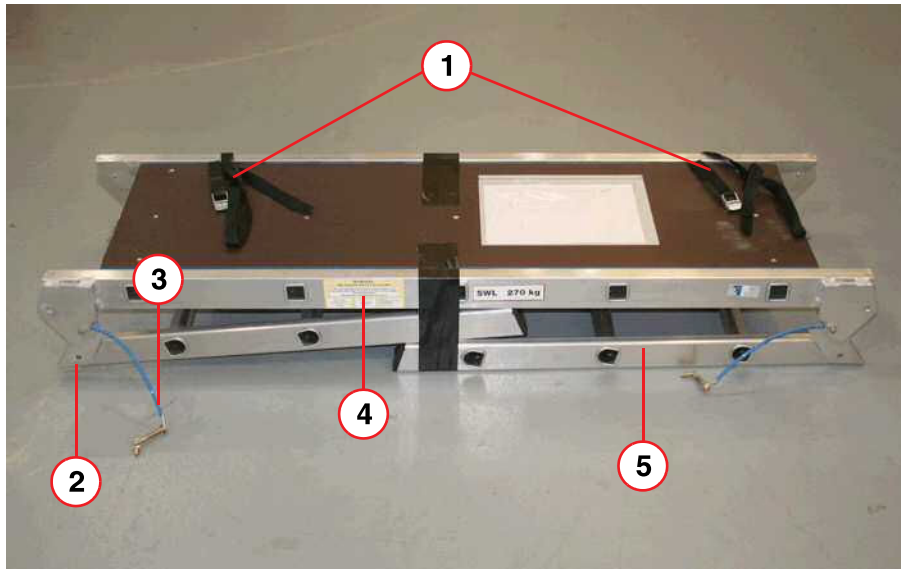


When the shorting links have been removed they can be discarded. They do not need to be fitted again after ramping. Remove from site.

2.1 Magnet

2.1.1 Service platform installation

Fig. 5: Service platform - as delivered



- (1) Safety straps (2 off)
- (2) Hinged locking pin location (each end)
- (3) Retaining wire and hinged locking pin (each end)
- (4) Warning label
- (5) Step legs (each end)

2.1.1.1 Service platform safety

Before using the service platform **read the following safety cautions** and perform the **required safety checks**.



CAUTION

Risk of personal injury

Failure to observe the following precautions may result in personal injury.

- » Always check the platform and hinge locking pins for damage.
- » Always check the hinge locking pins are secured.
- » Always secure the platform to the magnet frame or with the two long straps through the magnet bore.
- » Always ascend and descend (Walk UP or DOWN) the steps facing the platform.
- » Where possible use the frame or covers for additional hand support.
- » Never climb onto the platform from the side.
- » Never jump from the steps or top of the platform.

Fig. 6: Correct method - Up to platform



Fig. 7: Incorrect method - Down from platform

**CAUTION****Risk of personal injury**

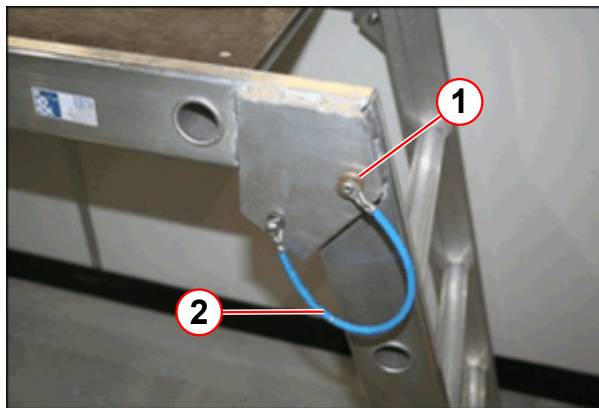
Carrying heavy components or tooling up or down the platform steps can lead to injury.

- » DO NOT carry heavy components up or down the platform steps.
- » Place heavy objects (i.e. cold head) onto the platform first.
- » Only lift the component when standing safely on the platform.

2.1.1.2 Platform assembly

- Remove any packaging.
- Unfold both step legs.

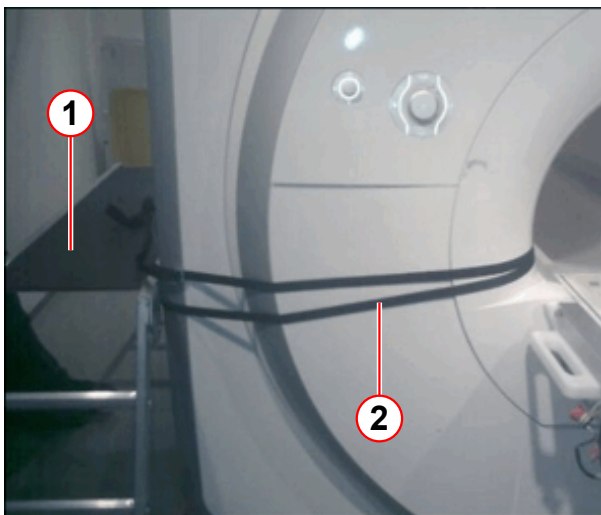
Fig. 8: Service platform - lock legs in position



- (1) Hinged locking pin
(2) Retaining wire

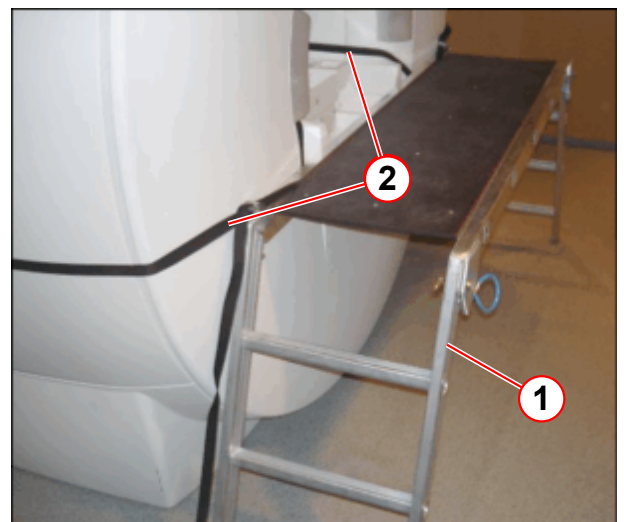
- Align the hole at the top of the leg with the **pin location hole** (pos 1).
- Push the **hinged locking pin** through the hole (pos 1) and fold back the hinge to secure.
- Repeat with the other leg.

Fig. 9: Securing the service platform (example)



- (1) Service platform
(2) Long straps (fitted through the magnet bore)

Fig. 10: Securing the service platform (example)



- (1) Service platform
(2) Long straps (fitted through the magnet bore)



Do not exceed the maximum load - the combined weight of 2 people must not exceed 270 kg.



The fiber optic connections in the top of RFIS are very fragile, so exercise due caution.

2.1.2 Depressurization of the helium vessel



CAUTION

Risk of asphyxiation and hypothermia if helium gas is vented into the magnet room

Failure to observe the following safety warning may result in dizziness or loss of consciousness for personnel.

- » Quench valve and venting system must be connected to the quench pipe.



CAUTION

Risk of injury

Danger of cold burns. Suitable protective clothing must be worn.

- » Do not undo joints on the HIGH pressure side until the system is fully depressurized.



Before opening up any of the turret ports, there must be a positive pressure of at least 7 mbarg (0.1 psig) and no more than 14 mbarg (0.2 psig) in the helium vessel, as displayed on the pressure gauge assembly.

Depressurization is performed via the **bypass valve assembly**, which bypasses the **quench valve**.

Depressurization of the helium vessel is required when:

- Helium filling - refer to the filling procedure described in the **System Startup**.
- Replacing parts on the venting system - **refer to Service - Replacement of Parts**.
- Replacing the cold head.

2.1.3 Helium filling procedure

Performing the procedure

Refer to the filling procedure described in the system start-up.

2.1.4 De-icing the Syphon Port

2.1.4.1 Prerequisites

General Information



This procedure must only be performed with the magnet RAMPED DOWN.

The service syphon is designed to be fully portable and compatible with all Siemens **4K** magnet products. By using a combination of the service syphon extension legs supplied, a helium top fill, bottom fill, or de-ice of the syphon port can be performed.

A de-ice may be required if:-

- A resistance is felt when inserting the syphon leg.
 - » This suggests the syphon cone is iced-up.
- Poor transfer efficiency.
- No movement on the helium level meter after 20 minutes.

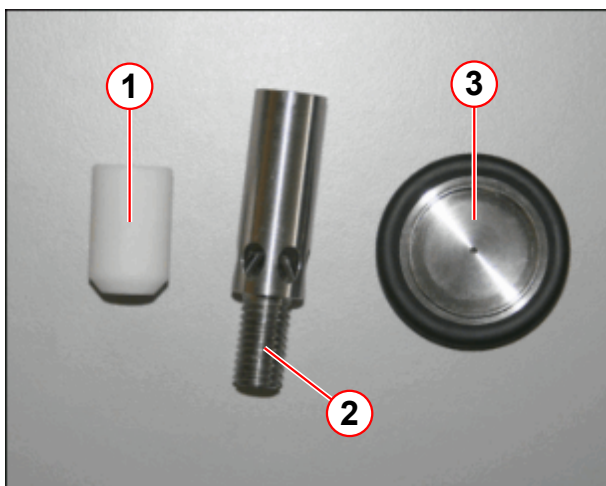
The de-ice tool is designed to blow helium gas (at room temperature), in defined directions within the syphon port dispersing any ice which has formed.

Requirements

Tab. 2 Service syphon kit and upgrades

Material number	Revision	Comment
08396363	00	Service syphon kit
10126691	00	Upgrade kit (200 mm extension leg added)
10129438	00	Upgrade kit (de-ice tooling)
10126651	01	Upgrade Service syphon kit (includes upgrade kits)

Fig. 11: Upgrade kit 10129438 - De-ice tooling



- (1) PTFE tip
- (2) De-ice helium fill diffuser
- (3) NW25 flow restrictor

Tooling

- 99.995% (minimum) helium gas (50 litre) cylinder and high pressure regulator.
- Pressure regulator 200 bar/ 0 - 1.5 bar.
- Gas bottle wrench.
- Protective clothing: gloves, protective goggles or face protection shield, protective clothing, protective shoes (non-magnetic).
- Heat gun.
- Paper toweling.
- 30 mm non-magnetic syphon wrench (MK2 Avanto and Espree turret syphon ports). (→ Fig. 16 Page 36)(→ Fig. 15 Page 36)
- Replacement copper seal for syphon port on the turret (material no **03108094**).
- NW25 hose connector (material number **08396298**)
- O₂ monitor.
- Leak check equipment capable of checking individual leaks down to 1×10^{-4} mbar l/s.
 - Use the Edwards hand-held leak detector (material number **10614850**).

Safety



CAUTION

Strong magnetic field

Failure to adhere to safety measures may result in serious bodily injury.

» The magnet must be **RAMPED DOWN** before attempting this procedure.

⚠ WARNING

Pressurized helium vessel.

Cold helium gas.

Failure to observe the following Safety Information may result in physical injury.

- » Do not disconnect any of the fittings without depressurization of the helium vessel.
- » Only trained and accredited CSEs must perform this procedure.

⚠ DANGER

Risk of asphyxiation.

Failure to observe the following Safety Information may result in dizziness and loss of consciousness.

- » Do not vent large volumes of helium gas directly into the magnet room.
- » Personal Oxygen monitors must be worn if a fixed room monitor has not been installed
- » In a storage facility, ensure that there is sufficient ventilation.

⚠ WARNING

Risk of cold burns

Failure to observe the following may result in injury to personnel.

- » Do not touch cold surfaces with bare hands.
- » Personal safety equipment must be worn.

2.1.4.2 De-ice Procedure

The de-ice diffuser tip is designed to direct the warm gas downwards past the syphon cone to clear ice from the lower regions of the turret.

When the syphon port is clear of ice the diffuser tip is able to fit into the syphon cone.

⚠ WARNING

Risk of quench

Failure to observe the following may result in the magnet quenching and loss of helium.

- » Ramp the magnet to zero field before deicing the syphon port.

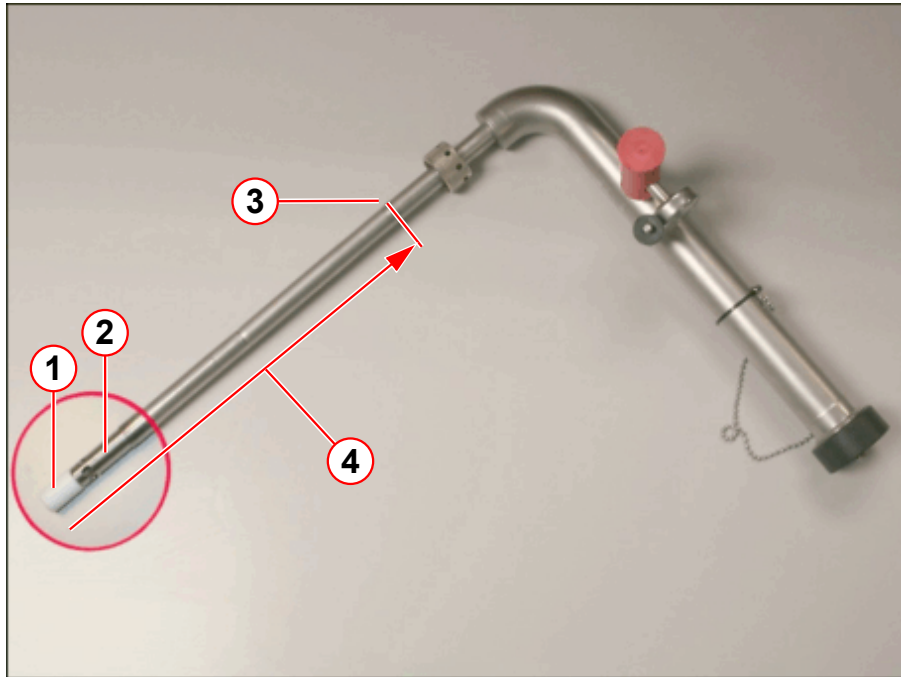
- Ramp the magnet to zero field.
- Deactivate the pressure heater.

- **Open** the magnet bypass valve to depressurize the helium vessel.

Assemble the Syphon

- Fit the required extension legs.
- Fit the de-ice diffuser tip to the syphon leg.
 - Check that the PTFE extension tip is screwed tightly onto the de-ice diffuser.

Fig. 12: Syphon assembly for de-icing the syphon port



- (1) PTFE tip
- (2) 6 hole de-ice diffuser
- (3) Syphon depth guideline
- (4) Depth to cone measurement

Tab. 3 Extension legs and de-ice diffuser requirements

Magnet Type	140 mm Leg	200 mm Leg	60 mm De-ice diffuser and PTFE tip	Depth to cone
Avanto (OR105)	2	1	1	485 mm
Esprea (OR122)	2	1	1	485 mm
Verio (OR97)	2	1	1	535 mm
MAGNETOM Es-senza (OR60)	2	1	1	505 mm
Aera (OR98)	2	1	1	554 mm

Magnet Type	140 mm Leg	200 mm Leg	60 mm De-ice diffuser and PTFE tip	Depth to cone
Skyra (OR99/103)	2	1	1	554 mm
Prisma (OR65)	2	1	1	550 mm

Refer to table above to determine the depth to cone measurement for required magnet type.

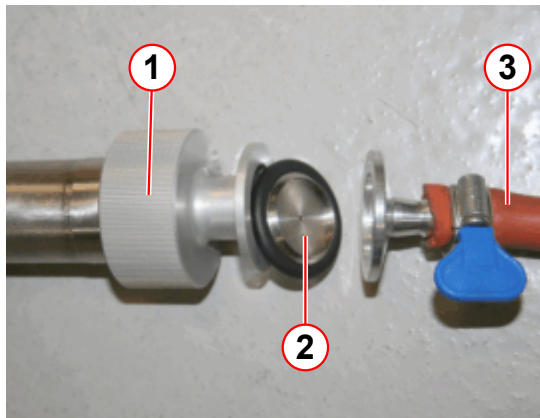
- Select the correct combination of extension legs.



The depth measurement from the top of the ring nut to the syphon cone is a guide. Add an additional 10 mm to ensure that the extension leg reaches the cone.

- Mark the syphon leg with a marker pen.
 - » Example: OR105: Mark a line at **495 mm** (485 mm + 10 mm).
- Connect the high pressure regulator to the helium gas bottle, (99.995% (minimum) helium gas).
- DO NOT turn on the helium gas supply.

Fig. 13: NW25 adaptor and flow restrictor



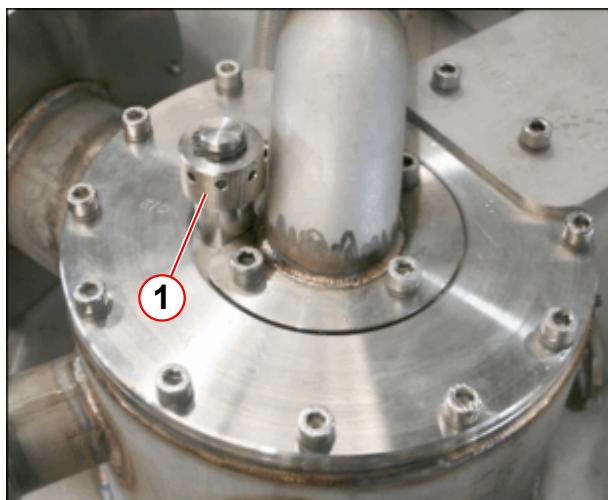
- (1) NW25 fill adaptor
- (2) NW25 flow restrictor
- (3) Helium supply line

- Remove the syphon end cap.
- Fit the NW25 hose adaptor to the coupling end of the syphon .
- Fit the NW25 flow restrictor, between the adaptor and helium supply line.
 - » The NW25 restrictor allows a calibrated flow of gas to pass through it for de-icing.

- Confirm that the magnet is depressurized completely.
- Warm and dry the syphon blanking plug using a heat gun.
- Remove the syphon blanking plug using the syphon 'C' wrench.

OR60/OR65/OR97/OR98/OR99/103/OR105/OR122
turret

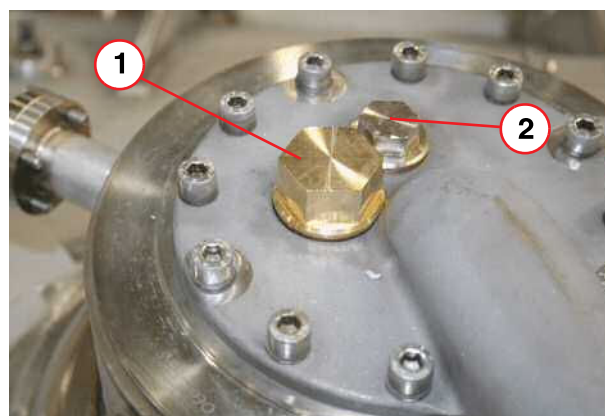
Fig. 14: Syphon entry port



(1) Syphon blanking plug

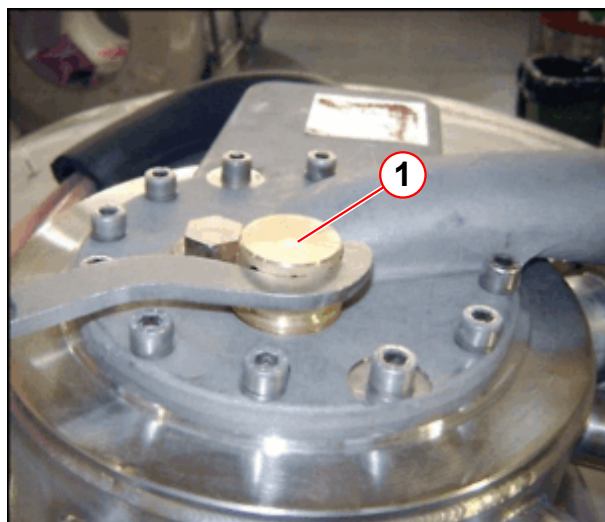
OR105/122/ turret

Fig. 15: Mk2 Syphon port entry



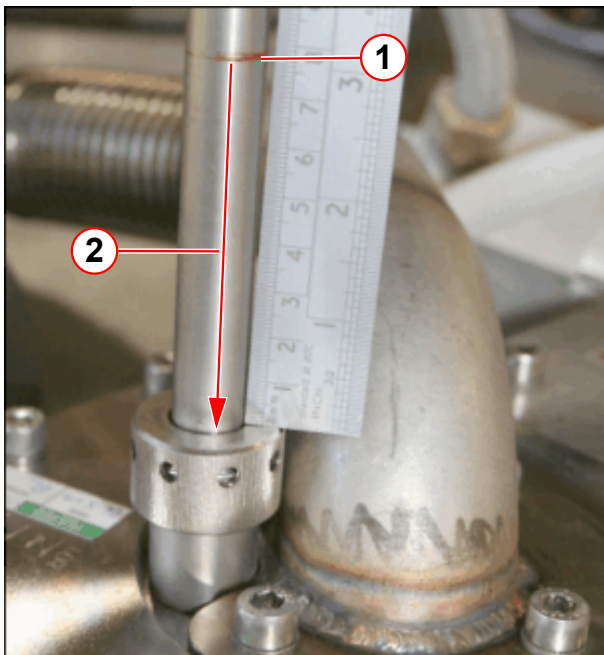
(1) Syphon blanking plug
(2) Sight glass

Fig. 16: Mk2 Syphon port



(1) Syphon blanking plug

Fig. 17: Measure the of depth of ice



- (1) Depth marker
- (2) Depth of ice in the turret

- Turn on the helium regulator and set the outlet pressure to **2 bar** on the gauge.
- Allow the helium gas to purge the syphon for 10 s.
- Insert the service syphon into the syphon port.
- Slowly lower the syphon until a resistance is felt.
 - » The extension tip is now touching the ice.
- The depth of ice in the syphon port is shown in (position 2).

- Wait **1 minute**:-
 - » Unscrew the syphon ring nut enough to allow the syphon to be pushed gently down 10 mm to 15 mm as the ice melts.
- Wait **1 minute**:-
 - » Repeat the process - push the syphon down a further 10 mm to 15 mm.

The internal magnet pressure gradually rises during the de-ice process.

- Continue until the syphon has engaged in the cone either:-
 - » by feeling the PTFE tip hitting the syphon cone.
 - » by reference to the mark made on the syphon extension leg.
(→ 1/ Fig. 17 Page 37)
- Continue the de-icing for a further 10 minutes. Turn off the helium gas pressure at the cylinder regulator.

It is possible to melt 10 mm to 15 mm of ice per minute. The process uses approximately one full (50 litre) cylinder of helium, taking 30 minutes to empty.



The internal magnet pressure quickly rise when there is a clear path through the ice into the helium vessel.

- Wait for the magnet pressure to return to atmosphere (zero).
- Remove the regulator line from the syphon end adaptor.
- Remove the NW25 flow constrictor and adaptor from the end of the syphon.
- Fit the syphon end cap.

- **CLOSE** the magnet bypass valve.
- Remove the syphon and fit the blanking plug or continue with helium filling.
 - Use standard filling procedures.

2.1.5 Venting leak check

2.1.5.1 General

For all recondensing systems, it is essential that the magnet system is leak tight to maintain a zero-loss cryogenic performance.

Individual leaks must be 1×10^{-4} mbar l/s or less.

Leak checking must be performed with an electronic leak detector capable of detecting leaks, at least as small as 1×10^{-4} mbar.

The magnet must be checked after any service operations have been performed, such as:

- Installation and Startup.
- After helium filling.
- After replacing a component on the high pressure side of the magnet. (cold head, venting).
- After a magnet quench. (Post quench).



CAUTION

Risk of icing

Small leaks may not show any change in pressure. Over time, they will contribute to air ingress and icing.

- » Leak check with the recommended equipment for small leaks greater than 1×10^{-4}

2.1.5.2 Safety



CAUTION

Pressurized helium vessel.

Risk of cold burns. Failure to observe the following safety information may result in injury.

- » Fittings on the high pressure side must not be disconnected without first depressurizing the helium vessel.



WARNING

Magnet at field.

Personal injury or damage to equipment may occur.

- » Do not use magnetic tools for repairing or tightening seals or screws.
- » Only non-magnetic equipment may be taken into the magnet room.

2.1.5.3 Tooling and Spare Parts

- Heat gun.
- MSUP-Display Unit.
- Paper towelling.
- Standard Service hand tools.
- Leak detection equipment:

Fig. 18: Edwards Leak detector (example)



- Use the Edwards hand-held leak detector (material number **10614850**). This is the recommended equipment for leaks greater than 1×10^{-4}

NOTE: The hand-held device is calibrated in ml/sec:-

1×10^{-4} mbar l/s = 1×10^{-4} ml/sec at normal pressure and temperature.

Refer to the equipment manufacturer's operating instructions.



The function of the Edwards leak detector can be compromised by magnetic fields if placed near the magnet bore.

Spare Parts

Helium leaks from the magnet pressure vessel are generally repaired by replacing a damaged seal or failed component.

- A number of seals are supplied in the spares kit. These must be replaced when one is used.

- Venting components will require ordering from CSML.

2.1.5.4 Venting Configurations

There are several configuration changes to the OR105 and OR122 magnet venting:

- Update to the 16 psi (A) magnet bypass valve assembly and its location.
- Reduction in size to the OVC vent and OVC burst disk.
- Removal of the OVC burst disk (DOP port).
- Pressure gauge updated to WIKA type.

Fig. 19: Mk1 OR105 Venting System



Fig. 20: OR122 Mk1 Venting System

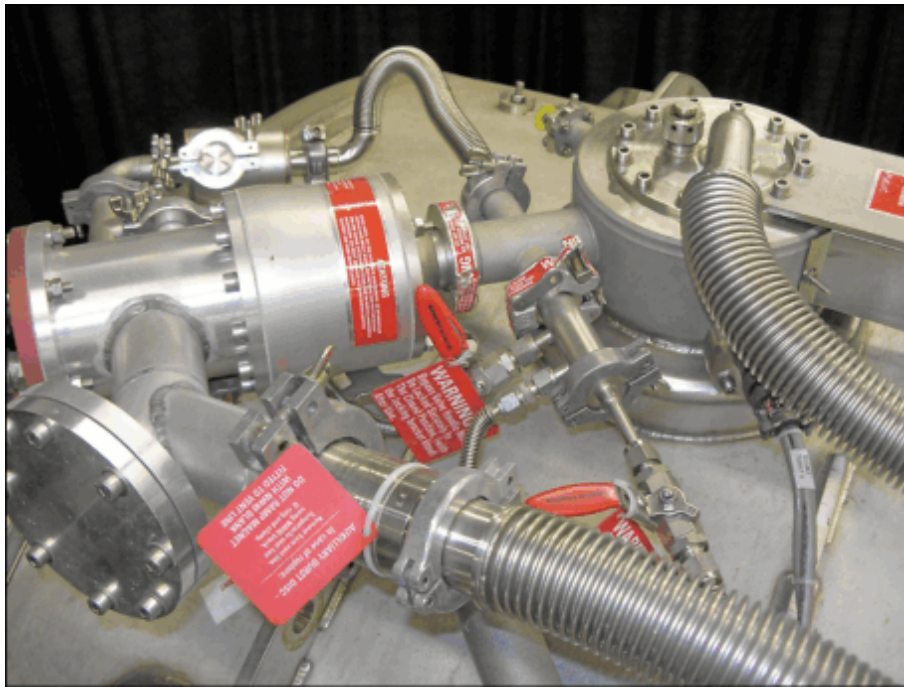


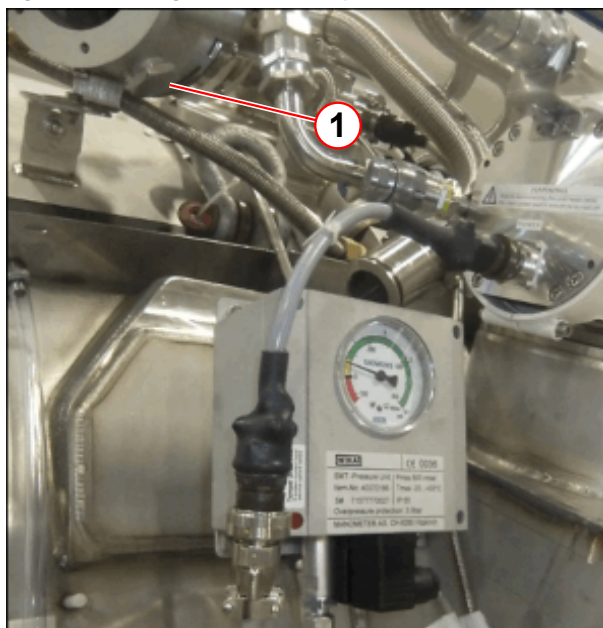
Fig. 21: Mk2 Venting System



Mk2 magnet bypass vent valve assembly new location.

Smaller OVC burst disk vent line and burst disk.

Fig. 22: Venting - No OVC vent port



(1) OVC vent port blank

Venting updated with the removal of the OVC burst disk and vent line to the quench valve.

WIKA pressure gauge fitted.

2.1.5.5 Leak Check Process at Installation and Service

Installation

- Check the system pressure on arrival. If the system has low or zero pressure then a gross leak has occurred during shipment.
- Zero pressure indicates a gross leak. Immediately check the burst disks.
- Check any joints (clamps and seals) that have been broken or damaged during transportation.
- If magnet pressure is still low, check for leaks through the magnet 17 psi (A) vent valve.

Service

- The turret, vent valve assembly and vent lines must be warmed up to room temperature after any service procedures have been performed, prior to leak check.
- Leak check all seals that have been subjected to low temperatures and the subsequent build up of frost. For example, magnets that have been helium filled, depressurized, or the magnet ramped.
 - Check joints broken and re-made during servicing. Replace damaged seals.
 - Check that all clamps and seals are tight. Do not over-tighten them, it may cause a leak, or make an existing leak worse.

Pre-requisites

- The helium vessel must be at normal operating pressure (15.5 psi (A)) prior to leak checking.
- Activate the pressure heater (if required) to raise pressure.
- If pressure heater activation continues for more than 30 min with NO increase in the magnet pressure:-
 - Stop heater activation and check for any obvious gross leaks such as the syphon ring nut or burst disks.
- All joints must be dry and at room temperature before starting a leak check.
 - » Warm all components up using a heat gun and dry. Moisture can cause false readings.

Leak Check Process

- Switch on the leak detector and allow it to warm up (refer to manufacturer's instructions).
- Check the equipment is working correctly (detecting helium). Check equipment before and after use.
 - Set the leak detector to its maximum sensitivity.
 - Check the response time of the detector.
- The background helium level in the magnet room must not be saturated with helium gas.
 - » **DO NOT** leak check immediately after a quench or helium filling.



The background must be at least one order of magnitude lower than the smallest leak expected, (i.e. leaks of 10^{-4} mbar l/s, require background of order 10^{-5} mbar l/s).

- Start at the lowest point.
- Hold the tip of the leak detector 5 mm above the surface of the component. **Keep the tip above water condensation.**
- Slowly move the detector around the seals or clamped joints.
- Hold the detector at a leak point until the reading has stabilized.
- Move the detector away and check that the indication disappears.
- Repeat the process to check that a leak is confirmed.
- Complete leak check of turret and venting.
- Fix all leaks (TSG) and repeat leak check of the component and seal.



Do not tighten any seals when the components are cold and covered in frost. Always warm up with a heat gun and dry with paper towelling.

Leak Check Requirements

The following points must be observed at all times:

- Before dismantling a joint, check that a replacement seal is available.
- Always **CLOSE** the magnet bypass valve as soon as the system is depressurized.
- Always plug open venting with paper towelling to prevent air ingress.
- **Do not** leave 2 joints open at the same time.
- **Do not** leave the system open for longer than is necessary to complete the repair.
- Check that the sealing surfaces are clean, dry, and undamaged.
- **Do not** replace metal seals with rubber or silicon seals.



Do not use any form of vacuum grease on O-rings and seals. Vacuum grease dries over time and may cause a seal to leak.

For all leaks, first try tightening the clamp or the securing screws.

- **Do not** overtighten clamps or screws.
 - » If tightening does not solve the leak, disassemble the joint and check for any damage to the sealing surfaces.
 - » Replace the high load seal.
- **DO NOT** re-use metal seals. Assemble the joint using a new seal. Ensure that the seal is undamaged and clean, and is seated correctly.
- After repair, leak test the entire venting system.

2.1.5.6 High Pressure Turret and Venting Components

The venting and components on the high pressure side of the magnet turret require leak checking to ensure that the re-condensing performance is achieved. This includes all connections too:

- Cold Head
- Turret top plate, including syphon port and venting
- 16 psi (A) Vent valve assembly
- Pressure gauge assembly
- Vent line connections.



If no external leak is found but the system pressure is low, then it is possible that one of the internal valves is leaking. This can be caused by a cracked or broken burst disk, or a broken O-ring.



Before a leak check, the venting components must be warmed to room temperature, and dried with paper towelling.

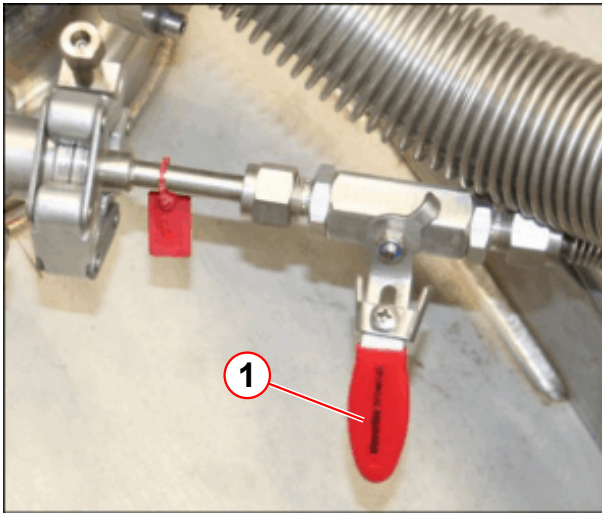
Cold Head

Leak check the cold head components:

- Check the hand-valve and cold head vent line seals at both ends.
- Cold head O-ring seals.
- Temperature sensor connector seals.

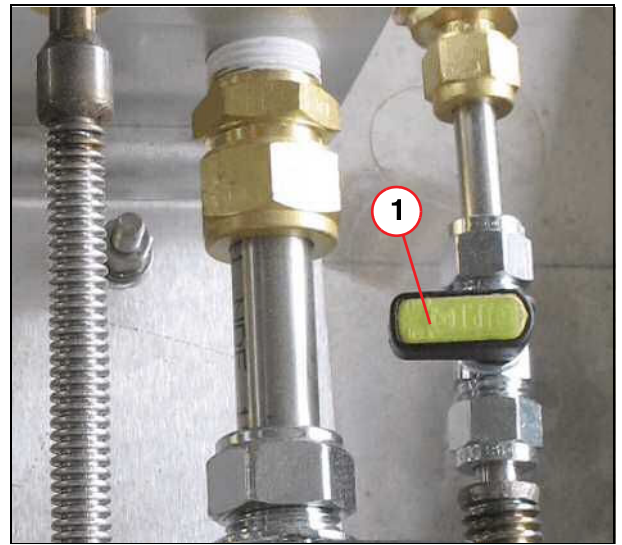
The cold head vent line is to remain connected with the hand-valve CLOSED during normal system operation.

Fig. 23: Mk1 Venting - Cold head vent line valve



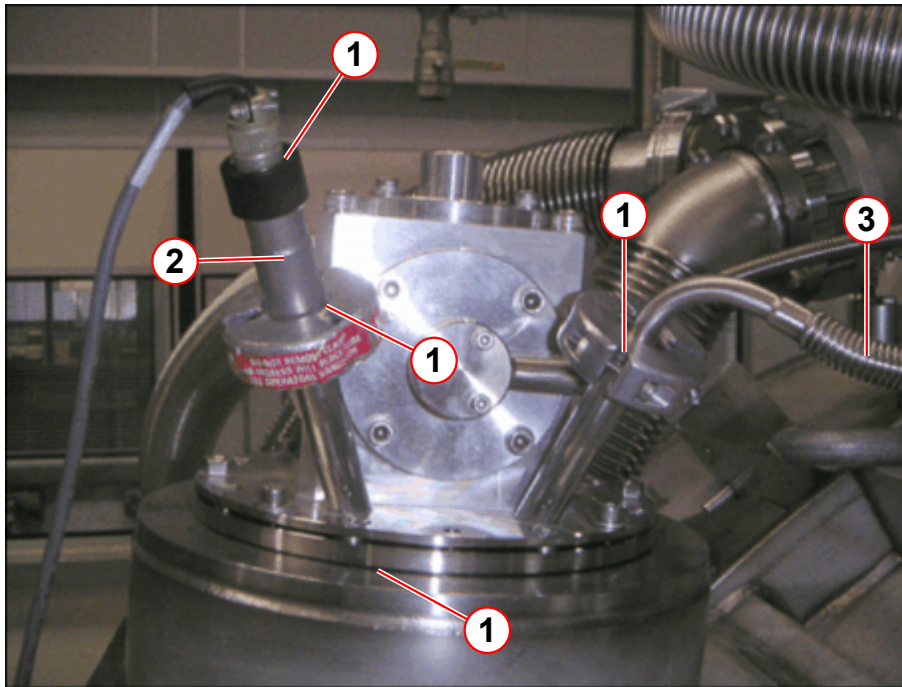
(1) Valve (shown closed)

Fig. 24: Cold head vent line valve is closed



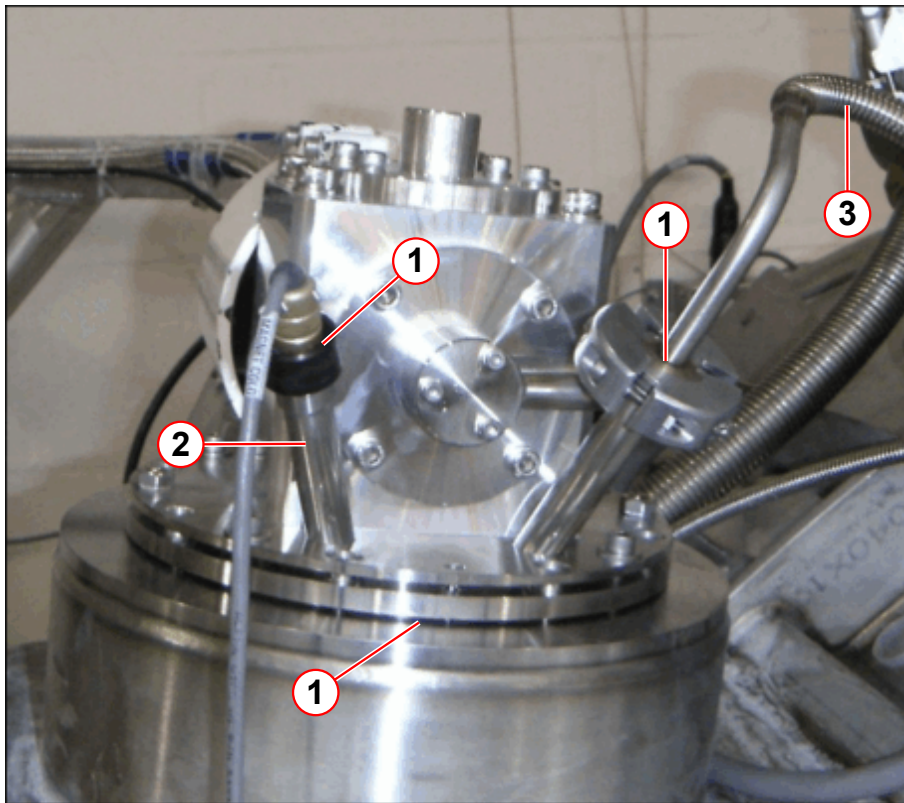
(1) Cold head vent line valve is closed

Fig. 25: Leak checking the cold head seals (early coldhead)



- (1) Cold head seals leak checkpoints
- (2) Temperature sensor feed-through
- (3) Cold head vent line

Fig. 26: Leak checking the cold head seals (later coldhead)



- (1) Cold head seals leak check points
- (2) Temperature sensor feedthrough
- (3) Cold head vent line



If a leak is found with the magnet at field, it is permissible to re-torque the 4 off cold head flange screws to a higher setting of 5 Nm.

If the leak persists, send a report to your local Uptime Service Center for repair by a specialist.

Leak Checking the Temperature Sensors



WARNING

Risk of cold burns. High helium gas pressure.

Risk of personal injury may occur.

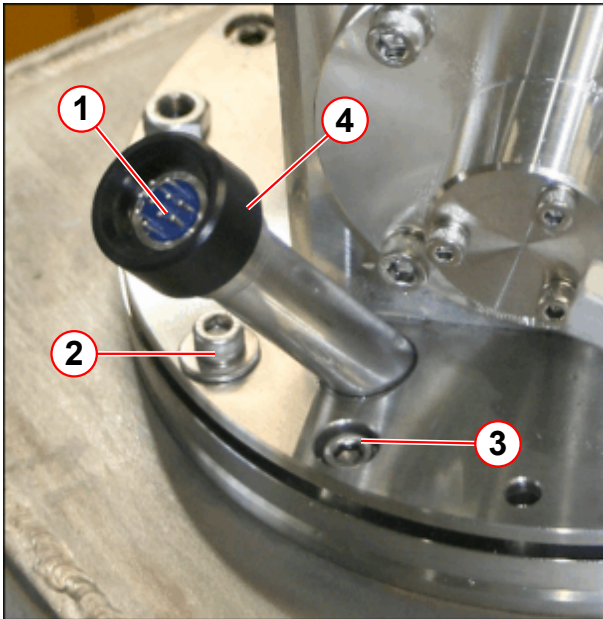
- » Do not unscrew or remove the black temperature sensor sealing cap.
- » Remove the temperature sensor cable with a straight pull away from the sealing cap.



Cold heads fitted from 2015 have a protective heat shrink and a warning label fitted around the temperature sensor connector.

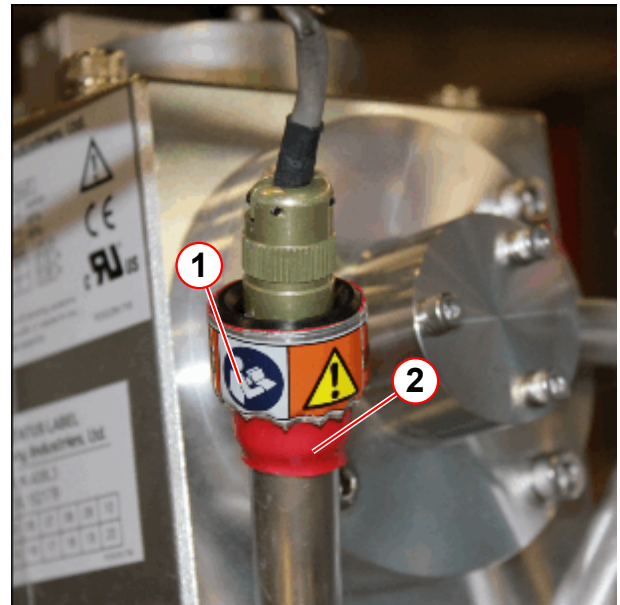
- **DO NOT** turn or try to remove the 'black' sealing cap (pos.4).
- Remove the temperature sensor cable with a straight pull away from the black sealing cap.
- Leak check the 10 pin seal connector,
- Immediately refit the cable if no leak is found.

Fig. 27: Cold head 10 pin seal



- (1) Leak check point
- (2) 4 off M6 x 25 mm cold head flange screws, torque to 4Nm
- (3) 8 off M6 x 16 mm skt hd screws O-ring clamp plate, torque to 8Nm
- (4) Black sealing cap

Fig. 28: Protective heat shrink & warning label



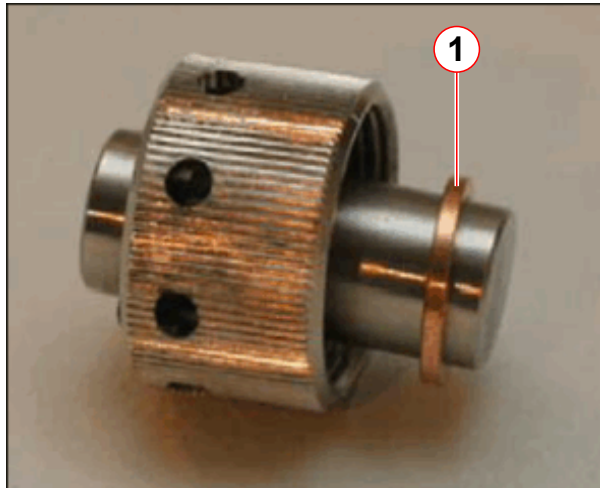
- (1) Warning label
- (2) Heat shrink

Syphon Port

There are several design revisions of the syphon port:

- Location remains on the turret top plate.
- Leak check and corrections are the same for each syphon port example.

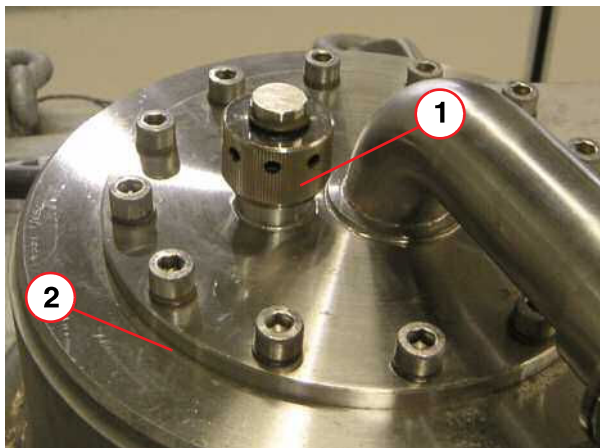
Fig. 29: Syphon blanking plug



(1) Copper seal

- After all helium fills warm up the syphon port. and fit a new, copper seal, (material number **03108094**).
- Tighten the ring nut firmly. Do not over- tighten the syphon plug.

Fig. 30: Mk1 Turret top plate



(1) Syphon entry
(2) Turret top plate seal

- Warm up and dry around the syphon port.
- Tighten the ring nut again. Do not over- tighten the syphon plug.
- Leak check. Repeat warm up of syphon plug and tightening of ring nut.
- If a leak persists, replace the copper seal.

Fig. 31: Syphon Brass Hex plug (example)

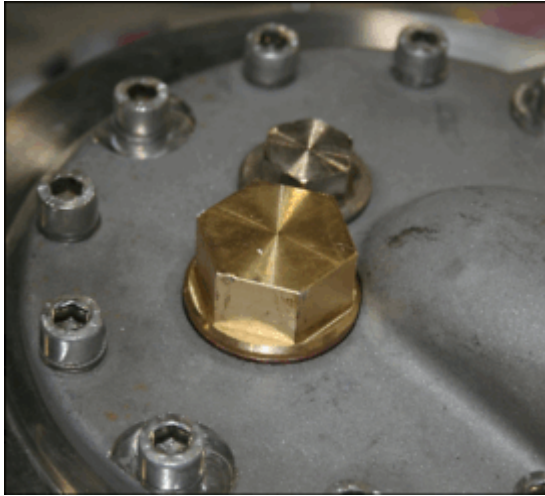


Fig. 32: Syphon plug (current 2017)

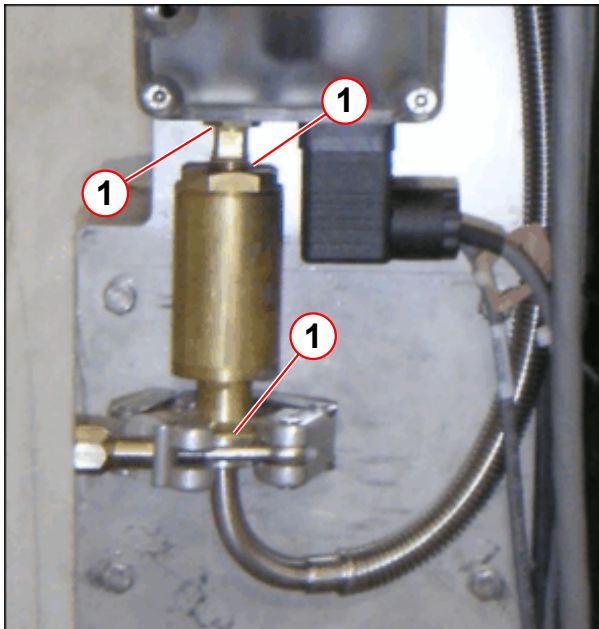


Pressure Gauge Assembly

Leak check the pressure switch transducer assembly. Note: there are different versions depending on system status level.

- Leak check around the fixings of the vent line to the pressure gauge.
- For leaks found at pipe couplings, first try tightening the clamp or securing screws.
BEWARE OF OVER-TIGHTENING!
- If a leak persists the seal must be replaced.

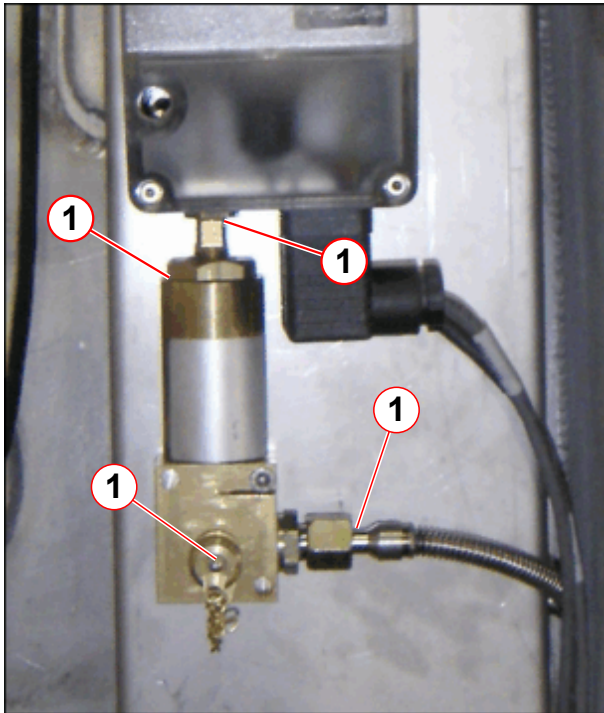
Fig. 33: Mk1 Pressure gauge - leak check



(1) Leak check points

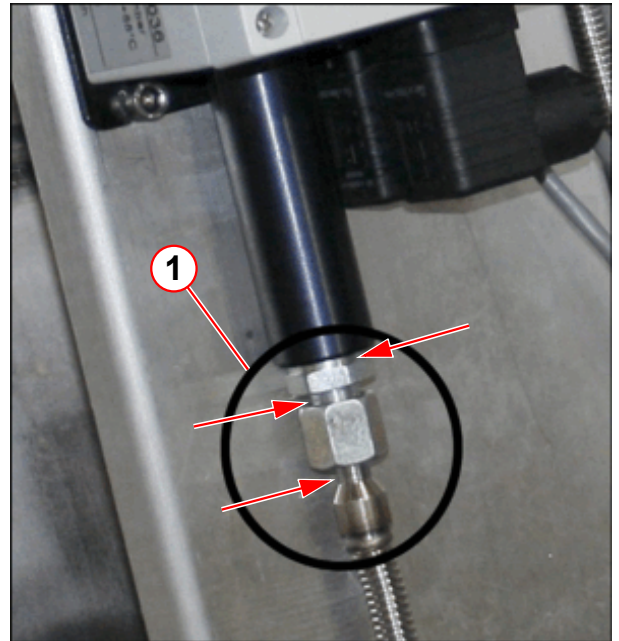
- Leak check around the high load seal and clamp at the pressure gauge.
- OR - leak check around the compression nut and high load seal.

Fig. 34: Pressure gauge - leak check



(1) Leak check points

Fig. 35: Pressure gauge - leak check



(1) Leak check points

FCL Lead Connector

The FCL lead connector passes through the turret in a ceramic interface. It is made leak tight by a high load seal. The FCL ceramic connector and seal should only be checked if no other leaks have been found.

- A broken ceramic seal will be seen as a gross leak.
- Report a damaged ceramic seal to the Magnet HSC. The FCL turret connector and internal wiring may need replacing.

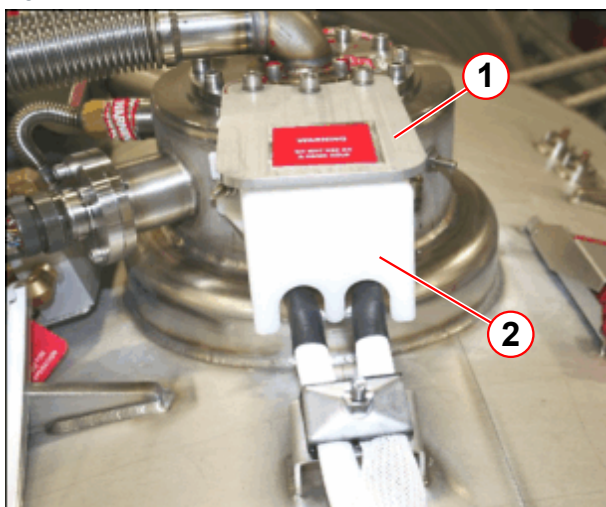
NOTICE

Air ingress

The ceramic seal is a fragile component

- » Take care when removing the FCL cover or working next to the ceramic seal.

Fig. 36: FCL cover and location



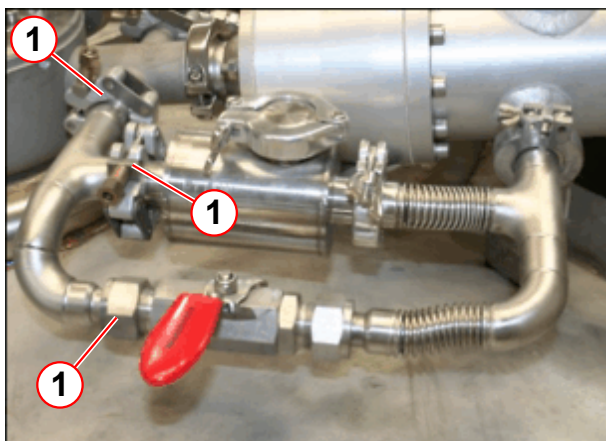
- (1) Protection plate
- (2) FCL cover

The FCL connector is located under the plastic protection cover.

- Only remove the cover if a leak is detected around this area.

16 psi (A) Vent Valve Assembly

Fig. 37: Mk1 Vent valve leak check

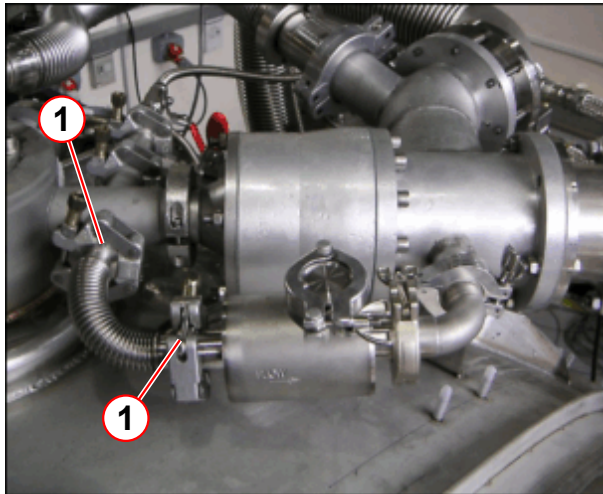


- (1) Leak check points

Mk1 valve and venting

- Check for a leak at the 16 psi (A) valve high pressure fittings.

Fig. 38: Vent valve leak check

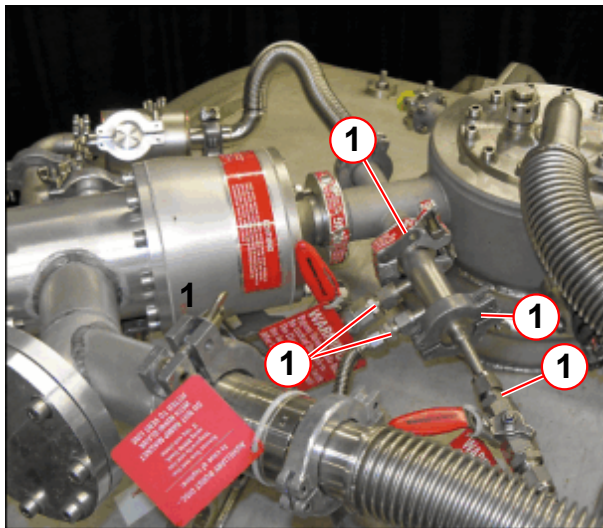


(1) Leak check points

Mk1 OR122 - valve and venting

- Check for a leak at the 16 psi (A) valve high pressure fittings (pos.1).

Fig. 39: Mk 1 Byass valve leak check

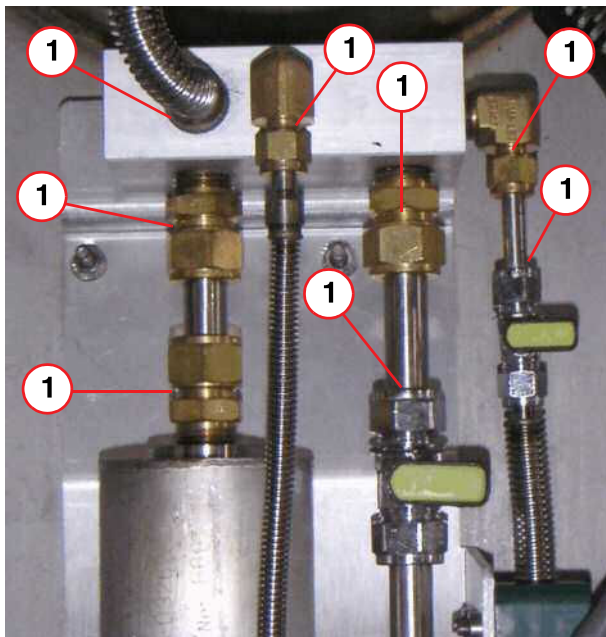


(1) Leak check points

Mk1 OR105 and OR122 - magnet by-pass valve venting.

- Check for a leak at the high pressure fittings (pos.1).

Fig. 40: Mk2 Venting 16 psi (A) Valve Assembly



(1) Leak test points

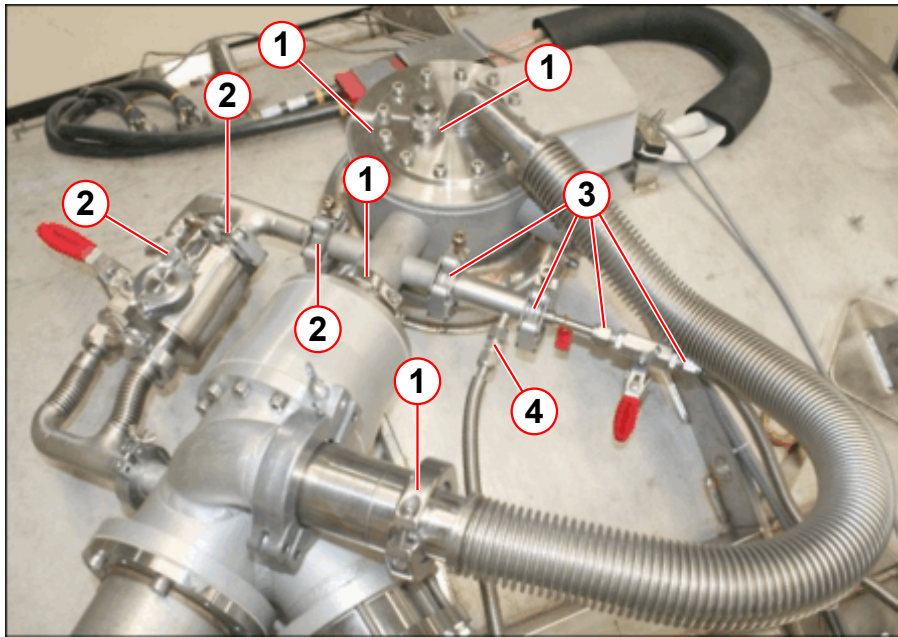
- Leak check the 16 psi (A) valve high pressure assembly.
- For leaks found at pipe couplings, first try tightening the clamp or securing screws. **DO NOT OVERTIGHTEN!**

Turret and Venting

The figures below show the high pressure leak check locations for MK1 and MK2 turret configurations.

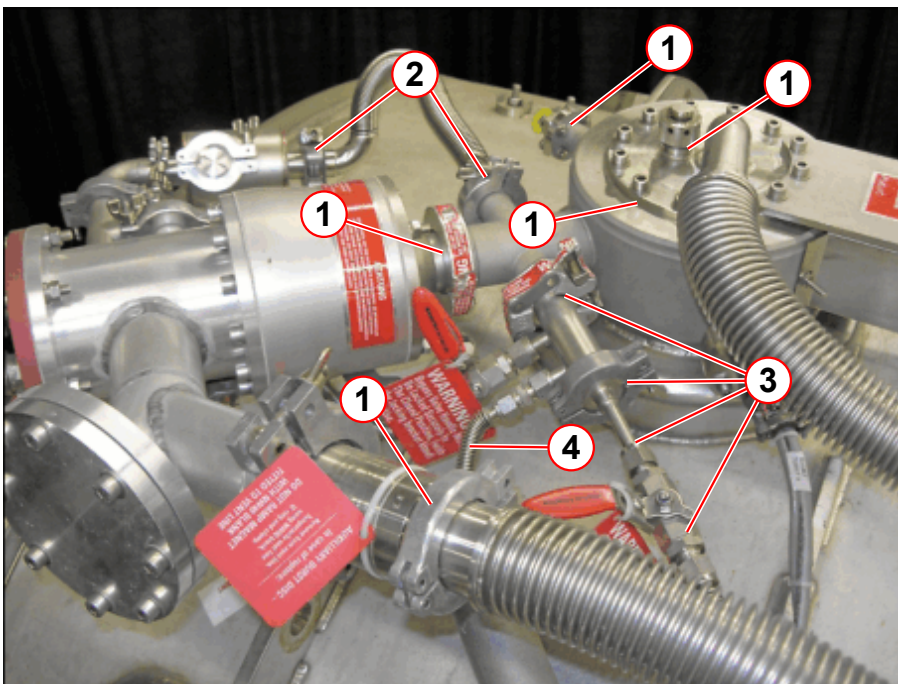
- Turret top plate, syphon port and O-rings (Pos. 1).
- Connections to the 16 psi (A) vent valve assembly (Pos. 2).
- Cold head line (Pos. 3).
- Pressure gauge line (Pos. 4).

Fig. 41: OR105 Mk1 - Turret and venting leak checks (high pressure)



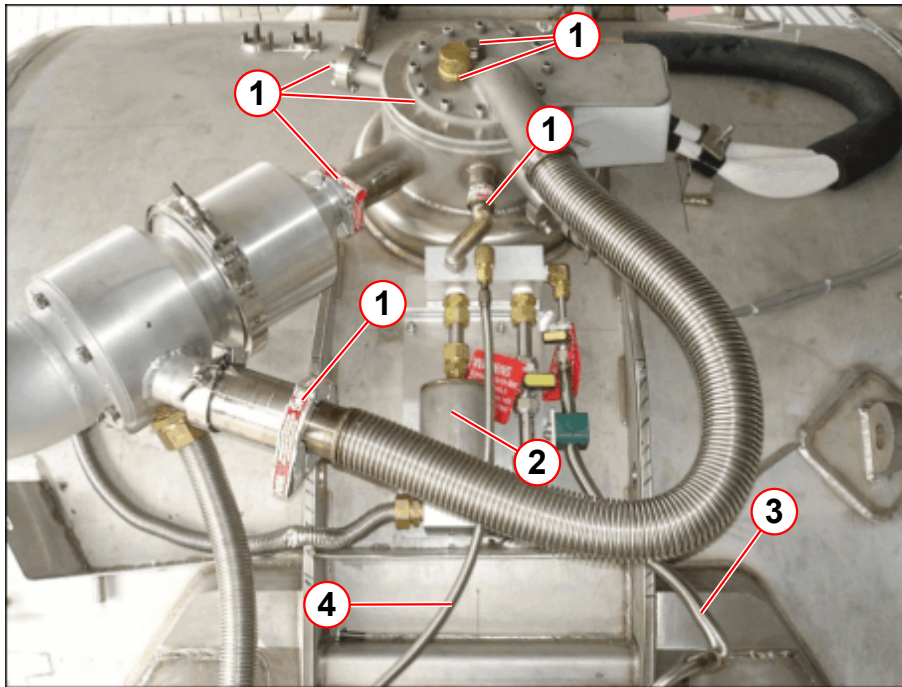
- (1) Turret top plate, syphon port and O-rings
- (2) 16 psi (A) vent valve
- (3) Cold head line
- (4) Pressure gauge line

Fig. 42: OR122 Mk1 - Turret and venting leak checks (high pressure)



- (1) Turret top plate, syphon port and O-rings
- (2) 16 psi (A) vent valve
- (3) Cold head line
- (4) Pressure gauge line

Fig. 43: OR105/122 Mk2 - Turret and venting leak checks (high pressure)



- (1) Turret top plate, syphon port and O-rings
- (2) 16 psi (A) vent valve
- (3) Cold head line
- (4) Pressure gauge line

2.1.5.7 Internal Leak Checks

Leaks can occur internally through the magnet 16 psi (A) vent valve or burst disks:

- 1.8 bar auxiliary metal bursting disk.
- Quench valve and graphite burst disk.
- 16 psi (A) vent valve assembly.

CAUTION

Internal leak checks.

Risk of air ingress causing internal icing of turret.

- » **Replace burst disk or valve immediately if found to be broken or leaking.**



The magnet must be at **ZERO FIELD** before leak checking inside the quench valve or auxiliary vent.

**WARNING****Risk of air ingress**

If a burst disk is broken there is a high risk of air ingress and ice.

- » **IMMEDIATELY** fit the replacement burst disk from the spare parts kit.
- » **Order a replacement burst disk.**

Auxiliary Burst Disk

The auxiliary burst disk should not break during a magnet quench.

- If necessary warm up the burst disk and venting with a heat gun.
 - Activate the pressure heater.
 - Allow the system to pressurize.
- Leak check externally around the 1.8 bar high pressure seal (pos.3).
- If there is an internal leak or ruptured burst disk the magnet pressure will be zero.

To check for any leak path through the auxiliary metal bursting disk:

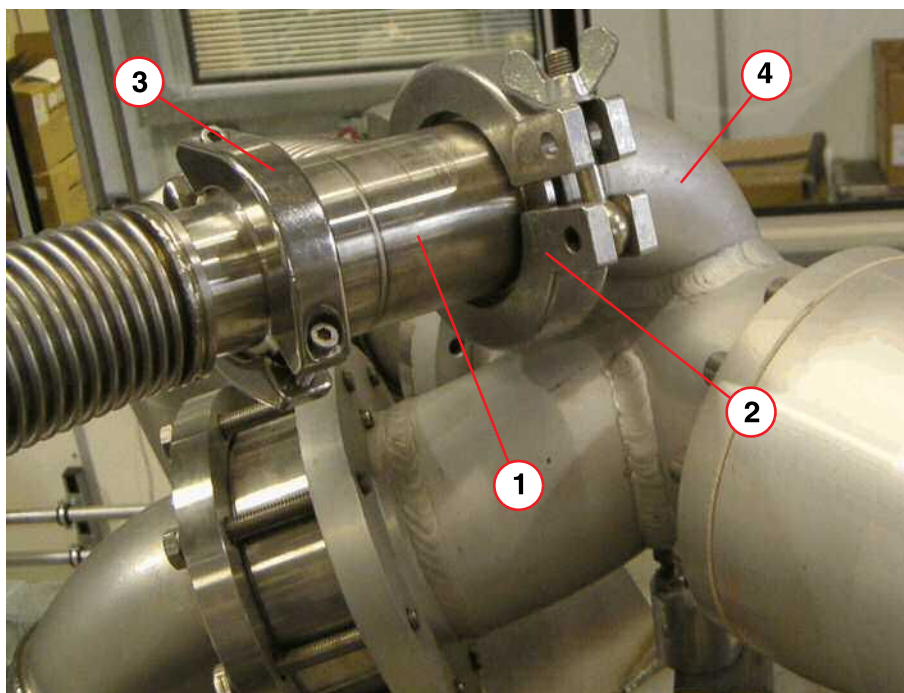
- Undo and remove the NW50 clamp on the low pressure side of the auxiliary burst disk (pos.2).



Do not touch the metal membrane; it is very fragile.

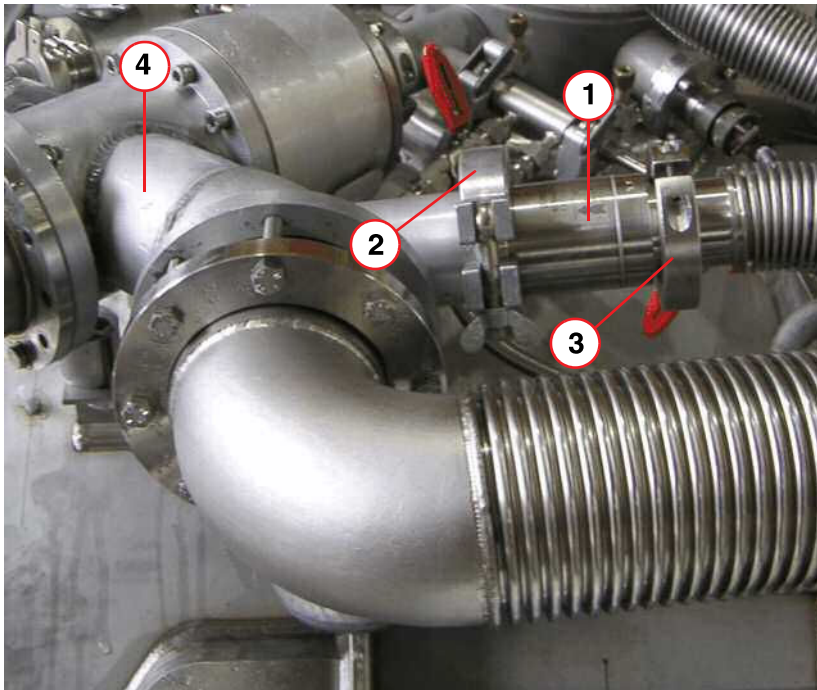
- Visually inspect the internal membrane of the burst disk for damage.
- Visually inspect the O-ring on the low pressure side.
- Replace the auxiliary metal burst disk if a leak is detected.

Fig. 44: OR105 Mk1 Auxiliary vent burst disc assy



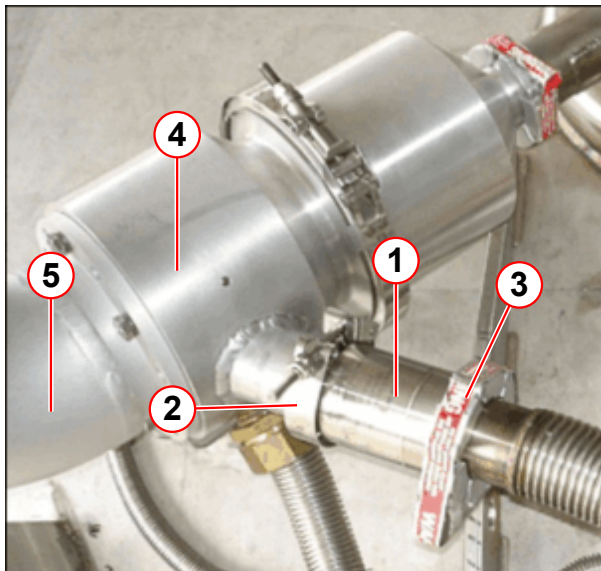
- (1) Aux vent burst disc
- (2) NW50 low load clamp (low pressure)
- (3) NW40 high load clamp (high pressure)
- (4) Low pressure manifold

Fig. 45: OR122 Mk1 Auxilliary vent burst disc assy



- (1) Aux vent Burst Disc
- (2) NW50 low load clamp (low pressure)
- (3) NW40 high load clamp(high pressure)
- (4) Low pressure manifold

Fig. 46: Mk2 Auxiliary burst disk assembly



- (1) Aux burst disk
- (2) V-clamp (low pressure)
- (3) NW40 clamp (high pressure)
- (4) Manifold
- (5) Quench elbow

To check for any leak path through the Mk 2 auxiliary bursting disk:

- Undo and remove the 'V' clamp from the low pressure side of the auxiliary burst disk.
- Visually inspect the internal membrane of the burst disk for damage.
- Visually inspect the O-ring on the low pressure side.

Quench Valve Burst Disk Assembly



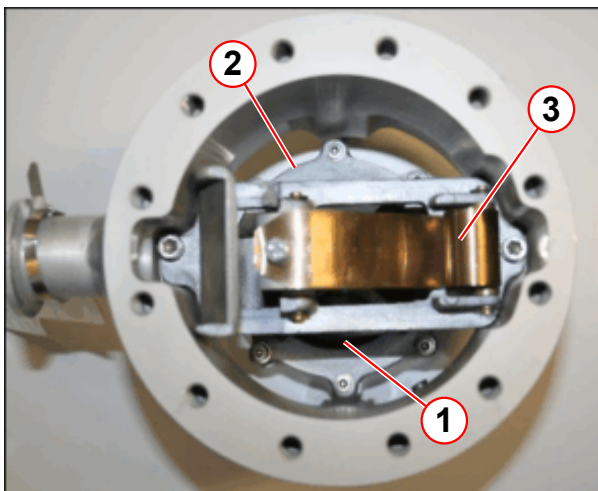
The magnet must be RUN TO ZERO before checking the quench valve.

If the magnet has a low gauge pressure and will not pressurize, it is possible that the quench valve has an internal leak.

The quench valve burst disk and spring plate must be leak checked at installation & Start Up and Post quench.

- Remove the quench valve elbow and manifold assembly from the turret to access the burst disk.

Fig. 47: Mk1 Quench valve assy

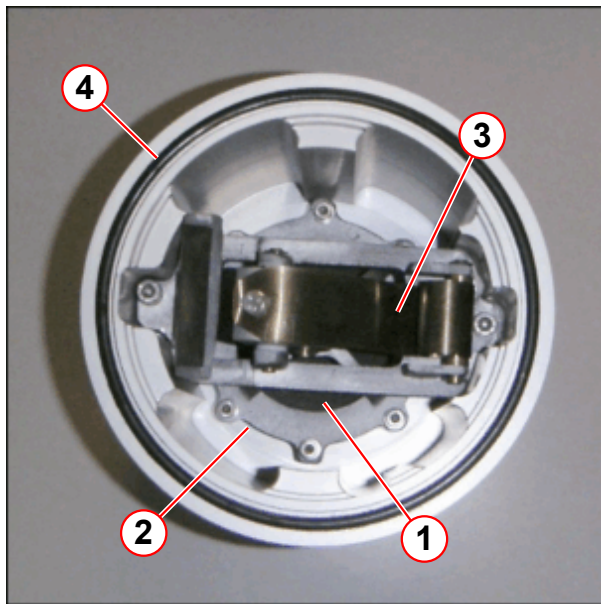


- (1) Graphite burst disk
- (2) O-ring sealing plate
- (3) Spring assy

Perform a leak check of:

- The graphite burst disk.
- Around the O-ring seal to the turret (pos.2).
- Check the condition of the spring and retaining screws.

Fig. 48: Mk2 Quench valve assy

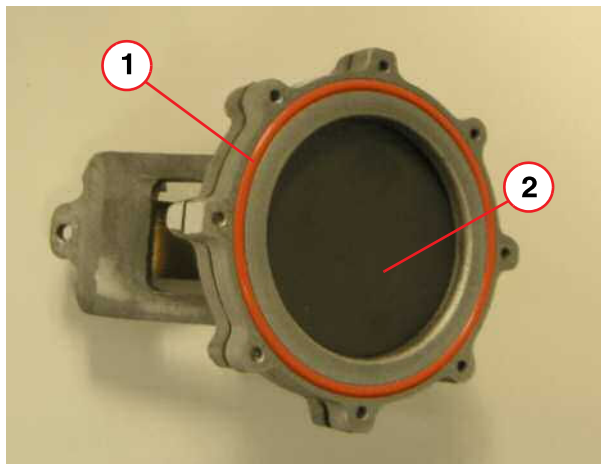


- (1) Graphite burst disk
- (2) O-ring sealing plate
- (3) Spring assy
- (4) O-ring (low pressure)

Perform a leak check of:

- The graphite burst disk.
- Around the O-ring seal to the turret (pos.2).
- Check the condition of the spring and retaining screws.

Fig. 49: Quench Valve Bursting Disk Assembly



- (1) O-ring
- (2) Burst disk

If a leak is seen around the burst disk or spring plate the magnet must be depressurized and the burst disk assembly removed.

- » Details of removing the quench valve is in the Replacement procedures.
- Inspect the burst disk.
 - » If the burst disk has ruptured or is cracked, it must be replaced.
- Inspect the burst disk O-ring.
 - » If the O-ring is split or damaged it must be replaced.

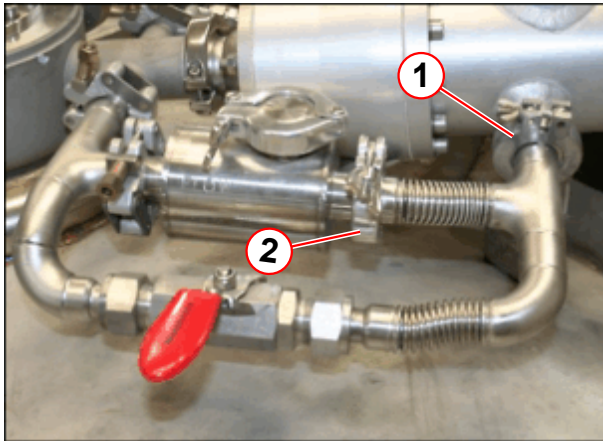


Do not use vacuum grease on the quench valve O-ring.

16 psi (A) Valve Assembly

If the magnet has a low gauge pressure and will not pressurize, it is possible that the vent valve assembly has a gross internal leak.

Fig. 50: Mk1 OR105 venting - internal leak checks

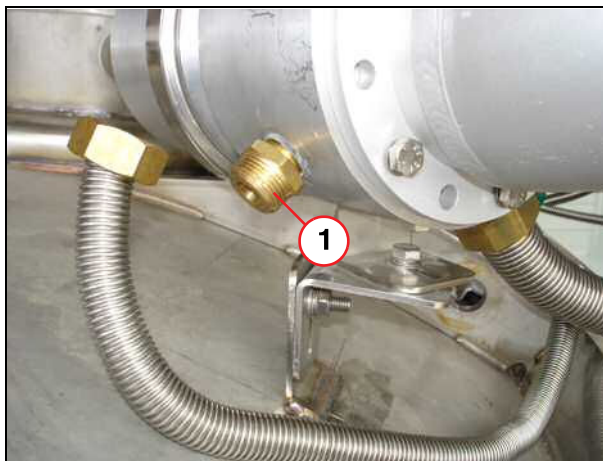


- (1) NW25 Clamp - quench valve port (low pressure)
- (2) NW25 Clamp - 16 psi (A) valve (low pressure)

Mk1 valve and venting

- Check for an internal leak through the 16 psi (A) valve.
- Disconnect the 16 psi (A) flexible vent line from the quench valve port.

Fig. 51: Mk2 16 psi (A) manifold line

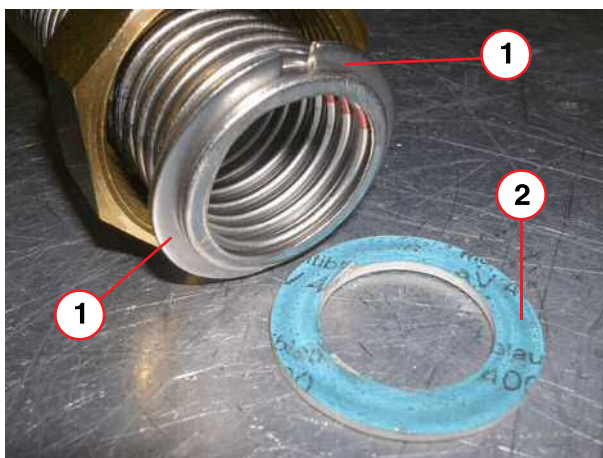


- (1) 16 psi (A) valve outlet

Mk2 valve and venting

- Disconnect the 16 psi (A) flexible vent line from the quench valve manifold.

Fig. 52: Mk2 16 psi (A) vent manifold connection

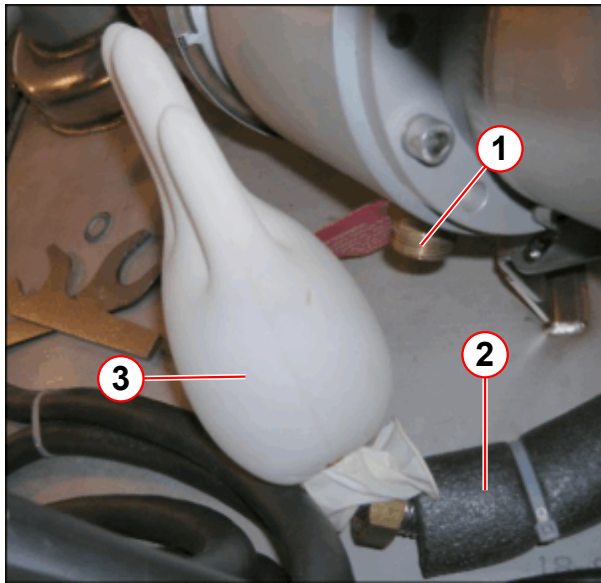


- (1) Split rings
- (2) Gasket

Take care not to lose the split rings during this process!

- Ensure the split rings do not fall out of position.
- Be careful to retain the gasket when disconnecting the vent line.
 - The gasket is not retained by the split rings and nut.

Fig. 53: 16 psi (A) valve - checking for gross internal leaks



- (1) 16 psi (A) quench valve outlet
- (2) 16 psi (A) vent line
- (3) Surgical glove

- Fit a surgical glove (or small airtight plastic bag) over the end of the open low pressure vent.
 - » Seal the glove with a cable tie.
 - » If the glove expands rapidly with the magnet pressure lower than 15.3 psi (A) (0.3 psi (G)), this indicates a gross leak.
 - » If the valve is not sealing, replace the bypass valve assembly.

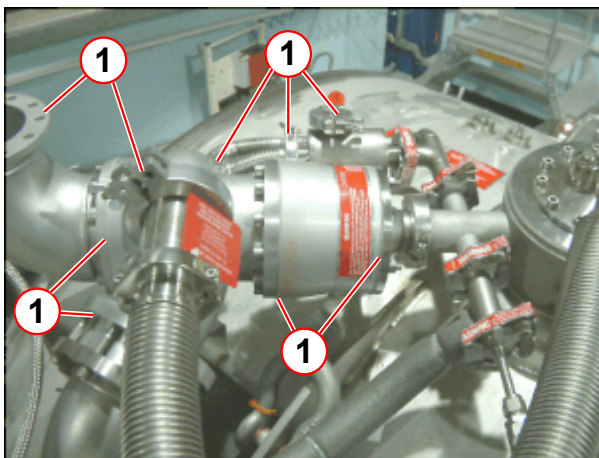
2.1.5.8 Low Pressure Turret and Venting Components



Low pressure venting leaks can only be checked visually during depressurization, helium filling or a magnet quench.

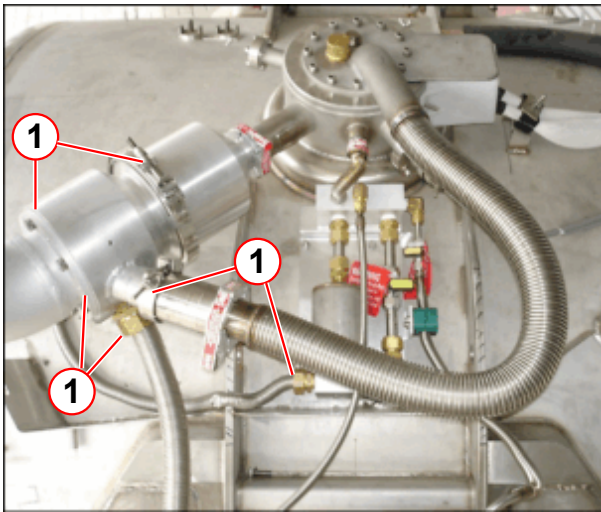
- Check visually for cold gas escaping around low pressure joints during:
 - depressurization, helium filling or a magnet quench.
- Check that all sealing nuts and flange screws on the venting are tight.
 - Warm venting and replace low pressure gasket seals and O-rings if found damaged.

Fig. 54: MK1 - Low pressure seals



- (1) Leak check points

Fig. 55: Low pressure seals (example)



(1) Leak check points

2.1.5.9 Final Work Steps

- After the leak check, ensure that the leak detector is working correctly:
 - » Check the equipment function with a known helium gas source.
- Ensure that all parts of the venting system have been connected and fitted.
- Refit all system covers.
- Order replacement seals if spares have been used.

2.1.6 Repairing leaks on the turret

2.1.6.1 General

For all recondensing systems, it is essential that the magnet system is leak tight to maintain a zero-loss cryogenic performance.

Individual leaks must be 1×10^{-4} mbar l/s or less.

Leak checking must be performed with an electronic leak detector capable of detecting leaks, as small as 1×10^{-4} mbar.



CAUTION

Risk of icing

Small leaks may not show any change in magnet pressure. Over time, they will contribute to air ingress and icing.

- » Leak check with the recommended equipment for small leaks greater than 1×10^{-4}

2.1.6.2 Safety

CAUTION

Pressurized helium vessel.

Risk of cold burns. Failure to observe the following safety information may result in injury.

- » Fittings on the high pressure side must not be disconnected without first depressurizing the helium vessel.

WARNING

Magnet at field.

Personal injury or damage to equipment may occur.

- » Do not use magnetic tools for repairing or tightening seals or screws.
- » Only non-magnetic equipment may be taken into the magnet room.

2.1.6.3 Tooling and Spare Parts

Tooling

- Heat gun.
- Paper towelling.
- Standard Service hand tools.
- Leak detection equipment.

Fig. 56: Edwards Leak detector (example)



- Use the Edwards hand-held leak detector (material number **10614850**). This is the recommended equipment for leaks greater than 1×10^{-4}

NOTE: The hand-held device is calibrated in ml/sec:-

1×10^{-4} mbar l/s = 1×10^{-4} ml/sec at normal pressure and temperature.

Refer to the equipment manufacturer's operating instructions.



The function of the Edwards leak detector can be compromised by magnetic fields if placed near the magnet bore.

Spare Parts

Helium leaks from the magnet pressure vessel are generally repaired by replacing a damaged seal.

- A number of seals are supplied in the spares kit. These must be replaced when used on the magnet.
- Venting components will require ordering from CSML.

2.1.6.4 Leak Check Process at Installation and Service

Installation

- Check the system pressure on arrival. If the system has low or zero pressure then a gross leak has occurred during shipment.
- Zero pressure indicates an gross leak. Immediately check the burst disks.
- Check any joints (clamps and seals) that have been broken or damaged during transportation.
- If magnet pressure is still low, check for leaks through the magnet 17 psi (A) vent valve.

Service

- The turret, vent valve assembly and vent lines must be warmed up to room temperature after any service procedures have been performed, prior to leak check.
- Leak check all seals that have been subjected to low temperatures and the subsequent build up of frost. For example, magnets that have been helium filled, depressurized, or the magnet ramped.
 - Check joints broken and re-made during servicing. Replace damaged seals.
 - Check that all clamps and seals are tight. Do not over-tighten them, it may cause a leak, or make an existing leak worse.

Pre-requisites

- The helium vessel must be at normal operating pressure (15.5 psi (A)) prior to leak checking.
- Activate the pressure heater (if required) to raise pressure.
- If pressure heater activation continues for more than 30 min with NO increase in the magnet pressure:-
 - Stop heater activation and check for any obvious gross leaks such as the syphon ring nut or burst disks.

- All joints must be dry and at room temperature before starting a leak check.
 - » Warm all components up using a heat gun and dry. Moisture can cause false readings.

Leak Check Process

- Switch on the leak detector and allow it to warm up (refer to manufacturer's instructions).
- Check the equipment is working correctly (detecting helium). Check equipment before and after use.
 - Set the leak detector to its maximum sensitivity.
 - Check the response time of the detector.
- The background helium level in the magnet room must not be saturated with helium gas.
 - » **DO NOT** leak check immediately after a quench or helium filling.



The background must be at least one order of magnitude lower than the smallest leak expected, (i.e. leaks of 10^{-4} mbar l/s, require background of order 10^{-5} mbar l/s).

- Start at the lowest point.
- Hold the tip of the leak detector 5 mm above the surface of the component. **Keep the tip above water condensation.**
- Slowly move the detector around the seals or clamped joints.
- Hold the detector at a leak point until the reading has stabilized.
- Move the detector away and check that the indication disappears.
- Repeat the process to check that a leak is confirmed.
- Complete leak check of turret and venting.
- Fix all leaks (TSG) and repeat leak check of the component and seal.



Do not tighten any seals when the components are cold and covered in frost. Always warm up with a heat gun and dry with paper towelling.

2.1.6.5 High Pressure Turret and Venting Components

There is considerable danger of allowing air into the system if the joint is on the high-pressure side.

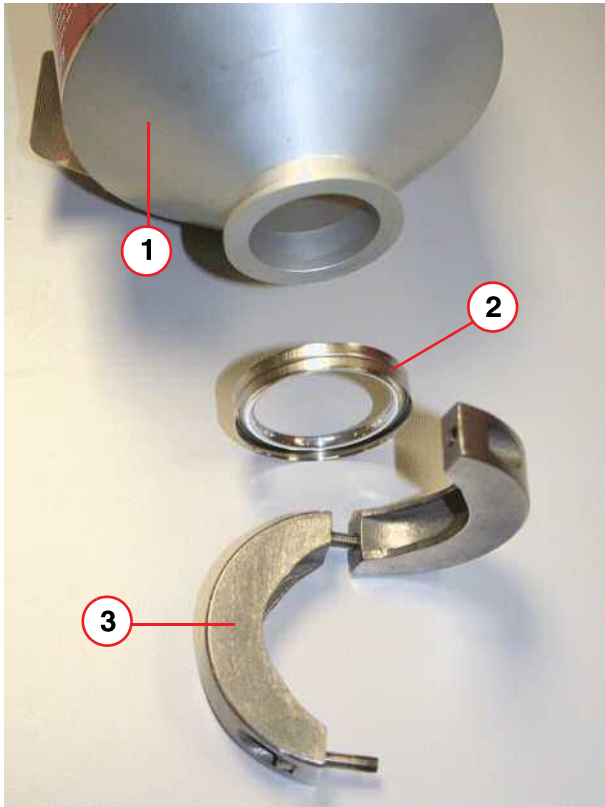
- **Always CLOSE** the magnet bypass valve as soon as the system is depressurized.
- **Always** plug the open venting with paper towelling to prevent air ingress.
- **Do not** have 2 joints open at the same time.
- **Do not** leave the system open for longer than is necessary to complete the repair.
- **Check** that the sealing surfaces are clean, dry and undamaged.

- **Do not** replace metal seals with rubber or silicon seals.



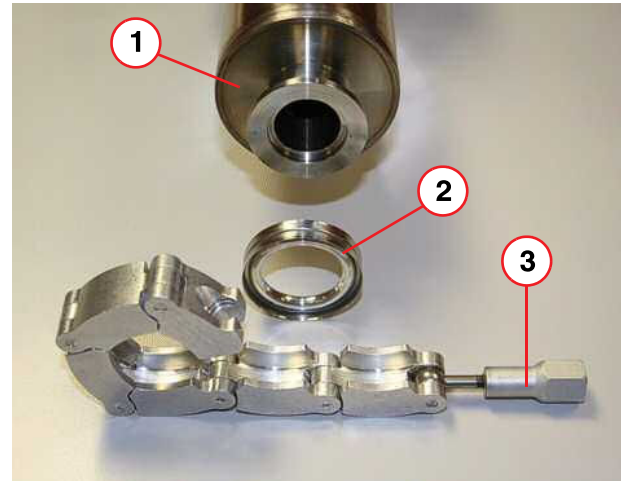
Do not use any form of vacuum grease on O-rings and seals. Vacuum grease dries over time and may cause a seal to leak.

Fig. 57: High Pressure - Quench valve joint



- (1) Quench valve
- (2) Metal seal
- (3) 2 part clamp

Fig. 58: High Pressure joint



- (1) Component
- (2) Metal seal
- (3) Chain clamp



Do not tighten any seals when the components are cold and covered in frost. Always warm up with a heat gun and dry with paper towelling.

For all leaks, first try tightening the clamp or the securing screws.

- **Do not** overtighten clamps or screws.
 - » If tightening does not solve the leak, disassemble the joint and check for any damage to the sealing surfaces.
 - » Replace the high load seal if necessary.
- Assemble the joint using a new seal. Ensure that the seal is undamaged and clean, and is seated correctly.
- After repair, leak test the entire venting system.

Cold Head

**WARNING**

Risk of cold burns. High helium gas pressure.

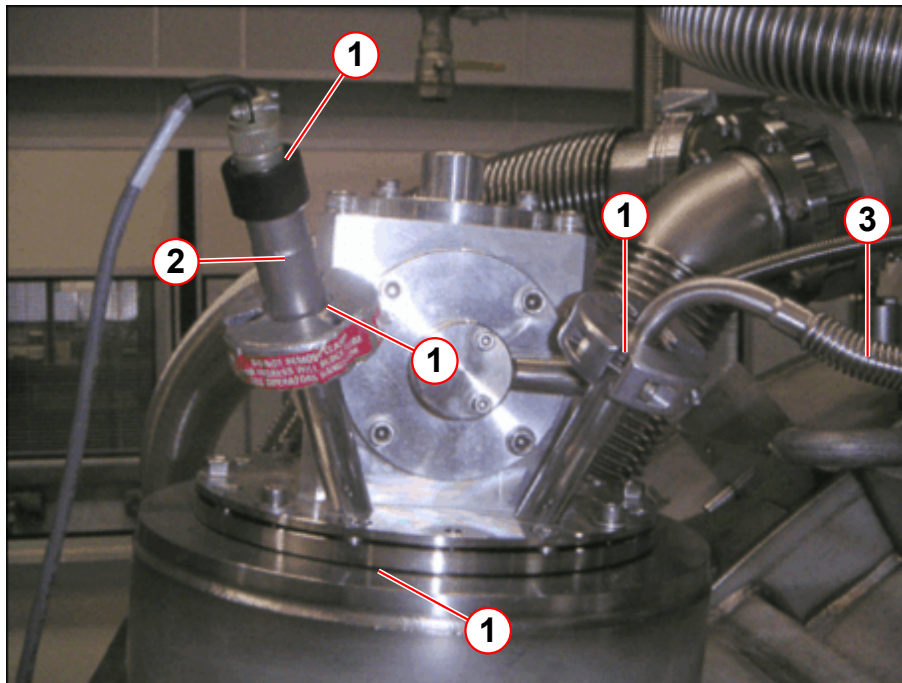
Risk of injury may occur.

- » Do not replace the NW16 clamp and seal without depressurizing the magnet.

Check all cold head connections:

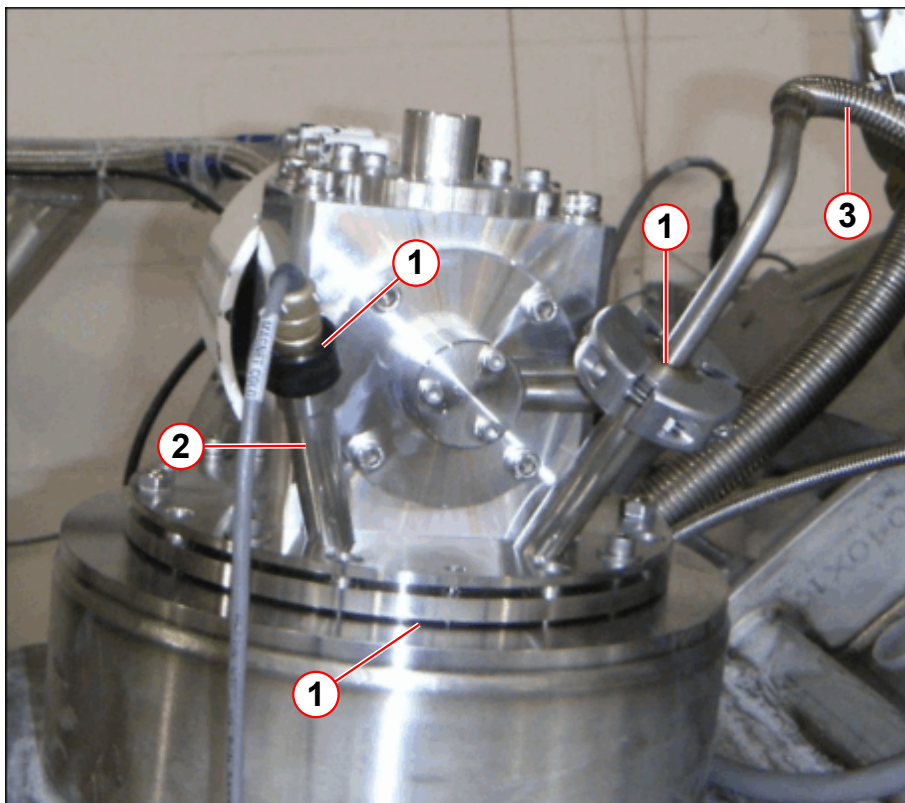
- NW16 clamp and temperature sensor seal.
- All vent pipe connections to the 16 psi (A) valve.
- Cold head O-ring seals (pos.1).

Fig. 59: Leak checking the cold head seals (early coldhead)



- (1) Cold head seals leak checkpoints
- (2) Temperature sensor feed-through
- (3) Cold head vent line

Fig. 60: Leak checking the cold head seals (later coldhead)



- (1) Cold head seals leak check points
- (2) Temperature sensor feedthrough
- (3) Cold head vent line

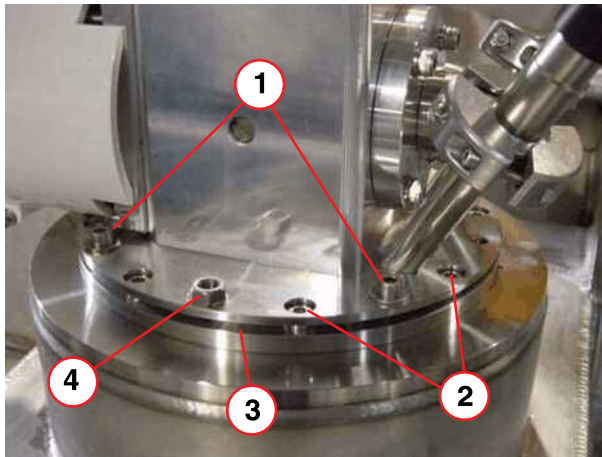
Replacement of the cold head O-ring interface seal requires a complete cold head replacement.



If a leak is found magnet at field, it is permissible to torque the 4 off cold head flange screws to a higher setting of 5 Nm.

If the leak persists, send a report to your local Uptime Service Center for repair by a specialist.

Fig. 61: 4K Cold head screw positions



- (1) M6 x 25mm skt hd screws & washers (4 off equi-distant)
- (2) M6 x 16mm skt hd screws (8 off)
- (3) Gap of 2-4mm
- (4) Cold head release screw position (2 off - parallel)

Check the torque of the cold head screws:

- 4 x M6 clamp ring screws, torque to 8 NM. (pos.1).
- 8 x M6 cold head flange screws torque to 4 NM (pos.2).

Leak Checking the Temperature Sensors



WARNING

Risk of cold burns. High helium gas pressure.

Risk of injury may occur.

- » Do not unscrew or remove the black temperature sensor sealing cap.
- » Remove the temperature sensor cable with a straight pull away from the sealing cap.

The cold heads are fitted with temperature sensors:

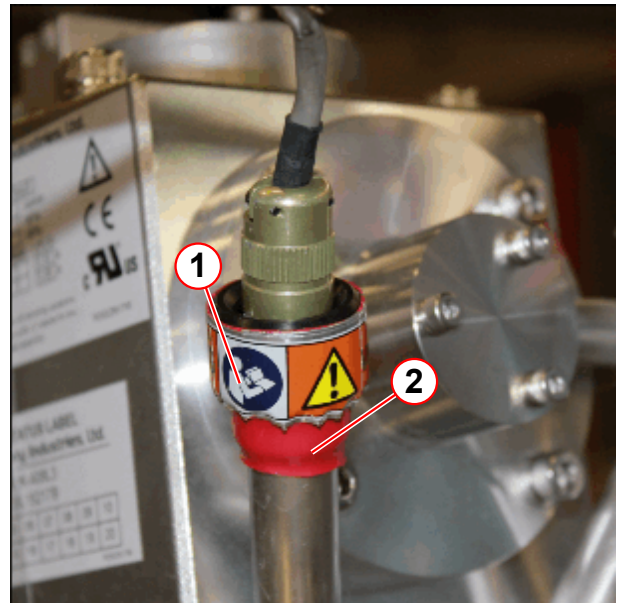
- The feedthrough for the wires is sealed with an O-ring.
- Check the feedthrough and cold head seals.
- **DO NOT** turn or try to remove the 'black' sealing cap (pos.2).
- Remove the temperature sensor cable with a straight pull away from the black sealing cap.
- Leak check the 10 pin seal connector,
- Immediately refit the cable if no leak is found.

Fig. 62: Cold head 10 pin seal



- (1) Leak check point
- (2) Black sealing cap

Fig. 63: Protective heat shrink & warning label



- (1) Warning label
- (2) Heat shrink

16 psi (A) Vent valve

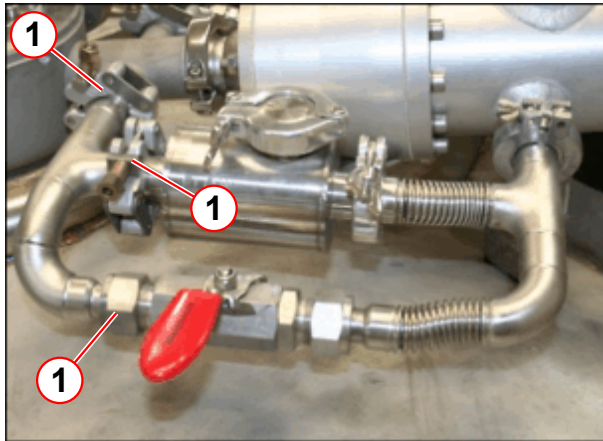
The magnet bypass vent valve has several high pressure vent components attached:

- Flexible vent line from the magnet pressure vessel.
- Cold head bypass vent lines and manifold.
- Magnet bypass vent line.
- Magnet and cold head hand valves (Mk2 valve).

If a leak cannot be fixed by tightening the clamp or compression nut:

- Depressurize the magnet.
- Warm up the components.
- Replace the high load metal seal or compression seal.

Fig. 64: Mk1 Vent valve leak check

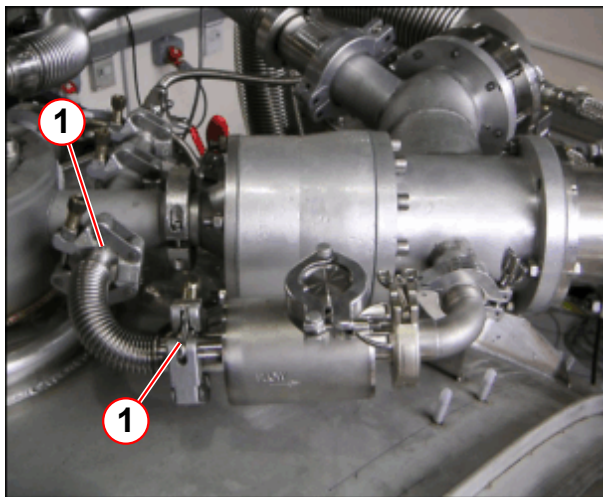


(1) Leak check points

Mk1 OR105 - valve and venting

- Check for a leak at the 16 psi (A) valve high pressure fittings (pos.1).

Fig. 65: Vent valve leak check

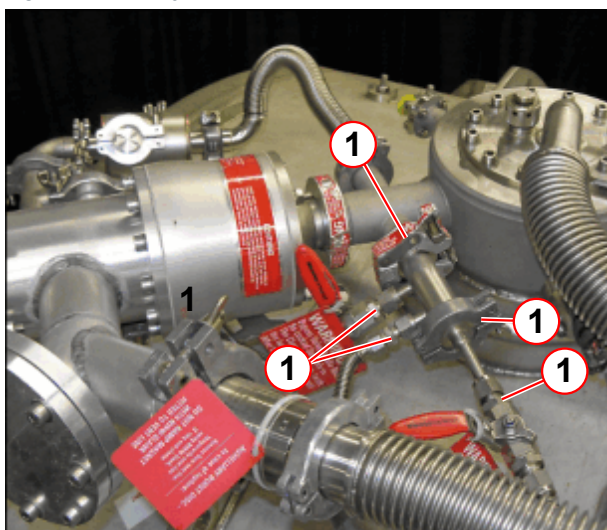


(1) Leak check points

Mk1 OR122 - valve and venting

- Check for a leak at the 16 psi (A) valve high pressure fittings (pos.1).

Fig. 66: Mk 1 Byass valve leak check

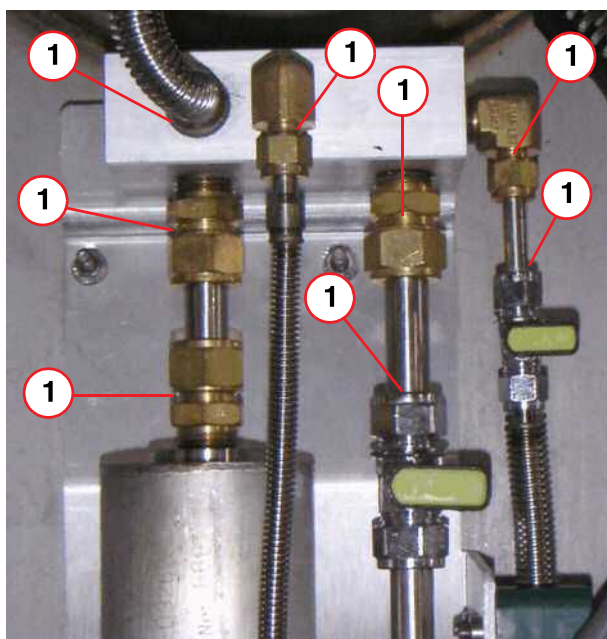


(1) Leak check points

Mk1 OR105 and OR122 - magnet by-pass valve venting.

- Check for a leak at the high pressure fittings (pos.1).

Fig. 67: Mk2 Venting 16 psi (A) Valve Assembly



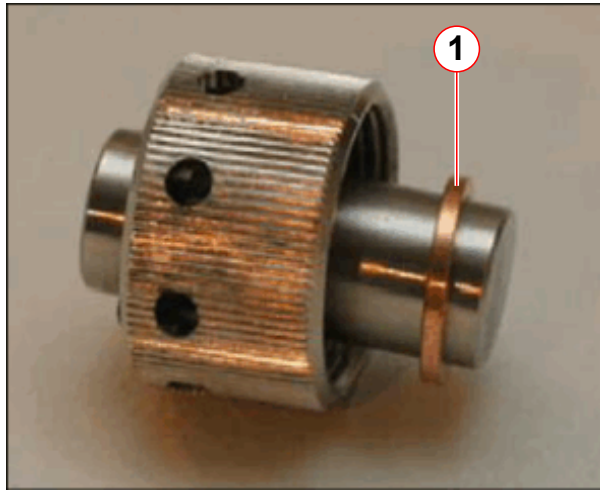
(1) Leak test points

High-pressure compression seals that continue to leak after tightening, must be replaced.

- **Do not overtighten** compression nuts.
- If tightening does not solve the leak, the copper vent lines and compression seals must be replaced.

Syphon Port

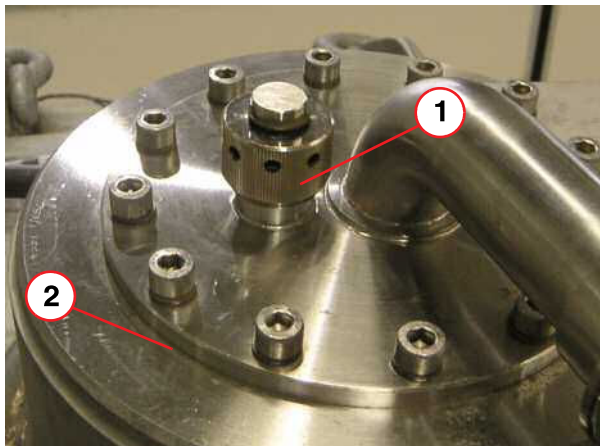
Fig. 68: Syphon blanking plug



(1) Copper seal

- After all helium fills warm up the syphon port. and fit a new, copper seal, (material number **03108094**).
- Tighten the ring nut firmly. Do not over- tighten the syphon plug.

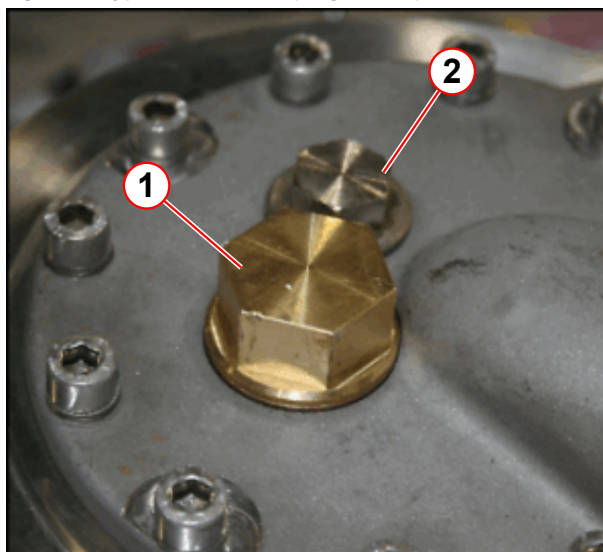
Fig. 69: Mk1 Turret top plate



(1) Syphon entry
(2) Turret top plate seal

- Warm up and dry the around the syphon port.
- Tighten the ring nut again. Do not over- tighten the syphon plug.
- Leak check. Repeat warm up of syphon plug and tightening of ring nut.
- If a leak persists the copper seal will need replacing.

Fig. 70: Syphon Brass Hex plug (example)



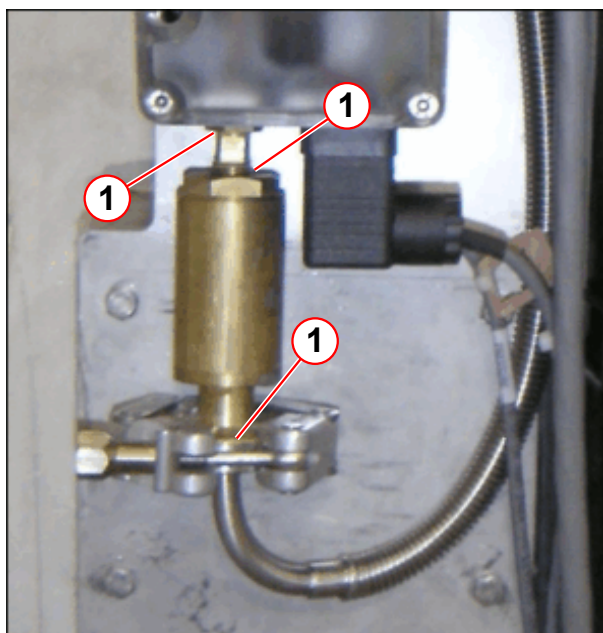
- (1) Syphon port blank
- (2) Sight port blank

Mk2 Syphon port blanking screw.

- Copper syphon port seal, (material number **10103250**).
- Copper sight port seal, (material number **10103162**).

Pressure Gauge

Fig. 71: Mk1 Pressure gauge - leak check



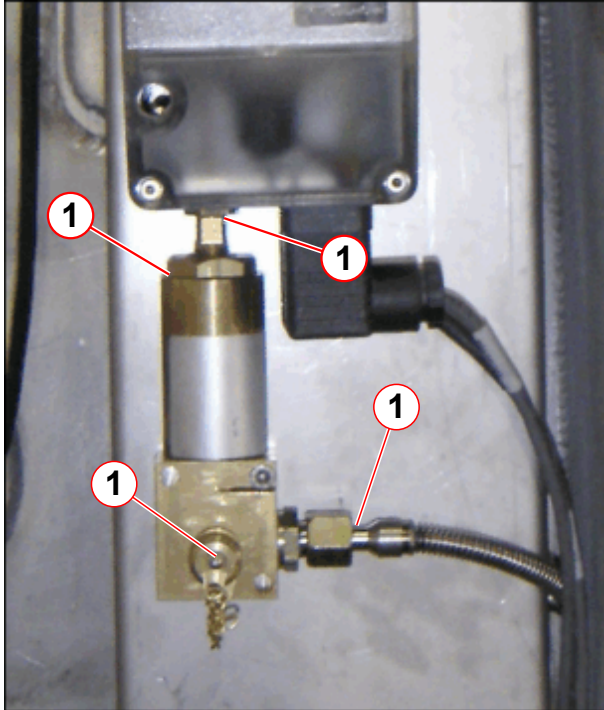
- (1) Leak check points

- **Mk1** - If the leak cannot be closed by carefully tightening the clamp replace the NW25 metal seal.

- Leak check the high pressure seal and compression nut.
- For a leaks found at pipe couplings, tighten the nut again.
 - » Do not overtighten!
- Check for damage to the copper pipe.

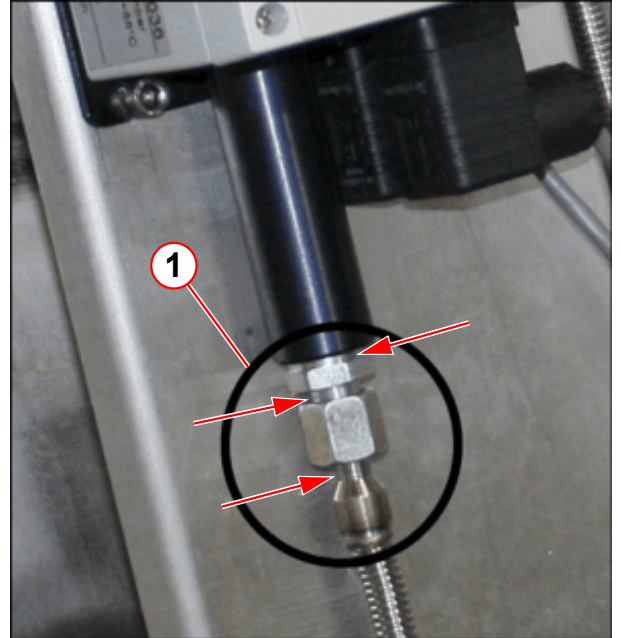
- If the leak cannot be closed by tightening the compression nut:
 - » Replace the pressure line, compression nut and ferrule.
- **Do not overtighten** the compression nut.
- **Do not** use vacuum grease or sealant on the new seal.

Fig. 72: Pressure gauge - leak check



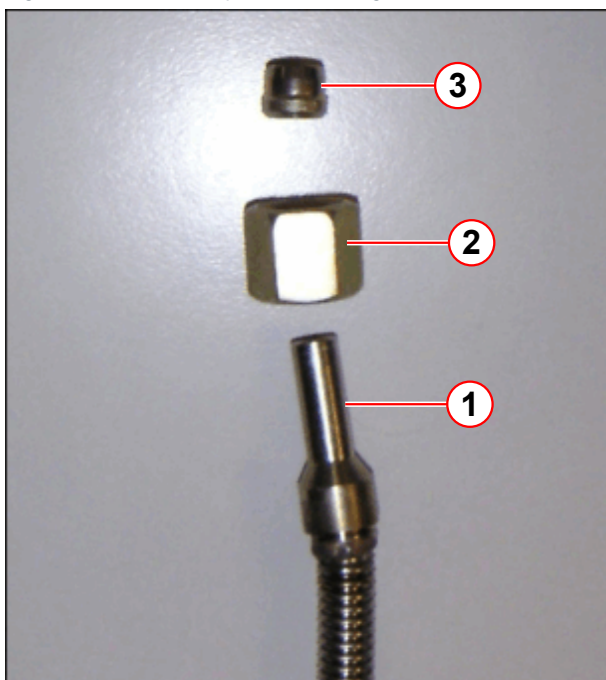
(1) Leak check points

Fig. 73: Pressure gauge - leak check



(1) Leak check points

Fig. 74: WIKA - Compression fittings



- (1) Vent line
- (2) Compression nut
- (3) Compression ferrule seal

Mk2 Compression Joint fittings

- Pressure Line - **10111085**
- Compression nut - **10113284**
- Compression ferrule - **10113289**

FCL Lead Connector

The FCL lead connector passes through the turret in a ceramic interface. It is made leak tight by a high load seal. The FCL ceramic connector and seal should only be checked if no other leaks have been found.

- A broken ceramic seal will be seen as a gross leak.
- Report a damaged ceramic interface to the Magnet HSC. The FCL turret connector and internal wiring may need replacing.

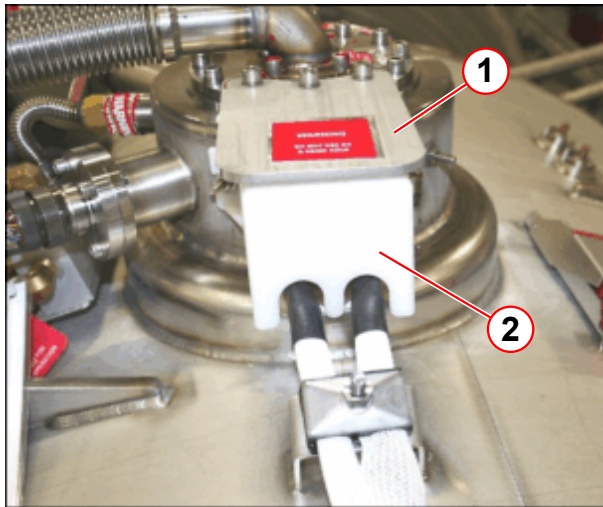
NOTICE

Air ingress

The ceramic seal is a fragile component

- » Take care when removing the FCL cover or working next to the ceramic seal.

Fig. 75: FCL cover and location

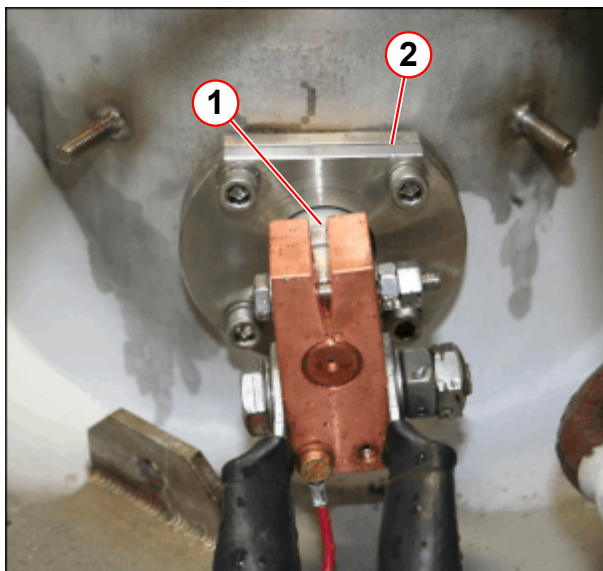


- (1) Protection plate
- (2) FCL cover

The FCL connector is located under the plastic protection cover.

- Only remove the cover if a leak is detected around this area.

Fig. 76: FCL ceramic connector



- (1) FCL Ceramic seal
- (2) Metal seal

Leak check around the FCL connector:

- Metal seal.
- Ceramic seal.
- Refit the plastic FCL cover.
- If a leak is detected on the ceramic seal, contact the Magnet HSC immediately.

2.1.6.6 Internal Leak Checks

Leaks can occur internally through the magnet 16 psi (A) vent valve or burst disks:

- 1.8 bar auxiliary metal bursting disk.
- Quench valve and graphite burst disk.

- 16 psi (A) vent valve assembly.

**CAUTION**

Internal leak checks.

Risk of air ingress causing internal icing of turret.

- » Replace burst disk or valve immediately if found to be broken or leaking.



The magnet must be at **ZERO FIELD** before leak checking inside the quench valve or auxiliary vent.

**WARNING**

Risk of air ingress

If a burst disk is broken there is a high risk of air ingress and ice.

- » IMMEDIATELY fit the replacement burst disk from the spare parts kit.
- » Order a replacement burst disk.

Auxiliary Burst Disk

The auxiliary burst disk should not break during a magnet quench.

- If necessary warm up the burst disk and venting with a heat gun.
 - Activate the pressure heater.
 - Allow the system to pressurize.
- Leak check externally around the 1.8 bar high pressure seal (pos.3).
- If there is an internal leak or breakage the magnet pressure will be zero.

To check for any leak path through the auxiliary metal bursting disk:

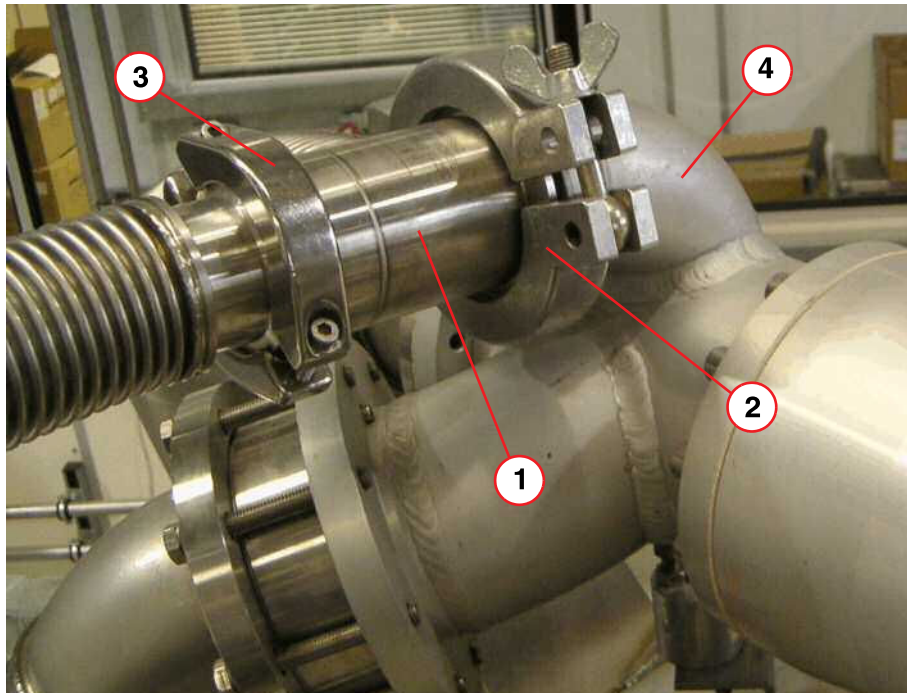
- Undo and remove the NW50 clamp on the low pressure side of the auxiliary burst disk (pos.2).



Do not touch the metal membrane; it is very fragile.

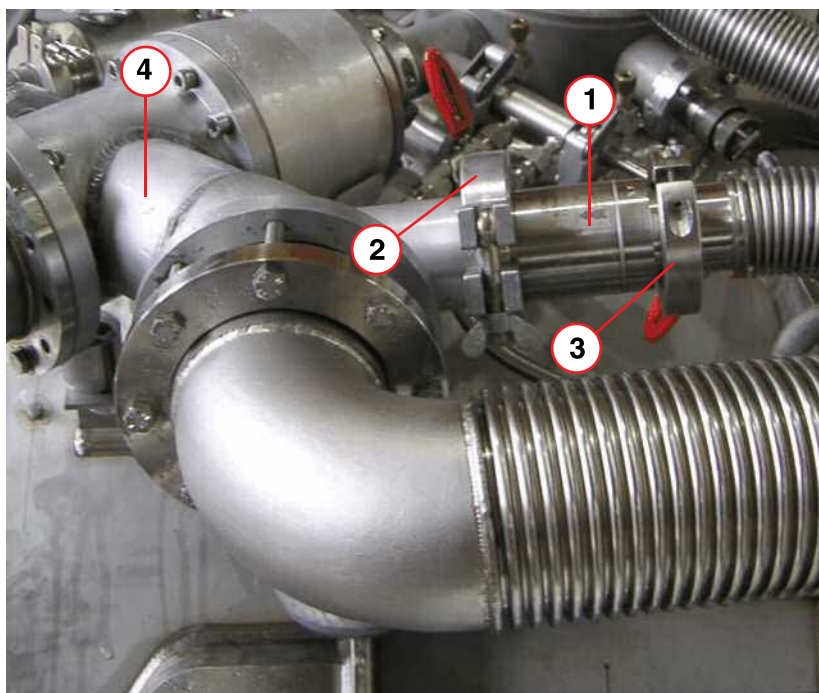
- Visually inspect the internal membrane of the burst disk for damage.
- Visually inspect the O-ring on the low pressure side.
- Replace the auxiliary metal burst disk if a leak is detected.

Fig. 77: OR105 Mk1 Auxiliary vent burst disc assy



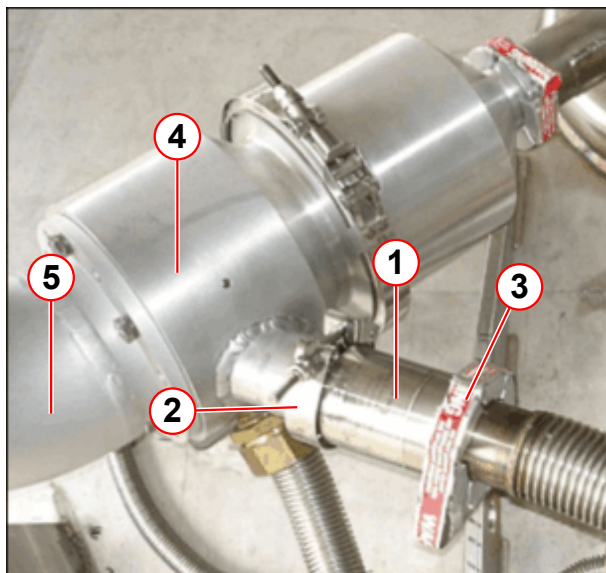
- (1) Aux vent burst disc
- (2) NW50 low load clamp (low pressure)
- (3) NW40 high load clamp (high pressure)
- (4) Low pressure manifold

Fig. 78: OR122 Mk1 Auxilliary vent burst disc assy



- (1) Aux vent Burst Disc
- (2) NW50 low load clamp (low pressure)
- (3) NW40 high load clamp (high pressure)
- (4) Low pressure manifold

Fig. 79: Mk2 Auxiliary burst disk assembly



- (1) Aux burst disk
- (2) V-clamp (low pressure)
- (3) NW40 clamp (high pressure)
- (4) Manifold
- (5) Quench elbow

To check for any leak path through the Mk 2 auxiliary bursting disk:

- Undo and remove the 'V' clamp from the low pressure side of the aux burst disk (Pos.2).
- Visually inspect the internal membrane of the burst disk for damage.
- Visually inspect the O-ring on the low pressure side.
- Replace the auxiliary metal burst disk or high load seal if a leak is detected.

Quench Valve Burst Disk Assembly



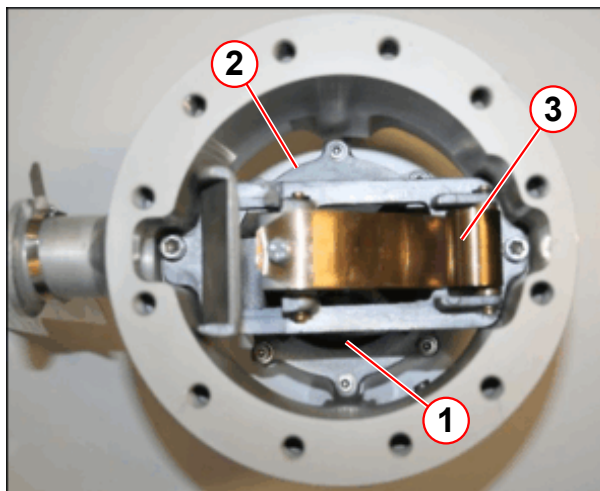
The magnet must be RUN TO ZERO before checking the quench valve.

If the magnet has a low gauge pressure and will not pressurize, it is possible that the quench valve has an internal leak.

The quench valve burst disk and spring plate must be leak checked at installation & Start Up and Post quench.

- Remove the quench valve elbow and manifold assembly from the turret to access the burst disk.

Fig. 80: Mk1 Quench valve assy

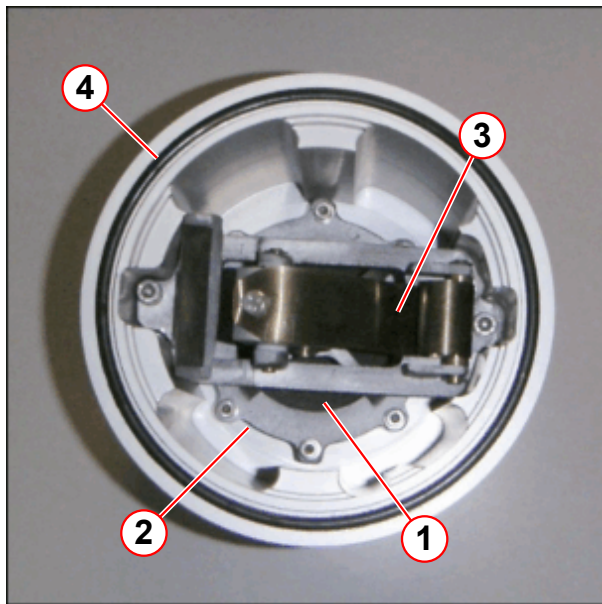


- (1) Graphite burst disk
- (2) O-ring sealing plate
- (3) Spring assy

Perform a leak check of:

- The graphite burst disk.
- Around the O-ring seal to the turret (pos.2).
- Check the condition of the spring and retaining screws.

Fig. 81: Mk2 Quench valve assy

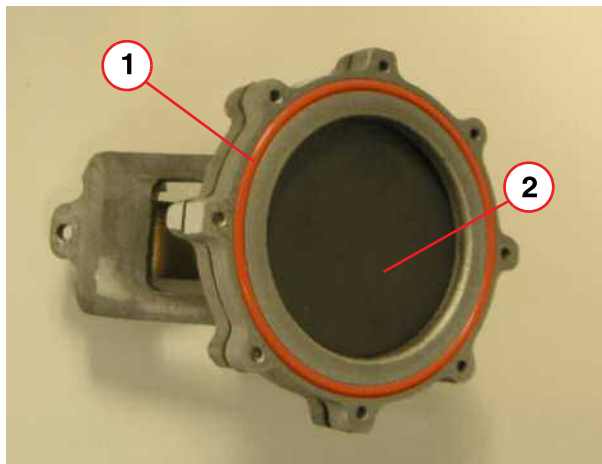


- (1) Graphite burst disk
- (2) O-ring sealing plate
- (3) Spring assy
- (4) O-ring (low pressure)

Perform a leak check of:

- The graphite burst disk.
- Around the O-ring seal to the turret (pos.2).
- Check the condition of the spring and retaining screws.

Fig. 82: Quench Valve Bursting Disk Assembly



- (1) O-ring
- (2) Burst disk

If a leak is seen around the burst disk or spring plate the magnet must be depressurized and the burst disk assembly removed.

- » Details of removing the quench valve is in the Replacement procedures.
- Inspect the burst disk.
 - » If the burst disk has ruptured or is cracked, it must be replaced.
- Inspect the burst disk O-ring.
 - » If the O-ring is split or damaged it must be replaced.

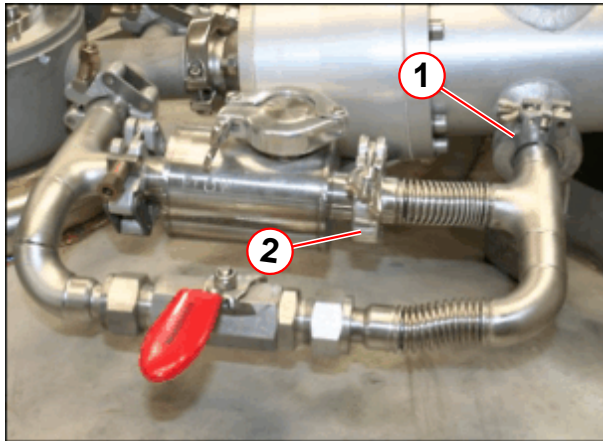


Do not use vacuum grease on the quench valve O-ring.

16 psi (A) Valve Assembly

If the magnet has a low gauge pressure and will not pressurize, it is possible that the vent valve assembly has a gross internal leak.

Fig. 83: Mk1 OR105 venting - internal leak checks

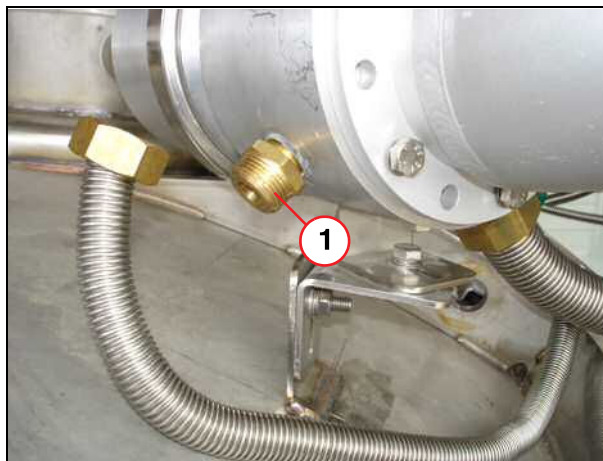


- (1) NW25 Clamp - quench valve port (low pressure)
 (2) NW25 Clamp - 16 psi (A) valve (low pressure)

Mk1 valve and venting

- Check for an internal leak through the 16 psi (A) valve.
- Disconnect the 16 psi (A) flexible vent line from the quench valve port.

Fig. 84: Mk2 16 psi (A) manifold line

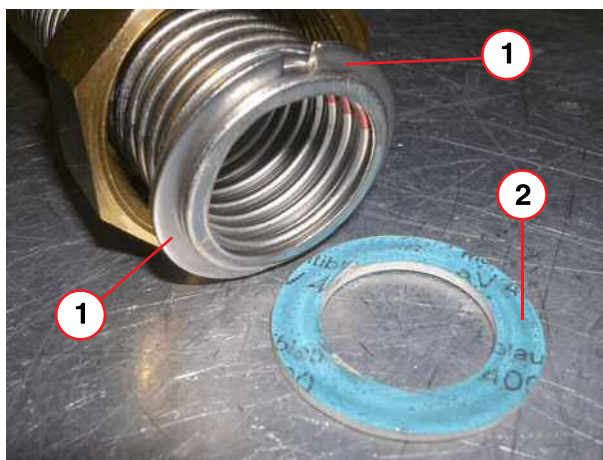


- (1) 16 psi (A) valve outlet

Mk2 valve and venting

- Disconnect the 16 psi (A) flexible vent line from the quench valve manifold.

Fig. 85: Mk2 16 psi (A) vent manifold connection

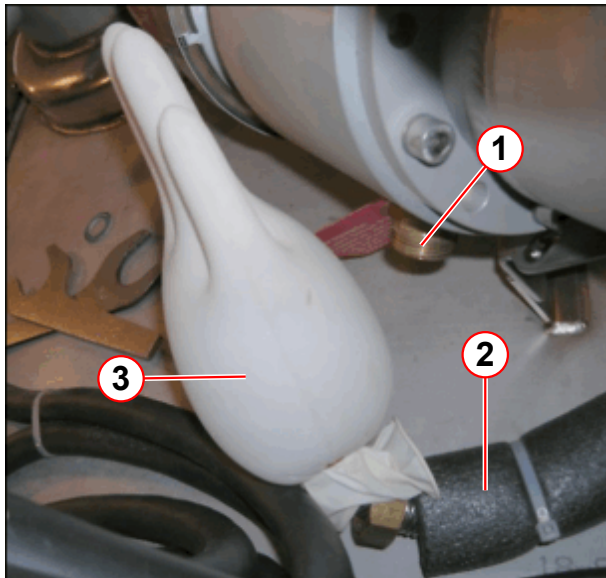


- (1) Split rings
 (2) Gasket

Take care not to lose the split rings during this process!

- Ensure the split rings do not fall out of position.
- Be careful to retain the gasket when disconnecting the vent line.
 - The gasket is not retained by the split rings and nut.

Fig. 86: 16 psi (A) valve - checking for gross internal leaks



- (1) 16 psi (A) quench valve outlet
- (2) 16 psi (A) vent line
- (3) Surgical glove

- Fit a surgical glove (or small airtight plastic bag) over the end of the open low pressure vent.
 - » Seal the glove with a cable tie.
 - » If the glove expands rapidly with the magnet pressure lower than 15.3 psi (A) (0.3 psi (G), this indicates a gross leak.
 - » If the valve is not sealing, replace the bypass valve assembly.

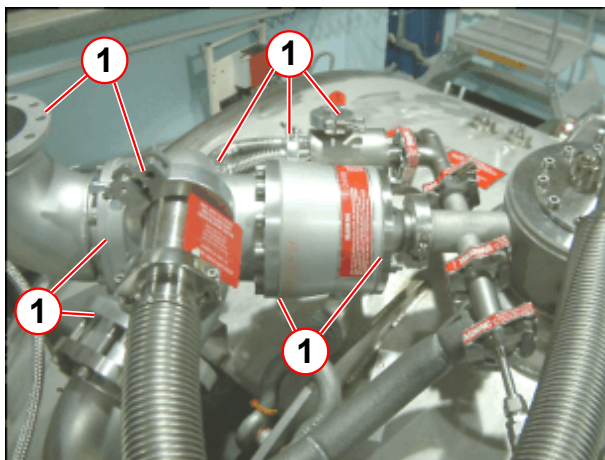
2.1.6.7 Low Pressure Turret and Venting Components



Low pressure venting leaks can only be checked visually during depressurization, helium filling or a magnet quench.

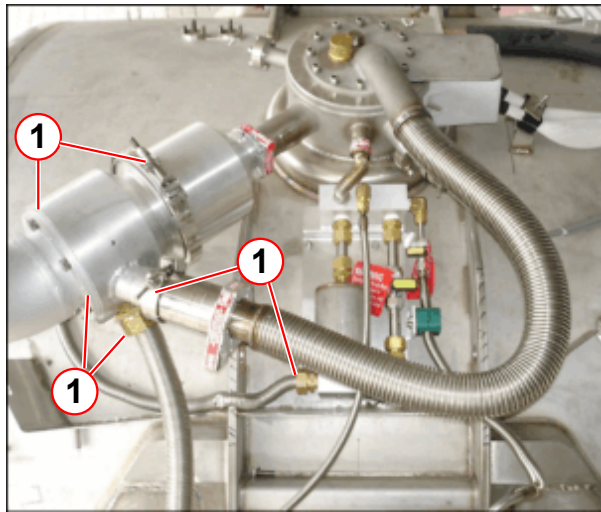
- Check visually for cold gas escaping around low pressure joints during:
 - depressurization, helium filling or a magnet quench.
- Check that all sealing nuts and flange screws on the venting are tight.
 - Warm venting and replace low pressure gasket seals and O-rings if found damaged.

Fig. 87: MK1 - Low pressure seals



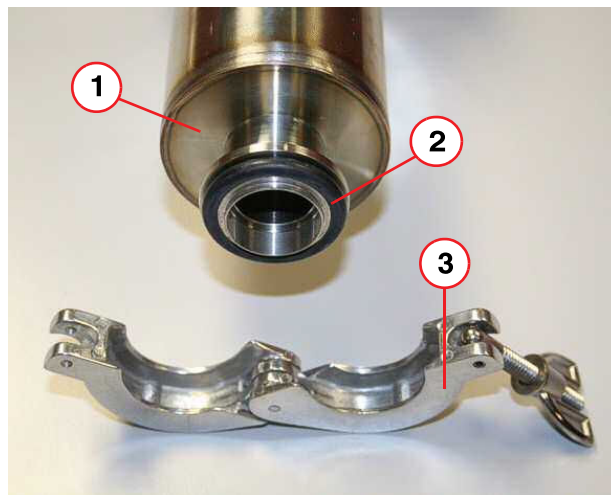
- (1) Leak check points

Fig. 88: Low pressure seals (example)



(1) Leak check points

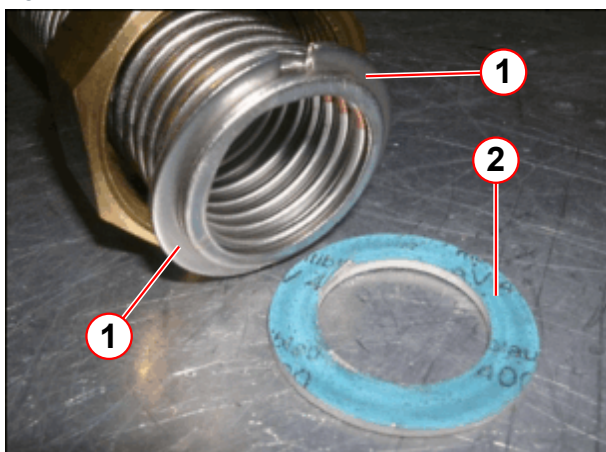
Fig. 89: Low pressure joint



(1) Component
(2) Rubber O-ring and carrier
(3) 2 part hinged clamp

- Joints on the low pressure side of the venting system are fitted with rubber (nitrile) O-rings or gasket seals.
- Always check that clamps and screws are tight.
- O-rings are re-usable. Check for any damage prior to fitting.
 - » Replace O-rings if necessary.

Fig. 90: Vent valve connection

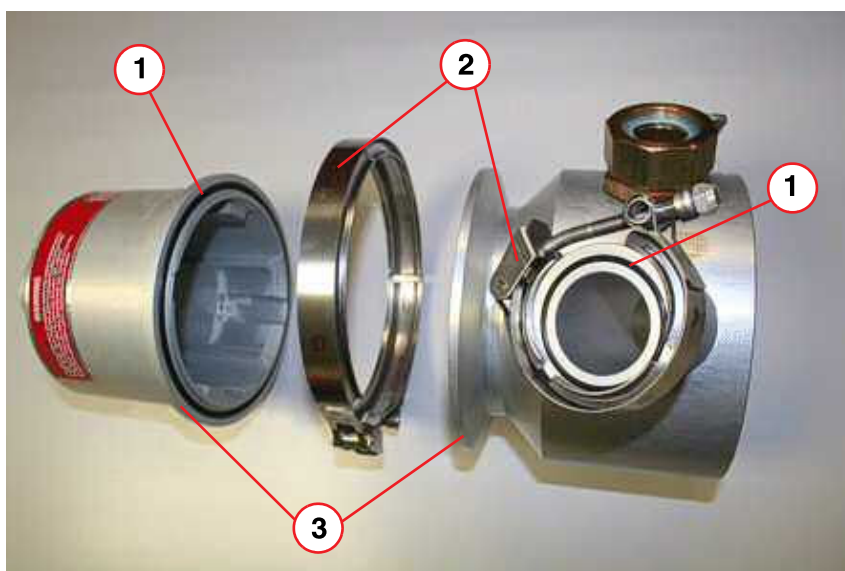


- (1) Split rings
(2) Gasket

- Gasket seals are re-usable. Check for any damage prior to fitting.
- Replace damaged gasket seals if necessary.

Example of O-ring seals fitted to Mk2 quench valve and manifold venting.

Fig. 91: Low Pressure - Quench Valve joints



- (1) Recessed rubber 'O'-rings
(2) Strap clamps
(3) 'V' Flanges

2.1.6.8 Final Work Steps

- After the leak check, ensure that the leak detector is working correctly :
 - » Check the equipment function with a known helium gas source.
- Ensure that all parts of the venting system have been connected and fitted.
- Refit all system covers.
- Order replacement seals if spares have been used.

2.1.7 What to do after a Quench (Avanto, Avanto Dot Upgrade, Espree with MSUP K2256)

2.1.7.1 General

If the magnet quenches, a large volume of helium gas is generated within the helium vessel. The quench valve opens, and stay open until the flow of gas subsides. The quench valve re-seals, preventing air (caused by cryo-pumping) from being drawn back into the magnet turret and helium vessel.

The MSUP automatically maintains the system pressure above atmospheric.



Magnet HSC must be notified of a quench (→ Siemens Healthineers Magnet WS CS HSC, <mailto:magnet.wscshsc.team@siemens-healthineers.com>)

Fill out the **Quench Report** - (→ MR Product Information, <https://productinformation.healthcare.siemens.com/ProductInformation.aspx?uuid=7abf6bb6-5daf-440c-9920-fab2d4640084>) - Links - MR - Quality Database (Quench Report)



The magnet helium level can be recovered if the magnet temperature is no warmer than nitrogen temperature.

Check the carbon resistor (CARBON RES 1;pins a-e) reading and compare to the calibration values on the Magnet Acceptance Document.

The magnet must be re-filled immediately with helium.

To re-fill the magnet from zero helium, or warm carbon resistor temperatures, perform a BOTTOM FILL procedure.

If proceeding with a helium fill, immediately place an order for 1000 liters of liquid helium.

NOTICE

If the magnet temperature warms up above 77K, a full nitrogen pre-cool is required.

Helium filling will not cool the magnet sufficiently to collect liquid helium.

» **A full nitrogen pre-cool will take 8 –10 days.**

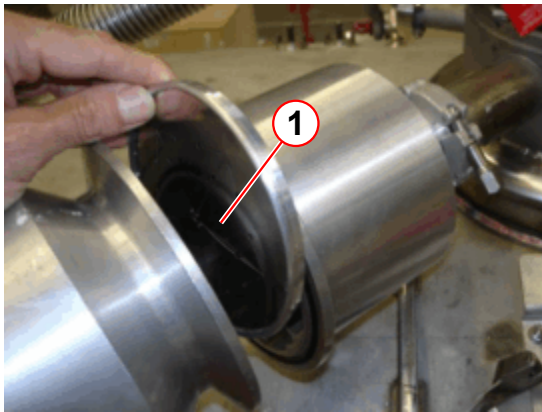
Mobile System

NOTICE**Risk of Air ingress**

The Mobile high altitude burst disk **WILL** rupture during a quench.

- » Post quench the high altitude 10 psi burst disk (10129467) and O-Ring (10113389) must be replaced.

Fig. 92: Removing the High Altitude disk



(1) Burst Disk

- Mobile systems are fitted with an additional High Altitude Burst Disk in front of the quench valve assembly.
 - » This disk will rupture if the magnet quenches.
 - » For replacement procedure: (→ Mobile Systems - High Altitude Burst Disk / Page 116)

Safety

Post quench procedures must be performed by CSE's that are trained to the MR2MAG-NET level. (or System level trained).



Switch off and isolate the magnet power supply from the mains before further action.

**WARNING****Risk of asphyxiation.**

Failure to observe the following safety warning may result in dizziness and loss of consciousness in personnel working with cryogenes.

- » Do not vent helium gas into the magnet room during helium transfer. As an exception, a small quantity of helium gas can be vented to cool the syphon.
- » Ensure the room is ventilated.
- » Ensure that the turret venting kit is connected to the quench pipe.
- » O₂ monitors must be used in the imaging suite.

**CAUTION****RISK OF ICING!!**

Air ingress will cause INTERNAL ICING OF THE TURRET.

- » Replacement of burst disk(s) must be performed by a qualified CSE.
- » Replace broken or damaged burst disk and O-ring seals IMMEDIATELY.
- » Open the bypass valve to check that there is a flow of gas. If there is no gas flow, there may be an ice blockage. Close the bypass valve.
- » If ice is suspected contact Magnet HSC immediately.

**CAUTION**

Cold helium gas.

Personal injury may occur.

- » Protective clothing (long sleeves), cryogenic gloves and eye protection must be worn.

2.1.7.2 Post Quench Steps

Follow the work steps in the table below for the different types of quench.

	Type of quench		
	Quench during ramping	Quench during He filling	Persistent mode quench or ERDU button pressed
Step 1	■ Isolate the MPSU from the magnet. ■ Download the 3600 MPSU ramp log files.	■ Stop filling.	■ Check that no ERDU buttons are pressed. ■ Re-set ERDU button.
Step 2	From the MSUP-Display Unit. ■ Check the liquid helium level. ■ Check the carbon resistor temperatures. (→ Checking the Carbon Resistors (CCR) Values / Page 95)		
Step 3	■ Order 1000 liters of liquid helium.¹		
Step 4	■ Warm up and leak check the quench valve and burst disk. (→ Checking the Quench Valve Burst Disk / Page 102) ■ Warm up and leak check the auxiliary vent burst disk. (→ Auxiliary Burst Disk / Page 111) ■ MOBILE system - replace the high altitude burst disk. (→ Mobile Systems - High Altitude Burst Disk / Page 116)		

	Type of quench		
	Quench during ramping	Quench during He filling	Persistent mode quench or ERDU button pressed
Step 5	After 1 hour : ■ Download the Magnet Supervisory log files. (→ Log File Capture / Page 94) ■ Helium fill. Fill to recommended helium ramp level.		
Step 6	■ Leak check the quench valve and venting. (→ Final Checks / Page 121)		
Step 7	■ Fill out the Quench report.		
Step 8	■ Email the (→ Magnet HSC, mailto:magnet.wscshsc.team@siemens-healthineers.com) with: » 3600 Power Supply logs. » Magnet Supervisory logs and screen capture. » Magnet serial number.		
Step	■ Ramp the magnet.		
Step 10	■ Perform the shim check. » If the shim check exceeds the specifications, a re-shim is necessary.		

1. The quantity of the order depends on the helium level remaining in the system after a quench, and volume already on site.

2.1.7.3 Pre-requisites

Tools Required

The post quench tools required:

- Heat gun and paper toweling.
- Service tool kit.
- Pressure service tool (material number **03861221**).
- Laptop and RS232 cable.
- O₂ monitor.
- Personal protection equipment. To include, cryogenic gloves, safety shoes, glasses and a long sleeved protective clothing.
- Helium leak detector.

Fig. 93: Edwards Leak detector (example)



- Use the Edwards hand-held leak detector (material number **10614850**). This is the recommended equipment for leaks greater than 1×10^{-4}

NOTE: The hand-held device is calibrated in ml/sec:-

1×10^{-4} mbar l/s = 1×10^{-4} ml/sec at normal pressure and temperature.

Spare Parts

Helium leaks from the magnet pressure vessel are generally repaired by replacing a damaged seal.

- A number of seals are supplied in the spares kit. These must be replaced when used on the magnet.
- Venting components will require ordering from CSML.

Mk1 Spares required:

- Graphite Burst disk 26 psi (material number 10129362).
- O-ring seal burst disk - (material number 10100140).
- Auxiliary burst disk (material number 08396090).
- High load seal (material number 10100171).

Mk2 Spares required:

- Graphite Burst disk 26 psi (material number 10129362).
- O-ring seal burst disk - (material number 10100140).
- Auxiliary burst disk (material number 10121913)
- High load seal (material number 10100171).
- O-ring seal 44.5mm ID (material number 10100151).



Always replace the burst disks and seals used during replacement procedures. They are critical components for reducing the risk of air ingress. A replacement must always be available for immediate use.

Unattended Quench

Unattended quench indicators are:

- If the Magnet STOP (ERDU) has been activated:
 - » the yellow WARNING LED lights on the alarm box.
 - » the alarm buzzer is activated.
- If the helium level is low:
 - » the yellow WARNING LED lights on the alarm box.
 - » the alarm buzzer is activated.
- If the magnet pressure is very low, the
 - » cold head switches off.
 - » the yellow WARNING LED lights on the alarm box.
 - » the alarm buzzer is activated.
- Imaging is not possible if there is no magnetic field.

Press the AUDIO ALARM OFF button on the alarm box to silence any audible alarms.

Log File Capture

Magnet Supervisory

- The **quench data is lost**, if:
 - a) The system is re-booted within 1 hour after a quench
 - b) The system is turned off within 1 hour after a quench
 - c) The data is downloaded within 1 hour after a quench
- Download the MSUP log files.

If the magnet quenches during ramping:-

- Download the Magnet Power Supply log file.



The MPSU log files **MUST** be downloaded **BEFORE** the next ramp is performed, otherwise previous data will be overwritten!

- Switch off and disconnect the MPSU from the magnet.

If the magnet quenches during helium fill:

- Stop the transfer.



Do not remove or disconnect the syphon as the magnet will be at high pressure.



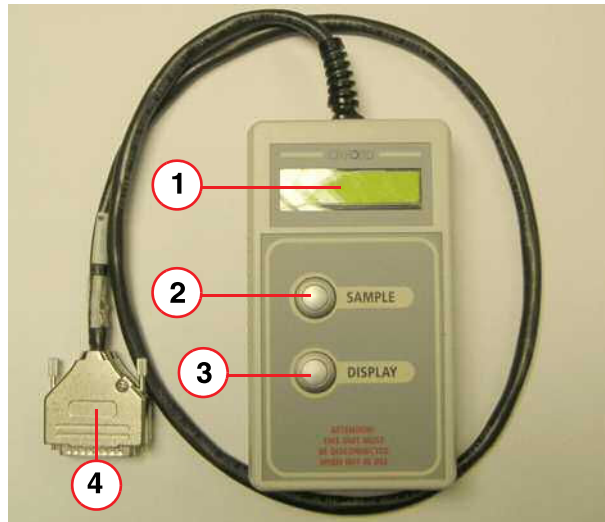
For all types of quench, allow up to 1 hour for the quench pressure to subside.

Use the MSUP Display unit:

- To check that the MSUP pressure control system is working. If a display unit is not available, use the SESO interface on the Host service screens.
- If the pressure control system is NOT working, the pressure inside the cryostat will fall to atmospheric levels. The compressor switches off.

Checking the Carbon Resistors (CCR) Values

Fig. 94: MSUP- Display Unit



- (1) LCD Display
- (2) Sample button
- (3) Display scroll button
- (4) 25 way 'D' connector

Using the MSUP-Display Unit:

- Connect the 25-pin 'D' connector to X9 on the MSUP.
- Activate the MDU by simultaneously pressing the SAMPLE button (2) and DISPLAY button (3).
- Press the Display button (3). The unit will automatically carry out a helium level test. The helium fill level is shown in the display (1).
- pressing the DISPLAY button, toggle through to Carbon resistor parameters.
- Check the carbon resistor values and compare the values with table with the expected values.



The magnet helium level can be recovered if the temperature of the carbon resistor (CARBON RES 1:Pins a-e) is still colder than nitrogen temperature.

The carbon resistor (CARBON RES 1) reading on the hand-held display must be in the range of 180 - 1500 ohms.

Confirm the readings, using the multimeter directly on the 41-pin connector.



Carbon resistors have been replaced with Ceramic Carbon resistors (CCRs), which have a different resistance range.



Carbon resistors have been replaced with Ceramic Carbon resistors (CCRs), which have a different resistance range.

- High readings = coldest temperature
- Low readings = warmest temperature
- Check temperatures with the figures on the Magnet Acceptance document.



If temperatures are below or equal to nitrogen temperature (77K), a BOTTOM FILL must be performed IMMEDIATELY.

Tab. 4 Expected Carbon Resistor Values

41-pin plug Measured in Ohms	Expected Values Room Temperature	Expected Values 77K Temperature	Expected Values 4.2K Temperature
Carbon One a - e (Bottom)	140 - 180	170 - 200	700 - 1500
Carbon Two b - e	120 - 160	130 - 180	700 - 1500
Carbon Three c - e	120 - 160	130 - 180	700 - 1500
Carbon Four d - e (Top)	120 - 160	130 - 180	700 - 1500
All Ceramic Carbon Resistors (CCRs)	880 - 1050	1150 - 1350	2300 - 3380

Fig. 95: OR105/122 Magnet Acceptance Document (CR)

SIEMENS		Magnet Acceptance Documentation Installation Data		Siemens Magnet Technology Quality Regulations	
Cryostat Number	21199812	SMT Reference	21201632 02.09		
System Description 1.494T Actively Shielded Superconducting Twin Turret Magnet System					
Measured Magnet Centre (Offset)		-1 (S) mm			
If magnetic Offset is outside the spec (+_5mm) the magnet will be marked (+_5mm) and Siemens Magnet Technology Ltd will be required to Shim the Magnet					
Operating Parameters					
Description	Specification value	Measured value			
Operating Current @ 1.494T	< 600 AMPS	509.95		AMPS	
Field/Current Ratio	N/A	12.48		kHz/O.1 Amp	
Carbon Resistor Values					
POSITION	Room Temperature	Helium Temperature			
Top	162.4	1140			
One	134.4	903			
Two	132.3	994			
TDC	140.4	943			
Helium Level Probe Resistance (When using a 250 mA supply)					
		Probe 1	Probe 2		
@ 0 Resistance	Ohms	85.36	85.36		
0 % Calibration		774	774		
100 % Resistance	Ohms	0.22	0.22		
100 % Calibration		2	2		
Diode Table No		-1			
Pressure Heater Res	Ohms	19			
Final Temperature Sensor Recon Values					
50kK	50kK Lin	Coldhead	Recon Margin		
48	45	42	K 467 Millivolts		
Checked by	K Suraina		Signature		
Date	02/02/2009				

Fig. 96: OR105/122 Magnet Acceptance Document

SIEMENS		Magnet Acceptance Documentation Installation Data	
MR Magnet Technology Quality Regulations			
Cryostat Number	21243234	SMT Reference	21245901 06.10
System Description			
1.494T Actively Shielded Superconducting Twin Turret Magnet System			OR105
Measured Magnet Centre (Offset)		-2 (S) mm	
If magnetic Offset is outside the spec (+_5mm) the magnet will be marked (+_5mm) and Siemens Magnet Technology Ltd will be required to Shim the Magnet			
Operating Parameters			
Description	Specification value	Measured value	
Operating Current @ 1.494T	> 600 AMPS	510.05	AMPS
Field/Current Ratio	N/A	12.48	kHz/O.1 Amp
Carbon Resistor Values			
POSITION	Room Temperature	Nitrogen(77K)	Helium Temperature
TDC d - e	878		2567
Top a - e	926		2938
Mid b - e	985		2715
Bottom c - e	877		2613
TDC Calibration Values:	850.4	1122.9	2731.7
Helium Level Probe Resistance (When using a 250 mA supply)			
		Probe 1	Probe 2
@ 0 Resistance	Ohms	85.36	85.36
0 % Calibration		774	774
100 % Resistance	Ohms	0.22	0.22
100 % Calibration		2	2
Diode Table No		-1	
Pressure Heater Res	Ohms	19	
Final Temperature Sensor Recon Values			
50kK	50kK Lin	Coldhead	Recon Margin
53	47	44	K 530 mW
Checked by		Signature	
K Suraina			
Date			
02/06/2010			
Page 1 of 1		25-AD EPS 000 01	

Check the Magnet Pressure

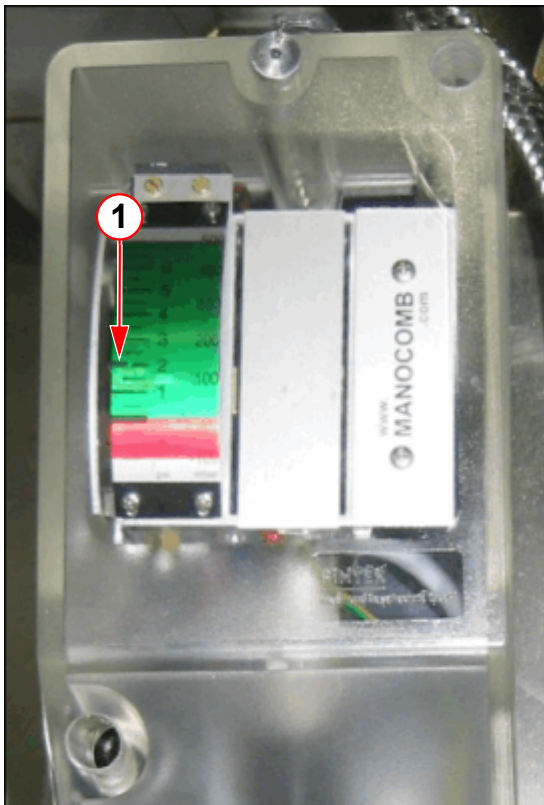
Check that the helium vessel has a positive pressure:

- Magnet pressure should read 15.3 psi (A) on the MSUP display unit.
- Check for positive pressure on the pressure gauge.
- If the pressure gauge is within the **RED zone**, the magnet is sub-atmospheric.
 - » **DO NOT** open the magnet bypass valve, or remove any high pressure fitting.
 - » If the magnet pressure does not stabilize via the pressure heater control circuit, manually induce pressure into the cryostat.



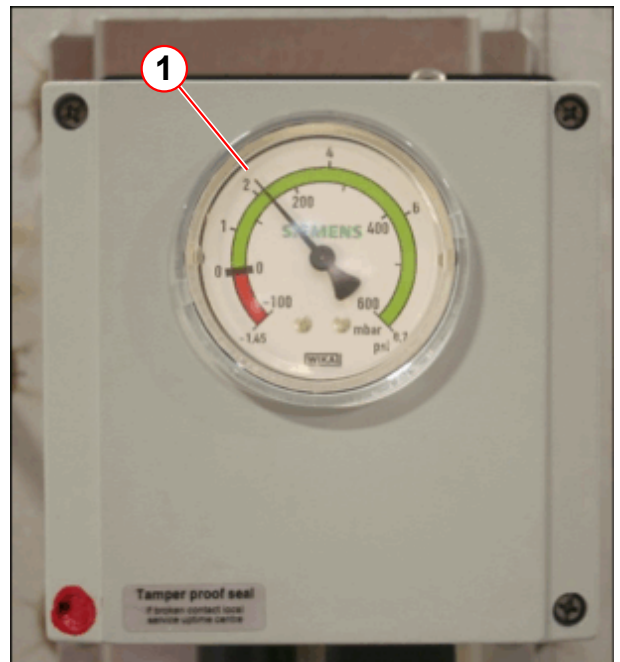
Wait until the magnet has gained a positive pressure before continuing with the burst disk and post quench leak checks.

Fig. 97: OR97 - Pressure gauge (expected pressure)



(1) Expected pressure 2.1 psi (G)

Fig. 98: OR97 - WIKA pressure gauge (expected pressure)



(1) Expected pressure 2.1 psi (G)

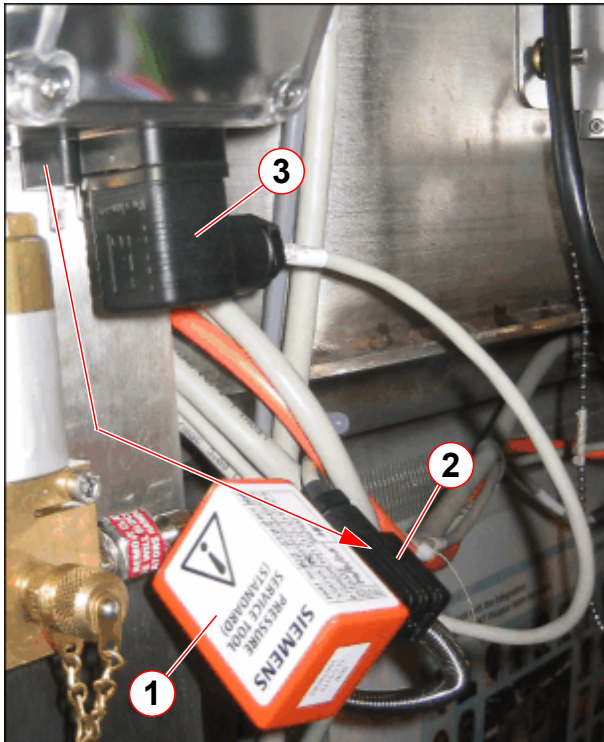
Deactivate the Pressure Heater

The Pressure Service Tool is used to deactivate the pressure heater control circuit. The electrical connectors are labeled pressure sensor (rear) and pressure switch (front).

- To fit the pressure service tool:
 - Disconnect the rear electrical connector on the pressure switch transducer.

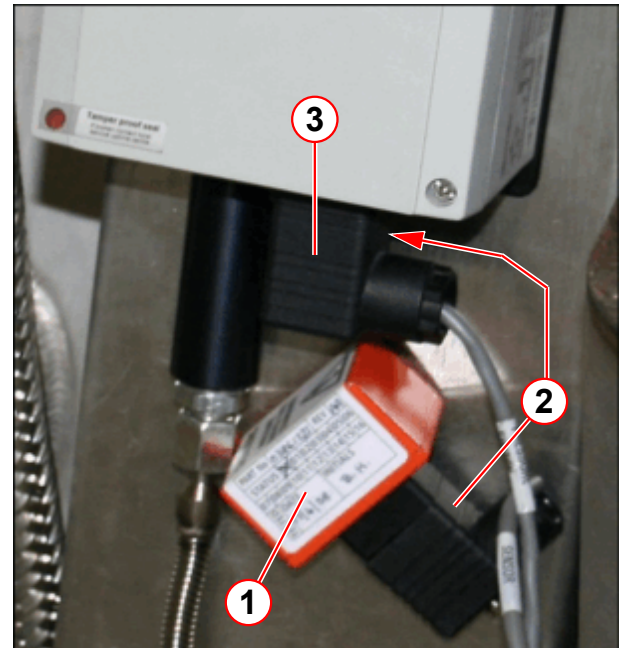
- Fit the pressure service tool to the connector lead.
- The Alarm Box will sound PRESSURE ERROR.
- Press the ACKNOWLEDGE button to cancel the alarm.
- The pressure reading will be set to ~18 psi (A), which will disable the pressure heater.
- To activate the pressure heater control circuit, connect the cable to the rear connector on the pressure switch transducer.
 - » Cancel any alarms.

Fig. 99: Fit the pressure service tool



- (1) Pressure service tool
- (2) Rear pressure sensor
- (3) Front pressure switch

Fig. 100: Fit the pressure service tool

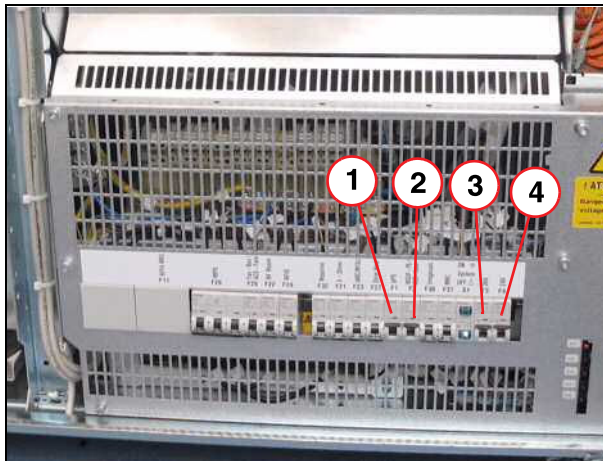


- (1) Pressure service tool
- (2) Rear pressure sensor
- (3) Front pressure switch



Do not leave the system unattended with the pressure service tool attached.

Fig. 101: ACC fuse (inside)



- (1) Fuse F1
- (2) Fuse F2
- (3) Fuse F3
- (4) Fuse F4

If the service pressure tool is not available:

- Switch OFF the MSUP at circuit breaker F3 (fuse) in the ACC.
 - » Pressure heater is powered down.
- The ERDU circuit is still functional.
- Helium level and magnet temperatures can still be obtained via the MSUP-Display Unit (via battery power only).

NOTE: On battery power, only 2 minutes of power is available.

Depressurize the Helium Vessel



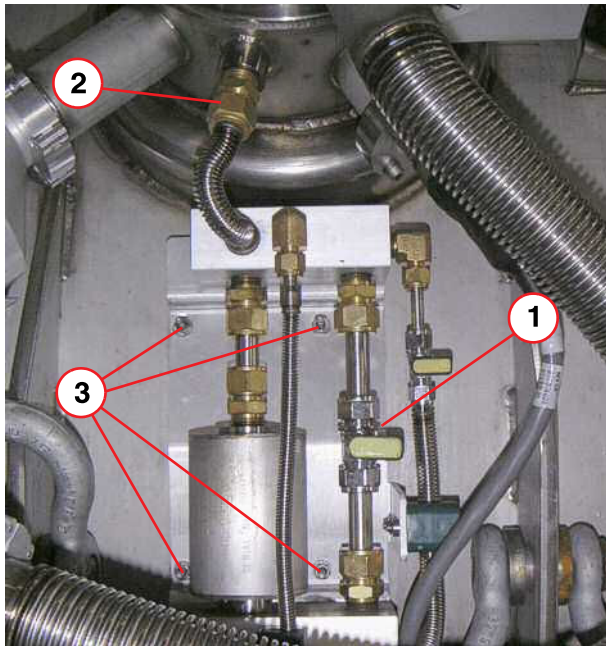
CAUTION

Risk of ice formation.

The magnet bypass valve is only OPENED for depressurization of the helium vessel. The bypass valve is CLOSED during normal operation.

- » To avoid air ingress, CLOSE the magnet bypass valve immediately after the helium vessel is depressurized.

Fig. 102: Mk2 Vent valve assembly



- (1) Bypass valve
- (2) Turret connection
- (3) Mounting screws (4 positions)

If depressurization of the helium vessel is required:

- Fully open the magnet bypass valve to depressurize the helium vessel.
- Wait until zero psi is displayed on the pressure gauge.
- The pressure switch will activate.
 - » the cold head will turn off.
- Close the magnet bypass valve to prevent air ingress.
- Warm up the vent valve assembly.
 - » Take care not to damage any components when using a heat gun.

2.1.7.4 Checking the Quench Valve Burst Disk

Post quench if there is no magnet pressure a burst disk has ruptured. Check the graphite (quench valve) burst disk.

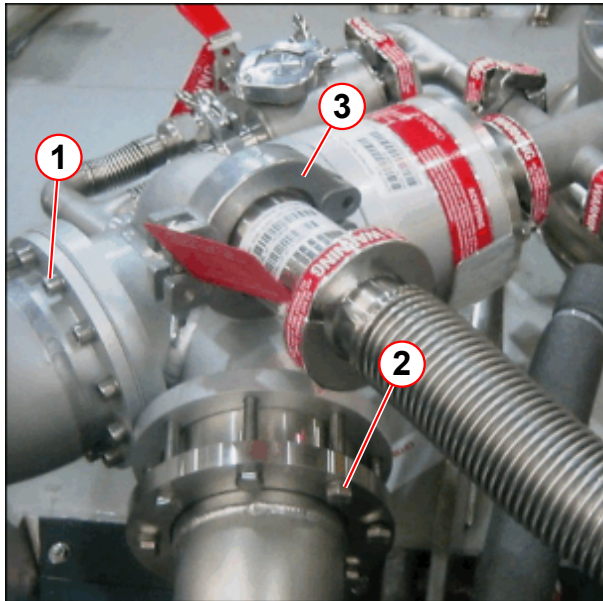
- Disable the pressure heater by fitting the pressure service tool.
- Warm up the quench valve and venting to room temperature, using a heat gun.
- The elbow assembly and manifold must be removed for inspection and leak check of the graphite burst disk and O-ring.



Check venting configuration fitted and follow applicable procedure for Mk1 or MK2 versions.

Mk1 Venting Configuration

Fig. 103: MK1 - Venting removal

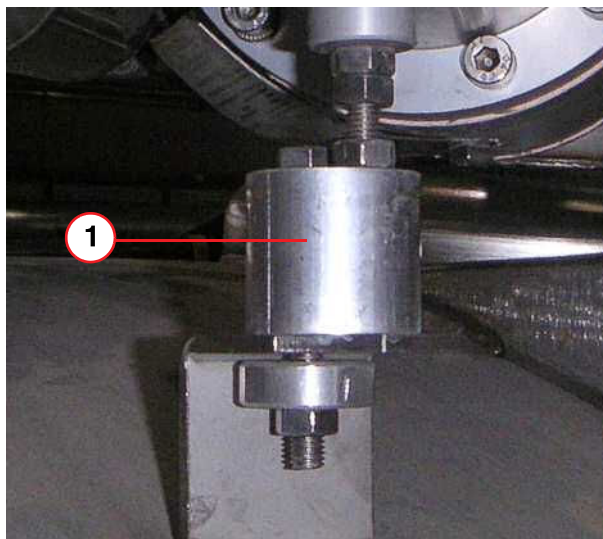


- (1) Elbow connection
- (2) OVC burst disk
- (3) Auxiliary burst disk

To access the quench valve and burst disk:

- Remove the 12 x M8 Cap hd screws and retaining the elbow assembly to the manifold venting (pos.1).
- Remove 8 x M8 Hex hd screws retaining the OVC burst disk to the manifold (pos.2).
- Disconnect the NW50 clamp (pos.3) and remove the auxiliary vent line.

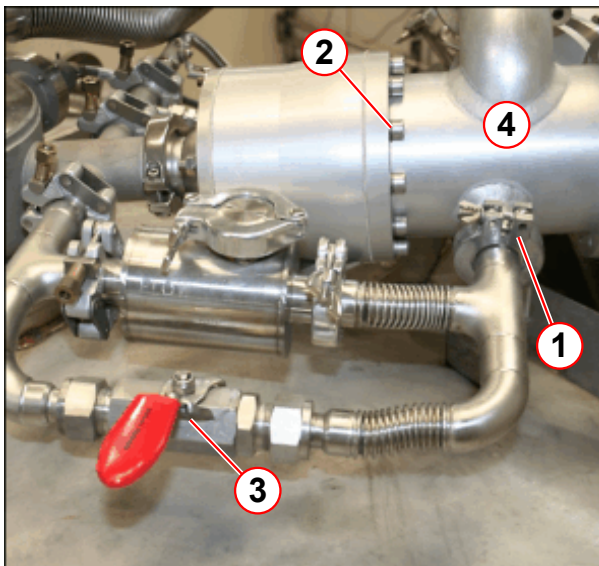
Fig. 104: Mk1 Support Bracket



- (1) Support Bracket

- Loosen the support bracket (1) screw underneath the quench valve interface assembly.

Fig. 105: Mk1 - Manifold removal

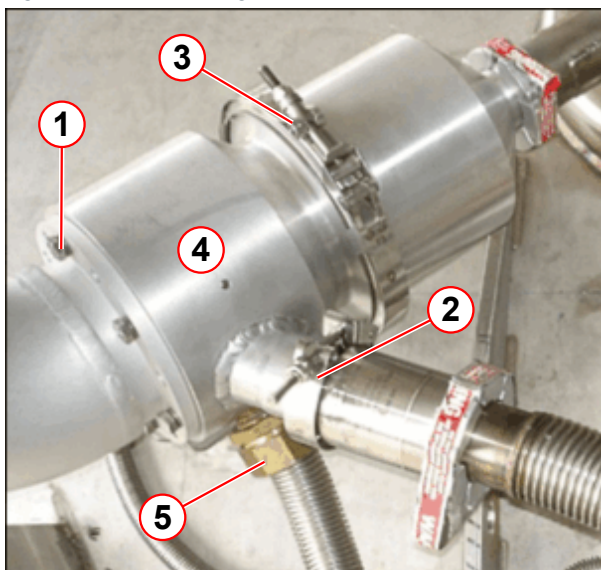


- (1) NW25 Bypass valve connection
- (2) Manifold screws (M8 x 12)
- (3) Bypass valve
- (4) Manifold

- Remove the NW25 (low pressure) clamp to release the 16 psi (A) valve assembly (pos.1).
- Remove the M8 Cap hd screws (x12) securing the manifold to the quench valve (pos.2).
 - » Place the manifold to one side, allowing access to the quench valve.
- Check that the magnet bypass valve is closed.

Mk2 Venting Configuration

Fig. 106: Mk2 Venting removal



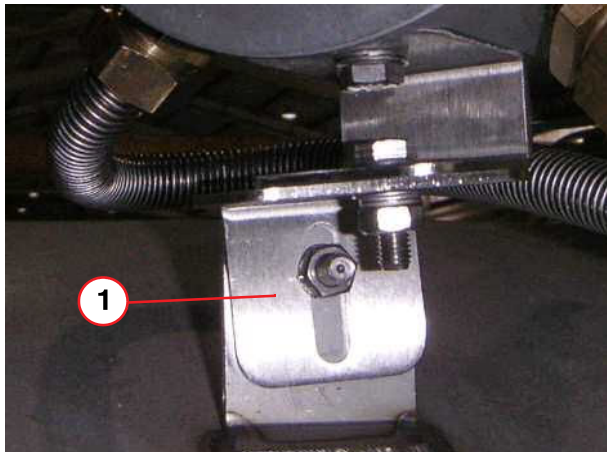
- (1) Elbow connection (M8 x 6)
- (2) V-clamp - Auxiliary vent
- (3) V-clamp manifold to quench valve
- (4) Manifold assembly
- (5) OVC vent connection

To access the quench valve and burst disk:

- Remove the 6 x M8 screws retaining the elbow assembly to the manifold (pos.1).
- Remove the V-clamp retaining the auxiliary vent to the manifold (pos. 2).
- Remove the V-clamp retaining the manifold to the quench valve (pos. 3).

NOTE: it should not be necessary to remove the OVC vent line connection (pos.5).

Fig. 107: Mk2 Support bracket



(1) Support bracket

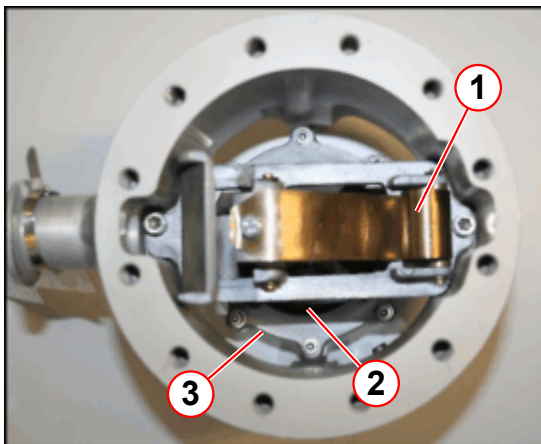
- Loosen the support bracket (1) screw underneath the quench valve interface assembly.
- Move the manifold assembly away from the quench valve.

The quench valve and graphite burst disk can now be accessed to check for leaks.



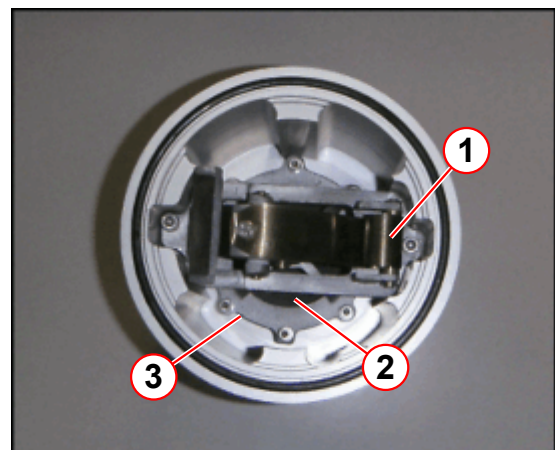
The spring plate and burst disk assembly is the same for MK1 and MK2 quench valves. The next steps show the MK2.

Fig. 108: Mk1 Quench valve assembly



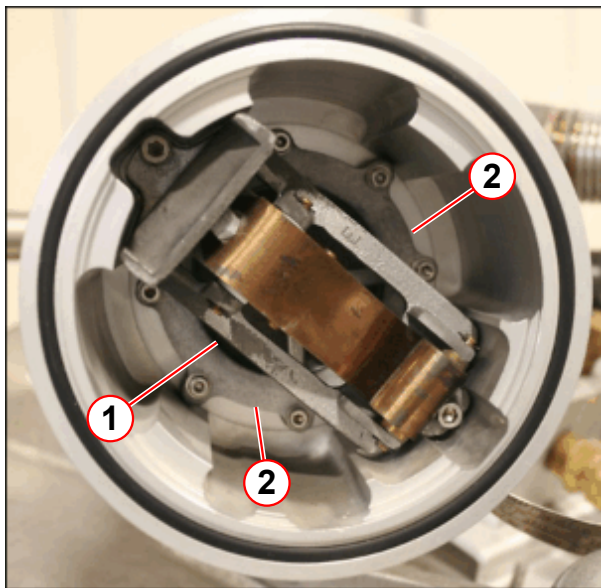
- (1) Spring plate assembly
- (2) Graphite burst disk
- (3) O-ring seal

Fig. 109: MK2 - Quench valve assembly



- (1) Spring plate assembly
- (2) Graphite burst disk
- (3) O-ring seal

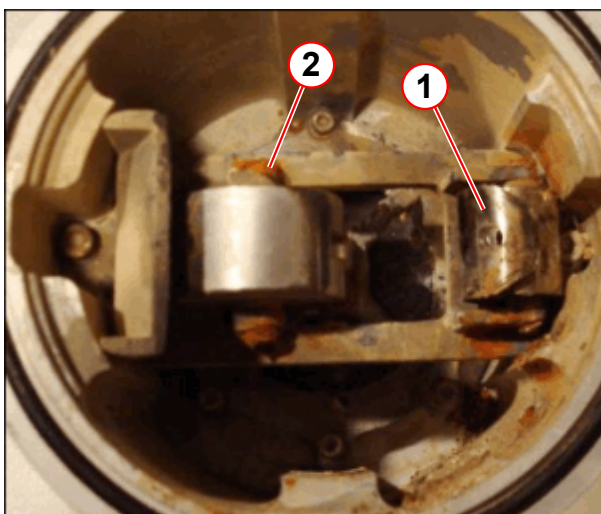
Fig. 110: Leak check



- (1) Graphite burst disk
- (2) Burst disk O-ring seal

- Check the graphite burst disk for breakage or cracks.
- If the graphite burst disk is broken or cracked, it must be replaced **immediately**.

Fig. 111: Water corrosion (example)



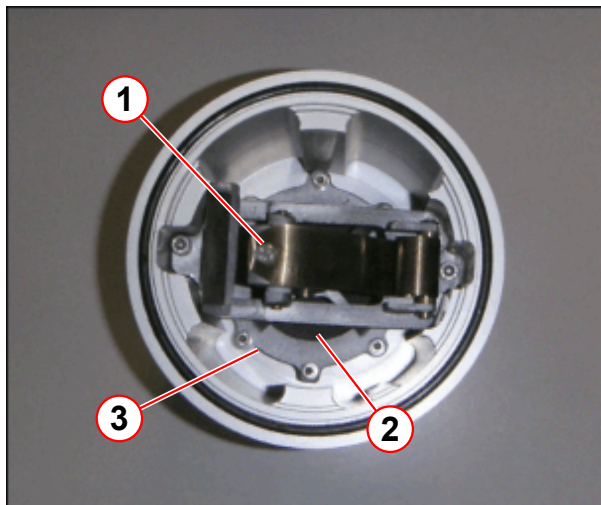
- (1) Spring broken
- (2) Corroded screws

Check for ice and water corrosion or breakage of:

- The spring plate assembly.
- All screws.
- Corroded quench valve components must be replaced.

- With no damage to the burst disk or spring plate visible, continue to leak check around the quench valve O-ring and retaining plate.
- Check that the magnet pressure is a minimum of 0.5 psi (G) for leak checking.
- Leak check around the burst disk and spring plate assembly.

Fig. 112: Quench Valve leak check

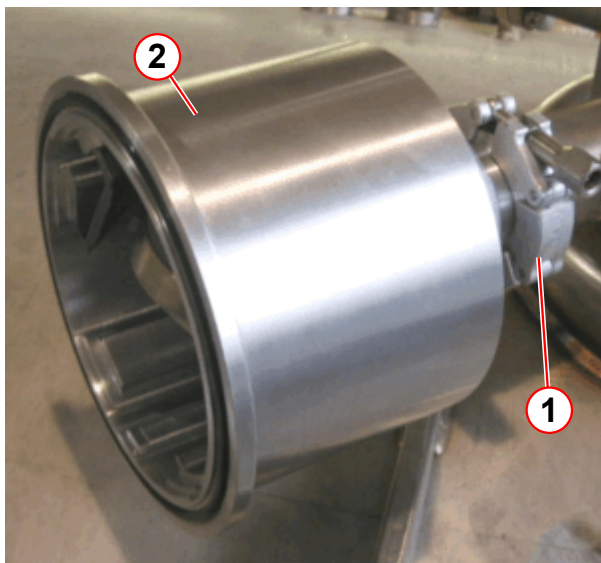


- (1) Spring plate assembly
- (2) Graphite burst disk
- (3) O-ring seal

- If there is no leak found the quench valve O-ring seal behind the spring plate is OK.
- There is no requirement to remove the burst disk spring plate.
- Continue to check the auxiliary burst disk and refit the quench valve.
- If a leak is detected the burst disk spring plate assembly will need removing to replace the O-ring.

Replacing the Burst Disk and O-ring Seal

Fig. 113: Fitting the quench valve



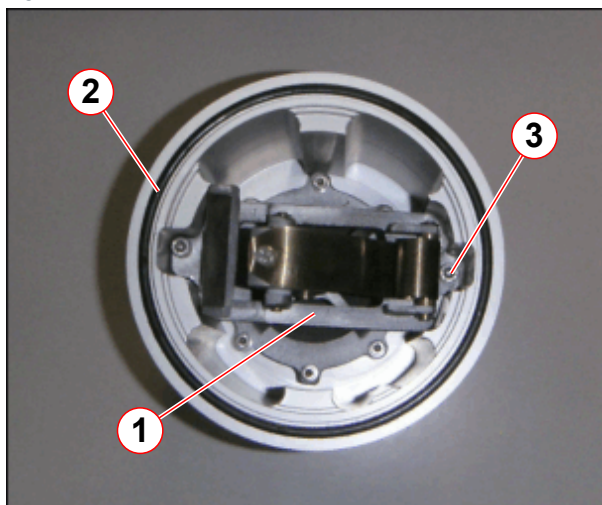
- (1) NW40 seal and clamp
- (2) Quench valve assembly

- Open the magnet bypass valve to depressurize (if necessary).
- De-ice the turret flange.
- Mark the orientation of the quench valve
- Remove the NW40 (high load) turret clamp and seal.
- Remove the quench valve from the turret flange.
- To minimize air ingress, immediately place a ball of paper toweling into the open turret port.
- Close the magnet bypass valve.



The spring exerts significant force on these screws.

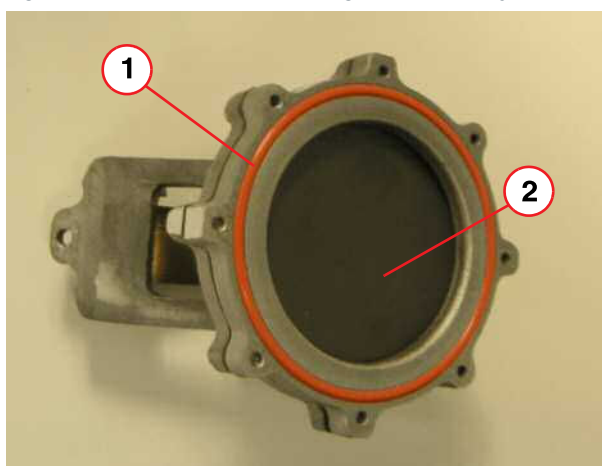
Fig. 114: Quench Valve



- (1) Spring plate assembly
- (2) O-ring
- (3) M5 cap screws (2 off)

- Unscrew and remove the (2 off) M5 cap head screws.
- Lift out the burst disk assembly from the quench valve body.

Fig. 115: Quench Valve Bursting Disk Assembly



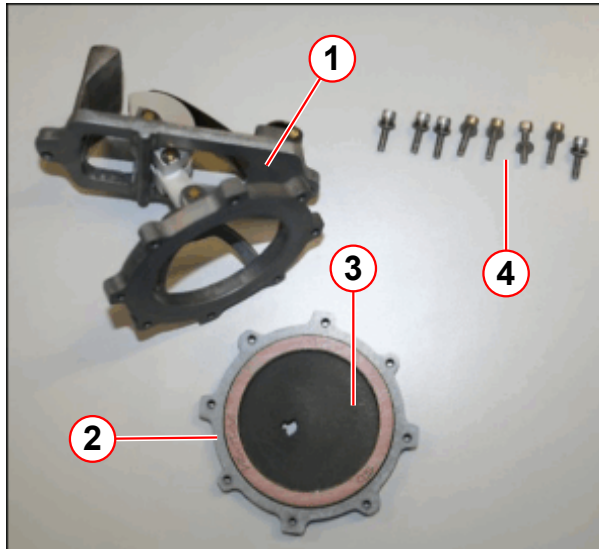
- (1) O-ring
- (2) Burst disk

- Inspect the burst disk 'O-ring and replace if damaged.
- Clean the O-ring groove and O-ring before re-fitting.
- Check that the O-ring is seated evenly around the groove.



The O-ring must be replaced each time the quench valve is removed. DO NOT apply grease to the O-ring.

Fig. 116: Burst disk assembly

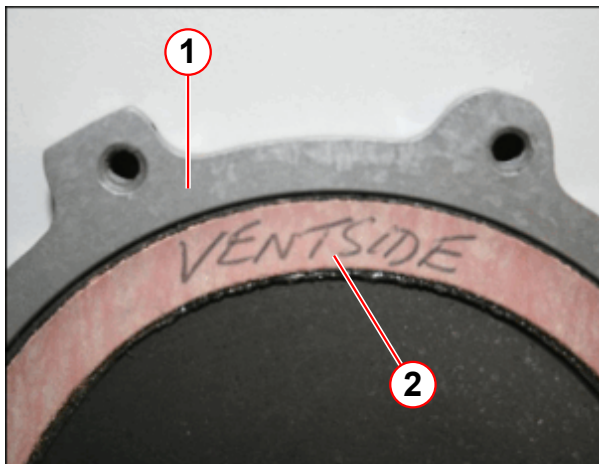


- (1) Spring plate assembly
- (2) Disk housing
- (3) Broken disk
- (4) M4 screws (8 off)

To replace the burst disk:

- Unscrew and remove the 8 x M4 cap screws.
- Remove the broken burst disk and discard.
- Check that the new burst disk and flange faces are clean and undamaged.

Fig. 117: Burst disk fitment



- (1) Burst disk housing
- (2) Burst disk (Vent side uppermost)

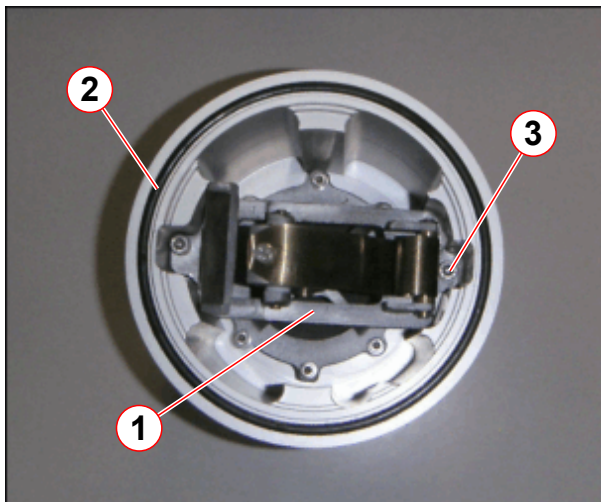
- Fit the replacement disk.
- The words '**Vent Side**' (written on the burst disk gasket) must be facing the spring plate assembly

Fig. 118: Thread locking compound



- Fit the burst disk assembly. (Do not apply grease).
- The (8 off) M4 cap screws and plain washers should be locked with threadlock compound..
- Diametrically torque the screws evenly to 0.8 Nm (0.6 ft. lbs).

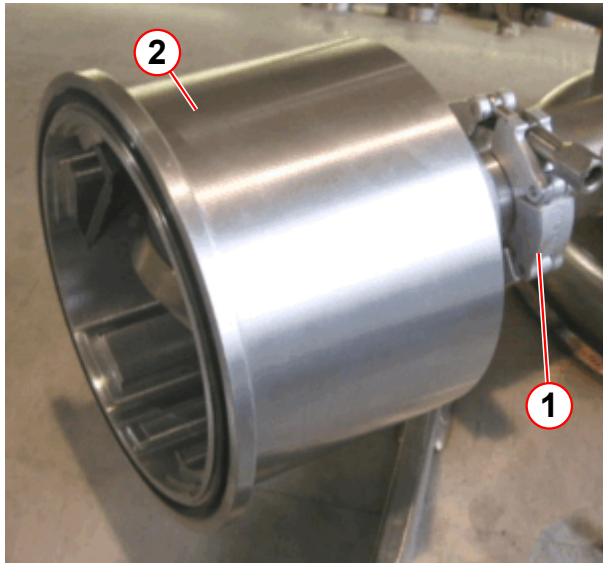
Fig. 119: Quench Valve



- (1) Spring plate assembly
- (2) O-ring
- (3) M5 cap screws (2 off)

- Re-fit the burst disk assembly into the quench valve body.
- Torque the 2 x M5 cap hd screws to 5 Nm. (3.8 ft.lbs).
- Check the quench valve manifold O-ring is in good condition (pos.2).
- Replace if damaged.

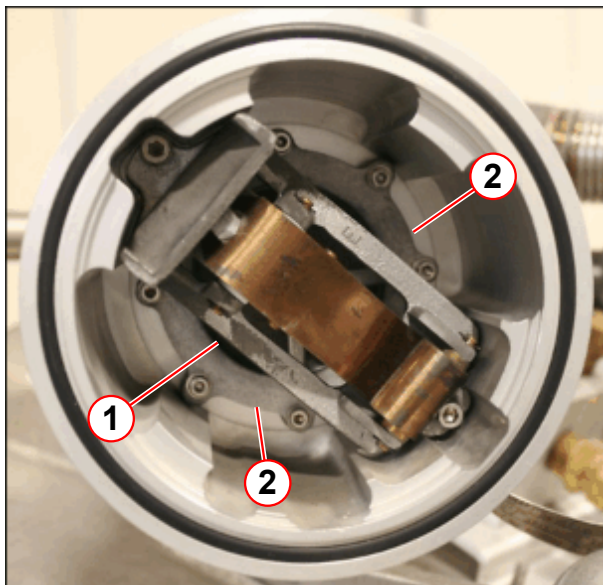
Fig. 120: Fitting the quench valve



- (1) NW40 seal and clamp
(2) Quench valve assembly

- **Remove the paper toweling** if used (or NW40 blank) from the turret port. Ensure the surfaces are clean and dry.
- Fit the quench valve assembly to the turret flange.
 - » Re-fit using a new NW40 high load seal (material number 10100171).
 - » Check the correct orientation.
 - » Warm with heat gun when tightening the NW40 clamp.

Fig. 121: Leak check



- (1) Graphite burst disk
(2) Burst disk O-ring seal

- Check that the magnet bypass valve is closed and pressurize the system.
- Leak check the burst disk and O-ring seal.
- Leak check the NW40 clamp and seal at the turret port.
- If no leaks are detected:
 - » Fit the quench valve elbow interface with the V-clamp to the quench valve.
 - » Fit the support bracket screw to the elbow interface assembly.
 - » Check the auxiliary burst disk.

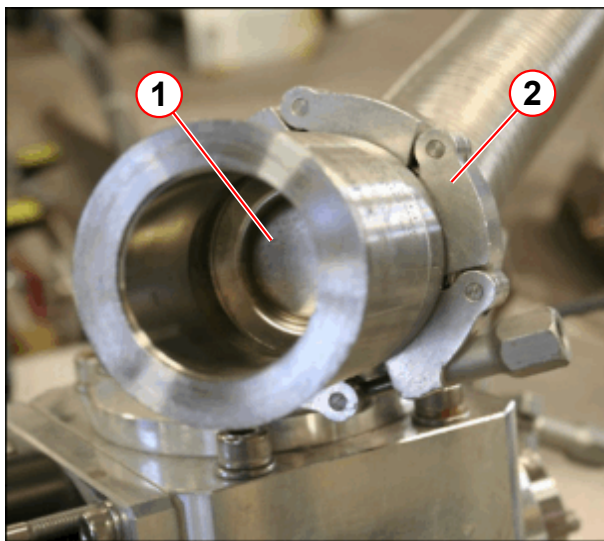
2.1.7.5 Auxiliary Burst Disk

Post quench the auxiliary burst disk vent line is removed when disconnecting the venting to leak check the quench valve.



Do not touch the metal membrane, it is very fragile.

Fig. 122: Auxiliary burst disk check



- (1) Burst disk
(2) High load seal

- Visually inspect the internal metal membrane of the burst disk for damage (pos.1).
- Leak check the high load seal (pos. 2).



WARNING

Risk of air ingress

If the auxiliary burst disk is broken, IMMEDIATELY fit the blanking gasket seal.

- » The blanking gasket seal is supplied in the magnet spares kit.
- » Order a replacement auxiliary burst disk and fit before ramping the magnet.



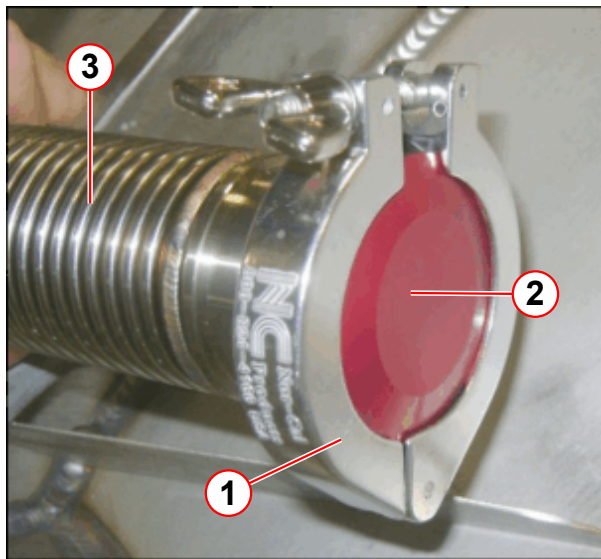
WARNING

Safety device removed

The magnet must not be ramped with the auxiliary burst disk removed or broken.

- » Replace the auxiliary vent burst disk before ramping.

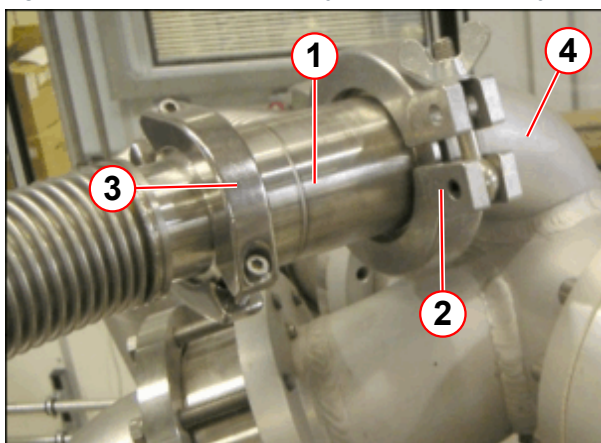
Fig. 123: Auxiliary vent - NW40 blanking plate



- (1) NW40 clamp
- (2) NW40 red blanking plate
- (3) Auxiliary vent line

- If the membrane has ruptured, disconnect the high load clamp and remove the burst disk from the vent line.
- Fit the red anodized blank immediately from the spares kit.
- Order a replacement auxiliary burst disk. Fit as soon as available.
- With the blank fitted allow magnet to pressurize.
- Leak check red blanking plate and seal on the end of the auxiliary vent line.

Fig. 124: OR105 Mk1 Auxiliary burst disk assembly

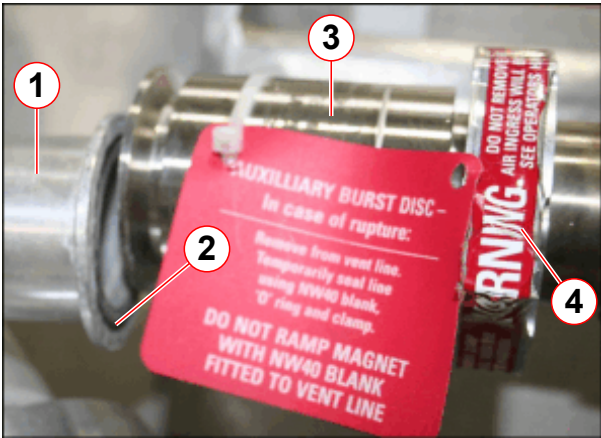


- (1) Aux burst disk assy
- (2) NW50 low load clamp (low pressure)
- (3) NW40 high load clamp (high pressure)
- (4) Low pressure manifold

For the Mk1 venting:

- Connect the auxiliary burst disk to the manifold assembly with a NW50 clamp and seal.

Fig. 125: Auxiliary vent interface - O-ring seal



- (1) Quench valve manifold
- (2) O-ring seal
- (3) Auxiliary bursting disk
- (4) NW40 clamp (high pressure)

- Check that the O-ring seal (Pos.2) is not damaged and that the surface is clean.
- If damaged, replace the O-ring.
- Connect the auxiliary burst disk to the manifold assembly with the V-clamp.

2.1.7.6 OVC Burst Disk

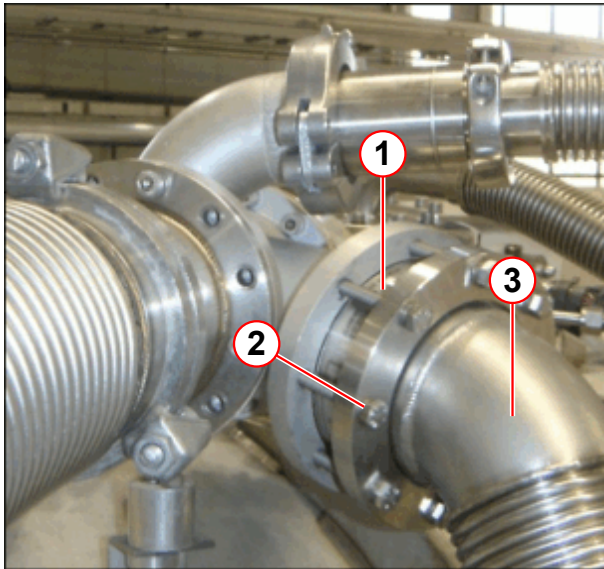


The OVC (DOP) port burst disk and vent line has been removed from the Mk2 magnet design. See Table for system serial number change.

Tab. 5 Venting configurations

Venting	OR105 Avanto serial #	OR122 Espree serial #
Mk1	Before System MR25575	Before System MR30125
Mk2	From System MR25575	From System MR30125
Mk 2 (No DOP)	From System MR57232	From System MR31427

Fig. 126: OR105 Mk1 OVC burst disk assembly



- (1) OVC burst disk
- (2) Screws (8 off)
- (3) OVC vent tube

Mk 1 OVC burst disk

A quench will not cause the OVC metal burst disk to rupture.

- During the post quench checks the burst disk can inspected when the venting removed to inspect the quench valve.
- Remove the (8 off) **screws** holding the OVC burst disk in place.



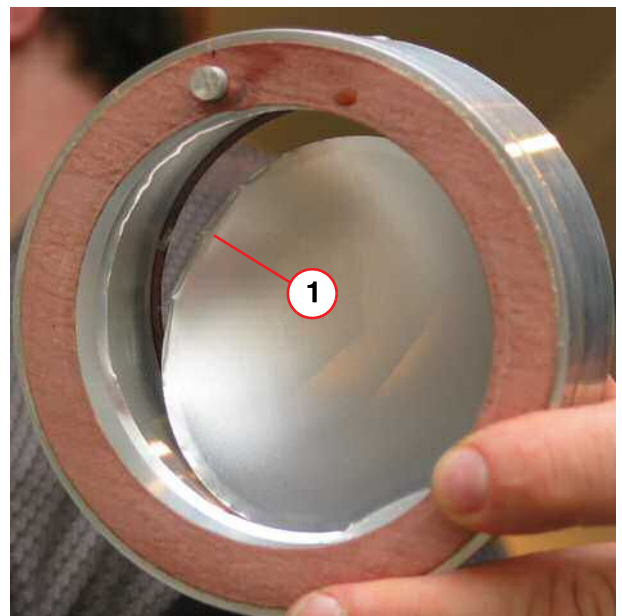
The membrane is very fragile - DO NOT TOUCH THE SURFACE.

Fig. 127: OVC burst disk



- (1) OVC burst disk

Fig. 128: OVC burst disk

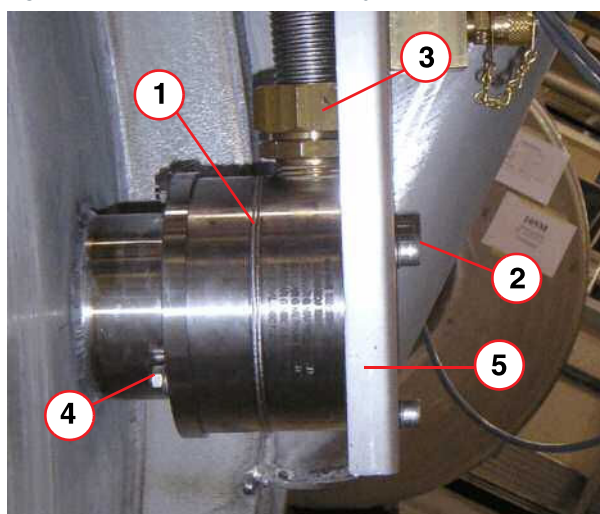


- (1) Broken OVC burst disk



The burst disk has a metal peg in one face to ensure correct orientation of the disk.

Fig. 129: Mk2 OVC Burst Disk Assy



- (1) OVC burst disk
- (2) Cap head screw (3 off)
- (3) OVC vent sealing nut
- (4) Nut (3 off)
- (5) Pressure gauge plate

Mk 2 OVC burst disk

A quench will not cause the OVC metal burst disk to rupture. The OVC burst disk requires checking:

- During routine safety checks and ice inspection of the venting.

2.1.7.7 Mobile Systems - High Altitude Burst Disk



Mobile systems are fitted with a high altitude burst disk, in addition to the quench valve burst disk.

If the magnet quenches, the high altitude disk will rupture and must be replaced.

Spare Parts

Post quench the high altitude burst disk will need to be replaced and the O-ring seal checked.

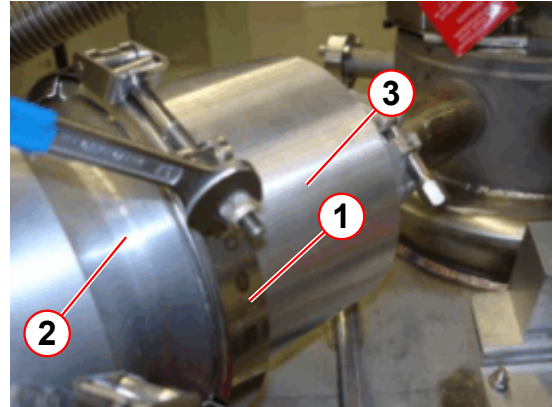
- 10 psi burst disk (Composite) - (material number **10129467**).
- O-ring seal - (material number **10113389**).

High Altitude Bursting Disk Removal

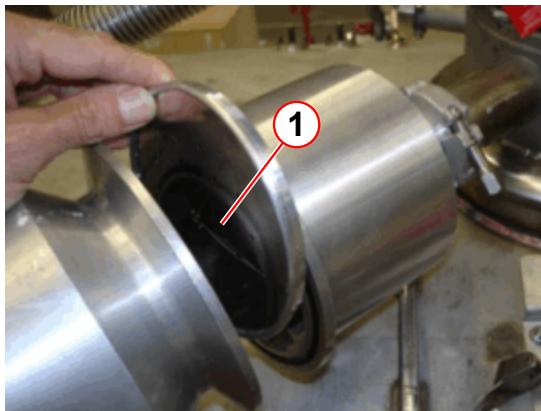
- Undo the M6 screw and nut and remove the retaining clamp on the manifold support bracket.
- Undo the M6 nut and release the V-clamp securing the manifold assembly to the quench valve.

Fig. 130: Manifold Support Bracket

(1) Support Clamp

Fig. 131: Manifold V Clamp

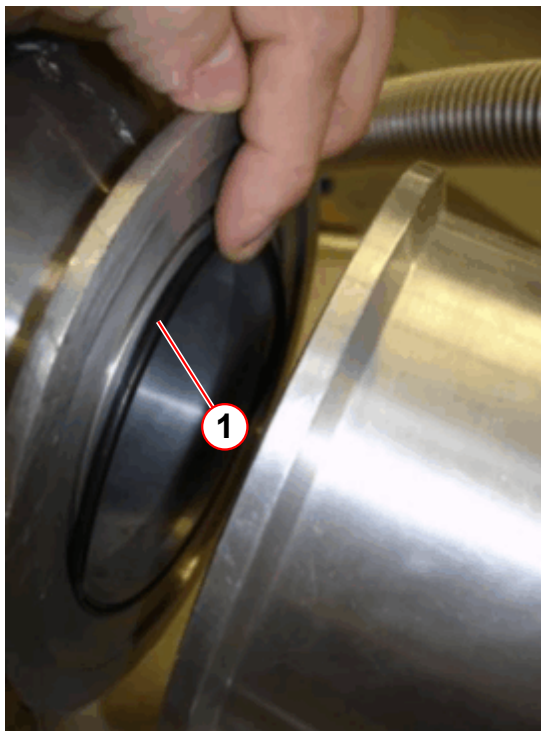
(1) V Clamp
(2) Manifold
(3) Quench Valve

Fig. 132: Removing the High Altitude disk

(1) Burst Disk

- Carefully pull the manifold away from the quench valve.
- The burst disk is held in position by the V-clamp seal. There are no screws.
- Remove the ruptured high altitude burst disk.

Fig. 133: O Ring Fitment



(1) O Ring and Groove

- Remove the O-ring in the manifold flange.
- Clean the flange mating surfaces.
- Complete the quench valve burst disk checks.



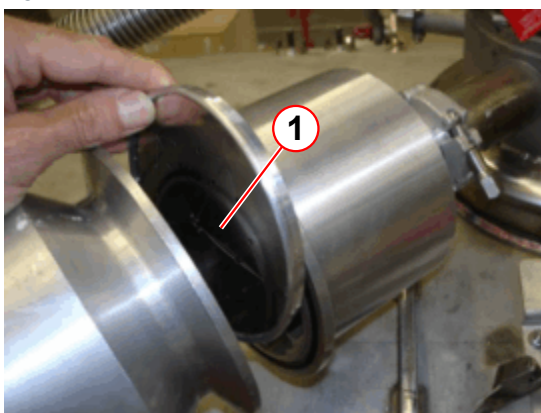
As part of the Post Quench checks at this stage the standard quench valve and graphite burst disk checks will be completed.

Fitting the High Altitude Bursting Disk



Take care not to touch the center of the disk. It can be damaged easily.

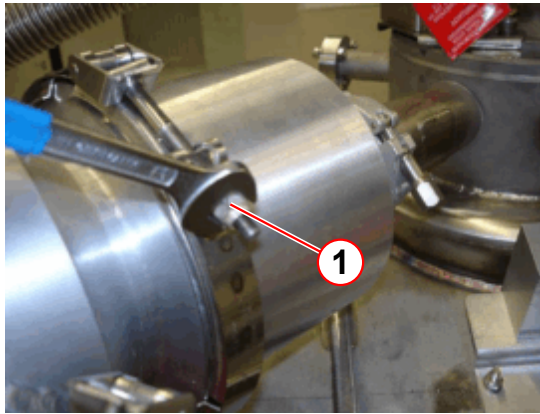
Fig. 134: Burst Disc Fitment



(1) Burst Disc Words "VENT SIDE"

- Check that the manifold O-ring is clean and not damaged.
 - » Replace if necessary.
- Check that the 2 flange surfaces are clean.
- On the disk, check that the wording 'Vent Side' is facing away from the quench valve.
- Fit the new high altitude burst disk between the manifold and quench valve flanges.

Fig. 135: V Clamp Fitment



(1) Self Locking Nut

- Hold the burst disk in position and join the manifold flange and burst disk to the quench valve.
- Check that the burst disk is still centered and refit the V-clamp.
- Tighten the nut until there is a sharp increase in resistance indicating that the self locking nut is fully tight.

Fig. 136: Manifold Support Bracket



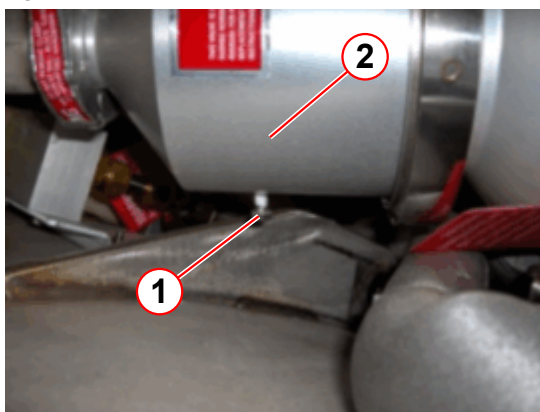
(1) M8 Nuts & Bolts

- Tighten the screw and M6 nut on the manifold support bracket.



At high altitudes the venting is only effective with the drain screw fitted in the quench valve.

Fig. 137: Drain Screw



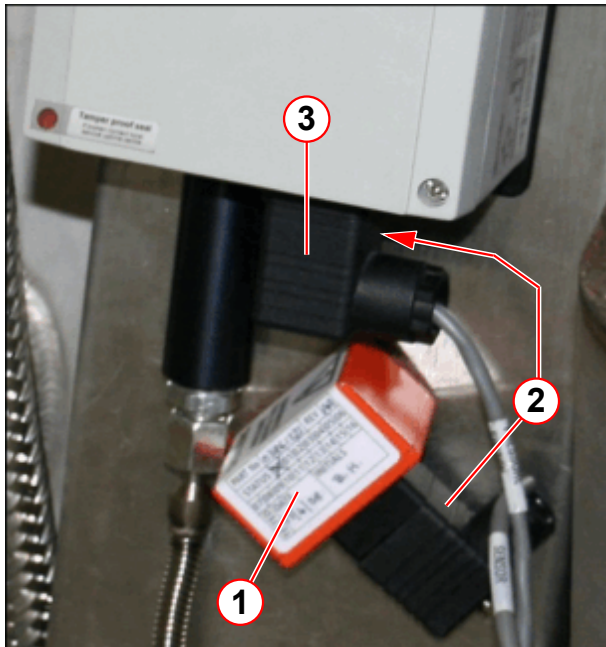
(1) Drain Screw
(2) Quench Valve

- Check that the drain screw is fitted in the quench valve body.

2.1.7.8 Final Work Steps

Reactivate the Pressure Heater

Fig. 138: Remove the pressure service tool



- (1) Pressure service tool
- (2) Rear pressure sensor
- (3) Front pressure switch

Re-configure the system to enable the pressure heater depending on method used:

1. Switch ON the MSUP at circuit breaker F3 (fuse) in the ACC.
2. Remove the pressure service tool from the pressure sensor cable. Re-fit the connector to the pressure display unit.

- Access the Service Software at the MRC (MRAWP) Host.
- Select the menu item Magnet & Cooling - Magnet Status.
- At MSUP Control - Set the pressure heater mode to **On 100%**.
- Select SAVE.
- Monitor the magnet pressure.
 - Select Update to refresh data until pressure reaches ≥ 15.3 psi (A).



Data is only accurate from the last refresh/update of the page, or from when the page was opened last.

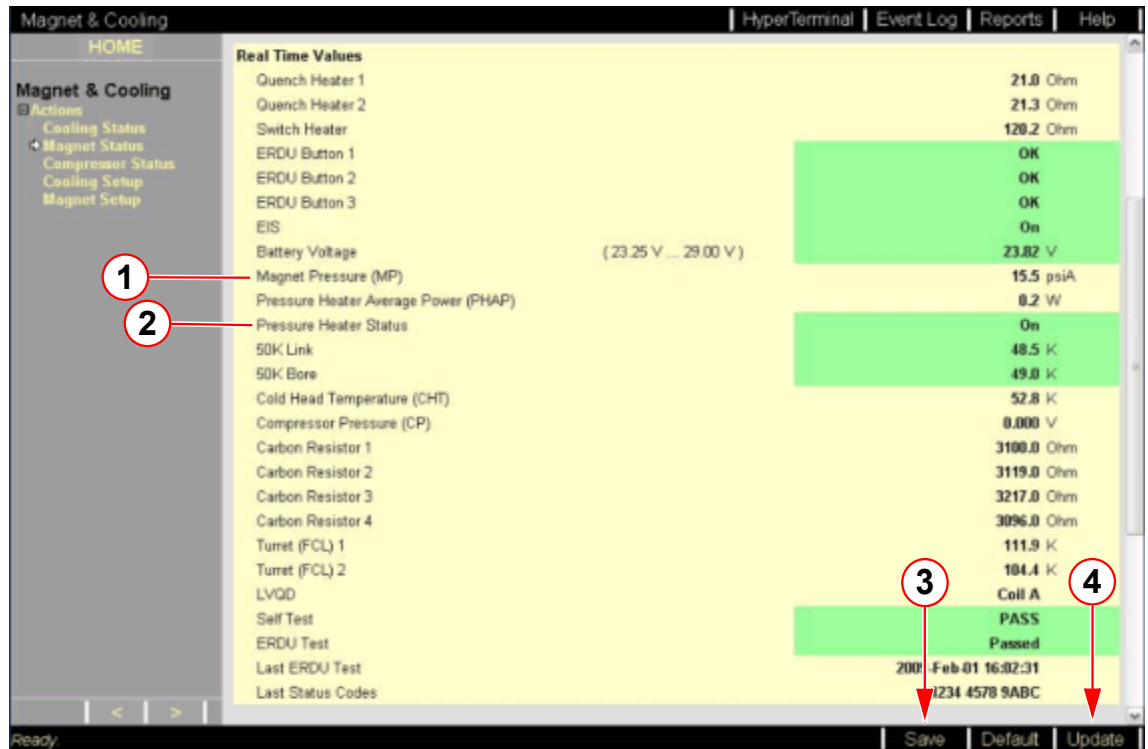


During pressurization, the cold head will switch on automatically.
If the cold head was turned off at MRC, select 'pressure heater status'.

- At MSUP Control - Set the pressure heater mode to Automatic mode.

- Select SAVE.

Fig. 139: Check magnet pressure level

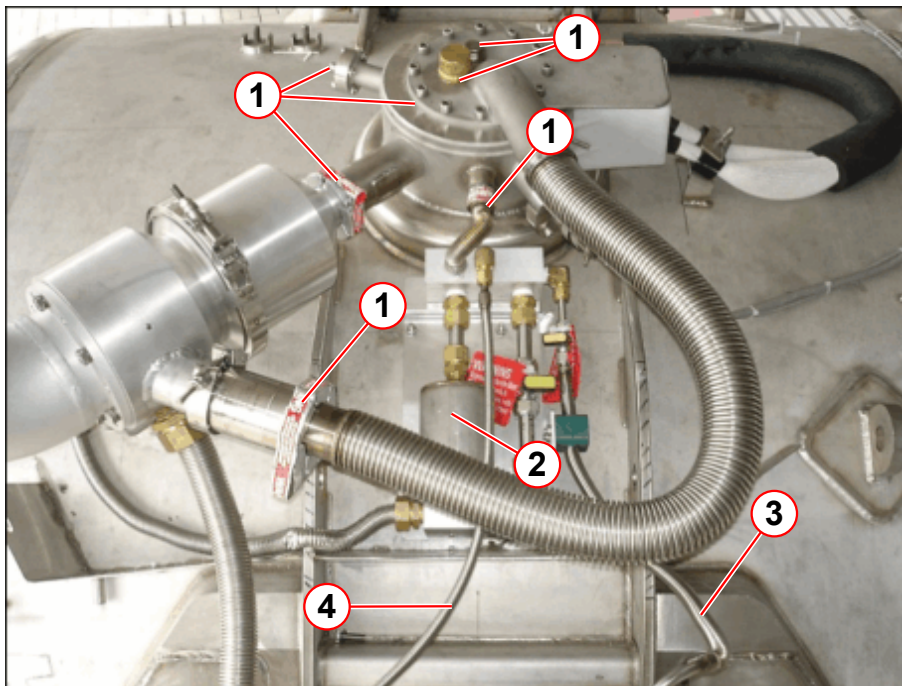


- (1) Magnet pressure (MP)
- (2) Pressure heater status
- (3) Save button
- (4) Update button

Final Checks

- Helium fill magnet to the required level for ramping the magnet.
- Order replacement components if the spare parts have been used (burst disk, seals).
- Perform a leak check of the turret venting and components.
 - » All vent pipes and flanges on the high pressure side of the helium vessel.

Fig. 140: OR105/122 - Turret and venting leak checks (Example Mk2)



- (1) Turret top plate, syphon port and O-rings
- (2) 16 psi (A) vent valve
- (3) Cold head line
- (4) Pressure gauge line



Allow a minimum of 2 hours after filling before ramping the magnet.

Quench Report

- Fill out the Quench report - MR - Quality Database and Quench Report.

TSG - Checking the Carbon Resistors

The Ceramic Carbon Resistors (CCR) values can be checked from the MSUP or manually by using a multimeter.

- The measurements are taken across pins **a-e**, **b-e**, **c-e**, **d-e** on the 41-pin turret connector.



CAUTION

Risk of asphyxiation.

Failure to adhere to safety precautions may result in serious bodily injury.

- » **The magnet must be RAMPED DOWN before attempting this procedure.**

Tooling requirements

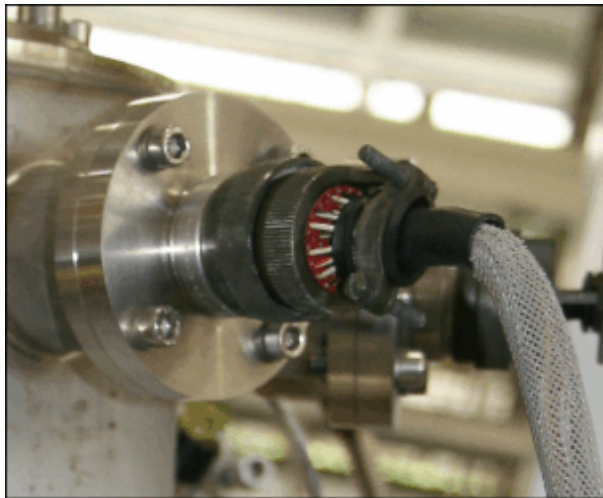
- Multimeter and pin leads.
- Laptop/PC with Terminal Emulation software, (PuTTY).
- RS232 cable.

Procedure

Using a multimeter:

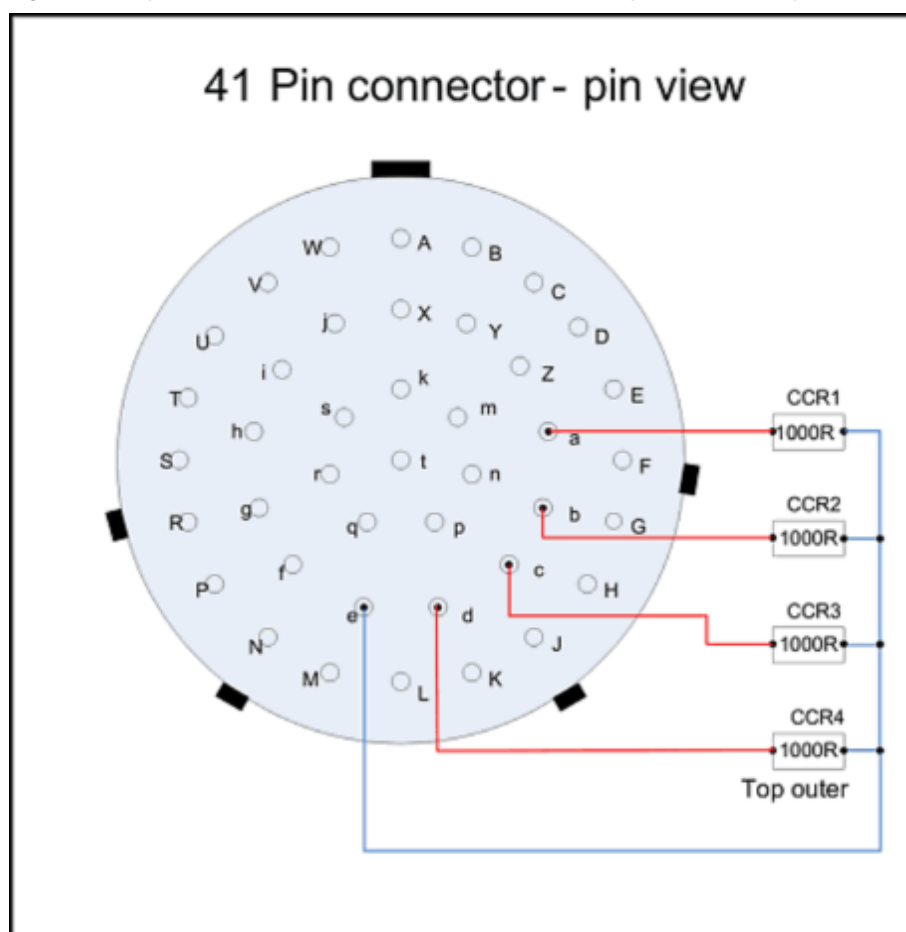
- Connect the pin leads.
- Turn the mode selector to resistor mode Ω .

Fig. 141: Turret - 41pin connector



- Disconnect the 41-pin connector.
- Measure across pins **a-e**, **b-e**, **c-e**, **d-e**
- Compare the values with the Magnet Acceptance Document.
- Re-connect the 41-pin connector.

Fig. 142: 41pin turret connector - ceramic carbon resistor pins (room temperature values)



2.1.8 What to do after a Quench (Avanto Dot Ex-factory, Espree with MSUP K2306)

2.1.8.1 General

What to do after a Quench (Avanto Dot Ex-factory, Espree with MSUP K2306)

If the magnet quenches, a large volume of helium gas is generated within the helium vessel. The quench valve opens, and stay open until the flow of gas subsides. The quench valve re-seals, preventing air (caused by cryo-pumping) from being drawn back into the magnet turret and helium vessel.

The MSUP should automatically maintain the system pressure above atmospheric.



Magnet HSC must be notified of a quench (→ Siemens Healthineers Magnet WS CS HSC, <mailto:magnet.wscshsc.team@siemens-healthineers.com>)

Fill out the **Quench Report** - (→ MR Product Information, <https://productinformation.healthcare.siemens.com/ProductInformation.aspx?uuid=7abf6bb6-5daf-440c-9920-fab2d4640084>) - Links - MR - Quality Database (Quench Report)

NOTICE

The magnet helium level can be recovered if the magnet temperature is between LN2 and He temperature.

The CCR (CARBON RES 1) reading must be in the range of between 1200 and 3400 ohms. Compare to the values on the Magnet Acceptance Document, or with HSC.

The magnet must be re-filled immediately with helium.

To re-fill the magnet from zero helium, or warm carbon resistor temperatures, perform a BOTTOM FILL procedure.

If proceeding with a helium fill, immediately place an order for 2000 liters of liquid helium.

» Contact Magnet HSC IMMEDIATELY before proceeding.

**CAUTION**

If the magnet is allowed to warm up above 77K, a full nitrogen pre-cool will be required.

Helium filling will not cool the magnet sufficiently to collect liquid helium.

A full nitrogen pre-cool takes about 10 days.

» Contact Magnet HSC IMMEDIATELY before proceeding.

Safety

Post quench procedures must be performed by CSE's that are trained to the MR2MAG-NET level. (or System level trained).



Switch off and isolate the magnet power supply from the mains before further action.

**WARNING**

Risk of asphyxiation.

Failure to observe the following safety warning may result in dizziness and loss of consciousness in personnel working with cryogens.

- » Do not vent helium gas into the magnet room during helium transfer. As an exception, a small quantity of helium gas can be vented to cool the syphon.
- » Ensure the room is ventilated.
- » Ensure that the turret venting kit is connected to the quench pipe.
- » O₂ monitors must be used in the imaging suite.

**CAUTION**

RISK OF ICING!!

Air ingress will cause INTERNAL ICING OF THE TURRET.

- » Replacement of burst disk(s) must be performed by a qualified CSE.
- » Replace broken or damaged burst disk and O-rings IMMEDIATELY.
- » Do not leave turret ports open to atmosphere.

**CAUTION**

Cold helium gas.

Personal injury may occur.

- » Protective clothing (long sleeves), cryogenic gloves and eye protection must be worn.

2.1.8.2 Post Quench Steps

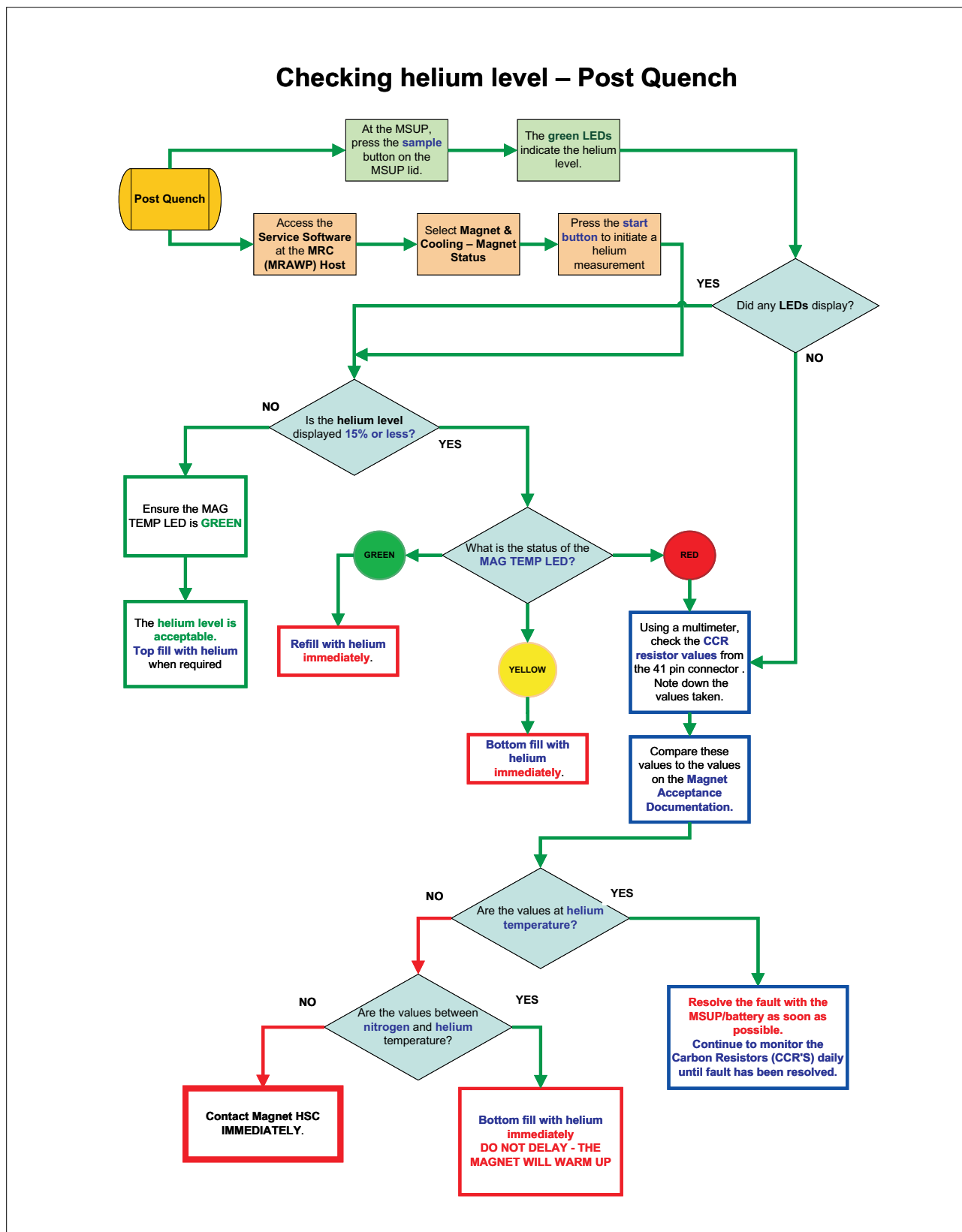
Follow the work steps in the table below for the different types of quench.

	Type of quench		
	Quench during ramping	Quench during He filling	Persistent mode quench or ERDU button pressed
Step 1	■ Isolate the MPSU from the magnet. ■ Download the 3600 MPSU ramp log files.	■ Stop filling.	■ Check that no ERDU buttons are pressed. ■ Re-set ERDU button.

	Type of quench		
	Quench during ramping	Quench during He filling	Persistent mode quench or ERDU button pressed
Step 2	- Check the liquid helium level. - Check the ceramic carbon resistor temperatures at the MRC Host. (→ Checking the Carbon Resistors (CCR) Values / Page 131)		
Step 3	Order 1000 liters of liquid helium.¹		
Step 4	- Warm up and leak check the quench valve. (→ Checking the Quench Valve Burst Disk / Page 137) - Warm up and leak check the auxiliary vent burst disk.(→ Auxiliary Burst Disk / Page 144)		
Step 5	After 1 hour : - Download the Magnet Supervisory log files. (→ Log File Capture / Page 130) - Helium fill. Fill to recommended helium ramp level.		
Step 6	Leak check the quench valve and venting. (→ Final Checks / Page 148)		
Step 7	Fill out the Quench report.		
Step 8	- Email the (→ Magnet HSC, mailto:magnet.wscshsc.team@siemens-healthineers.com) with: » 3600 Power Supply logs. » Magnet Supervisory logs and screen capture. » Magnet serial number.		
Step 9	Ramp the magnet.		
Step 10	- Perform the shim check. » If the shim check exceeds the specifications, a re-shim is necessary.		

1. The quantity of the order depends on the helium level remaining in the system after a quench, and volume already on site.

Fig. 143: Check helium level - Post Quench



2.1.8.3 Pre-requisites

Tools Required

The post quench tools required:

- Heat gun and paper toweling.
- Service tool kit.
- Laptop and RS232 cable.
- O₂ monitor.
- Personal protection equipment. To include cryogenic gloves, safety shoes, glasses and a long sleeved protective clothing.
- Helium leak detector.

Fig. 144: Edwards Leak detector (example)



- Use the Edwards hand-held leak detector (material number **10614850**).

This is the recommended equipment for leaks greater than 1×10^{-4}

NOTE: The hand-held device is calibrated in ml/sec:-

1×10^{-4} mbar l/s = 1×10^{-4} ml/sec at normal pressure and temperature.

Spare Parts

Helium leaks from the magnet pressure vessel are generally repaired by replacing a damaged seal.

- A number of seals are supplied in the spares kit. These must be replaced when used on the magnet.
- Venting components will require ordering from CSML.

Mk2 Spares required:

- Graphite Burst disk 26 psi (material number 10129362).
- O-ring seal burst disk - (material number 10100140).
- Auxiliary burst disk (material number 10121913)
- High load seal (material number 10100171).
- O-ring seal 44.5mm ID (material number 10100151).



Always replace the burst disks and seals used during replacement procedures. They are critical components for reducing the risk of air ingress. A replacement must always be available for immediate use.

Unattended Quench

Unattended quench indicators are:

- If the Magnet STOP (ERDU) has been activated:
 - » the yellow WARNING LED lights on the alarm box.
 - » the alarm buzzer is activated.
- If the helium level is low:
 - » the yellow WARNING LED lights on the alarm box.
 - » the alarm buzzer is activated.
- If the magnet pressure is very low, the
 - » cold head switches off.
 - » the yellow WARNING LED lights on the alarm box.
 - » the alarm buzzer is activated.

- Imaging is not possible if there is no magnetic field.

Press the AUDIO ALARM OFF button on the alarm box to silence any audible alarms.

Log File Capture

Magnet Supervisory

- The **quench data is lost**, if:
 - a) The system is re-booted within 1 hour after a quench
 - b) The system is turned off within 1 hour after a quench
 - c) The data is downloaded within 1 hour of after a quench
- Download the MSUP log files.

If the magnet quenches during ramping:-

- Download the Magnet Power Supply log file.



The MPSU log files **MUST** be downloaded **BEFORE** the next ramp is performed, otherwise previous data will be overwritten!

- Switch off and disconnect the MPSU from the magnet.

If the magnet quenches during helium fill:

- Stop the transfer.



Do not remove or disconnect the syphon as the magnet will be at high pressure.



For all types of quench, allow up to 1 hour for the quench pressure to subside.

Use the MSUP Display unit:

- To check that the MSUP pressure control system is working. If a display unit is not available, use the SESO interface on the Host service screens.
- If the pressure control system is NOT working, the pressure inside the cryostat will fall to atmospheric levels. The compressor switches off.

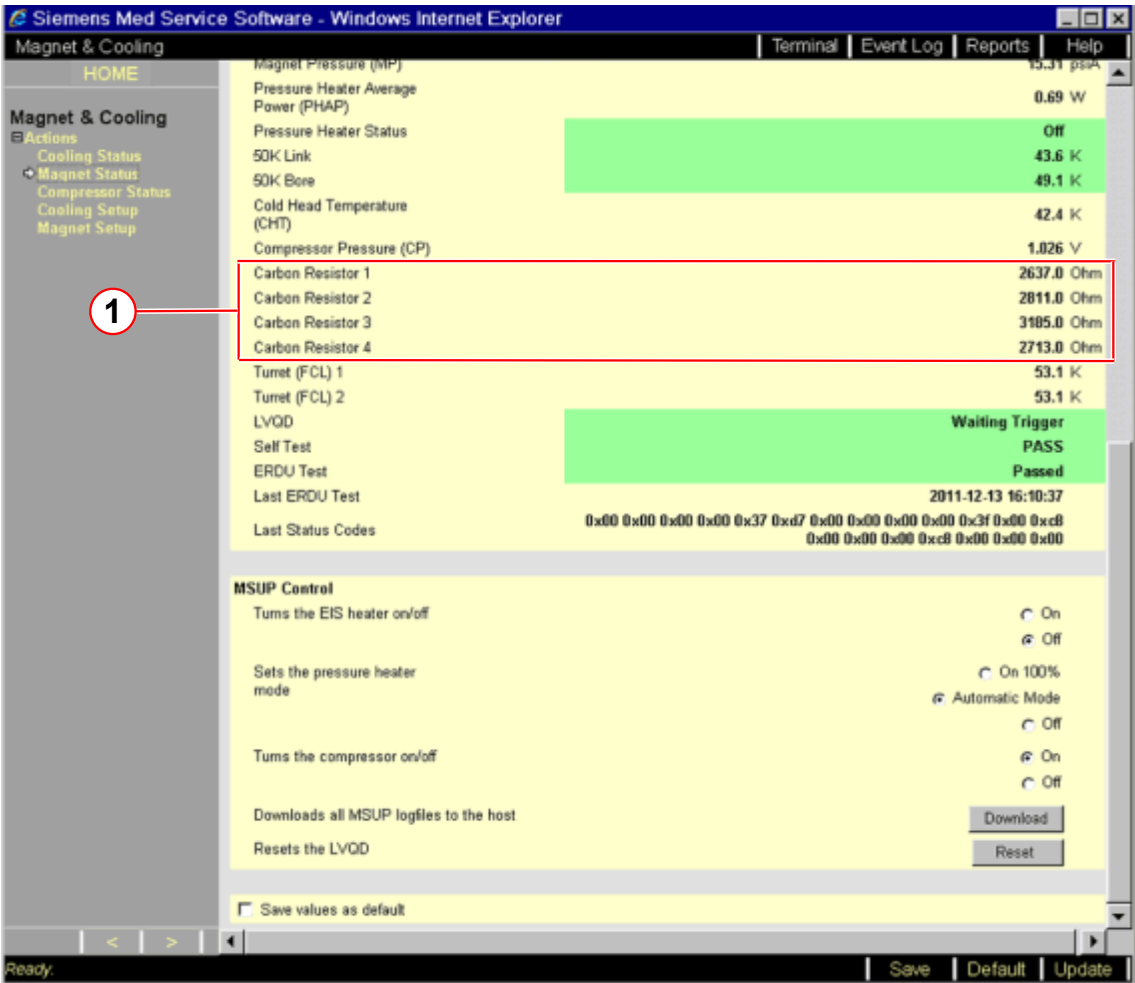
Checking the Carbon Resistors (CCR) Values

Compare the temperatures with the values on the Magnet Acceptance Document.

- Access the Service Software at the MRC (MRAWP) Host.

- Select the menu item Magnet & Cooling - Magnet Status.

Fig. 145: Checking the Ceramic Carbon Resistors



(1) Resistor values

Tab. 6 Expected CCR resistor values

41 way plug, measured in Ohms	Expected values room temperature	Expected values 77 K temperature	Expected values 4.2 K temperature
All ceramic carbon resistors (CCRs)	880 - 1050	1150 - 1350	2800 - 3380

If the host (or laptop), is not available, check CCR values directly from the pins (a-e) on the 41 pin connector on the turret.



The magnet helium level can be recovered if the temperature of the CCR is still colder than nitrogen temperature.

The CCR (CARBON RES 1) reading must be in the range of between 1200 and 3400 ohms.



If temperatures are between N2 and helium temperature (77K), a BOTTOM FILL must be performed IMMEDIATELY.

Fig. 146: OR105 Magnet Acceptance Document

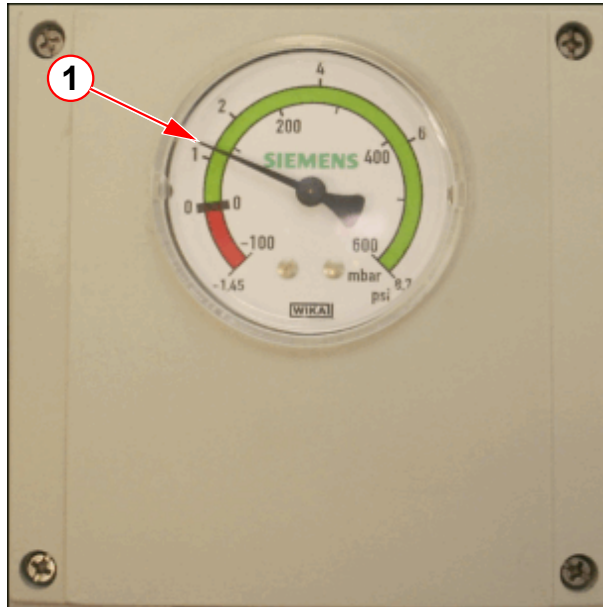
SIEMENS		Magnet Acceptance Documentation	
		Installation data	
		MR Magnet Technology Quality Regulations	
Cryostat number	21326022	SMT reference	21330107 12.11
System description	1.494T Actively Shielded Superconducting Magnet System OR105		
Measured magnet centre (Offset)	+1 (P) mm		
If magnetic offset is outside the spec ($\pm 5\text{mm}$), the magnet will be marked ($\pm 5\text{mm}$).			
 This system requires 0 Z2 rings and 2.92 kg of iron to shim. This is a theoretical calculation that may change due to site conditions.			
Operating parameters			
Description	Specification value	Measured value	
Operating current @ 1.494T	> 500 Amps	510.01 Amps	
Field/Current Ratio	N/A	12.48 kHz/0.1 Amp	
Magnet type:	OR105-01		
TDC calibration values		Carbon resistor values	
Room temperature (23 °C)	1009 Ohms	Calibration values	
Liquid nitrogen temperature (77K)	1309 Ohms		
Liquid helium temperature (4.2K)	2994 Ohms		
Position	Room temperature	Helium temperature	
TDC d - e	1034 Ohms	2837 Ohms	
Top a - e	980 Ohms	3014 Ohms	
Middle b - e	1110 Ohms	2681 Ohms	
Bottom c - e	1006 Ohms	2930 Ohms	
Helium level probe resistance (when using a 250mA supply)			
	Probe 1	Probe 2	
@ 0 resistance	85.36 Ohms	85.36 Ohms	
0% calibration	774	774	
100% resistance	0.22 Ohms	0.22 Ohms	
100% calibration	2	2	
Diode table no.	-1		
Pressure heater resistance	20.5 Ohms		
Final temperature sensor recon values			
Shield bore	Shield Link	Coldhead	Recon margin
47K	47K	42K	658mW
Checked by	V Turner		
Signed	<i>AN Other</i>		
Date	23/12/2011		
Page 1 of 1		25-AD EPS 000 01	

Check the Magnet Pressure

Check that the helium vessel has a positive pressure:

- Magnet pressure should read 15.3 psi (A) on the MSUP display unit.
- Check for positive pressure on the pressure gauge.

Fig. 147: WIKA pressure switch - expected pressure



(1) Expected pressure 1.3 psi (G)

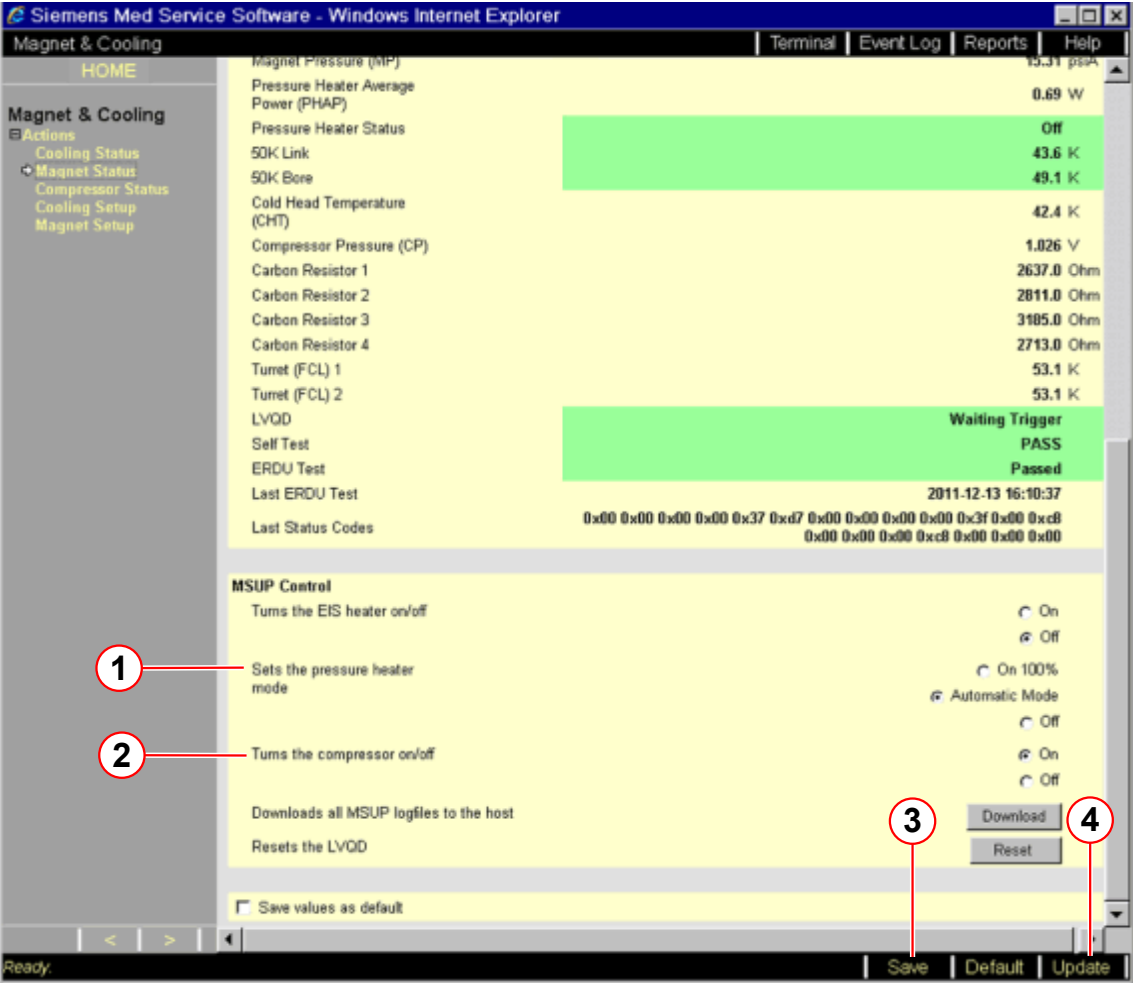
- If the pressure gauge is within the **RED zone**, the magnet is sub-atmospheric.

» **DO NOT** open the magnet bypass valve, or disturb any high pressure fitting.

Deactivate the Pressure Heater

- Access the Service Software at the MRC (MRAWP) Host.
- Select the menu item Magnet & Cooling - Magnet Status.
- At MSUP Control - Sets the pressure heater mode select the OFF button.
- Select SAVE.

Fig. 148: Service screen Mag&Cooling - Magnet Status - MSUP Control



- (1) Sets the pressure heater mode
- (2) Sets the compressor on/off
- (3) Save
- (4) Update



The pressure heater will reactivate within 60 min.

Depressurize the Helium Vessel

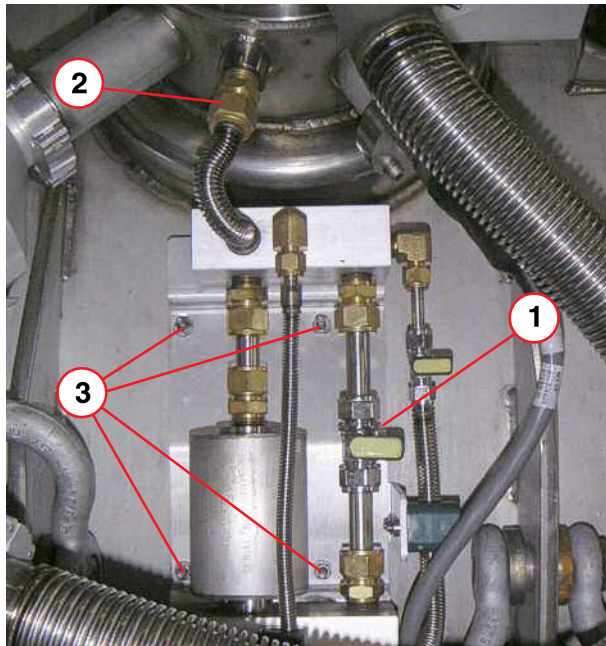
 **CAUTION**

Risk of ice formation.

The magnet bypass valve is only OPENED for depressurization of the helium vessel. The bypass valve is CLOSED during normal operation.

» To avoid air ingress, CLOSE the magnet bypass valve immediately after the helium vessel is depressurized.

Fig. 149: Mk2 Vent valve assembly

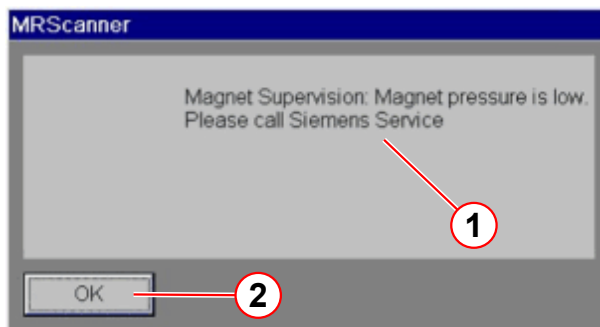


- (1) Bypass valve
- (2) Turret connection
- (3) Mounting screws (4 positions)

If depressurization of the helium vessel is required:

- Fully open the magnet bypass valve to depressurize the helium vessel.
- Wait until zero psi is displayed on the pressure gauge.
- The pressure switch will activate.
 - » the cold head will turn off.
- Close the magnet bypass valve to prevent air ingress.
- Warm up the vent valve assembly.
 - » Take care not to damage any components when using a heat gun.

Fig. 150: MSUP - Magnet pressure is low



- (1) Pop-up message
- (2) Click OK to clear

On the MR Scanner:

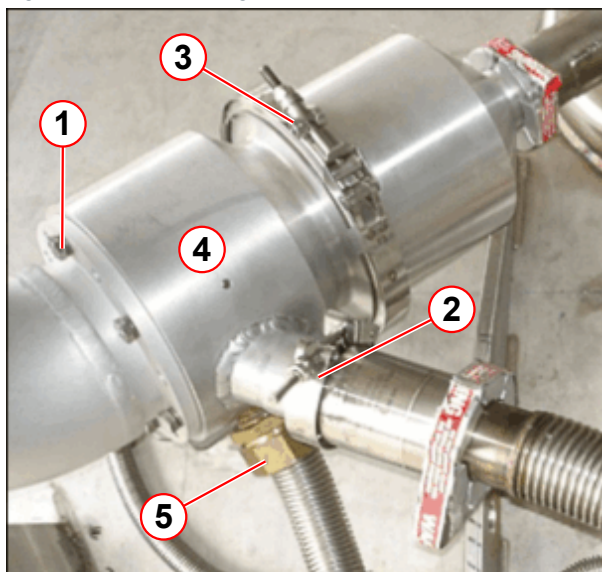
- A pop-up alarm message is displayed.
 - » Magnet pressure is low.
 - » Click OK to clear the alarm.
- The alarm box will sound when pressure falls to alarm limits.
 - » The yellow WARNING LED lights on the alarm box.
 - » The alarm buzzer is activated
 - » Press AUDIO ALARM OFF to silence the alarm.

2.1.8.4 Checking the Quench Valve Burst Disk

Post quench if there is no magnet pressure a burst disk has ruptured. Check the graphite (quench valve) burst disk.

- Disable the pressure heater by fitting the pressure service tool.
- Warm up the quench valve and venting to room temperature, using a heat gun.
- The elbow assembly and manifold must be removed for inspection and leak check of the graphite burst disk and O-ring.

Fig. 151: Mk2 Venting removal



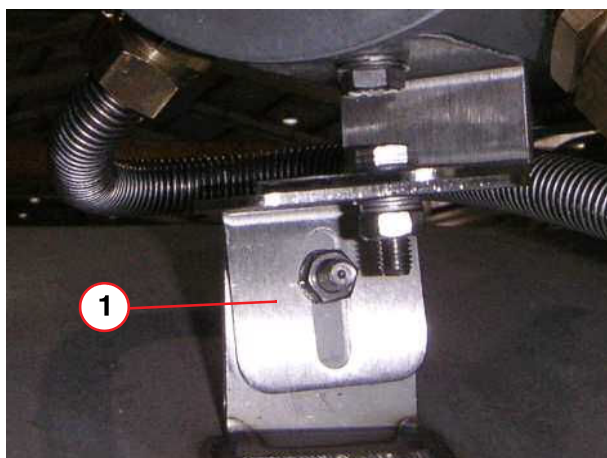
- (1) Elbow connection (M8 x 6)
- (2) V-clamp - Auxiliary vent
- (3) V-clamp manifold to quench valve
- (4) Manifold assembly
- (5) OVC vent connection

To access the quench valve and burst disk:

- Remove the 6 x M8 screws retaining the elbow assembly to the manifold (pos.1).
- Remove the V-clamp retaining the auxiliary vent to the manifold (pos. 2).
- Remove the V-clamp retaining the manifold to the quench valve (pos. 3).

NOTE: it should not be necessary to remove the OVC vent line connection (pos.5).

Fig. 152: Mk2 Support bracket

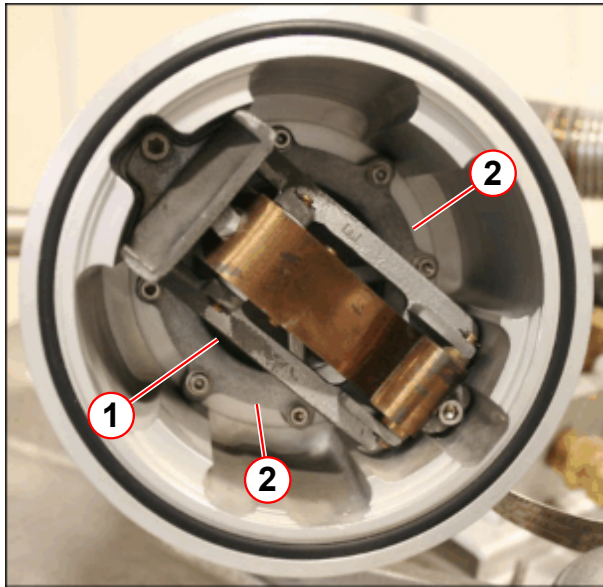


- (1) Support bracket

- Loosen the support bracket (1) screw underneath the quench valve interface assembly.
- Move the manifold assembly away from the quench valve.

The quench valve and graphite burst disk can now be accessed to check for leaks.

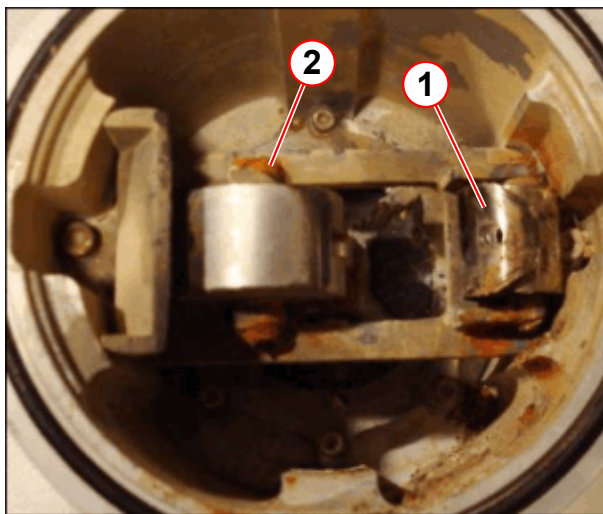
Fig. 153: Leak check



- (1) Graphite burst disk
- (2) Burst disk O-ring seal

- Check the graphite burst disk for breakage or cracks.
- If the graphite burst disk is broken or cracked, it must be replaced **immediately**.

Fig. 154: Water corrosion (example)



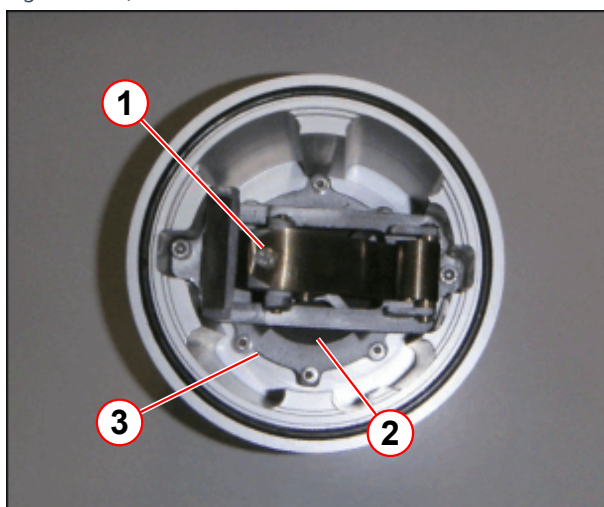
- (1) Spring broken
- (2) Corroded screws

Check for ice and water corrosion or breakage of:

- The spring plate assembly.
- All screws.
- Corroded quench valve components must be replaced.

- With no damage to the burst disk or spring plate visible, continue to leak check around the quench valve O-ring and retaining plate.
- Check that the magnet pressure is a minimum of 0.5 psi (G) for leak checking.
- Leak check around the burst disk and spring plate assembly.

Fig. 155: Quench Valve leak check

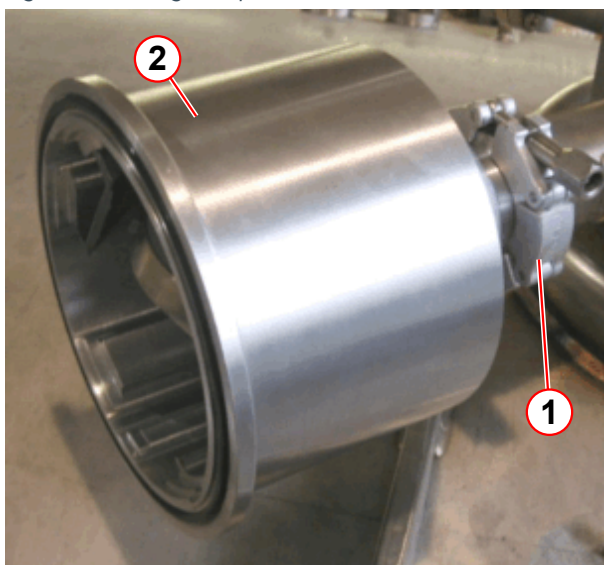


- (1) Spring plate assembly
- (2) Graphite burst disk
- (3) O-ring seal

- If there is no leak found the quench valve O-ring seal behind the spring plate is OK.
- There is no requirement to remove the burst disk spring plate.
- Continue to check the auxiliary burst disk and refit the quench valve.
- If a leak is detected the burst disk spring plate assembly will need removing to replace the O-ring.

Replacing the Burst Disk and O-ring Seal

Fig. 156: Fitting the quench valve



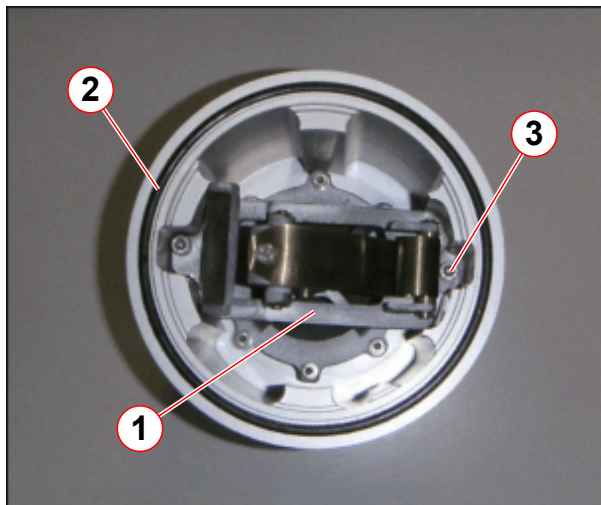
- (1) NW40 seal and clamp
- (2) Quench valve assembly

- Open the magnet bypass valve to depressurize (if necessary).
- De-ice the turret flange.
- Mark the orientation of the quench valve
- Remove the NW40 (high load) turret clamp and seal.
- Remove the quench valve from the turret flange.
- To minimize air ingress, immediately place a ball of paper toweling into the open turret port.
- Close the magnet bypass valve.



The spring exerts significant force on these screws.

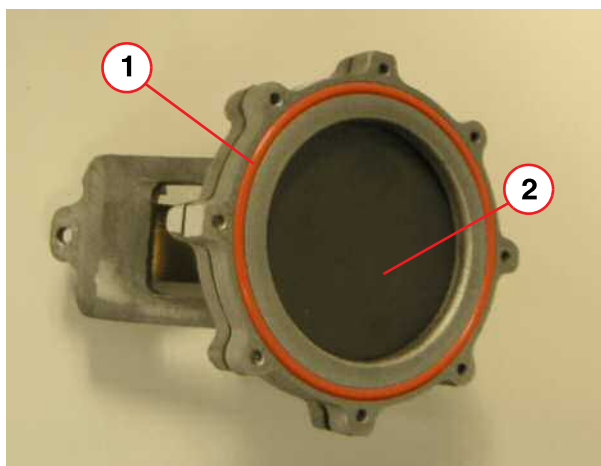
Fig. 157: Quench Valve



- (1) Spring plate assembly
- (2) O-ring
- (3) M5 cap screws (2 off)

- Unscrew and remove the (2 off) M5 cap head screws.
- Lift out the burst disk assembly from the quench valve body.

Fig. 158: Quench Valve Bursting Disk Assembly



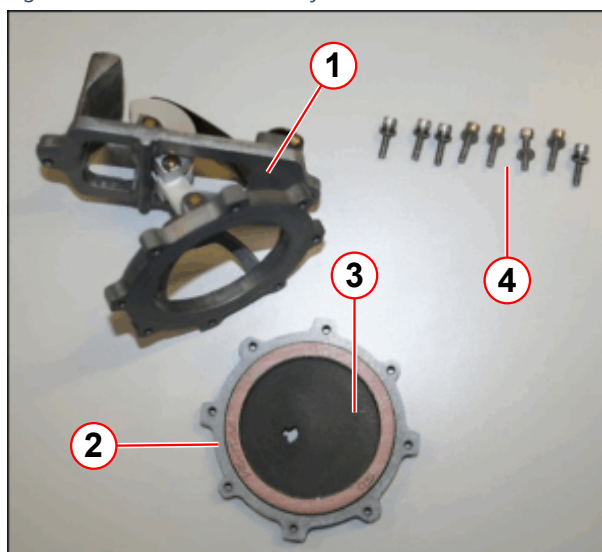
- (1) O-ring
- (2) Burst disk

- Inspect the burst disk 'O-ring and replace if damaged.
- Clean the O-ring groove and O-ring before re-fitting.
- Check that the O-ring is seated evenly around the groove.



The O-ring must be replaced each time the quench valve is removed. DO NOT apply grease to the O-ring.

Fig. 159: Burst disk assembly

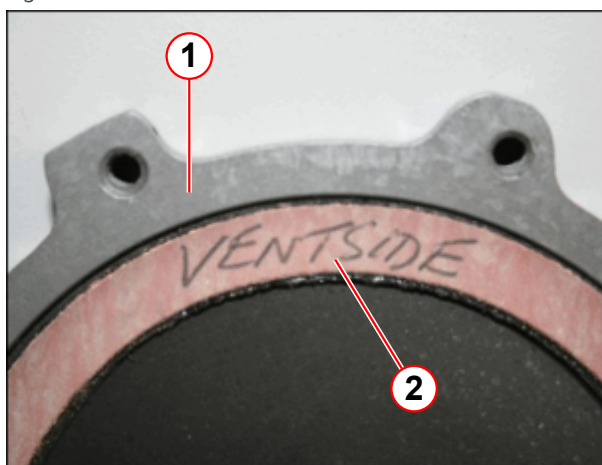


- (1) Spring plate assembly
- (2) Disk housing
- (3) Broken disk
- (4) M4 screws (8 off)

To replace the burst disk:

- Unscrew and remove the 8 x M4 cap screws.
- Remove the broken burst disk and discard.
- Check that the new burst disk and flange faces are clean and undamaged.

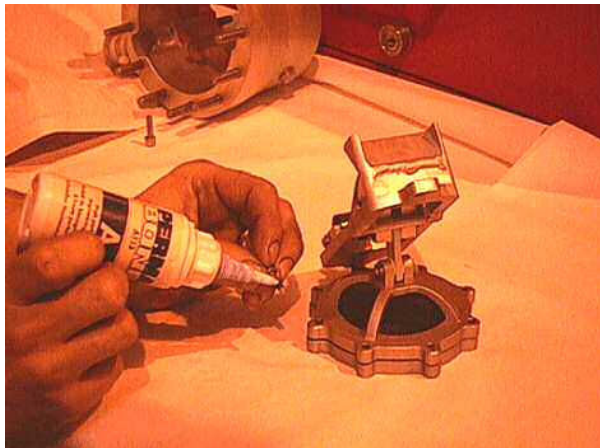
Fig. 160: Burst disk fitment



- (1) Burst disk housing
- (2) Burst disk (Vent side uppermost)

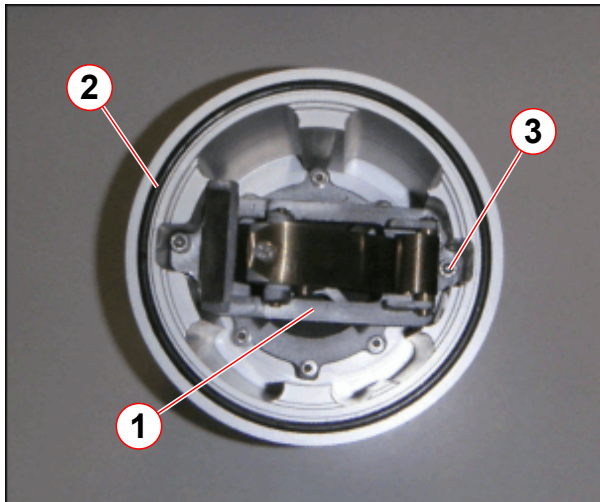
- Fit the replacement disk.
- The words '**Vent Side**' (written on the burst disk gasket) must be facing the spring plate assembly

Fig. 161: Thread locking compound



- Fit the burst disk assembly. (Do not apply grease).
- The (8 off) M4 cap screws and plain washers should be locked with threadlock compound..
- Diametrically torque the screws evenly to 0.8 Nm (0.6 ft. lbs).

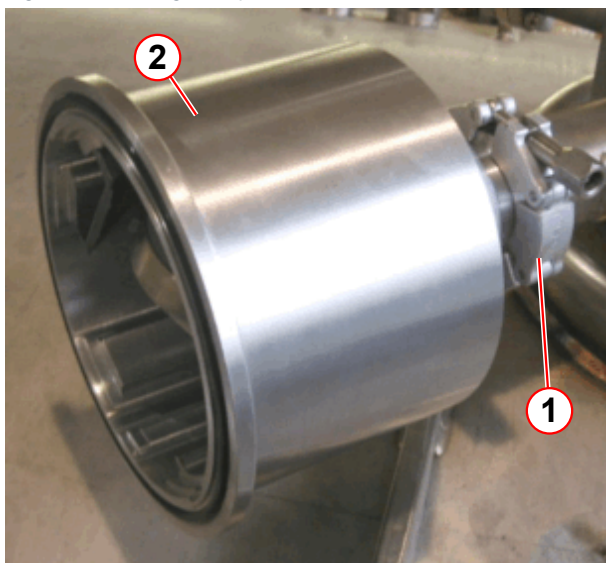
Fig. 162: Quench Valve



- (1) Spring plate assembly
- (2) O-ring
- (3) M5 cap screws (2 off)

- Re-fit the burst disk assembly into the quench valve body.
- Torque the 2 x M5 cap hd screws to 5 Nm. (3.8 ft.lbs).
- Check the quench valve manifold O-ring is in good condition (pos.2).
- Replace if damaged.

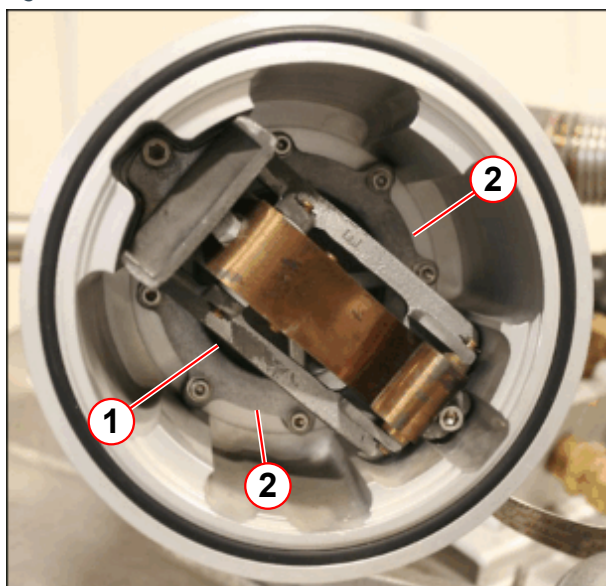
Fig. 163: Fitting the quench valve



- (1) NW40 seal and clamp
(2) Quench valve assembly

- **Remove the paper toweling** if used (or NW40 blank) from the turret port. Ensure the surfaces are clean and dry.
- Fit the quench valve assembly to the turret flange.
 - » Re-fit using a new NW40 high load seal (material number 10100171).
 - » Check the correct orientation.
 - » Warm with heat gun when tightening the NW40 clamp.

Fig. 164: Leak check



- (1) Graphite burst disk
(2) Burst disk O-ring seal

- Check that the magnet bypass valve is closed and pressurize the system.
- Leak check the burst disk and O-ring seal.
- Leak check the NW40 clamp and seal at the turret port.
- If no leaks are detected:
 - » Fit the quench valve elbow interface with the V-clamp to the quench valve.
 - » Fit the support bracket screw to the elbow interface assembly.
 - » Check the auxiliary burst disk.

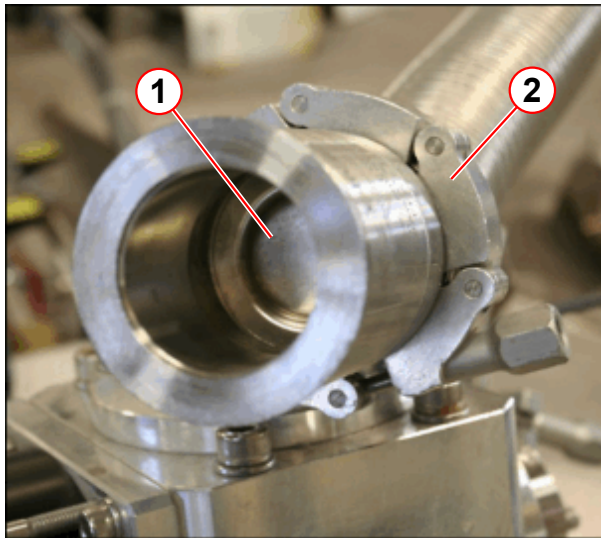
2.1.8.5 Auxiliary Burst Disk

Post quench the auxiliary burst disk vent line is removed when disconnecting the venting to leak check the quench valve.



Do not touch the metal membrane, it is very fragile.

Fig. 165: Auxiliary burst disk check



- (1) Burst disk
- (2) High load seal

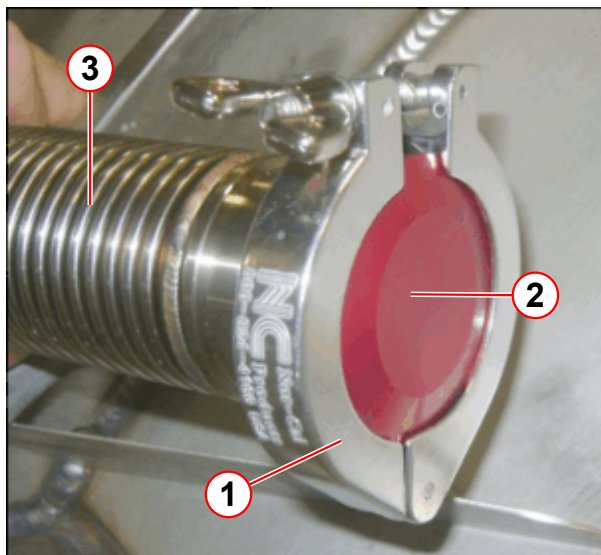
- Visually inspect the internal metal membrane of the burst disk for damage (pos.1).
- Leak check the high load seal (pos. 2).

**WARNING****Safety device removed**

The magnet must not be ramped with the auxiliary burst disk removed or broken.

» Replace the auxiliary vent burst disk before ramping.

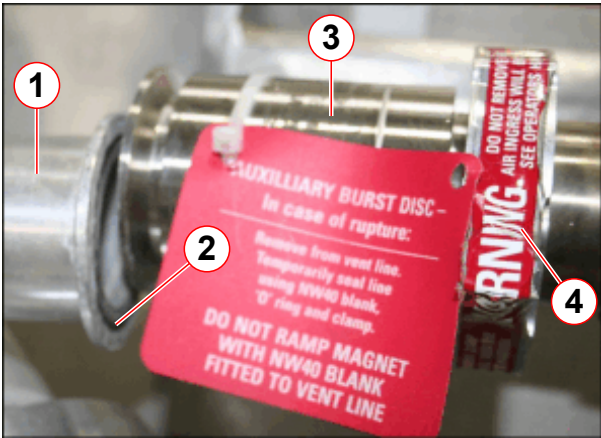
Fig. 166: Auxiliary vent - NW40 blanking plate



- (1) NW40 clamp
- (2) NW40 red blanking plate
- (3) Auxiliary vent line

- If the membrane has ruptured, disconnect the high load clamp and remove the burst disk from the vent line.
- Fit the red anodized blank immediately from the spares kit.
- Order a replacement auxiliary burst disk. Fit as soon as available.
- With the blank fitted allow magnet to pressurize.
- Leak check red blanking plate and seal on the end of the auxiliary vent line.

Fig. 167: Auxiliary vent interface - O-ring seal



- (1) Quench valve manifold
- (2) O-ring seal
- (3) Auxiliary bursting disk
- (4) NW40 clamp (high pressure)

- Check that the O-ring seal (Pos. 2) is not damaged and that the surface is clean.
- If damaged, replace the O-ring.
- Connect the auxiliary burst disk to the manifold assembly with the V-clamp.

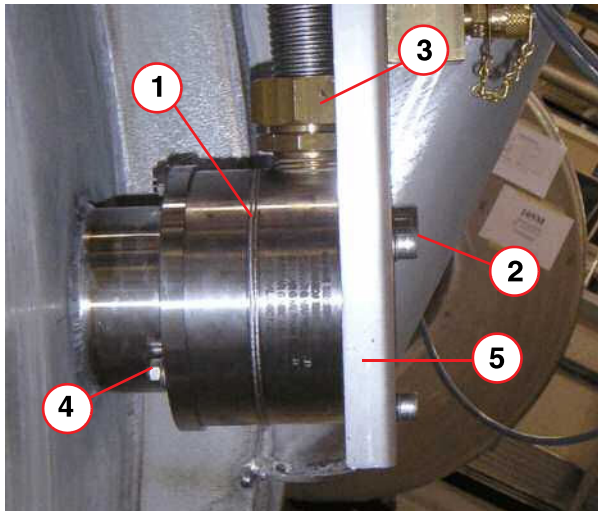
2.1.8.6 OVC Burst Disk

The OVC (DOP) port burst disk and vent line has been removed from the Mk2 magnet design. See Table for system serial number change.

Tab. 7 Venting configurations

Venting	OR105 Avanto serial #	OR122 Espree serial #
Mk2	From System MR25575	From System MR30125
Mk 2 (No DOP)	From System MR57232	From System MR31427

Fig. 168: Mk2 OVC Burst Disk Assy



- (1) OVC burst disk
- (2) Cap head screw (3 off)
- (3) OVC vent sealing nut
- (4) Nut (3 off)
- (5) Pressure gauge plate

OVC burst disk

A quench will not cause the OVC metal burst disk to rupture. There is no requirement to check the condition of the OVC burst disk 'Post quench'.

2.1.8.7 Final Work Steps

Reactivate the Pressure Heater

- Access the Service Software at the MRC (MRAWP) Host.
- Select the menu item Magnet & Cooling - Magnet Status.
- At MSUP Control - Set the pressure heater mode to **On 100%**.
- Select SAVE.
- Monitor the magnet pressure.
 - Select Update to refresh data until pressure reaches ≥ 15.3 psi (A).



Data is only accurate from the last refresh/update of the page, or from when the page was opened last.

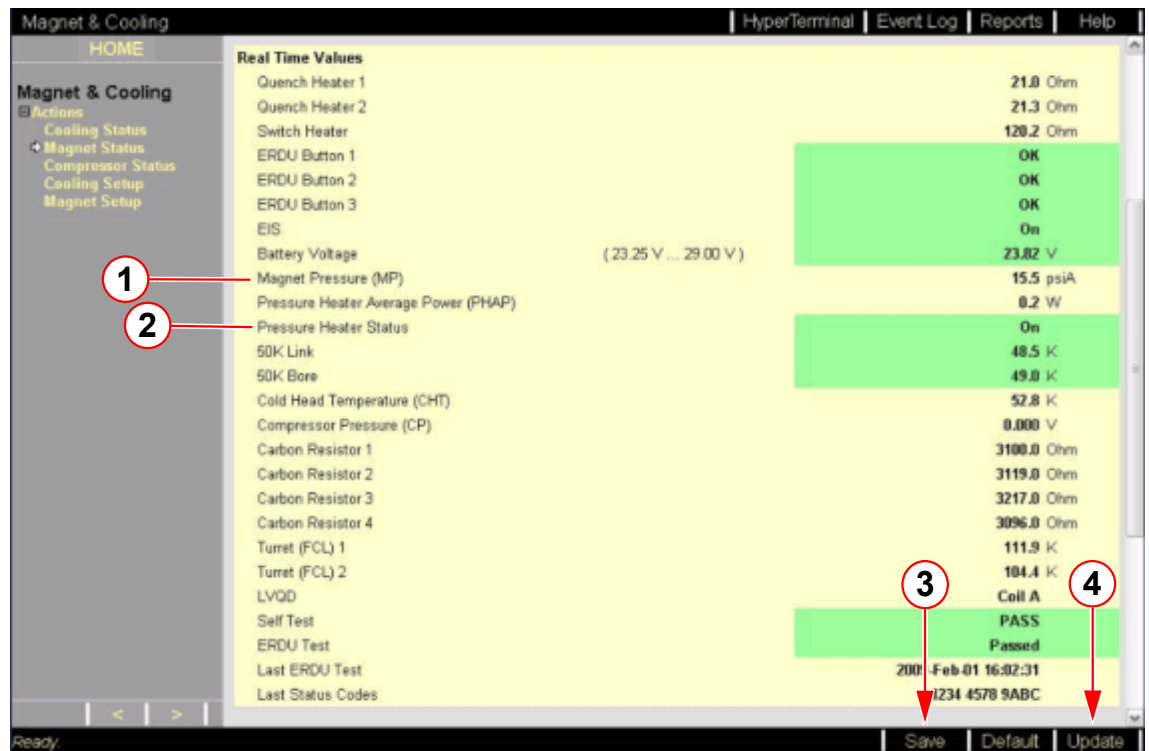


During pressurization, the cold head will switch on automatically.
If the cold head was turned off at MRC, select 'pressure heater status'.

- At MSUP Control - Set the pressure heater mode to Automatic mode.

- Select SAVE.

Fig. 169: Check magnet pressure level

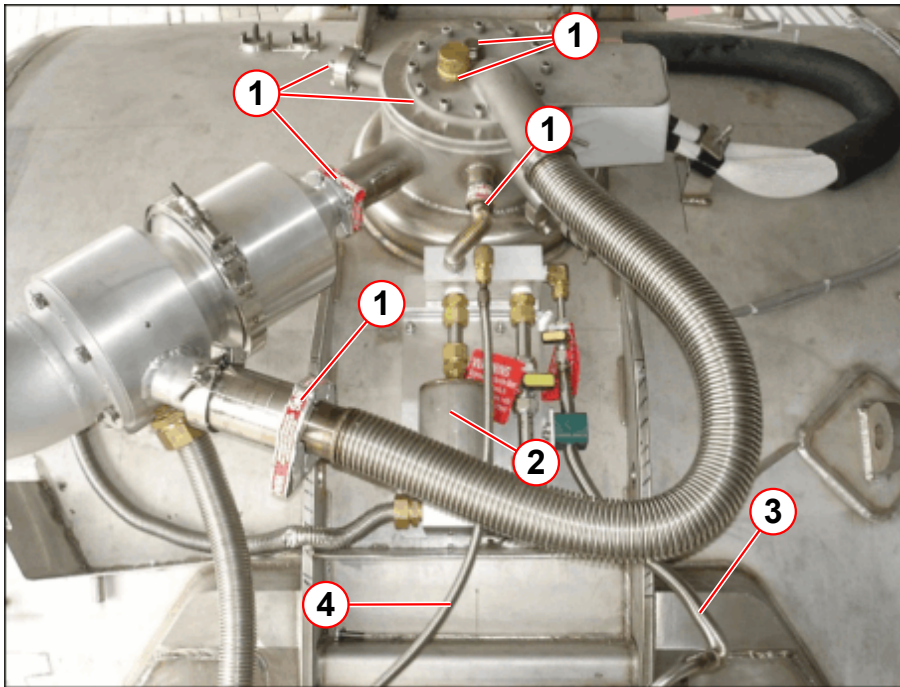


- (1) Magnet pressure (MP)
- (2) Pressure heater status
- (3) Save button
- (4) Update button

Final Checks

- Helium fill magnet to the required level for ramping the magnet.
- Order replacement components if the spare parts have been used (burst disk, seals).
- Perform a leak check of the turret venting and components.
 - » All vent pipes and flanges on the high pressure side of the helium vessel.

Fig. 170: OR105/122 - Turret and venting leak checks (Example Mk2)



- (1) Turret top plate, syphon port and O-rings
- (2) 16 psi (A) vent valve
- (3) Cold head line
- (4) Pressure gauge line



Allow a minimum of 2 hours after filling before ramping the magnet.

Quench Report

- Fill out the Quench report - MR - Quality Database and Quench Report.

TSG - Checking the Carbon Resistors

The Ceramic Carbon Resistors (CCR) values can be checked from the MSUP or manually by using a multimeter.

- The measurements are taken across pins **a-e**, **b-e**, **c-e**, **d-e** on the 41-pin turret connector.



CAUTION

Risk of asphyxiation.

Failure to adhere to safety precautions may result in serious bodily injury.

- » **The magnet must be RAMPED DOWN before attempting this procedure.**

Tooling requirements

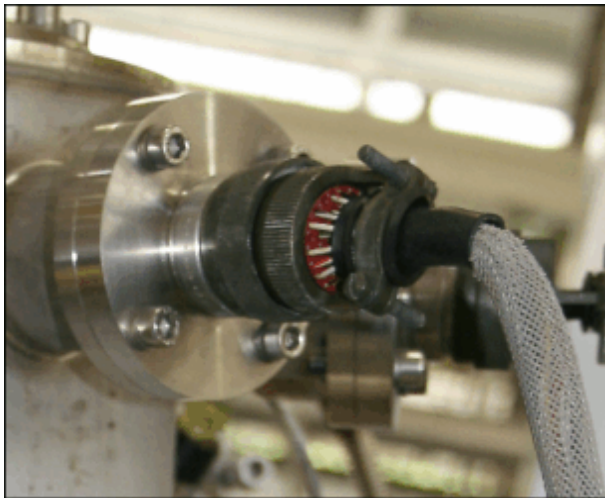
- Multimeter and pin leads.
- Laptop/PC with Terminal Emulation software, (PuTTY).
- RS232 cable.

Procedure

Using a multimeter:

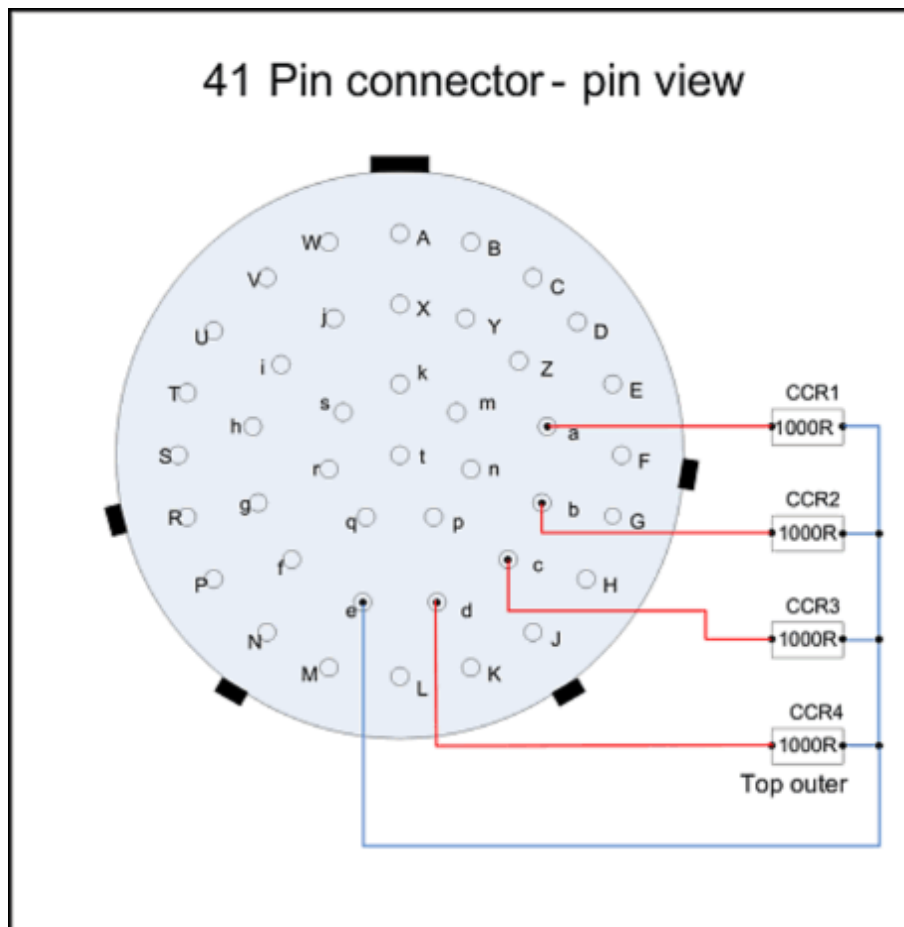
- Connect the pin leads.
- Turn the mode selector to resistor mode Ω .

Fig. 171: Turret - 41pin connector



- Disconnect the 41-pin connector.
- Measure across pins **a-e**, **b-e**, **c-e**, **d-e**
- Compare the values with the Magnet Acceptance Document.
- Re-connect the 41-pin connector.

Fig. 172: 41pin turret connector - ceramic carbon resistor pins (room temperature values)



2.2 MSUP 2306 download instructions



Ensure a safe distance from the magnet for troubleshooting and service actions which require the use of a laptop with extended cabling.

Log files can be downloaded via either:

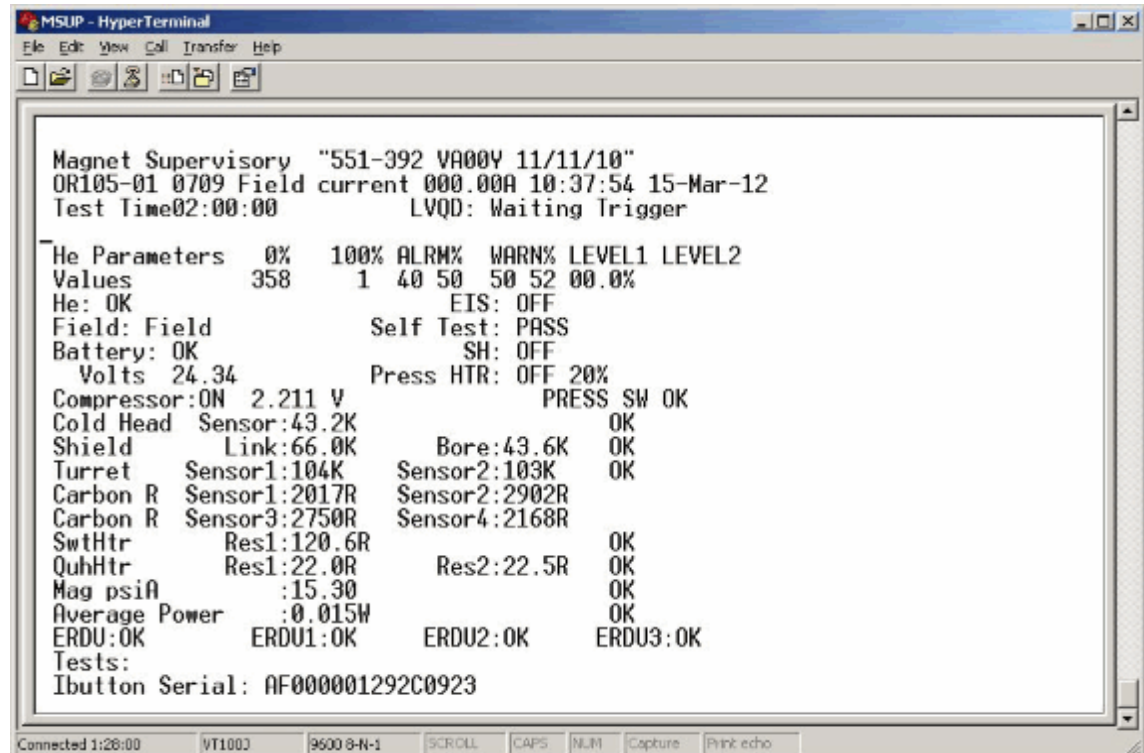
- **Direct connection** to the MSUP via RS232 (**X6**) connection.
- Service software at the **MRC (MRAWP) Host computer**.

2.2.1 Downloading via the RS232 connection

1. Turn on the **laptop** and open the **MSUP Terminal Emulation Software**.
 - » Set bits per second to 9600
 - » Set data bits to 8
 - » Set parity to 1
 - » Set flow control to none
 - » Set Emulation to VT100 (File: Properties:Settings)
2. **Connect** the laptop to connector **X6** on the MSUP **via RS232 cable**.
3. **Press ESC**.
4. Type **L0 6932** <enter>.
5. Type **R** <enter>

6. Press **ESC** to display the **MSUP Process display screen**.

Fig. 173: MSUP Process display screen (Example)



7. **Capture the screenshot.** (→ Capturing a screenshot / Page 157)
8. **Press ESC** to display the **MSUP Command input screen**.
9. Click **Transfer** on the **MSUP Terminal Emulation Screen** tool bar.
10. Click **Capture text**.
11. Click **Browse** and select a directory and a name for the file to be stored. The **Name** should contain the **magnet serial number**.
 - e.g. C:\templmsn 12345678log.txt
12. Click **Start**.
13. Type the following:
 - **VA** <enter>
 - **VQ** <enter>
 - **VM** <enter>
 - **VP** <enter>
 - **VX** <enter>
14. Wait until data has been captured.
15. Click **Transfer, capture text, Stop**.
16. Type **R** <enter>.
17. **Email** the saved files to (→ Magnet HSC, <mailto:magnet.wscshsc.team@siemens-healthineers.com>)



After the download is complete, the MSUP must be reset.

This can be done via

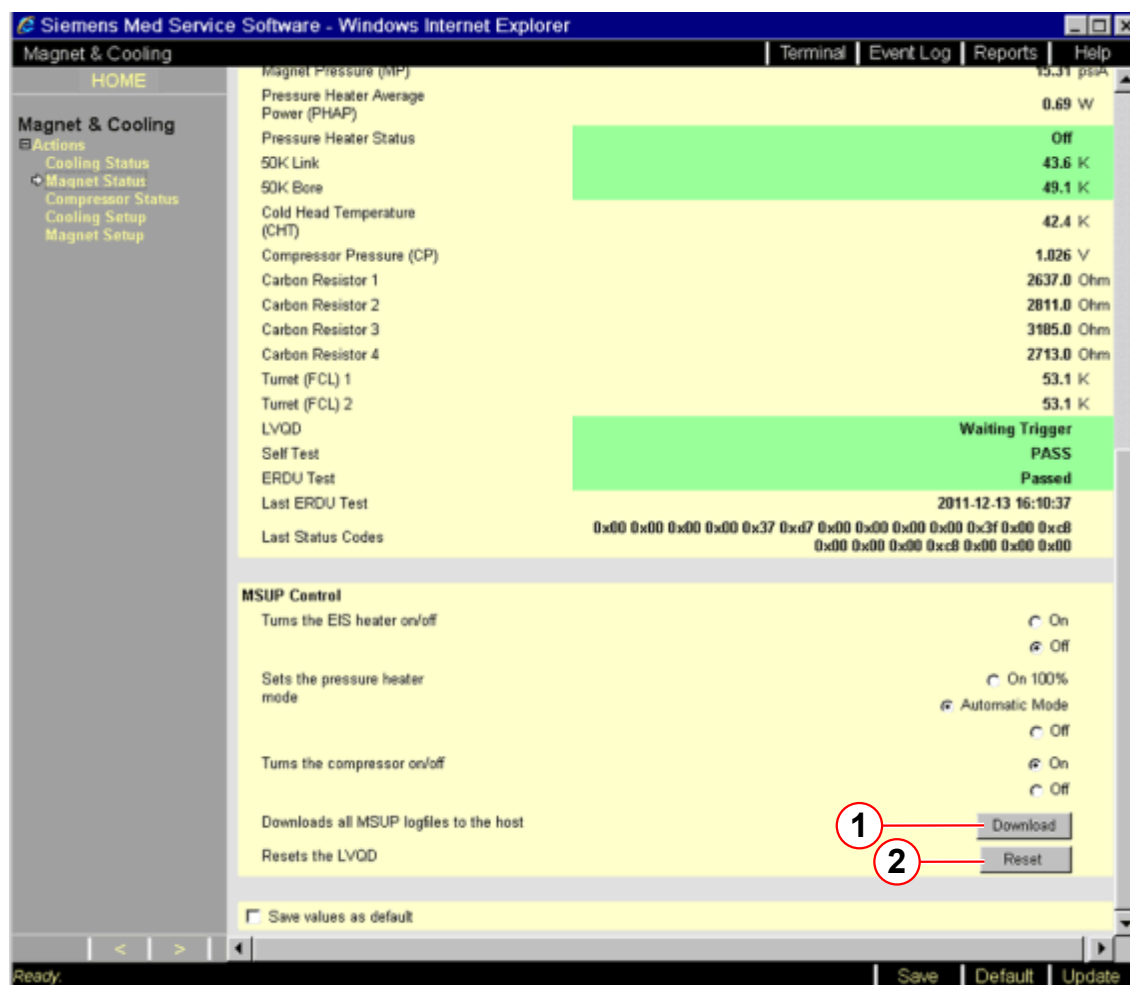
- rebooting the MRC (MRAWP) Host.
- cycling the power to the MSUP.
- allow the MSUP to complete an overnight sample.

2.2.2 Downloading the MSUP 2306 log file data to the MRC via SESO

Access the **Service software** at the **MRC (MRAWP) Host**

- Select menu item **"Magnet & Cooling."**
- Select **"Magnet Status"** in the **"Action"** menu.
- Click **"Download"** to download all the **MSUP logfiles** to the **MRC**.
 - » logfiles are **saved** to the MRC 'C' Drive, under **C:\MEDCOM.LOG** (→ Fig. 175 Page 155)

Fig. 174: Downloading MSUP logfiles to the Host

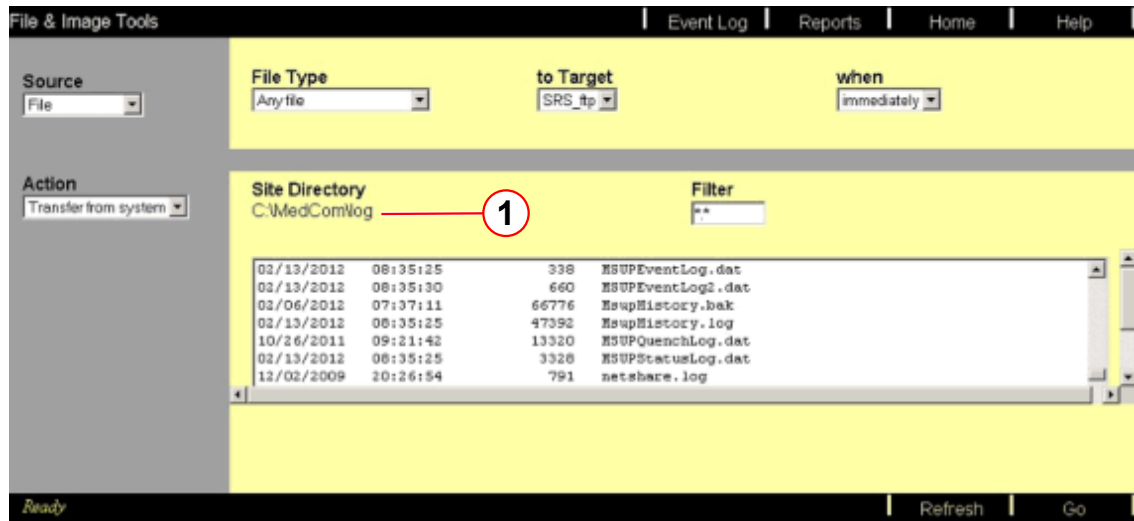


(1) Download button

(2) Reset button

- Click **"Reset"** (pos 2).

Fig. 175: Host MRC MSUP logfiles location



(1) Site Directory

- **Email** the downloaded log files to (→ Magnet HSC, <mailto:magnet.wscshsc.team@siemens-healthineers.com>).

2.2.3 Capturing the iButton menu on the MSUP using a laptop

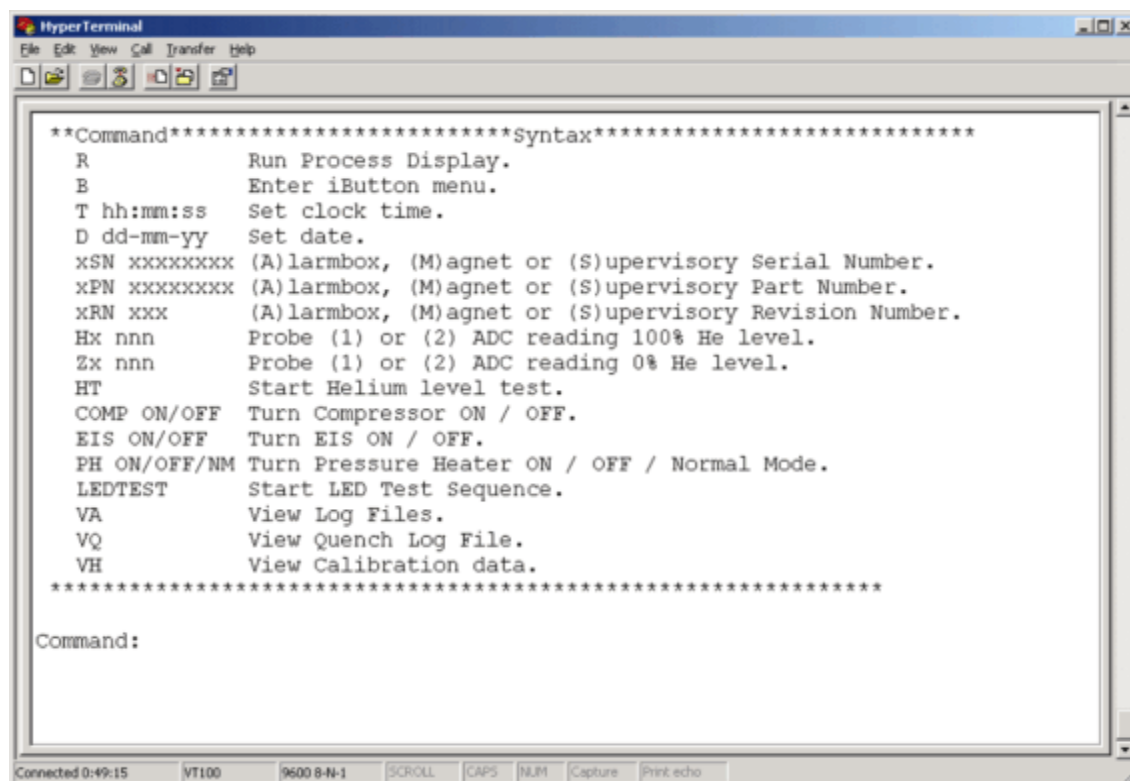


Ensure a safe distance from the magnet for troubleshooting and service actions which require the use of a laptop with extended cabling.

1. Turn on the laptop and open the **MSUP Terminal Emulation Software**.
 - » Set bits per second to 9600
 - » Set data bits to 8
 - » Set parity to 1
 - » Set flow control to none
 - » Set Emulation to VT100 (File:Properties:Settings)
2. **Connect** the laptop to connector **X6** on the MSUP via **RS232 cable**.
3. Press **ESC**.
4. Type **L0 6932** <enter>.
5. Type **R** <enter>.

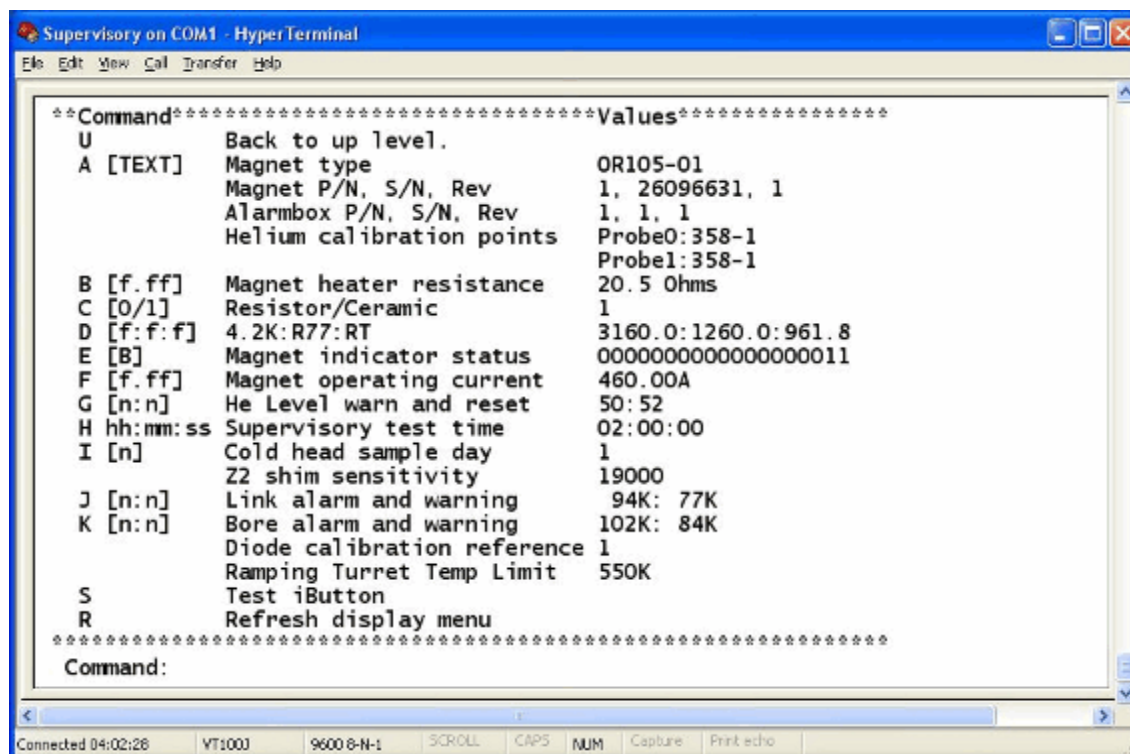
6. Press **ESC** to display the **MSUP Command input screen**.

Fig. 176: MSUP Command Input screen (example)



7. Type **B<enter>** this displays the **iButton menu**.

Fig. 177: MSUP 2306 iButton menu (example)



8. **Capture the screenshot.**(→ Capturing a screenshot / Page 157)

9. Type **U** <enter>.
10. Type **R** <enter>.
11. **Email** the saved files to (→ Magnet HSC, <mailto:magnet.wscshsc.team@siemens-healthineers.com>).

2.2.3.1 Capturing a screenshot

- Capture a **screenshot** of the **MSUP Terminal Emulation Screen**:
 - a) Press **Ctrl:Alt:Prt Sc**.
 - b) Go to: **Start:Programs:Accessories**:and open Paint.
 - c) Press **Ctrl:V** to paste screenshot.
 - d) **Name** and save as a **.jpg** or **.bmp**.
Name the file DD (**Day**), MM (**Month**), YYYY (**Year**), TTTT (**Time:24hour format**).
i.e. iButtonDDMMYYYYTTTT: MSUPDDMMYYYYTTTT.
- E-mail the saved file to (→ Magnet HSC, <mailto:magnet.wscshsc.team@siemens-healthineers.com>).

2.2.4 Restoring system calibration parameters

2.2.4.1 Downloading MSUP 2306 data to the MRC

The **MRC (MRAWP) Host** must be updated with the **system parameters**.

Check the **serial/revision/part numbers** are correct for the **Alarm box, MSUP 2306, and Magnet**.

Access the Service software at the MRC (MRAWP) Host.

- Select menu item **"Magnet & Cooling."**

- Select **"Magnet Setup"**.

Fig. 178: MSUP 2306 - Entering system calibration parameters

Siemens Med Service Software - Windows Internet Explorer

Magnet & Cooling | Terminal | Event Log | Reports | Help

HOME

Magnet & Cooling

- Actions
- Cooling Status
- Magnet Status
- Compressor Status
- Cooling Setup
- Magnet Setup**

Magnet & Cooling - Magnet Setup

Magnet

Helium Level Probe 1	Min. (0%)	85.5	Ohm
	Max. (100%)	0.2	Ohm
Helium Level Probe 2	Min. (0%)	85.5	Ohm
	Max. (100%)	0.2	Ohm
Helium Level Probe Selection		☒ Probe 1 ☐ Probe 2 ☐ Both Probes	
Helium Level Warning	Min.	50	%
Shield Temperature Warning 50K Link	Max.	77.0	K
Shield Temperature Warning 50K Bore	Max.	84.0	K
CCR Calibration	Room Temperature	980.0	Ohm
	Nitrogen Temperature	1259.0	Ohm
	Helium Temperature	2772.0	Ohm
Pressure Heater Resistance		20.5	Ohm
Time Setting	Daily at	02:00	O'Clock
MSUP	Serial Number	1599	
	Revision Number	7	
	Part Number	10523587	
Alarmbox	Serial Number	1508	
	Revision Number	3	
	Part Number	10523563	
Magnet	Magnet Type	OR105-01	
	Serial Number	21259456	
	Revision Number	43	
	Part Number	7758167	
Magnet Current		509.87	A

MSUP Data

Downloads the magnet setup/calibration data to the host

Uploads the magnet setup/calibration data from the host to the MSUP

☐ Save values as default

Ready. | Save | Default | Update

- (1) Download button (MSUP data stored on MRC)
- (2) Save button - (transfer calibration values to MSUP)
- (3) Update button (refreshes on-screen data)

- **Compare** the values with the **Magnet Acceptance Document**. (→ Fig. 179 Page 160)
- **Correct** MSUP parameters as necessary. Compare **acceptance data** to the **serial data** captured **from the MSUP**.

- Click **"UPDATE"**
 - » This **refreshes** the on-screen values.(→ 3/Fig. 178 Page 158)
- Update the **serial/revision/part** numbers (if required).
- Click **"SAVE"** to transfer calibration values **to the MSUP**. (→ 2/Fig. 178 Page 158)
- Click **"DOWNLOAD"**
 - » This takes the **MSUP data** and stores it in a file on the **MRC**.
(→ 1/Fig. 178 Page 158)

Fig. 179: OR105 Magnet Acceptance Document

SIEMENS		Magnet Acceptance Documentation Installation data MR Magnet Technology Quality Regulations	
Cryostat number	21326022	SMT reference	21330107 12.11
System description	1.494T Actively Shielded Superconducting Magnet System OR105		
Measured magnet centre (Offset)	+1 (P) mm		
If magnetic offset is outside the spec (± 5mm), the magnet will be marked (± 5mm).			
 This system requires 0 Z2 rings and 2.92 kg of iron to shim. This is a theoretical calculation that may change due to site conditions.			
Operating parameters			
Description	Specification value	Measured value	
Operating current @ 1.494T	> 500 Amps	510.01 Amps	
Field/Current Ratio	N/A	12.48 kHz/0.1 Amp	
Magnet type: OR105-01			
TDC calibration values		Carbon resistor values	
Room temperature (23 °C)	1009 Ohms	Calibration values	
Liquid nitrogen temperature (77K)	1309 Ohms		
Liquid helium temperature (4.2K)	2994 Ohms		
Position	Room temperature	Helium temperature	
TDC d - e	1034 Ohms	2837 Ohms	
Top a - e	980 Ohms	3014 Ohms	
Middle b - e	1110 Ohms	2681 Ohms	
Bottom c - e	1006 Ohms	2930 Ohms	
Helium level probe resistance (when using a 250mA supply)			
	Probe 1	Probe 2	
@ 0 resistance	85.36 Ohms	85.36 Ohms	
0% calibration	774	774	
100% resistance	0.22 Ohms	0.22 Ohms	
100% calibration	2	2	
Diode table no.	-1		
Pressure heater resistance	20.5 Ohms		
Final temperature sensor recon values			
Shield bore	Shield Link	Coldhead	Recon margin
47K	47K	42K	658mW
Checked by V Turner	Signed <u>AN Other</u>		
	Date 23/12/2011		
	Page 1 of 1 25-AD EPS 000 01		

2.3 Magnet Supervision (MSUP)

The MSUP is a **Field Replaceable Unit**. The fuses are the only elements which can be replaced if required.

This procedure can be used to fault find the MPSU. An understanding of Tim systems is required to use the following information.

2.3.1 MSUP fuses

For location of these fuses, refer to (→ Fig. 180 Page 163).

Fuse	Rating	Title
F1	125 mA F/250 mA F*	EIS control
F2	4 A T	+36 V DC input supply
F3	500 mA T	He probe
F4	5 A T	Battery supply
F5	3 A F	Quench heater 1
F6	3 A F	Quench heater 2
F7	3 A F	Pressure heater control
F8	375 mA T/1A F*	Switch heater

*Higer value fitted after Rev 14.



Fuse F7 will blow if pressure heater is on for 150 minutes continuously.

2.3.2 MSUP test points

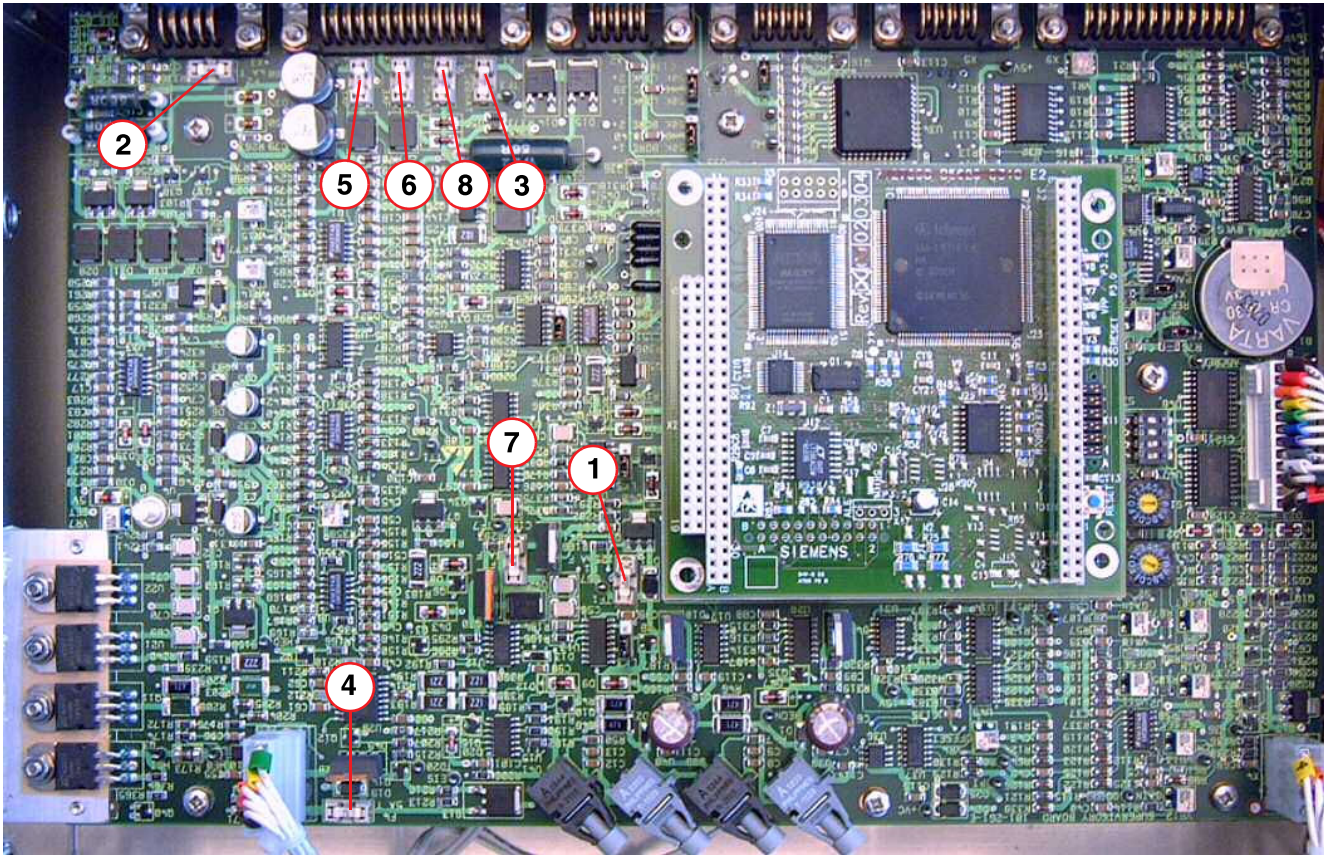
For location of these items, refer to PCB diagram (→ Fig. 181 Page 164).

Ref	Function
+36	Incoming +36V supply (ref 0 test point)
P	Magnet pressure signal to ADC
P0	Pressure monitor 0V reference test point
+VC	+VCONT supply to +15V regulator (ref 0 test point)
PI	Magnet pressure transducer signal (ref P0 test point)
+15V	15 V supply (ref 0 test point)
EIS	EIS output current signal (ref 0 test point)

Ref	Function
HE0	Helium probe 0V reference test point
HEI	Helium probe current signal (ref HE0 test point)
+5VREF	5V reference supply (ref P0 test point)
0	0V star point test point
PS	Pressure transducer supply (ref P0 test point)
+5V	5V supply (ref 0 test point)
+B	Battery voltage (ref 0 test point)
0V(S)	Sensor 0V test point
+5VREF2	5V reference supply #2
+20V	20V sensor supply (ref 0V (S) test point)

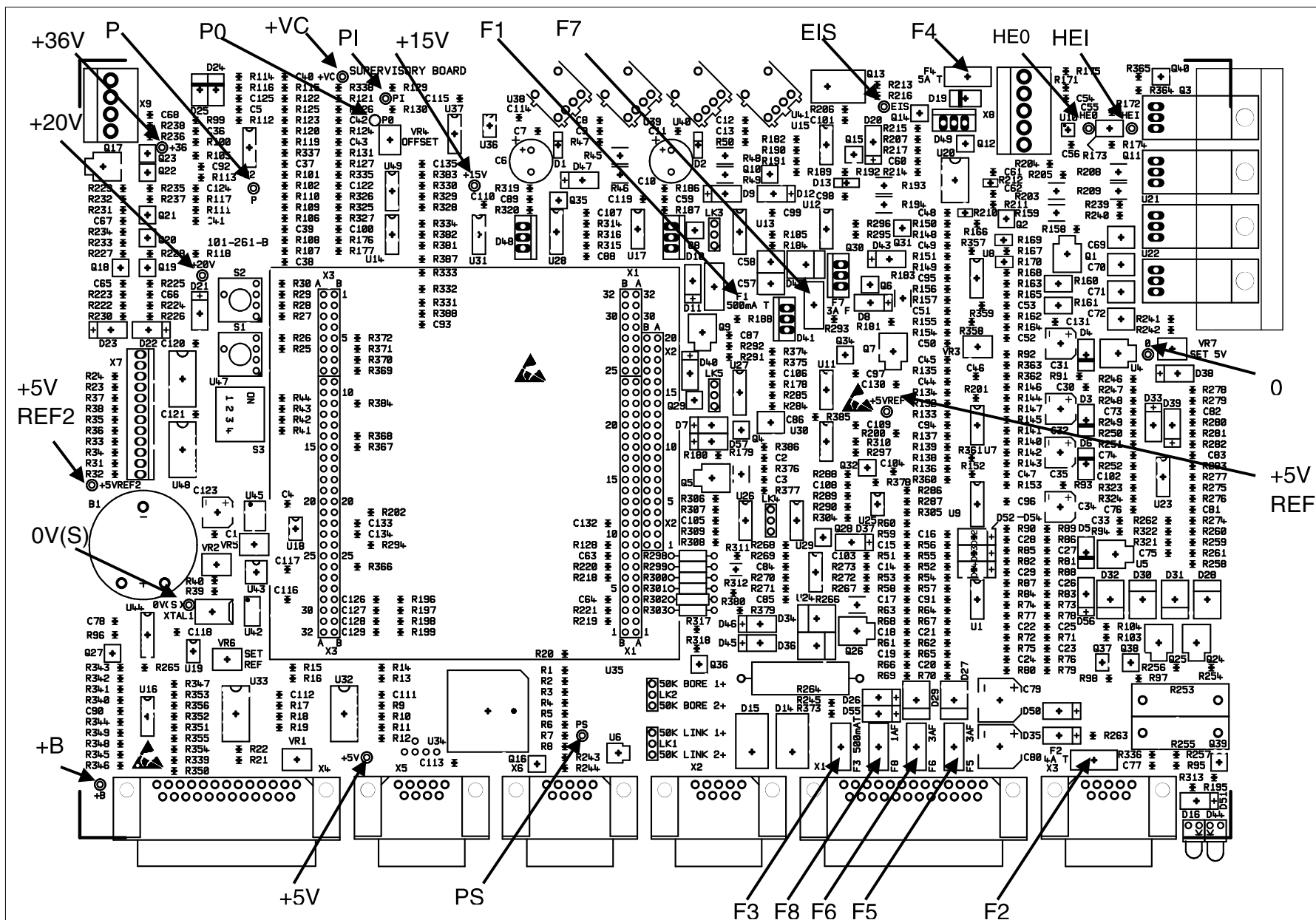
2.3.2.1 Fuses

Fig. 180: MSUP Fuses



- (1) Fuse F1, 125 mA, EIS Control (250 mA for MSUP with Rev. ≥ 14)
- (2) Fuse F2, 4 AT, +36 V MSUP supply
- (3) Fuse F3, 500 mA, LHe level probe
- (4) Fuse F4, 5 AT, Battery Supply
- (5) Fuse F5, 3 AF, Quench Heater 1
- (6) Fuse F5, 3 AF, Quench Heater 2
- (7) Fuse F7, 3 AF, Pressure Heater Control
- (8) Fuse F8, 375 mA, Switch Heater (1 AF for MSUP with Rev. ≥ 14)

Fig. 181: Fuses location on PCB



2.3.3 MSUP Jumper links

For location of these items, refer to PCB diagram (→ Fig. 181 Page 164).

Link	Normal Operating Position	Description
LK1	50K LINK 1+ Normally linked pin1/pin2	Link Temp Sensor 1 or 2 Fit a link pin1/pin2 to enable Link Temp Sensor 1+ Fit a link pin2/pin3 to enable Link Temp Sensor 2+
LK2	50L BORE 1+ Normally linked pin1/pin2	Bore Temp Sensor 1 or 2 Fit a link pin1/pin2 to enable Bore Temp Sensor 1+ Fit a link pin2/pin3 to enable Bore Temp Sensor 2+
LK3	Normally linked pin2/pin3	Test or Normal operate delay EIS Crowbar. Fit a link pin1/pin2 to Test EIS Crowbar. Fit a link pin2/pin3 to enable Normal EIS Crowbar operation.
LK4	Normally linked pin1/pin2	Enable or Disable Hardware Pressure Control. Fit a link pin1/pin2 to Test Pressure Heater Crowbar. Fit a link pin2/pin3 to Disable Hardware Pressure Control.
LK5	Normaly linked pin2/pin3	Test or Normal operation of Pressure Heater Crowbar. Fit a link pin1/pin2 to Test Pressure Heater Crowbar. Fit a link pin2/pin3 to enable Normal Pressure Heater Crowbar operation.
LK6	Normally linked pin1/pin2	Internal or External Magnet Field Sensor. Fit a link pin1/pin2 to select an Internal Magnet Field Sensor. Fit a link pin2/pin3 to select an External Magnet Field Sensor.
X10	Normally linked	For Development purposes only. U3 U43 Chip Select Line. Disables data logging to EPROM.
X11	Normally linked	For development purposes only. Disables the ADC reference voltage.

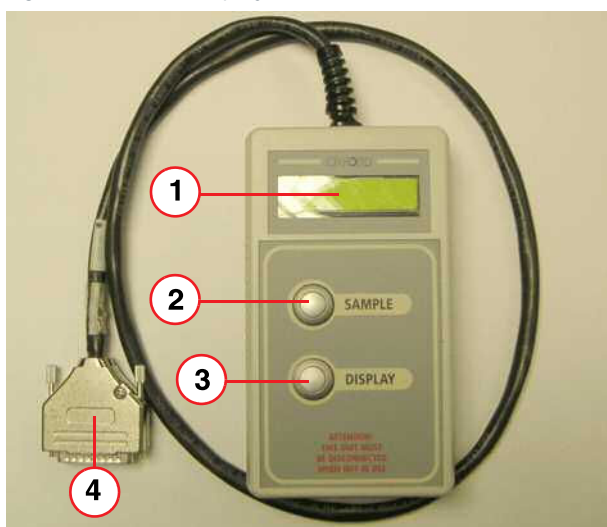
2.3.4 MSUP-Display Unit

The only operations that can be performed independently of the controlling host computer are through the **MSUP-display unit**.

Required tools

MSUP-display unit

Fig. 182: MSUP- Display Unit



- (1) LCD Display
- (2) Sample button
- (3) Display scroll button
- (4) 25 way 'D' connector

Performing the procedure

- Connect the display unit to **X9** on the MSUP via the 25 way 'D' connector (4)
- Ensure that power has been applied to the MSUP.
- Switch on the hand held unit by simultaneously pressing the **Sample** button (2) and **Display** button (3).
- Press **Display** (3).
- The unit will **automatically** carry out a helium level test. The **helium fill level** is shown in the display (1).

- Scroll through the parameters by pressing the **DISPLAY** button, one parameter for each press.

The parameters in order of presentation are:

1. He level (press sample to recheck level)
2. Battery volts
3. 50K link, bore (K, K)
4. Carbon resistor 1, 2 (ohm, ohm)
5. Carbon resistor 3, 4 (ohm, ohm)
6. Turret 1, 2 (K, K)
7. Switch heater (Ohm, status)
8. Quench heaters 1, 2 (Ohm)
9. Cold head (K)
10. ERDU buttons (status)
11. Magnet pressure (psiA)
12. Pressure heater (Off/On, %) (press Sample to switch on, 50 min timeout)
13. Pressure heater (average power over last hour of previous measurements)
14. EIS (Off, Sample to switch On, 60 min timeout)
15. Self-test



Disconnect the display unit from the MSUP after test is finished.

2.3.5 MSUP download instructions

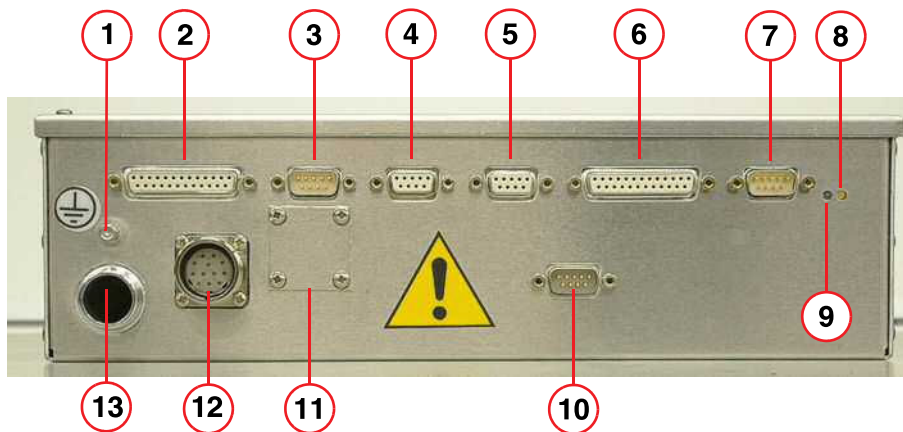


Ensure a safe distance from the magnet for troubleshooting and service actions which require the use of a laptop with extended cabling.

Instruction on how to download the **log file** of the MSUP

1. Turn on the laptop and open the hyper terminal application. Set to the following settings. Select the appropriate Com Port.
 - » Set bits per second to 9600
 - Set data bits to 8
 - Set parity to none
 - Set stop bits to 1
 - Set flow control to none
 - In the file, properties, settings tab, set Emulation to VT100.
2. Connect the laptop via an **RS232 cable** to connector **X5** on the MSUP (pos 3).

Fig. 183: MSUP 2657 Indicators and input/output connectors



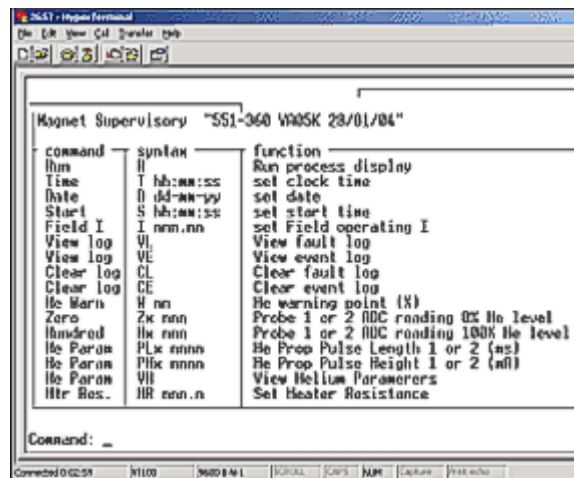
- (1) Earth stud (6mm)
- (2) X9 Display
- (3) X5 RS232
- (4) X6 ERDU/Field
- (5) X2 CH/DIAG/PRESS
- (6) X1 Turret
- (7) X3 Power/SL/UPS
- (8) Pressure heater LED (yellow)
- (9) Power ON LED (green)
- (10) X12 Battery
- (11) ERDU switch cover
- (12) X7 Magnet ID
- (13) CAN Fiber optics wave guide

3. Press **Esc** and then type **R.** (→ Fig. 185 Page 168) should be displayed.

Fig. 184: Download



Fig. 185: Download



4. Press **Esc.** (→ Fig. 184 Page 168) should be displayed.
5. Press **Esc** again.
6. Click "**Transfer**"
7. Click "**Capture text**"
8. Click "**Browse**" and select a directory and a name for the file to be stored. Name should contain the magnet serial number.
e.g. C:\Temp\MSN 12345678log.txt
9. Click "**Start**"
10. Type the following
 - » **VL** <enter>
 - VE** <enter>
 - VC** <enter>
 - VH** <enter>
 - VQ** <enter>
 - VD** <enter>
 - VP** <enter>
 - VX** <enter>
11. Click "**Transfer**", "**capture text**", "**Stop**" after downloaded data has been transferred.
12. Capture a **screenshot** of the **MSUP terminal emulation screen**:
 - a) Press **Ctrl:Alt:Prt Sc**
 - b) Go to: **Start:Programs:Accessories:**and open Paint
 - c) Press **Ctrl:V** to paste screenshot
 - d) Name and save as a .jpg or .bmp.

13. E-mail the saved file and screenshot to (→ Magnet HSC, <mailto:magnet.wscshsc.team@siemens-healthineers.com>)

2.4 Magnet Supervision (MSUP 2657B)

The MSUP is a **Field Replaceable Unit**. The fuses are the only elements which can be replaced if required.

This procedure can be used to fault find the MSUP. An understanding of Tim systems is required to use the following information.

2.4.1 MSUP fuses

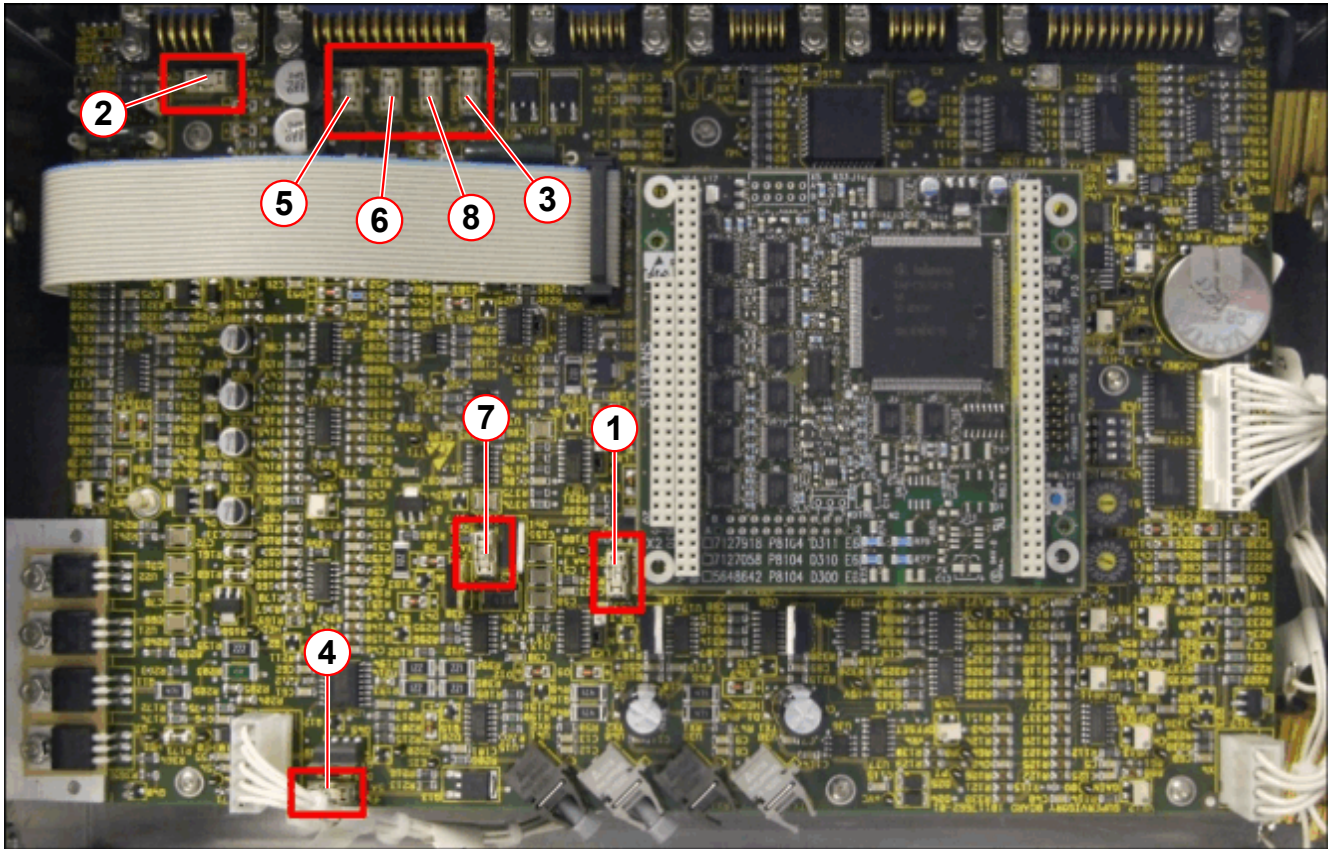
For location of these fuses, refer to (→ Fig. 186 Page 171).

Fuse	Rating	Title
F1	125 mA F/250 mA F*	EIS control
F2	4 A T	+36 V DC input supply
F3	500 mA T	He probe
F4	5 A T	Battery supply
F5	3 A F	Quench heater 1
F6	3 A F	Quench heater 2
F7	3 A F	Pressure heater control
F8	375 mA T/1A F*	Switch heater

*Higher value fitted after Rev 14.

2.4.2 Fuses

Fig. 186: MSUP 2657B - Fuses



- (1) Fuse F1, 125 mA, EIS Control (250 mA for MSUP with Rev. = 14)
- (2) Fuse F2, 4 AT, +36 V MSUP supply
- (3) Fuse F3, 500 mA, LHe level probe
- (4) Fuse F4, 5 AT, Battery Supply
- (5) Fuse F5, 3 AF, Quench Heater 1
- (6) Fuse F5, 3 AF, Quench Heater 2
- (7) Fuse F7, 3 AF, Pressure Heater Control
- (8) Fuse F8, 375 mA, Switch Heater (1 AF for MSUP with Rev. = 14)

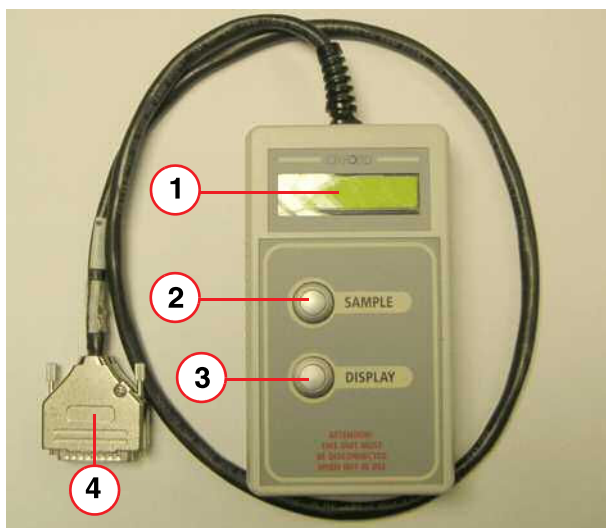
2.4.3 Hand-held MSUP Display-Unit

The only operations that can be performed independently of the controlling host computer are through the **hand-held MSUP Display-Unit**.

Required tools

Hand-held MSUP Display-Unit.

Fig. 187: MSUP- Display Unit



- (1) LCD Display
- (2) Sample button
- (3) Display scroll button
- (4) 25 way 'D' connector

Performing the procedure

- Connect the display unit to **X9** on the MSUP via the 25 way 'D' connector (4)
- Ensure that power has been applied to the MSUP.
- Press **SAMPLE** button (2).
 - The unit will carry out a helium level and battery test.
 - The helium fill level is shown in the display (1).
- Press **DISPLAY** (3), and **SAMPLE** (2) simultaneously to unlock the scroll function.

- Scroll through the parameters by pressing the **DISPLAY** button, one parameter for each press.

The parameters in order of presentation are:

1. He level (press sample to recheck level)
2. Battery volts
3. 50K link, bore (K, K)
4. Carbon resistor 1, 2 (ohm, ohm)
5. Carbon resistor 3, 4¹ (ohm, ohm)
6. Turret 1, 2 (K, K)
7. Switch heater (Ohm, status)
8. Quench heaters 1, 2 (Ohm)
9. Cold head (K)
10. ERDU buttons (status)
11. Magnet pressure (psiA)
12. Pressure heater (Off/On, %) (press Sample to switch on, 50 min time-out)
13. Pressure heater (average power over last hour of previous measurements)
14. EIS (Off, Sample to switch On, 130 min time-out)
15. Self-test
16. Compressor status (OFF/ON,P,T)
17. LVQD (Waiting trigger/quench location) Time & Date
 - Time and date

1. Carbon resistor 4 is shown in Kelvin for CCR type magnets.

- Magnet type
- Part no and software version



Disconnect the display unit from the MSUP after test is finished.

2.4.4 MSUP download instructions



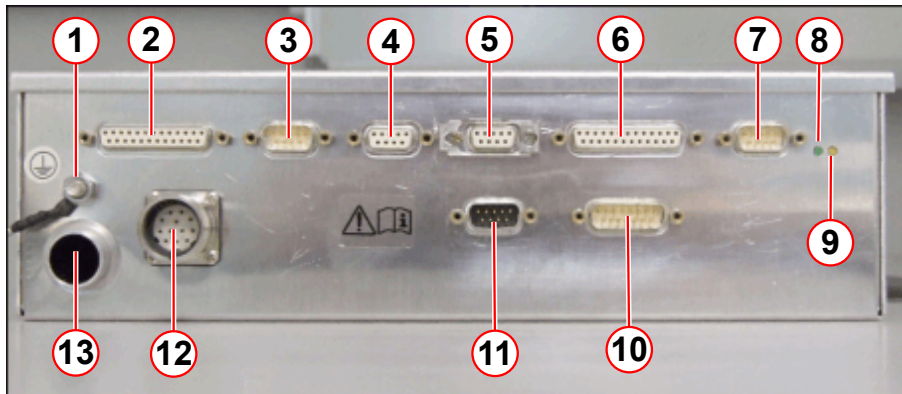
Ensure a safe distance from the magnet for troubleshooting and service actions which require the use of a laptop with extended cabling.

Instruction on how to download the **log file** of the MSUP

1. Turn on the laptop and open the hyper terminal application. Set to the following settings. Select the appropriate Com Port.
 - » Set bits per second to 9600
 - Set data bits to 8
 - Set parity to none
 - Set stop bits to 1
 - Set flow control to none
 - In the file, properties, settings tab, set Emulation to VT100.

2. Connect the laptop via an **RS232 cable** to connector **X5** on the MSUP (pos 3).

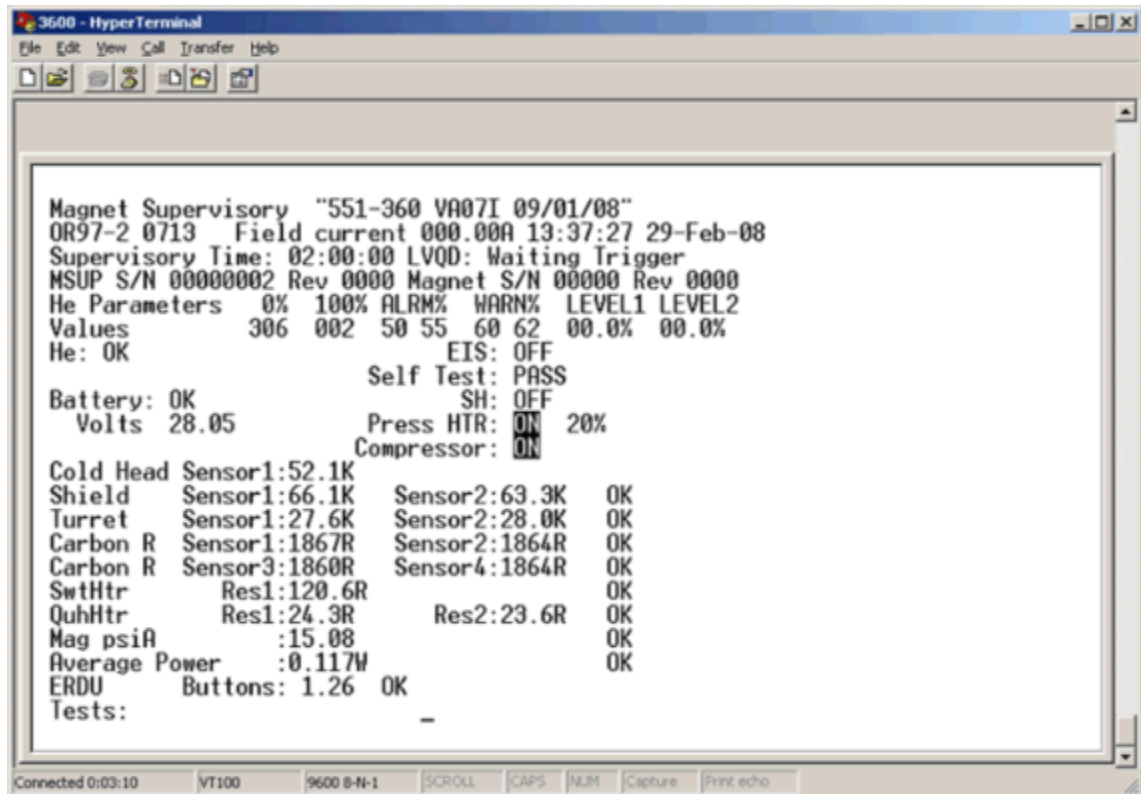
Fig. 188: MSUP 2657B - Indicators and input/output connections



- (1) Earth stud (6mm)
- (2) X9 DISPLAY
- (3) X5 RS232
- (4) X6 ERDU
- (5) X2 CH/DIAG/PRESS
- (6) X1 TURRET
- (7) X3 POWER/SL/UPS
- (8) Power ON LED (green)
- (9) Pressure heater LED (yellow)
- (10) X14 LVQD
- (11) X12 BATTERY
- (12) X7 Magnet ID
- (13) CAN Fiber optics wave guide

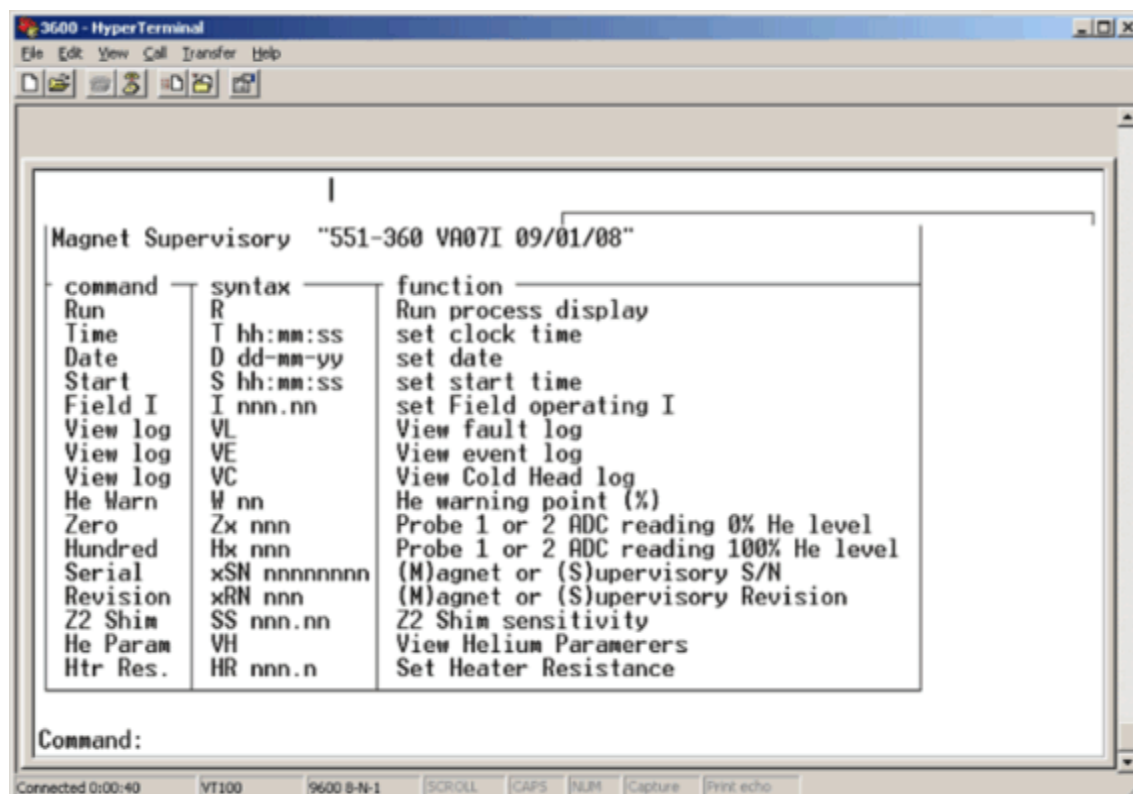
3. The **MSUP Process Display** is displayed.

Fig. 189: MSUP 2657B - Process Display screen (typical)



4. Press **Esc.** to display the **MSUP Command Input** screen.

Fig. 190: MSUP 2657B - Command Input screen (typical)



5. Click "**Transfer**"
6. Click "**Capture text**"
7. Click "**Browse**" and select a directory and a name for the file to be stored. Name should contain the **magnet serial number**.
e.g. C:\Temp\MSN 12345678log.txt
8. Click "**Start**"
9. Type the following
 - » VL <enter>
 - VE <enter>
 - VC <enter>
 - VH <enter>
 - VQ <enter>
 - VD <enter>
 - VP <enter>
 - VX <enter>
10. Click "**Transfer**", "**capture text**", "**Stop**" after downloaded data has been transferred.
11. Capture a **screenshot** of the **MSUP terminal emulation screen**:
 - a) Press **Ctrl:Alt:Prt Sc**
 - b) Go to: **Start:Programs:Accessories:**and open Paint

- c) Press **Ctrl:V** to paste screenshot
 - d) Name and save as a .jpg or .bmp.
12. E-mail the saved file and screenshot to (→ Magnet HSC, <mailto:magnet.wscshsc.team@siemens-healthineers.com>)

Tab. 8 MSUP 2657B Input Codes

Description	Command	Operation/Example
Time	T hh:mm:ss<R>	Set time in real time clock 1 hour 30mins 30secs - enter as T 13:30:30<R>
Date	D dd-mm-yy<R>	Set date in real time clock 1st Jan 2008 - enter as D 01-01-08<R>
Start test time	S hh:mm:ss<R>	Set supervisory test time 2 hour 45mins 45secs - enter as T 2:45:45<R>
Magnet operating current	I nnn.nn<R>	Set Mag Operating Current Operating current 685.02 - enter as I 685.02<R>
Helium warn point	W nn<R>	Set Helium Warning point 50% enter as W 50<R>
0% Calibration (probe 1)	Z1 nnnn<R>	Set He Probe 1 0% Probe1: 88.8 ohms -enter as Z1 805<R>
0% Calibration (probe 2)	Z2 nnnn<R>	Set He Probe 2 0% Probe2: 88.8 ohms - enter as Z2 805<R>
100% Calibration (probe 1)	H1 nnnn<R>	He Probe 1 100% Probe1: 1.8 ohms - enter as H1 16<R>
100% Calibration (probe 2)	H2 nnnn<R>	He Probe 2 100% Probe2: 1.8 ohms - enter as H2 16<R>
Magnet Serial No.	MSN nnnnnnnn<R>	Adjust Magnet Serial No Enter as MSN12345678<R>. Password required
Magnet Revision No.	MRN nnn<R>	Adjust Mag Rev No Magnet Rev 4 - enter as MRN4<R> Password required
Supervisory Serial No.	SSN nnnnnnnn<R>	Adjust Supervisory Serial No. Supervisory Serial No S0014 - enter as SSN S0014<R>
Supervisory Revision No.	SRN nnn<R>	Adjust Supervisory Revision No Supervisory Revision No 14 - enter as SRN14<R> Password required
Pressure heater resistance	HR nnn.n<R>	Adjust Pressure Heater Res 19.4 ohms - enter as HR 19.4<R> Password required
Shield bore warning	L4 2947<R> SBW nnn<R>	Shield Bore Warning Shield Bore warning of 77K - enter as SBW 77<R>

Description	Command	Operation/Example
CCR Calibration	SR <R> XXX.X <R> YYYY.Y <R> ZZZZ.Z <R>	Adjust CCR calibration to RT (XXX.X), 77K (YYYY.Y), 4.2K (ZZZZ.Z) Password required Refer to Replacement of Parts for Calibration of CCRs .
Run process image	R<R>	Refresh the screen
View status log	VL<R>	View status log
View error log	VE<R>	View error log
View month log	VM<R>	View month log
View cold head log	VC<R>	View cold head log
View quench log	VQ<R>	View quench log
View LVQD log	VD<R>	View LVQD log
View helium log	VH<R>	View the helium calibrations
View helium pulse log	VP<R>	View the helium pulse log
Notes: ■ <R> = 'Enter' or 'Return'. ■ Where a password is required, type L 9912<R> before the command is input. ■ CCR Calibration - The MSUP will prompt the user to input values after the command SR has been used. - Refer to the Magnet Acceptance Document for TDC CCR4 Calibration Values.		

2.5 Refrigerator

2.5.1 Helium gas charging CSW-71



Make sure the refrigerator is powered off and compressor and cold head temperatures are almost at room temperature before adjusting the helium gas pressure.

If the compressor has been running, switch off and leave for a minimum of 30 minutes before checking static pressure.



CAUTION

High pressure helium gas.

Failure to observe the following safety warning may result in personal injury.

- » Protective clothing, gloves and eye protection must be worn.

Required tools

Charge and vent adapter (gas charging tool)(part number **08396660**)

Bleed adapter assembly (part number **10125874**)

High pressure regulator for helium gas bottle.

30bar Helium pressure hose (charging line).

Helium gas bottle, purity of 99.999% .



Use only helium of purity 99.999% or better.

3/4" (19mm) Open-end wrench/spanner.

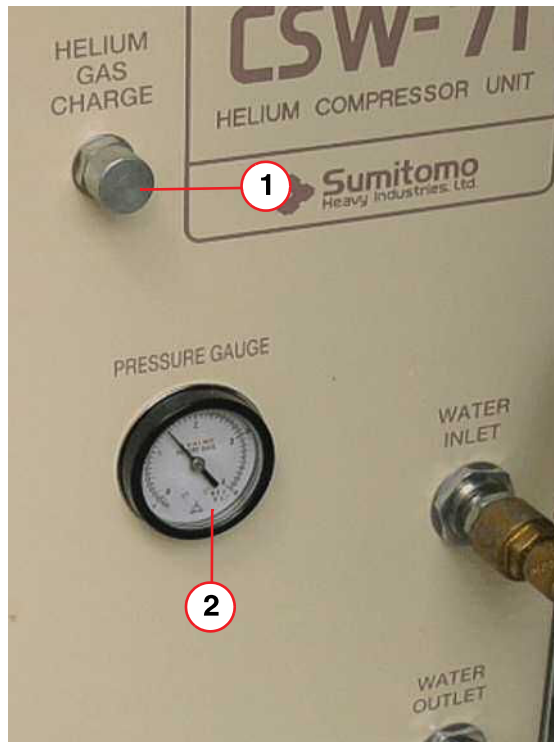
5/8" (16mm) Open-end wrench/spanner.

Performing the procedure

- Turn **off** the compressor at the **main power switch**.
 - » The Alarm Box will display a red **COMPRESSOR** LED, and an audible alarm will sound.
 - » Press the **ACKNOWLEDGE** button to cancel the audible alarm

- Wait for at least **30 minutes** to allow the static pressure to stabilize.

Fig. 191: Compressor He Gas charge & pressure gauge



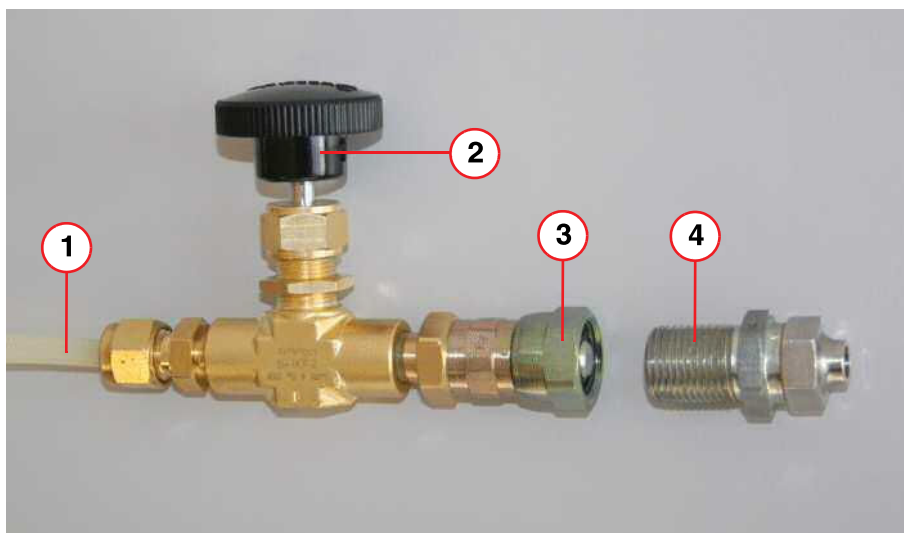
- Remove the **protective cap (1)** from the **helium gas charge coupling** on the compressor unit.
- Check the **rubber aeroquip gasket** for dirt or dust and ensure it is correctly fitted. Replace if required (part no **03861494**)

NOTE: Incorrect fitment may cause a helium gas leak.

- (1) Helium gas charge
(2) Helium pressure gauge

- Connect a **high pressure regulator** and **charging line** to the helium gas bottle (purity 99.999%).

Fig. 192: He compressor - charge and vent adaptor (gas charging tool)



- (1) Charging line (30bar)
(2) Charge and vent adaptor valve
(3) Aeroquip coupling (female)
(4) Bleed adaptor assembly

- **Open** the valve on the **helium gas bottle**.
 - Ensure the **regulator valve** is closed.
 - Ensure that **30bar** of helium gas is available.
- Screw the **charging line** (→ 1/Fig. 192 Page 181) into the **gas charging tool**.
- Screw the **bleed adaptor assembly**(→ 4/Fig. 192 Page 181) into the **gas charging tool**.
- **Purge the charging line** of any air contaminants by setting the high pressure regulator to 10psi for 30 seconds and **opening** the valve on the **gas charging tool**.
- Unscrew and remove the **bleed adaptor assembly**.
- **Close** the **gas charging tool valve** (→ 2/Fig. 192 Page 181)
- Set the **gas pressure regulator** to:
 - » **16.5 - 17 bar or 239 - 246 psig**
- Using two wrenches, connect the **gas charging tool** to the **helium gas charge coupling** on the compressor unit.
- Slowly **open** the gas charging tool valve.

Fig. 193: Helium Compressor



(1) Gas charging tool with charging line

- Wait 60 seconds for gas pressure to **stabilize** on the compressor gauge.
- If the compressor pressure still indicates **lower** than the required:-
 - Adjust the gas pressure regulator until the compressor gauge reads **16.0 - 16.5 bar**.
 - Wait 60 seconds for gas pressure to **stabilize** on the compressor gauge.
- When pressure is achieved, **close** the **gas charge tool valve**.



It is possible to overpressurize the compressor. To adjust pressure, refer to (→ Adjusting the equalization pressure / Page 183).

- **Disconnect** the **gas charging tool** from the **compressor**.
 - Take care not to dislodge the aeroquip gasket seal.
- Replace the **protective cap** on the **helium gas charge coupling**.
- **Close** the valve on the **helium gas bottle**.
- Screw the **bleed adaptor assembly** into the end of the **charging line**.
- Slowly open the **gas charging tool valve** to release pressure from the **charging line**.
 - » The regulator gauge should read **zero**.
- **Unscrew** and remove the **bleed adaptor assembly**.
- **Disconnect** the high pressure **regulator** from the **gas bottle**.
- Store the **charge line** to keep it clean.
- Re-start the compressor.

2.5.1.1 Adjusting the equalization pressure

If the compressor has been **overpressurized**, the excess helium gas must be vented to atmosphere.



The compressor should not be operated above the specified pressure. Overpressure will cause the compressor to run in continuous bypass mode.

- Turn **off** the compressor at the **main power switch**.
 - » The Alarm Box will display a red **COMPRESSOR** LED, and an audible alarm will sound.
 - » Press the **ACKNOWLEDGE** button to cancel the audible alarm.
- Wait for at least **30 minutes** to allow the static pressure to stabilize.
- **Remove** the **protective cap** from the **helium gas charge coupling** on the compressor unit (→ 1/Fig. 191 Page 181).
- Make sure the **gas charge tool valve** is **closed** (→ 2/Fig. 192 Page 181).
- Using two wrenches, connect the **gas charging tool** to the **helium gas charge coupling** on the compressor unit.
- **Slowly open** the **gas charging tool valve**.
- **Vent** the **helium gas** until the required **equalization pressure** is achieved.
- **Close** the **gas charging tool valve**.
- **Disconnect** the **gas charging tool** from the **compressor**.
 - Take care not to dislodge the aeroquip gasket seal.
- Replace the **protective cap** on the **helium gas charge coupling**.
- Re-start the compressor.

2.5.2 Helium gas charging F70



Make sure the refrigerator is powered off and compressor and cold head temperatures are almost at room temperature before adjusting the helium gas pressure.



CAUTION

High-pressure helium gas

Failure to observe the following safety warning may result in personal injury.

- » Protective clothing, gloves, and eye protection must be worn.

Required tools

- **Charge & Vent tool Kit** (part number **10125858**)
 - Adaptor valve (gas charging tool) (part number **10125873**)
 - Bleed adapter assembly (part number **10125874**)
- High pressure regulator for helium gas bottle
- 30bar Helium pressure hose (charging line)
- Helium gas bottle, purity of 99.999%



Use only helium of purity 99.999% or better.

3/4" (19 mm) Open-end wrench/spanner.

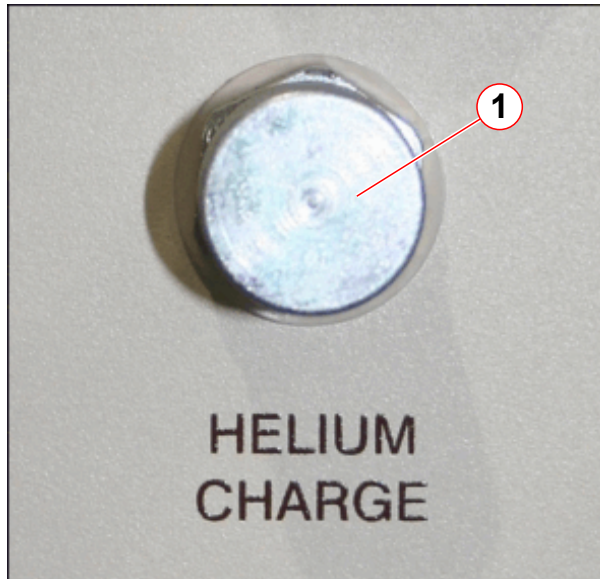
5/8" (16 mm) Open-end wrench/spanner.

Performing the procedure

- Turn **off** the compressor at the main power switch.
 - » The Alarm Box will display a warning LED, and an **audible** alarm will sound.
 - » Press the **ACKNOWLEDGE** button to cancel the audible alarm

- Wait for at least **30 minutes** to allow the static pressure to stabilize.

Fig. 194: F70 Compressor - Helium charge port



(1) Protective cap

- Remove the **protective cap (1)** from the **helium gas charge coupling** on the compressor unit.

Fig. 195: F70 Compressor - Helium Charge fill port gasket seal



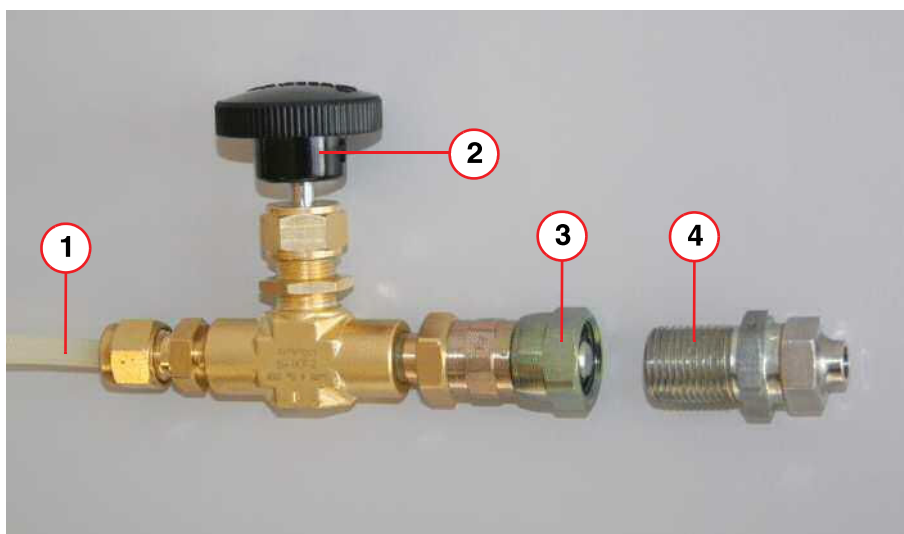
(1) Aeroquip gasket seal

- Check the **rubber aeroquip gasket** for dirt or dust and ensure it is correctly fitted. Replace if required (part number **03861494**).

NOTE: Incorrect fitment may cause a helium gas leak.

- Connect a **high pressure regulator** and **charging line** to the helium **gas bottle** (purity 99.999%).

Fig. 196: He compressor - charge and vent adaptor (gas charging tool)



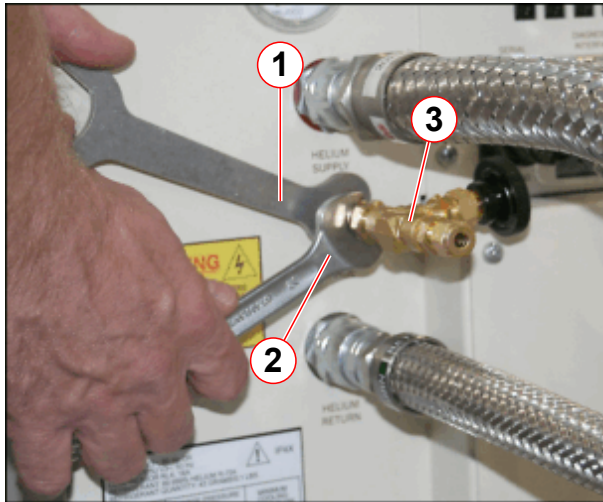
- (1) Charging line (30bar)
- (2) Charge and vent adaptor valve
- (3) Aeroquip coupling (female)
- (4) Bleed adaptor assembly

- **Open** the valve on the **helium gas bottle**.
 - Ensure the **regulator valve** is closed.
 - Ensure that **30bar** of helium gas is available.
- Screw the **charging line** (→ 1/ Fig. 196 Page 186) into the **adaptor valve**.
- Screw the **bleed adaptor assembly** (→ 4/ Fig. 196 Page 186) into the **adaptor valve**.
- **Purge the charging line** of any air contaminants by setting the **high pressure regulator** to 10 psi for 30 seconds and **opening** the valve on the **adaptor valve**.
- Unscrew and remove the **bleed adaptor assembly**.
- **Close** the **adaptor valve**.
- Set the **gas pressure regulator** as required - see table below:-

Tab. 9 Helium gas pressures

Cold head	Static pressure	Dynamic pressure
4 K - 50/60 Hz	13.5 - 14.0 bar (13.6 nominal) 196 - 203 psi (G) 1350 - 1400 kPa	21 - 22 bar 304 - 319 psi (G) 2100 - 2200 kPa

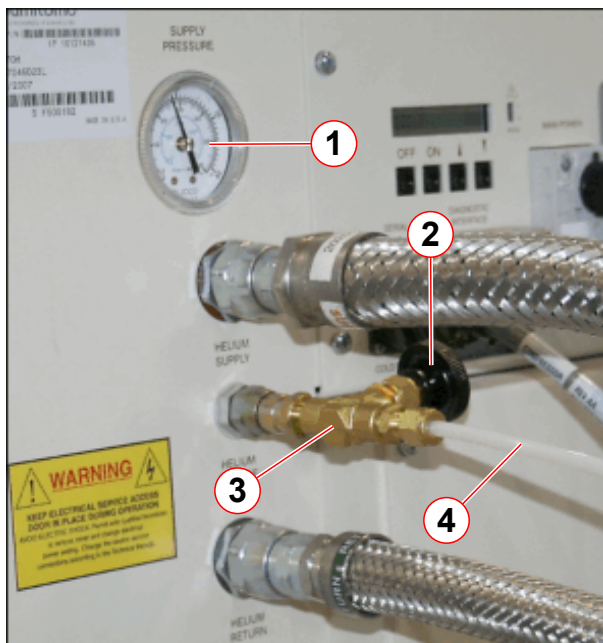
Fig. 197: F70 Compressor - Fit/remove charge & vent adaptor (gas charging tool)



- (1) Turn 5/8" (16mm) spanner/wrench
- (2) Hold 3/4" (19mm) spanner/wrench
- (3) Charge & vent adaptor (gas charging tool)

- Using two wrenches, connect the **adaptor valve** to the **helium gas charge coupling** on the compressor unit.

Fig. 198: F70 - Charge & vent adaptor (gas charging tool)



- (1) Pressure gauge
- (2) Valve
- (3) Charge and vent adaptor (gas charging tool)
- (4) Charging line

- **Slowly open** the adaptor valve 2.

- Wait **60 seconds** for gas pressure to **stabilize** on the compressor gauge.
- If the compressor pressure still indicates **lower** than the required:-
 - Slowly open the **high pressure regulator** until the required pressure has been achieved.
 - Wait 60 seconds for gas pressure to stabilize on the compressor gauge.

- When pressure is achieved, **close** the **adaptor valve**. (→ 2/Fig. 198 Page 187)



It is possible to overpressurize the compressor. To adjust pressure, refer to Adjusting the equalization pressure.

- **Disconnect** the **adaptor valve** from the **compressor**.
 - Take care not to dislodge the aeroquip gasket seal.
- Replace the **protective cap** on the **helium gas charge coupling**.
- **Close** the valve on the **helium gas bottle**.
- Screw the **bleed adaptor assembly** into the end of the **charging line**.
- **Slowly** open the **adaptor valve** to release pressure from the **charging line**.
 - » The regulator gauge should read **zero**.
- **Unscrew** and remove the **bleed adaptor assembly**.
- **Disconnect** the high pressure **regulator** from the **gas bottle**.
- Store the **charge line** to keep it clean.
- Restart the compressor.

2.5.2.1 Adjusting the equalization pressure

If the compressor has been **over pressurized**, the excess helium gas must be vented to atmosphere.

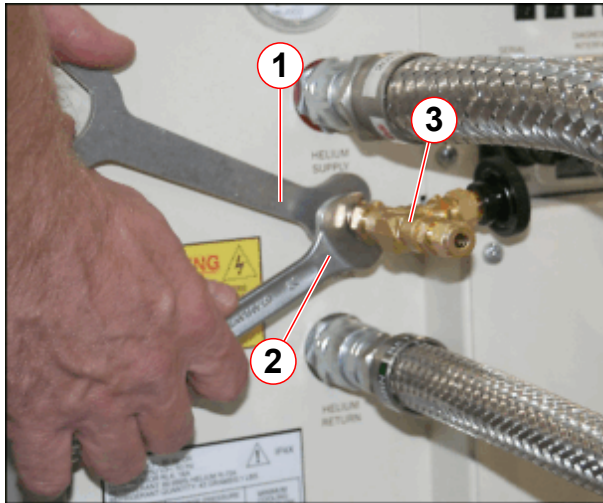


The compressor should not be operated above the specified pressure. Overpressure will cause the compressor to run in continuous bypass mode.

- Turn **off** the compressor at the **main power switch**.
 - » The Alarm Box will display a red warning LED, and an audible alarm will sound.
 - » Press the **ACKNOWLEDGE** button to cancel the audible alarm.
- Wait for at least **30 minutes** to allow the static pressure to stabilize.
- **Remove** the **protective cap** from the **helium gas charge coupling** on the compressor unit.

- Make sure the **gas charge tool valve** is closed.

Fig. 199: F70 Compressor - Fit/remove charge & vent adaptor (gas charging tool)



- (1) Turn 5/8" (16mm) spanner/wrench
- (2) Hold 3/4" (19mm) spanner/wrench
- (3) Charge & vent adaptor (gas charging tool)

- Using two wrenches, connect the **adaptor valve** to the **helium gas charge coupling** on the compressor unit.

- **Slowly open the adaptor valve.**
- **Vent the helium gas** until the required **equalization pressure** is achieved.
- **Close the adaptor valve.**
- **Disconnect the adaptor valve** from the **compressor**.
 - Take care not to dislodge the aeroquip gasket seal.
- Replace the **protective cap** on the **helium gas charge coupling**.
- **Restart** the compressor.

2.5.3 Error corrections

An error condition will cause either a system alarm or a shutdown. The following table lists the alarm and the shutdown errors that are monitored. Overload trip of the Mains Power switch is not monitored or reported.

Tab. 10 F70 Error Conditions

LCD Display First Line	Type	Error	Indication
Helium Temp - ERR	Shutdown	High helium discharge temperature	Temperature > 93°C (200°F)
Water Temp - ERR	Alarm	High water supply temperature	Temperature > 35°C (95°F)
Water Flow - ERR	Alarm	Low water flow	Temperature > 46°C (115°F)

LCD Display First Line	Type	Error	Indication
Motor Temp - ERR	Shutdown	High compressor motor winding temp (internal protector open)	Return Pressure >140psig while compressor is running
Phase Seq - ERR	Shutdown	Phase sequence or open fuse	Monitored by microprocessor
Return Pressure- ERR	Shutdown	Loss of gas charge	Return pressure < 15psig
Oil Level - ERR	Shutdown	Loss of circulating oil	Oil level switch opens
DB-25 Config 1 - ERR (or Config 2)	Shutdown	Remote cable does not match switch setting.	Monitored by microprocessor

If the compressor has been shut down by one of these interlocks, do not restart until the problem has been found and corrected. Refer to the Troubleshooting Guide to identify the problem.

In the event of a shutdown error, the compressor motor and cold head valve motor are turned off. The microprocessor will annunciate the error condition via signals at the System Diagnostics Connector and via the LCD Display.

The Mains Power switch trips when steady-state current exceeds 1.2 x the front panel set point. When tripped, the switch knob locates halfway between the on and off positions.

2.5.3.1 Clearing Error Conditions



Errors can only be cleared locally when configuration mode is set for Mode 1. When configuration mode is set for Mode 2, errors must be cleared using the DB25 diagnostic interface.

When an error is corrected, the message(s) can be cleared from the display and the system diagnostics connector by any of the following procedures:

- Press the **DISPLAY UP** and **DISPLAY DOWN** (arrow) buttons simultaneously.
- Turn the **MAIN POWER SWITCH** to **OFF**, pause briefly and switch back **ON**.

The MSUP will automatically apply a reset signal via the DB25 diagnostic interface.

2.5.3.2 Restarting after an Error Condition

When the shutdown is caused by helium discharge high-temperature, the compressor will be able to restart only after it has cooled enough for the thermistor to read a temperature below the cut-off point. After waiting for the compressor to cool, clear the error message then push the ON button to restart. Should the compressor fail to start, allow more cooling time. Repeat the restart procedure. Check cooling water temperature and flow. Compare with Specifications.

If the shutdown is due to a motor high temperature, the motor winding high temperature relay will automatically reset in approximately 30 minutes. To restart the compressor, clear the error message then push the ON button. If the compressor fails to start, allow more cooling time. Repeat the restart procedure.

If the motor circuit protector opens, the knob moves part way toward OFF. Reset it by turning the knob completely to OFF, then clockwise to ON. The compressor and the cold head should start automatically.

2.5.4 F70 Compressor download instructions

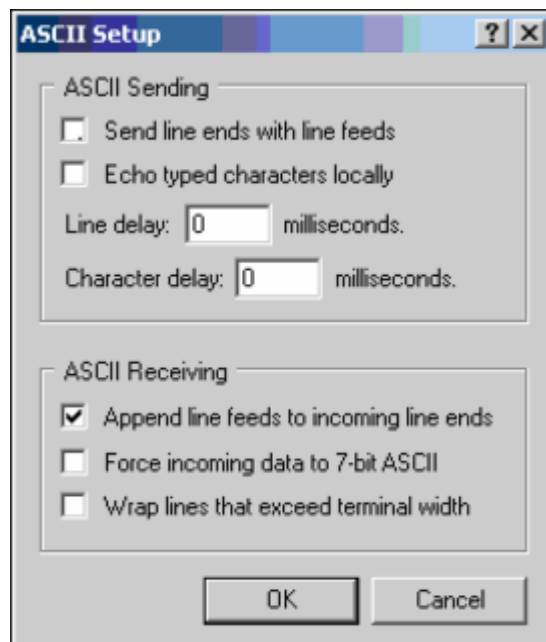
2.5.4.1 Tooling required

- RS232 Cable
- Laptop

2.5.4.2 Download Instructions

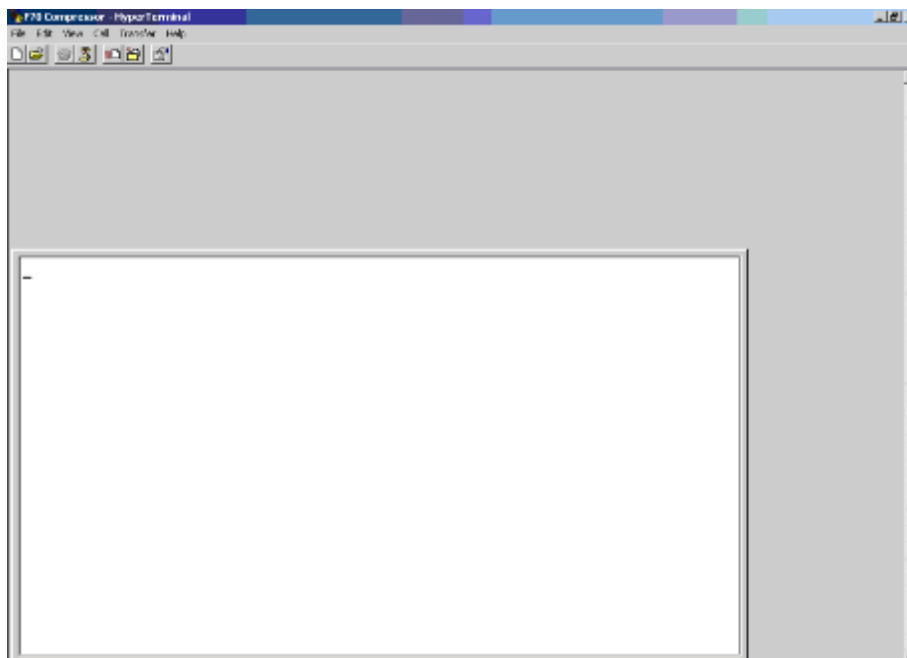
1. **Turn off** the compressor **main breaker**.
2. Turn on laptop and open the Terminal emulation software (e.g HyperTerminal) application. Set to the following settings. Select the appropriate **Com Port**.
 - » Set bits per second to 9600
 - » Set data bits to 8
 - » Set parity to none
 - » Set stop bits to 1
 - » Set flow control to none
 - » Set Emulation to VT100 (File: Properties:Settings)
 - » Click OK
 - » Set ASCII setup (File: Properties:Settings)
 - » Click **OK**

Fig. 200: ASCII Setup



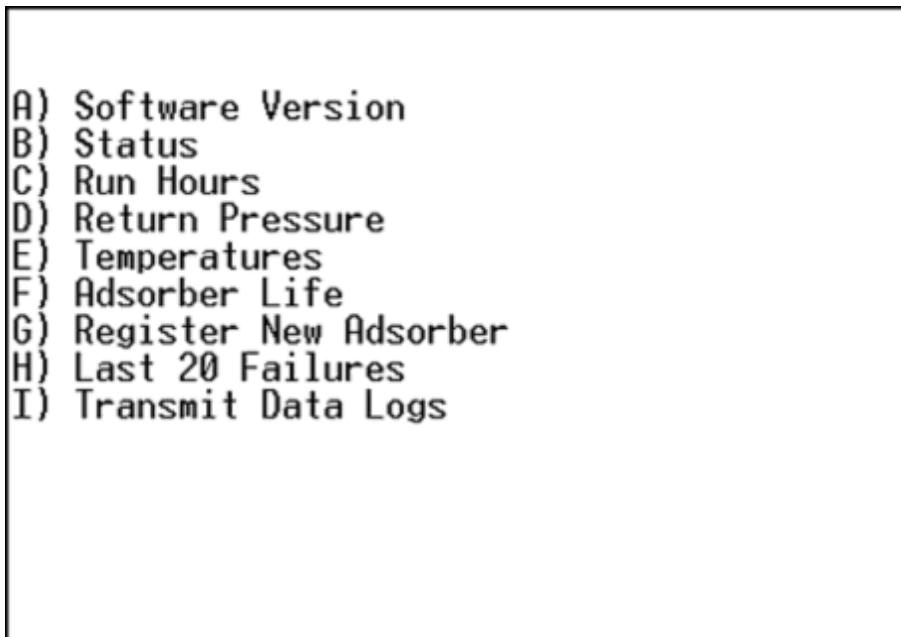
3. Connect the **laptop** via an **RS232 cable** to **serial port** on front of F70 compressor.
4. Turn **on** compressor **main breaker**.
5. Open **terminal emulation program**, the screen below is displayed.

Fig. 201: F70 Terminal emulation program



6. Press letter '**M**' <Enter> to open menu. See below:.

Fig. 202: Select from the menu



7. Click '**Transfer**'.
8. Click '**Capture text**'.
9. Click '**Browse**'. Select a directory and a name for the file to be stored. Name should contain **Compressor serial number** e.g. **C:\Temp\Comp 12345678log.tx**.
10. Click '**Start**'.
11. Type the following:-



Allow 10 seconds between each letter to ensure complete download of data.

- A <Enter>
- B <Enter>
- C <Enter>
- D <Enter>
- E <Enter>
- F <Enter>
- G <Enter>
- H <Enter>
- I <Enter>

12. Click '**Transfer**' '**Capture text**' '**Stop**' after completing transfer.
13. Turn **off** the compressor.
14. **Disconnect** the RS232 cable from the serial port of the F70.
15. Turn **on** the compressor **main breaker**.

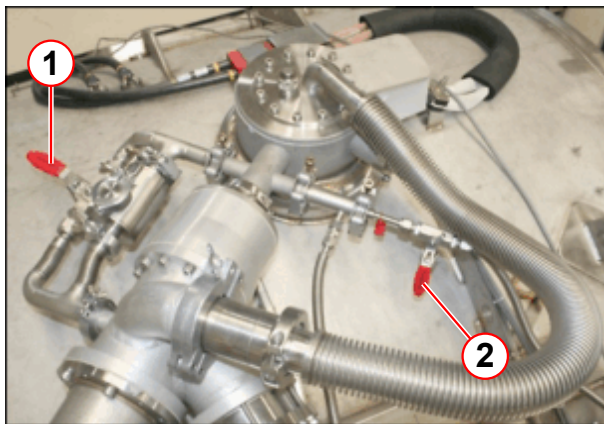
16. **Email** captured file to (→ Magnet HSC, <mailto:magnet.wscshsc.team@siemens-healthineers.com>).

2.5.5 Refrigeration Failures with Magnet at Field

Failure of the refrigeration system when the magnet is at field, will see increased magnet pressure with the loss of liquid helium.

To reduce helium loss and risk of the magnet quenching the cold head bypass must be opened after **30 hours**, (not before).

Fig. 203: Avanto - Espree Mk1 vent valves



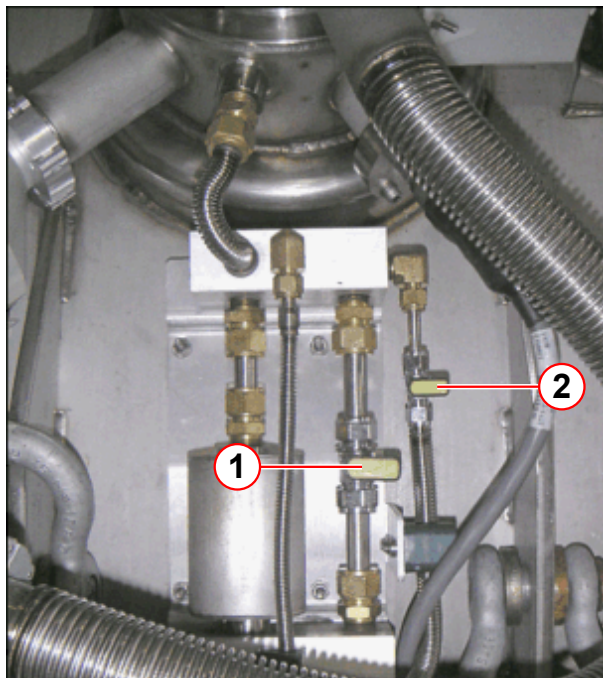
- (1) Magnet bypass valve (closed)
(2) Cold head bypass valve (closed)

MK1 Venting

For long periods of refrigeration down time, greater than **30 hours**:

- Open the cold head valve to reduce helium loss.
- Close cold head valve as soon as the refrigeration is running correctly.

Fig. 204: Avanto - Espree Mk2 vent valves



- (1) Magnet bypass valve (closed)
 (2) Cold head bypass valve (closed)

MK2 Venting

For long periods of refrigeration down time, greater than **30 hours**:

- Open the cold head valve to reduce helium loss.
- Close cold head valve as soon as the refrigeration is running correctly.

2.5.6 F70 Hours run clock display

2.5.6.1 Counters / Timers

Compressors running software version 1.7 have 3 counters. Each one can help with service operations.

- **Display Run Time.**

Fig. 205: Display Run Time



This timer is the run hours that are displayed on the liquid crystal display. The run hours can also be viewed on a compressor download.

The Display Run Time is a counter that displays the total running time (in hours) since the compressor has last visited a Sumitomo service center

- **Master Run Time.**

This timer can only be viewed on a compressor download.

The master run time is a timer that displays (in hours) the total run time of the compressor since it was first manufactured. An example of this will read '000844.8 Master Run Time'.

- **Adsorber Life.**

This timer can only normally be viewed on a compressor download. It is a count down of when the next Adsorber on the compressor should be changed.

After an Adsorber is changed in the compressor the time must be reset, this takes the timer to 30,000 hours. The timer will then count down while the compressor is running and will even become negative once zero hours is passed. At the next adsorber change the timer is reset back to 30,000 hours. An example of this will read '29155 HRS Adsorber Life'.

The Adsorber Life timer can be reset through the **front panel control buttons**:-

1. With the Main Power switch turned **OFF**, press and hold **all four control buttons** below the LCD window.
2. Keep these buttons held and turn the Main Power Switch **ON** to enter "**Service Level**" mode. Release the buttons.
3. Scroll using the arrow control buttons until the display shows "**NEW ADS? ON=Yes.**"
4. Press the **ON** control button. Display will show "**SAVE? ON = Yes OFF = NO.**"
5. Press the **ON** control button again to complete. (Press the **OFF** control button to back out of the function). When **ON** is pressed, display will show: "**Saving New ADS.**" When function is complete, display will again show "**NEW ADS? ON =YES.**"
6. Turn the **main power** switch to OFF to leave "**Service Level**".
7. Switch **ON** main power breaker.
8. Scroll **UP/DOWN** arrows, ADS Life = **30,000Hrs.**
9. The new adsorber is now **registered**.

2.5.7 Fuse Location F70



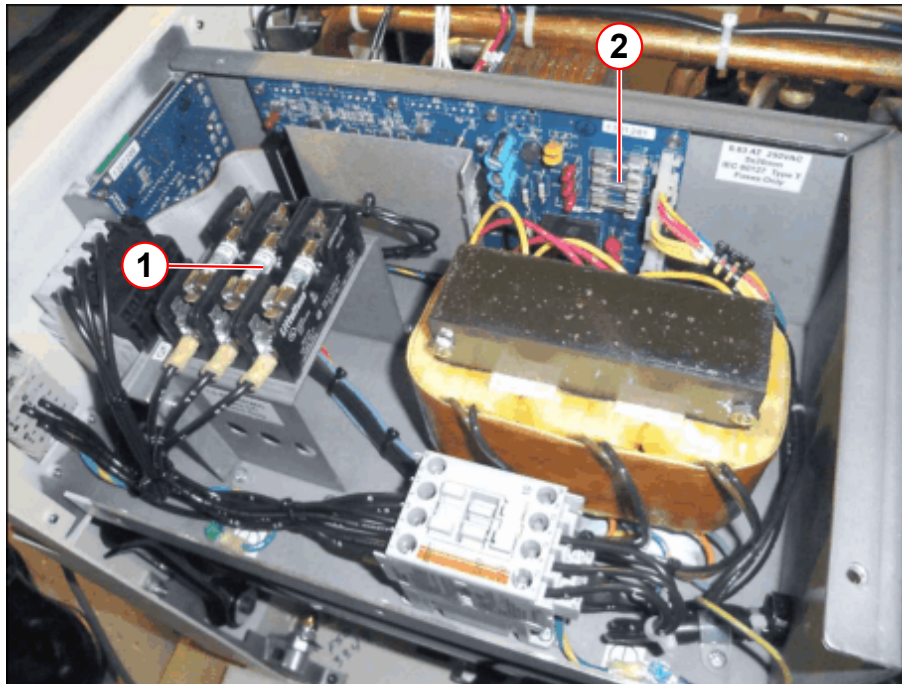
WARNING

AVOID ELECTRIC SHOCK

Failure to observe this warning can result in serious injury.

- » Disconnect the power to the compressor before troubleshooting the electrical components.
- » Permit only qualified electrical technicians to open electrical enclosures.

Fig. 206: F70 - Electrical Chassis



- (1) Fuses, FU-1, FU-2, FU-3
- (2) PCB 1 with Fuses, FU-4, FU-5, FU-6

2.5.8 Static Compressor Pressure Test

2.5.8.1 General



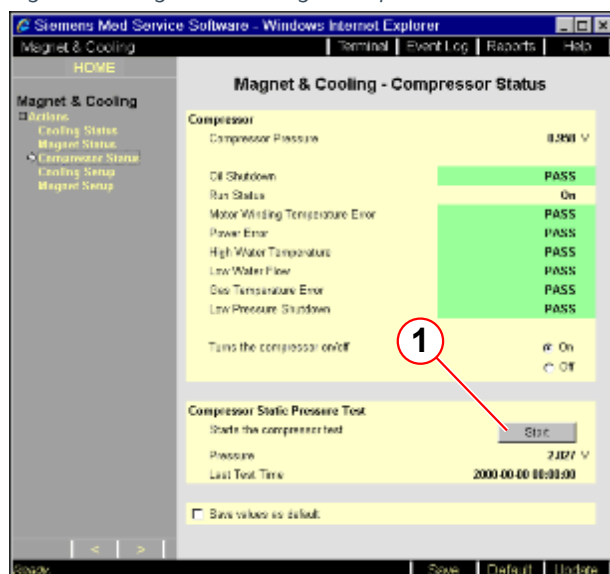
Static Compressor Pressure Test process can only be completed by DOT / Fit systems

On the MRC host there is an option to complete a Compressor Static Pressure Test. This can only be completed on site or with remote full access to the system. The voltage read on the system is equal to the static pressure reading on the compressor.

2.5.8.2 Procedure

- Log on to the MRC host (gaining full access if required).
- Navigate to 'Magnet and Cooling', then on to the Compressor Status.

Fig. 207: Magnet & Cooling - Compressor Status

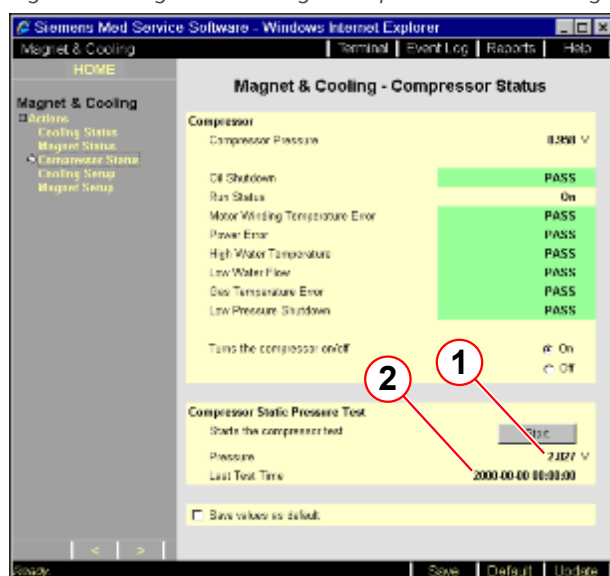


(1) Start compressor test button

- Click the start button. (under Compressor Static Pressure Test) This will initialize a compressor test.

The first step of the automatic test is the compressor is turned off, this in turn turns off the cold head.

Fig. 208: Magnet & Cooling - Compressor Status Voltage



(1) Voltage reading
(2) Logged time and date

- After approximately one hour the voltage reading is taken and then displayed with the time and date of when the reading was taken.

Once the system has finished taking and recording the readings and logged them the cold head will restart automatically.

2.5.8.3 Converting the voltage reading to a compressor static pressure.

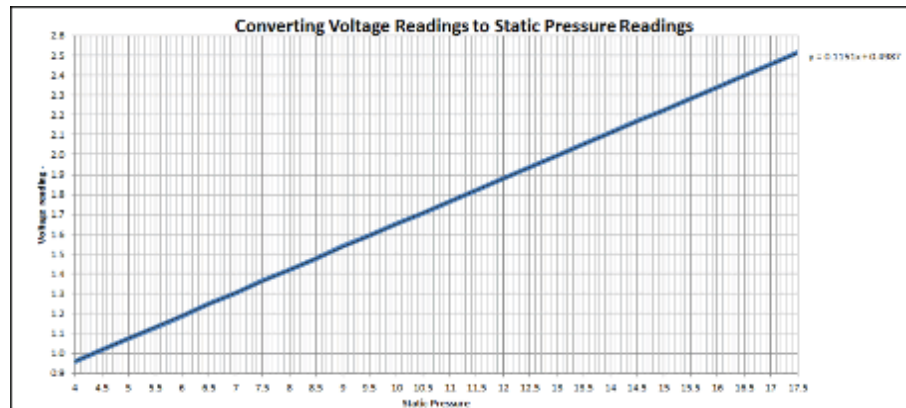
Once the system has produced a voltage reading a pressure can be obtained.



The voltage reading obtained using the method is OK to convert to a static pressure reading.

DO NOT USE any other voltage reading to convert to a pressure as the reading will be incorrect.

Fig. 209: Converting voltage readings to static pressure readings



The graph above can be used to convert this voltage to a Static pressure. Or this equation can be used.

Static Pressure (in Bar) = (Voltage reading – 0.4987) / 0.1151

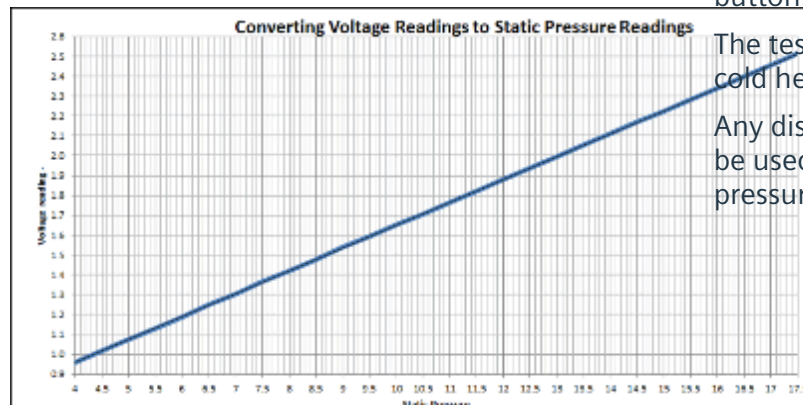
The calculated pressure has an approximate tolerance +/- 0.3 Bar

2.5.8.4 Aborting the test.

The test runs and completes automatically.

Fig. 210: Converting voltage readings to static pressure readings

- To abort the test click the **Abort** button.



The test will be stopped and the cold head will restart.

Any displayed voltage should not be used to work out compressor pressures.

2.6 Mobile 298 Shock-log™

2.6.1 Setting up the 298 Shock-log™

To set up the Shock-log™ ready for Service:

Equipment required

- Laptop PC + USB cable.
- USB extenders (41135M).
- RJ45 LAN cable (V192M).
- Communications Software for the 298 Shock-log™ provided by IMC- Group.

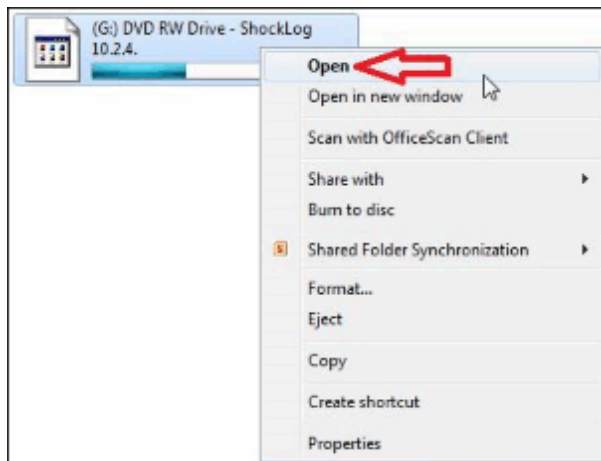
2.6.1.1 Installing the Operating Shock-log™ Software

- Install the software as 'administrator' following the Installation Instructions:



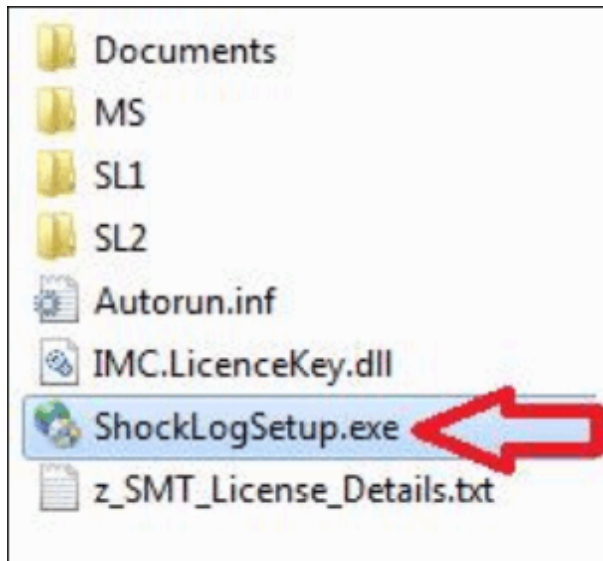
Skip this section for the first installation.

Fig. 211: Open directory



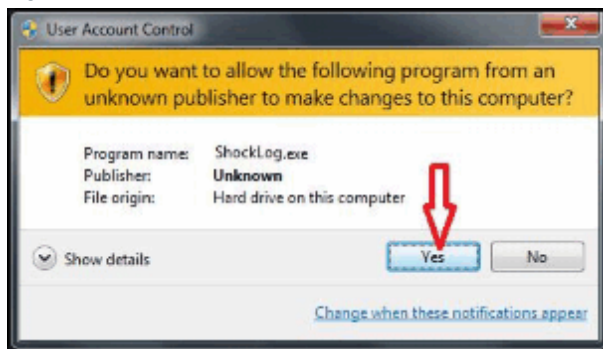
- Open CD Drive folder.

Fig. 212: ShockLogSetup file



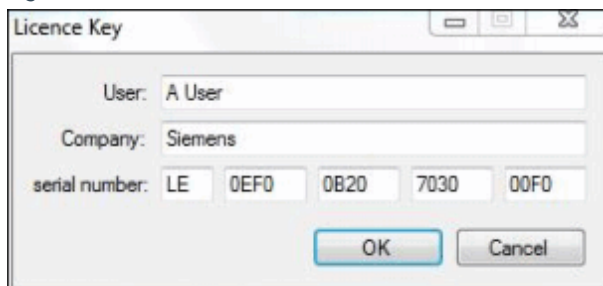
- Run "Shock-log™Setup.exe" as administrator.

Fig. 213: Select Yes



- If the following window appears select "Yes".

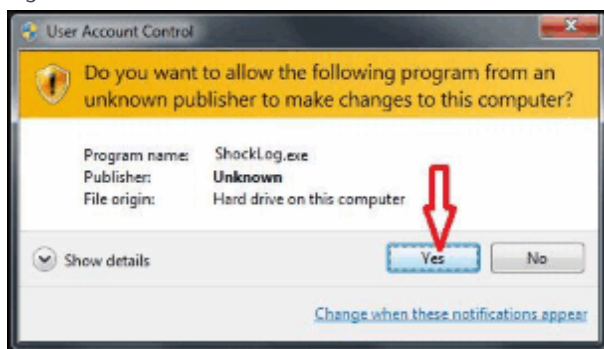
Fig. 214: Serial number



- Enter the following serial number when prompted:
 - » LE 0EF0 0B20 7030 00F0.

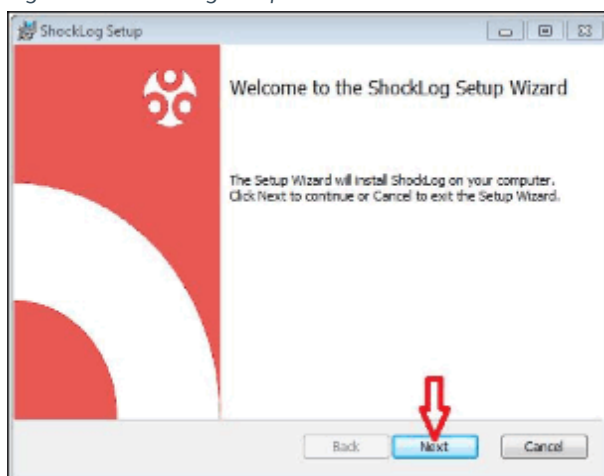
NOTE: 0 = zero

Fig. 215: Select Yes



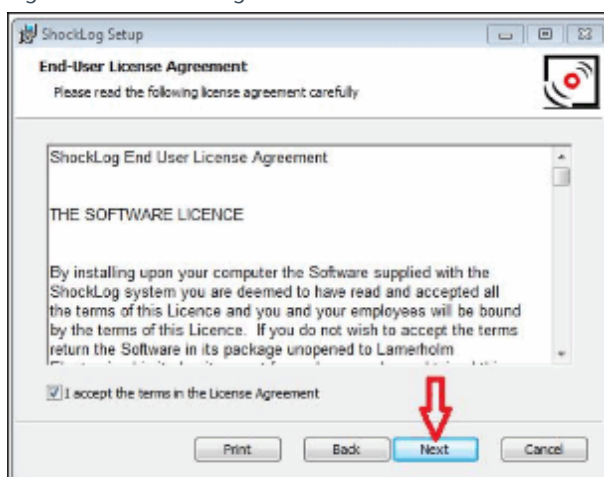
- If the following window appears select "Yes".

Fig. 216: ShockLog setup



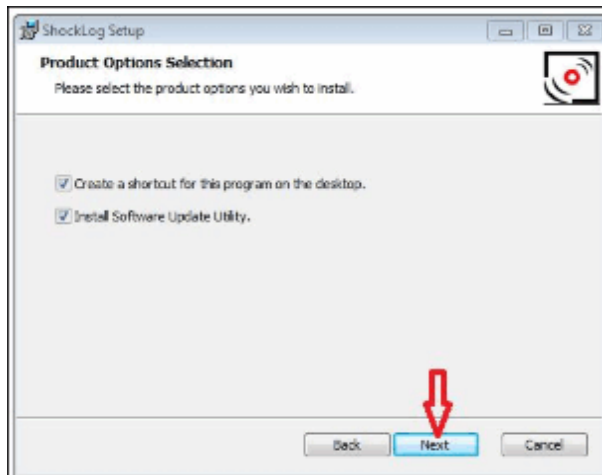
- Select 'Next'.

Fig. 217: Software agreement



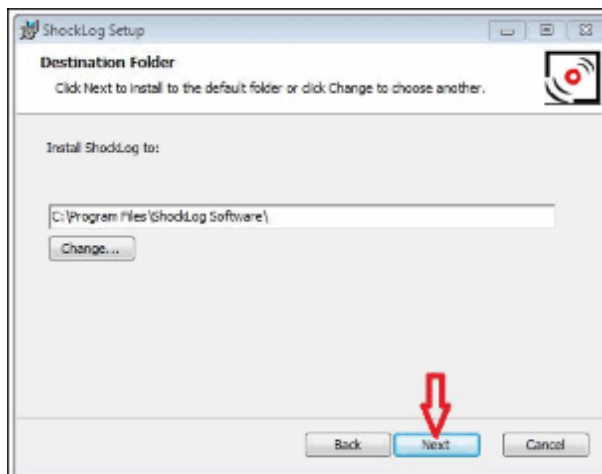
- Tick the box "I accept the terms in the License Agreement"
- Select 'Next'.

Fig. 218: Product Options



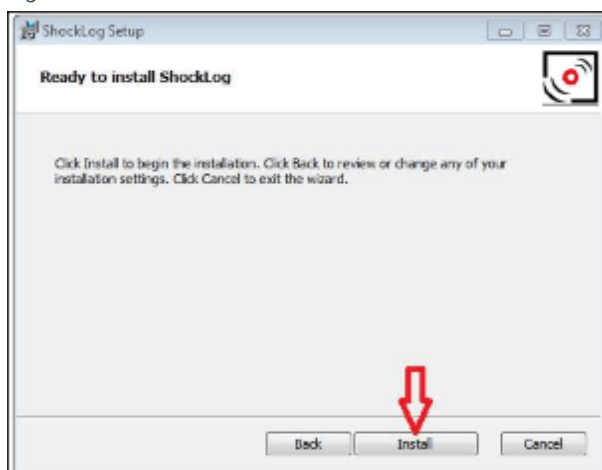
- Select 'Next'.

Fig. 219: Destination folder



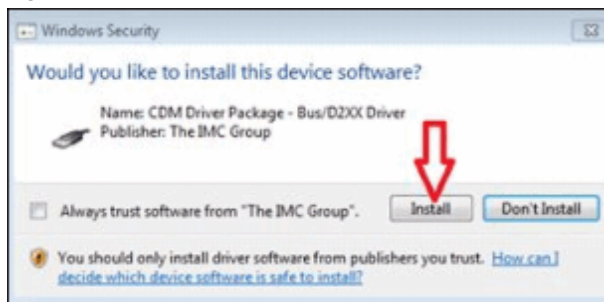
- Select 'Next'.

Fig. 220: Software installation



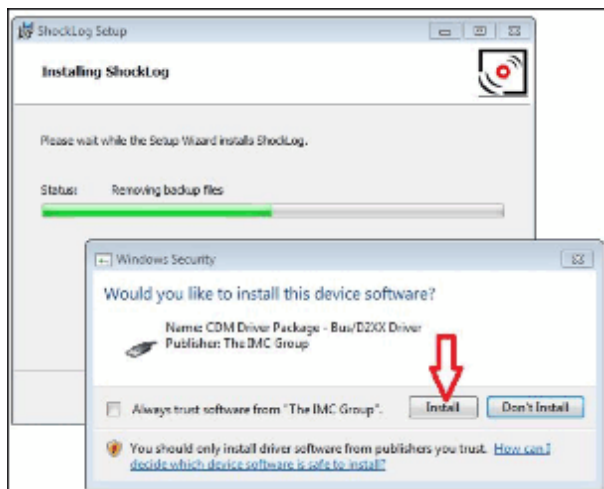
- Select "Install". Installation may take a few minutes.

Fig. 221: Software install



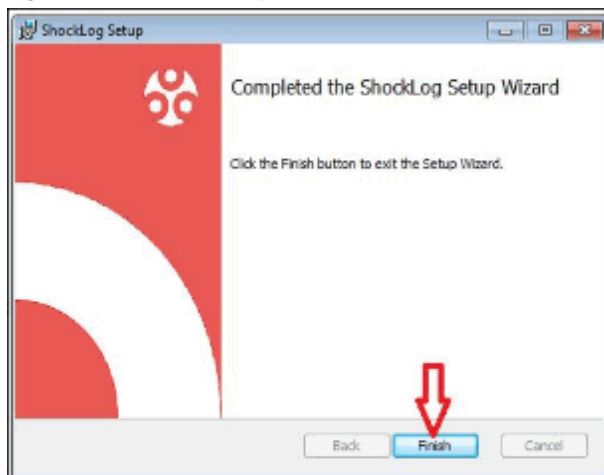
- If you get the following Windows Security windows, select "Install".

Fig. 222: Software install



- Repeat – press install for each new setup window

Fig. 223: Finish of setup



- Once the installation is complete, press "Finish"

2.6.2 Downloading the 298 Shock-log™ Data



CAUTION

Strong Magnetic Field

Damage to laptop PC and equipment.

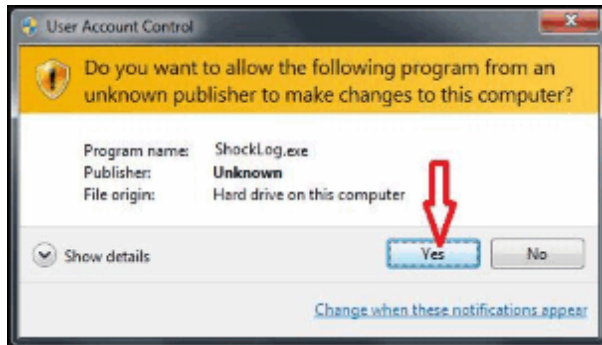
- » When downloading data from the ShockLog™ always use a suitable length USB cable. Keep Laptop PC away from the magnet.

Fig. 224: ShockLog & Software



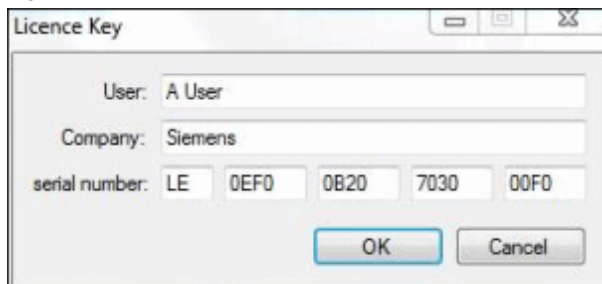
- Connect the Shock-log™ to your computer with the USB cable. If necessary, use the USB extender and the RJ45 patch cable.

Fig. 225: Select Yes



- Right mouse click the icon and select **"Run as administrator"**.
- If the following window appears select **"Yes"**.

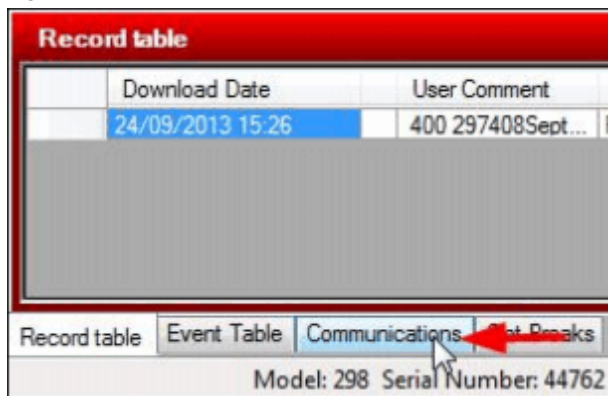
Fig. 226: Serial number



- Enter the following serial number when prompted:
 - » LE 0EF0 0B20 7030 00F0.

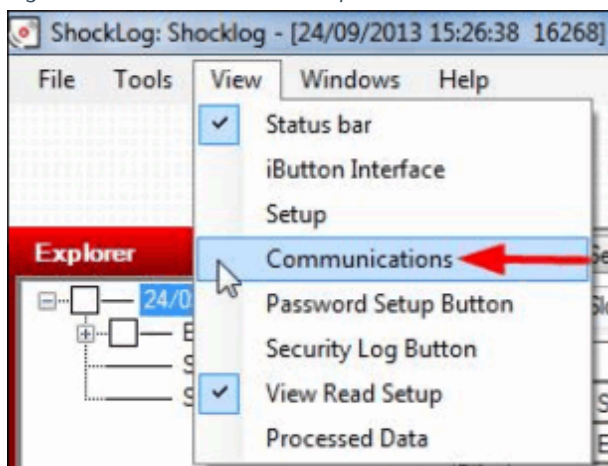
NOTE: 0 = zero

Fig. 227: Communication window



- Open "Communications" window

Fig. 228: Communications drop down



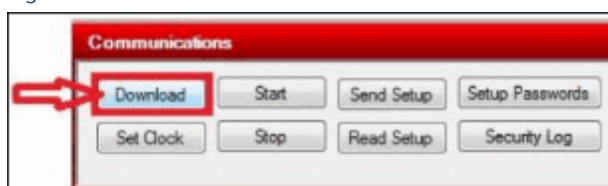
- The "Communications" window can also be accessed from the 'View' drop-down.

Fig. 229: Communications



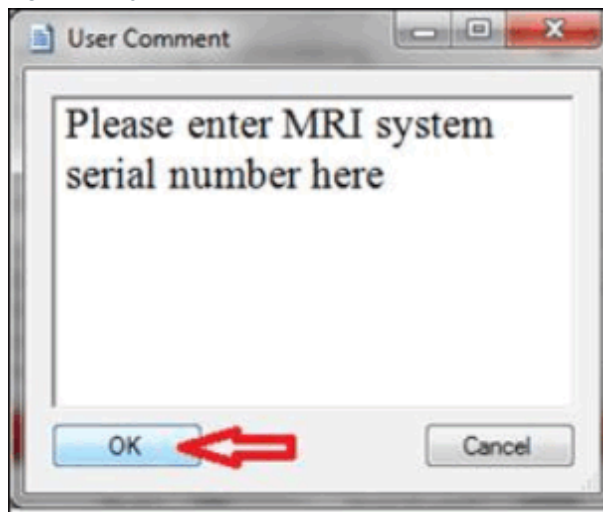
- In communication window select "Stop" button.

Fig. 230: Communication 'Download'



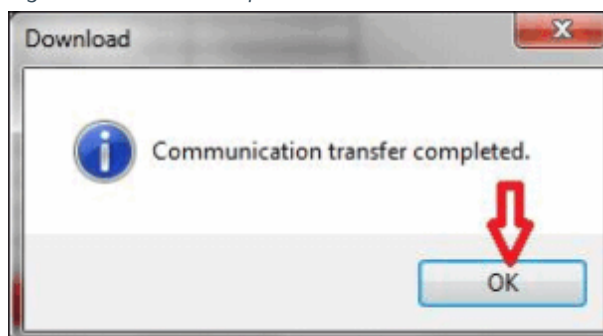
- After "State" changes to "Stopped" select 'Download' button.

Fig. 231: System serial number



- Enter MRI system serial number in the following window, Press "OK".

Fig. 232: Transfer completed



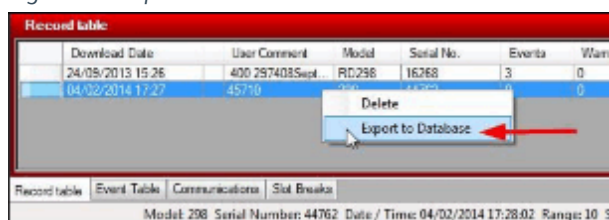
- After downloading the process is completed - Press "OK".

Fig. 233: Communications 'record table'



- Open "Record table" tab.

Fig. 234: Export to database



- Find downloaded file, right-click on it and select "Export to Database".

- Select 'Desktop' and then press 'Save' button.

- Make sure that you have a new “*.accdb” file saved on your desktop.
- Close Shock-log™ software and unplug Laptop.

2.7 Mobile RD 298 ShockLog

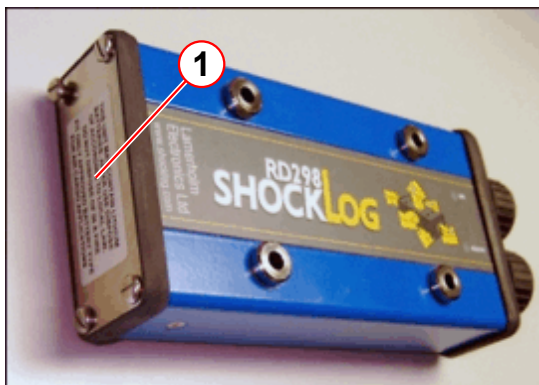
2.7.1 RD298 (Type 1) ShockLog Set up and Download

2.7.1.1 General Information

The procedure covers the installation and set up of the original 'Blue type' Lamerholm RD298 ShockLog.

Any impacts during transportation will be recorded on the ShockLog providing a record of the time and date of the event.

Fig. 235: RD298 ShockLog (Blue type)



Safety



CAUTION

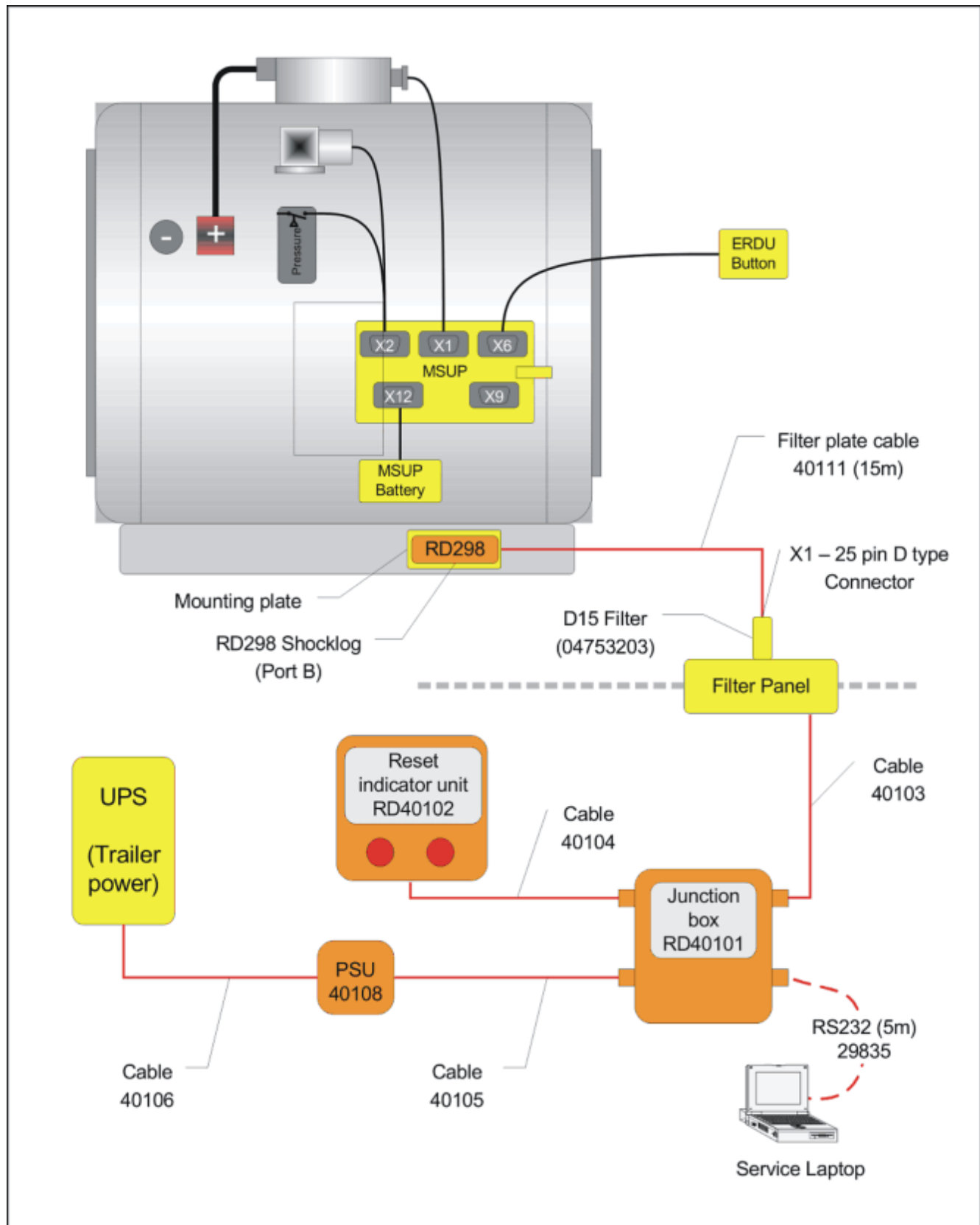
Personal injury

Electrical shock during installation

- » Ensure that electrical power to the CCA cabinet is switched OFF.

2.7.1.2 Component Set Up

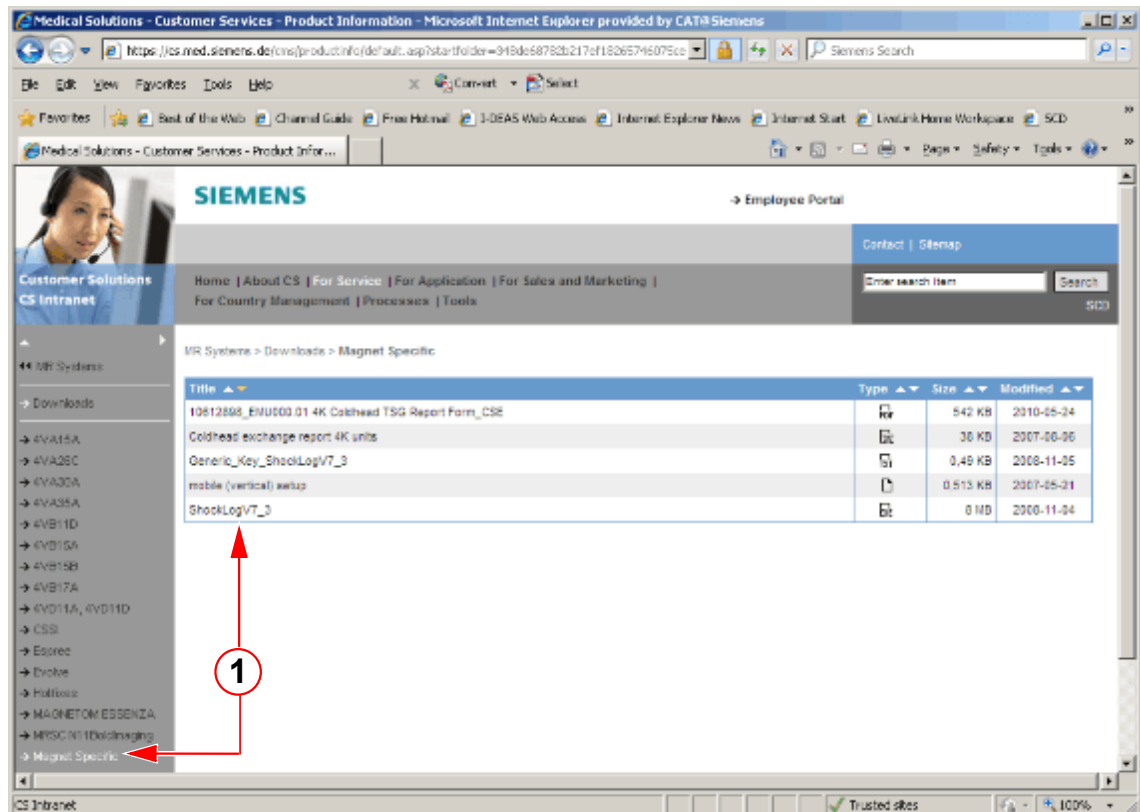
Fig. 236: RD298 Shock logger Setup



2.7.1.3 ShockLog Operating Software

- Download the Lamerholm operating software from the CD supplied in the ShockLog kit onto a Service laptop.
- Install the RD298 ShockLog setup file '**mobile (vertical).setup.exe**' on the service laptop. This can be downloaded from:-
 - **Med CS Intranet:** For Service/Product information/MR Systems/Downloads/Magnet Specific

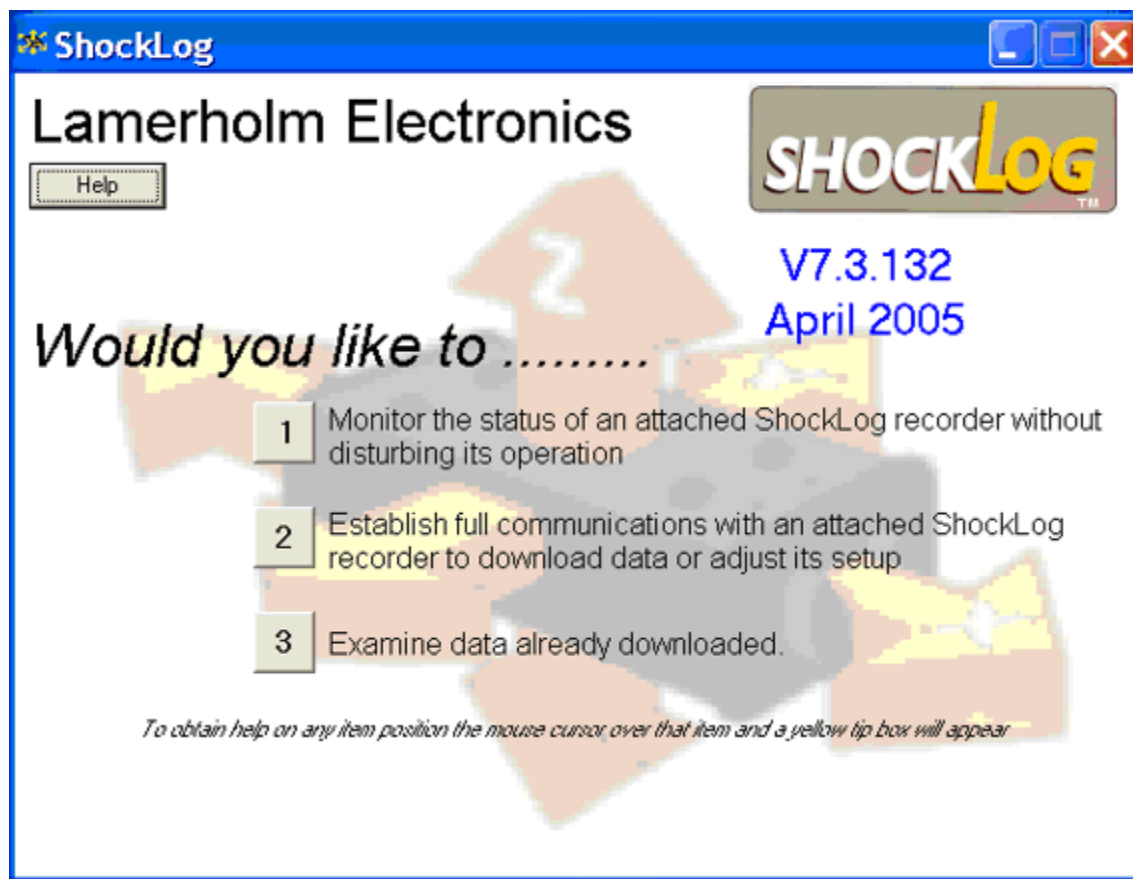
Fig. 237: ShockLog download location



(1) ShockLog download location

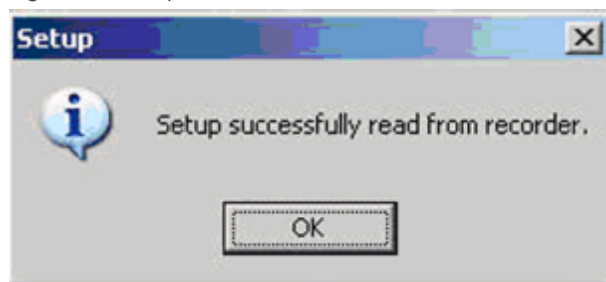
- Save the file **mobile (vertical) setup.exe** (C:/Programs/shocklog/setups).
- Upload the file '**mobile (vertical).setup.exe**. Use the help file in the ShockLog program (if necessary) for guidance.
- Select button 2 - 'Establish full communications....'

Fig. 238: RD298 Shock-log software interface page



- From the Shocklog startup screen, select **option 2** 'Establish full communications'.

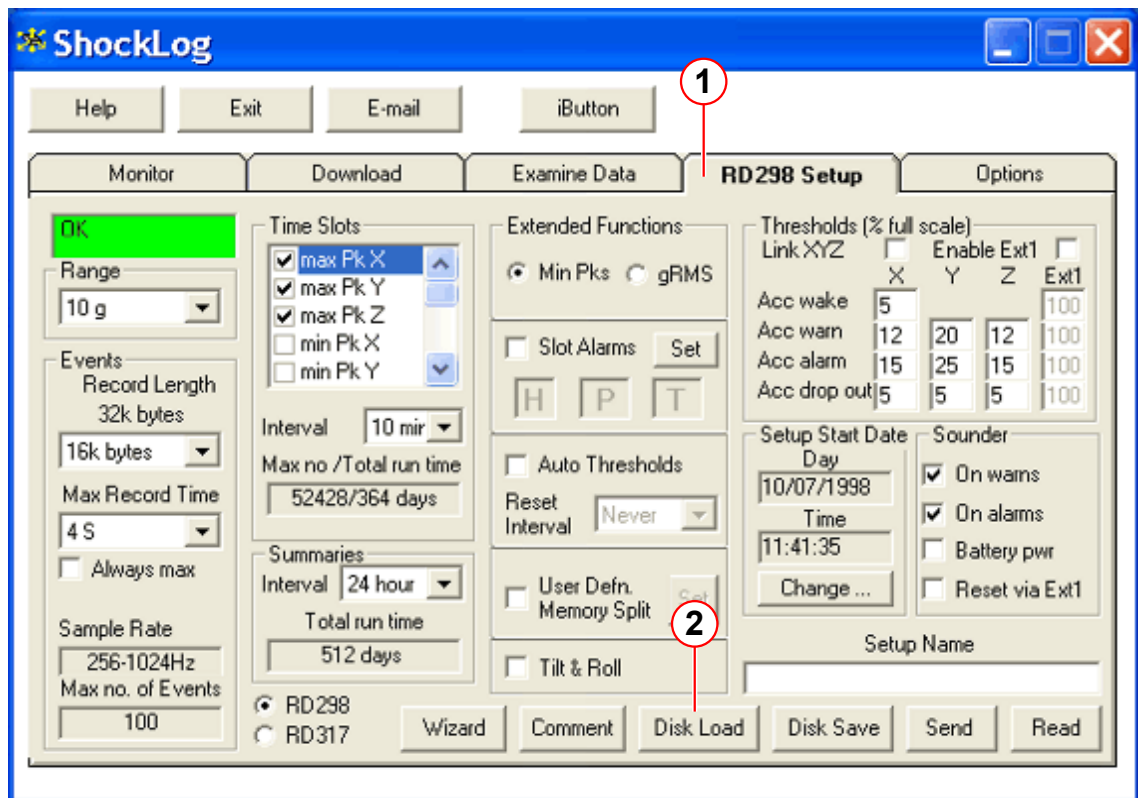
Fig. 239: Setup successful



The laptop will attempt to connect to the Shocklog

- The message window '**Attaching to Shocklog recorder**' will appear and count down connection time.
- This should be less than 10 seconds.
- Communication is successful.
 - » Click OK
- From the setup screen:
 - » select RD298 setup Tab.
 - » click on the Disk Load button.
- Double-click on the file mobile (vertical) setup.

Fig. 240: RD298 Setup tab



(1) RD298 Setup tab

(2) Disk Load

- Double click on the file mobile (vertical) setup.

Fig. 241: Mobile (vertical) setup

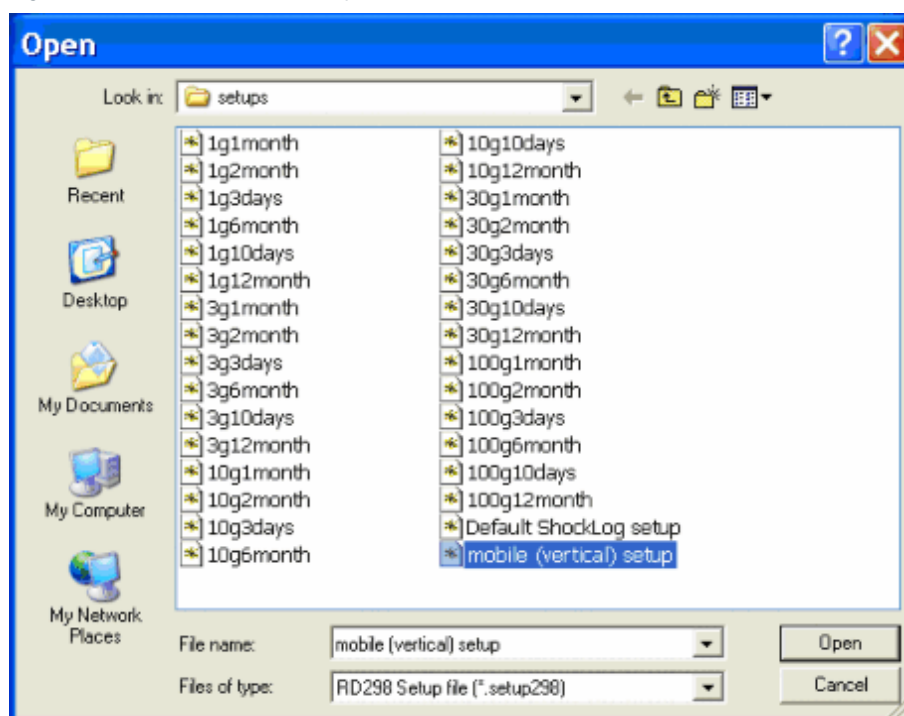
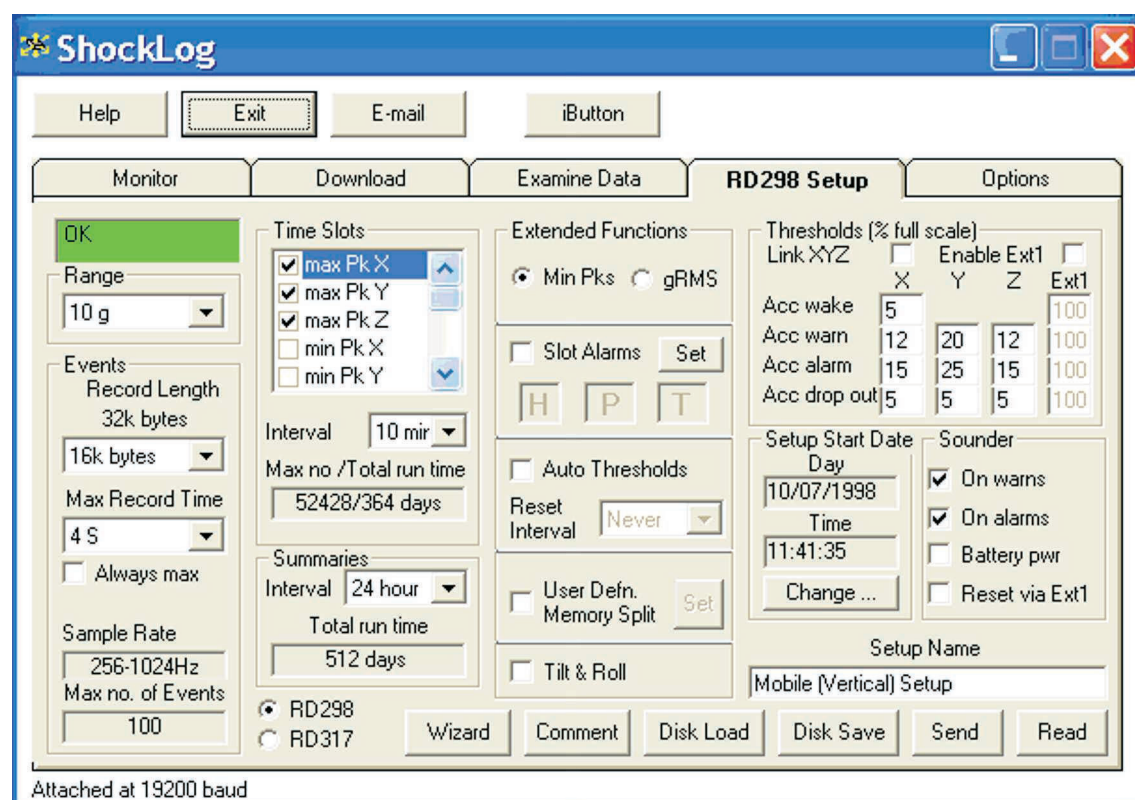


Fig. 242: Attaching to the ShockLog recorder

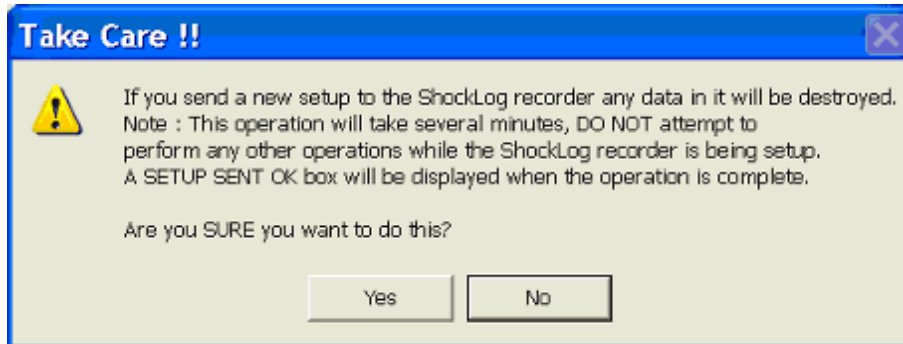


- (1) Send button
- (2) Baud rate

- On the Shocklog set up screen click on the 'Send' button.

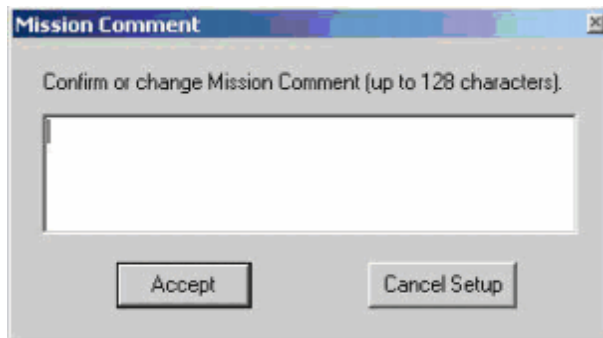
- The message box, 'Attaching to shocklog recorder' will appear for 10 seconds
- The message box 'Take Care' will appear.

Fig. 243: Take Care!! Warning



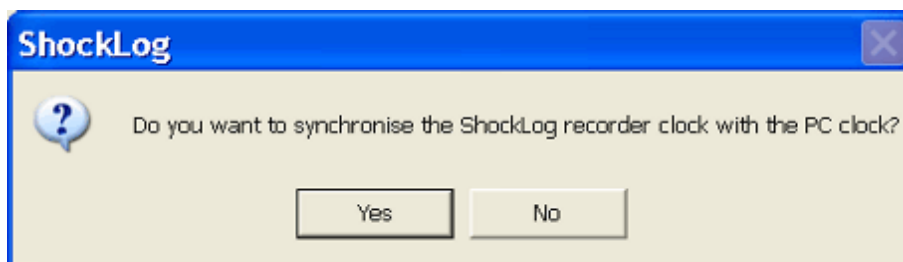
- Click 'Yes'.

Fig. 244: Mission Comment screen



- Click on Accept

Fig. 245: Synchronise the ShockLog recorder



- Click on Yes



The communication checks and setup will take a few minutes to complete.

Fig. 246: Synchronising ShockLog recorder with PC Clock

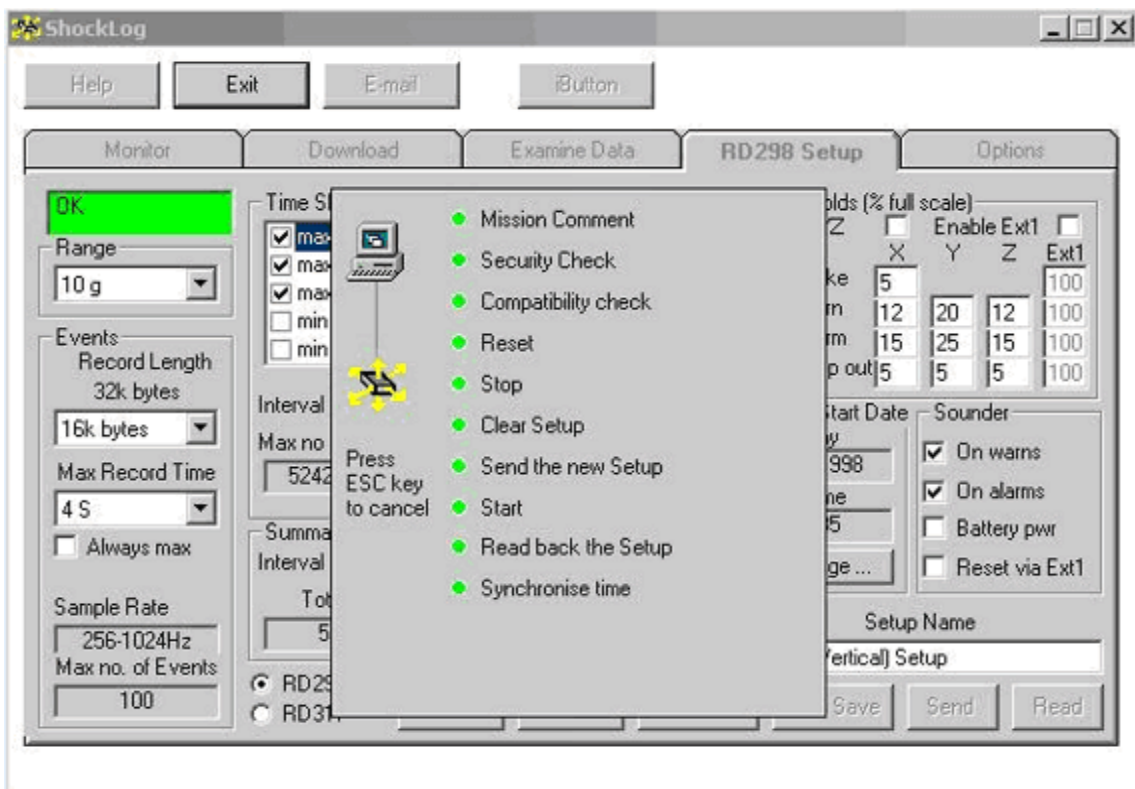
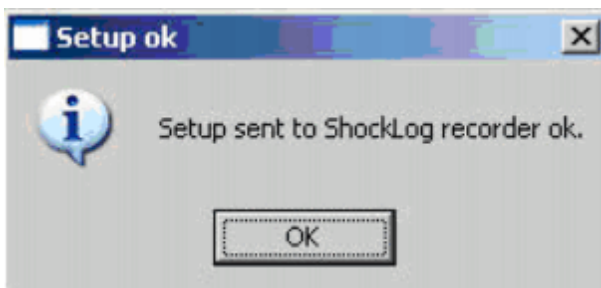
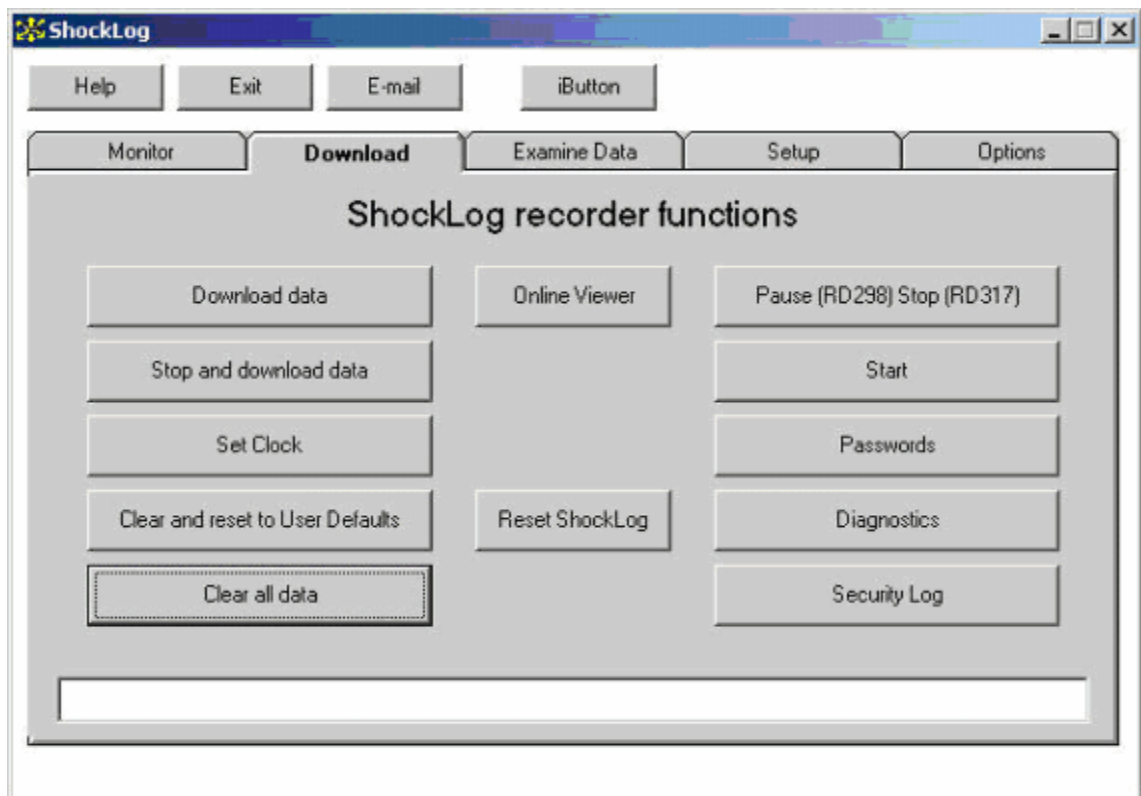


Fig. 247: Setup sent to recorder



- Click OK.
- Select the Download Tab and click on the Start button.

Fig. 248: ShockLog recorder functions



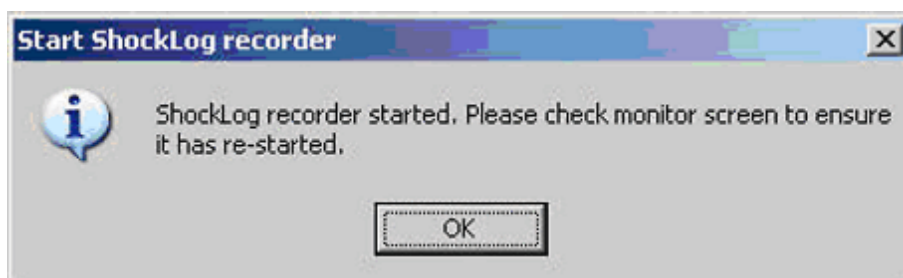
- Click 'Yes' to start the Shocklog recorder.

Fig. 249: Start Shocklog recorder



- The ShockLog will time down to start for 10 seconds until the Start ShockLog Recorder box appears.
- Click the 'OK' button.

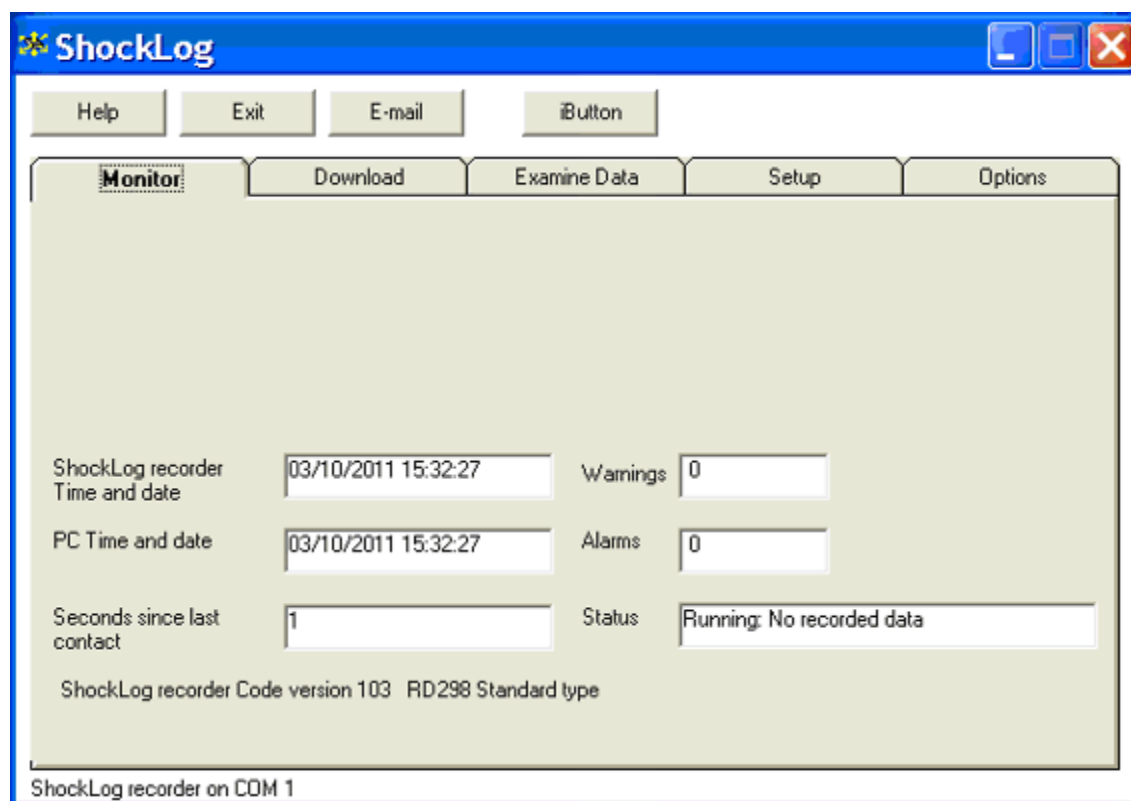
Fig. 250: Start ShockLog recorder started



2.7.1.4 ShockLog Monitor Display

- Select the Monitor Tab.
- The box seconds since last contact will count through 10 sec and continue to repeat the count.
- The Status box will show Running: No recorded data.

Fig. 251: ShockLog set up time and date display



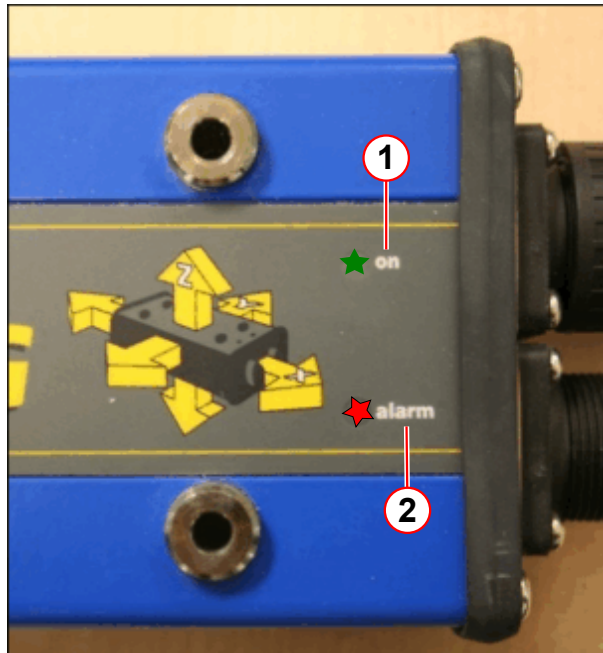
- The ShockLog set up time and date will be displayed.
- The Warning and Alarm windows will display zero (0)

The ShockLog is now set up and recording. The green 'ON' LED will be flashing every 5 seconds

- Click on the Exit button.

ShockLog LED Status

Fig. 252: ShockLog LED status



- (1) OK status :No alarms = Green LED flashes
- (2) Warning status: Red and green LEDs flash
- (3) Alarm status: Red LED flashes

The ShockLog has two LEDs visible when fitted.

- On = **Green** LED
- Alarm = **Red** LED

There are 3 status levels:

- **OK Status** - no alarms **green LED** flashes every 5 seconds.
- **Warning status** - red and green LEDs will flash.
- **Alarm status** - the red LED flashes:
 - » at 5 second intervals - the ShockLog will continue to record.
 - » Every 10 seconds, the ShockLog is no longer recording.



Both LEDs will flash every 1 second if any internal faults on the ShockLog are detected. In this instance remove the battery for 10 seconds to re-set the ShockLog.

2.7.1.5 Connecting the Laptop Computer to the RD298 ShockLog

- Connect the 9-pin 'D' type or USB connector onto the Filter plate cable (40111) marked 'ShockLog'. Connect to the serial or USB port on the PC or Laptop.
- **Open** the ShockLog software on the Laptop PC. Select:
 - » Start
 - » Programs
 - » ShockLog
- Open the ShockLog startup screen, (or click on the ShockLog desktop icon).
 - » The screen loads with the Monitor tab selected.
 - » The message 'ShockLog on COM 1/2/USB' is displayed at the bottom of the screen.

- If the PC and ShockLog recorder are communicating correctly, the monitor display will update every 10 secs.
 - » The PC will 'beep' to indicate that the display has updated.
 - » The box 'Seconds since last contact' will be set to zero.



If the update process is not operating correctly, identify and correct the problem before attempting any other communication functions.



Downloaded data can also be examined when the PC and ShockLog are not communicating.

- From the Monitor tab, note the number of Wakeup and Alarms in the system log.

Clearing the data from the ShockLog memory

- Activate the 'Clear all data' button to clear the memory.

2.7.1.6 Service Instructions

Service engineers are required to check, maintain and download the data recorded from the ShockLog:-

- At regular maintenance intervals.
- On a Service call if a shock event has occurred.

Spare Parts

The spare parts required for maintenance and downloading data are:-

- Service Laptop
- USB to RS232 converter cable.
- Lamerholm communication software CD.
- 3.6 volt lithium battery. Size 'C' (LR14) - (LS26500:SAFT), **10100429**.



Alkaline batteries must not be used as a substitute for lithium. The lifetime of a alkaline battery is < 1 month.

Remote Indicator Unit (RD401)

Fig. 253: Remote Indicator unit



When the **'Wake-up'** or **'Alarm levels'** are exceeded, the indicator light will be activated.

- Record the date, time and circumstances of the alarm activation.
- Download data from the ShockLog.
- Report activation to the local RSC.
- Cancel alarm by pressing the 'reset' button..



Instructions are on the front label of the RD401 indicator unit.

2.7.1.7 Downloading RD298 ShockLog Data



The RF filter may cause download communications problems from the ShockLog to the laptop. Use the short (10ft) serial cable and connect directly between the ShockLog and laptop.



The ShockLog data must be downloaded and sent to the RSC in the event of the indicator LED being found 'ON' during a maintenance visit or customer call out.

- In the equipment room, connect the 9-pin 'D' type or USB connector on the lead to the serial port on the Laptop USB port; the connector is labelled 'ShockLog'.
- Run the ShockLog software.
- Open the start-up screen and select Option 1.
 - The ShockLog working screen will load, with the Monitor Tab selected.
 - The status message at the bottom of the screen will say 'ShockLog on Com 1/2 - USB'.

If the Service laptop and ShockLog are communicating correctly, the monitor display will be updated every 10 secs. The laptop will 'beep' to indicate that the display has updated.

The display screen box labelled 'seconds since last contact' will be set to zero.



Not all Laptop computers will make a 'beep' sound when communicating and updating from an attached device.

If the communication and updating process is not working then the problem must be corrected before continuing.

Data already downloaded and saved can still be viewed from the system log.

- Select 'tab 3' from the main screen, 'examine data already downloaded'.

Downloading the Shock-log data

- Select the 'Download' tab and click on the 'Download data from Shock-log' button.
 - The dialogue box 'Enter user file name' will appear.
- Enter a file name, system number, date and the CSE's name or initials, up to a maximum of 16 characters.
- Press the 'RETURN' key or click the 'OK' button.
 - The download checklist will appear and the PC will start to download.



The status box displays the download progress. The communications status may appear in the status line at the bottom of the Shock-log window.

When the download is complete, a 'Data Downloaded' message box will appear, reporting the file name and location of the data. The size of the file will vary between 30k and 500k/bytes, depending on how long the ShockLog has been operating.



When sending the downloaded files to the RSC or to the Magnet HSC, both files will be required. See example below of the file names downloaded:-

20090623094201173846446EJAS6232009-.sl_slots

20090623094201173846446EJAS6232009-.sl_data



Check data integrity using the 'examine data' tab, before clearing the memory.

- Clear the logger memory using the 'Clear all data' button.
- Ensure the logger is 'running' after download is complete.
 - » This may take a few minutes. The logger may require a 'restart'.



Check the lithium battery if the ShockLog appears to be operating normally but will not download or clear correctly. The battery may need to be replaced.



In the event of a tripped ShockLog a further review of the data is required. Send the downloaded files to the magnet HSC.

Re-Setting the Clock

- At the start-up menu screen:
 - » Select the download tab.
 - » Select the 'Set Clock' button.
 - » Follow the on-screen instructions.

Replacing the Battery

The battery must be replaced regularly as part of the maintenance process.

Before replacing the battery:-

- As a precaution download any ShockLog data that is required.



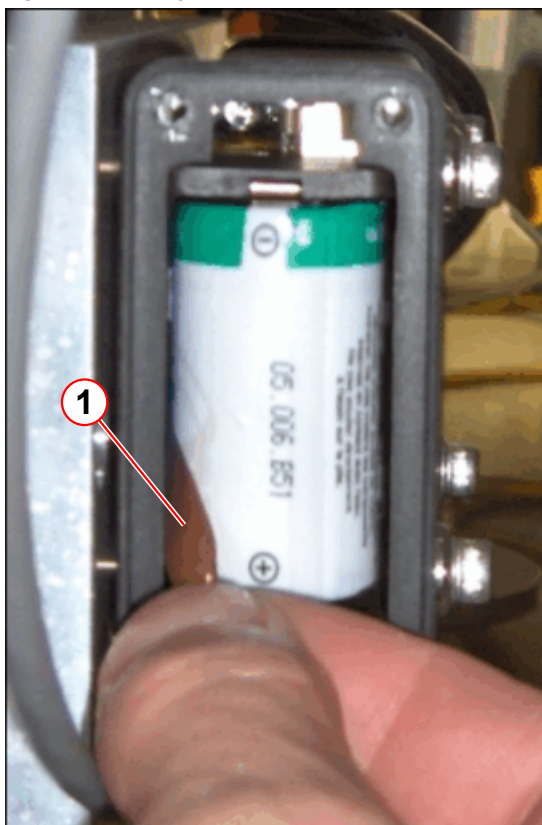
Recorded data is maintained when the battery is removed.



If the battery is fitted in the wrong polarity lithium battery will discharge rapidly. If this happens another battery replacement will be required.

- Download and then clear ShockLog data.
- Remove the system side cover to access the ShockLog on the magnet.
- Locate the battery compartment at the end of the ShockLog.
- Remove the 4 x screws retaining the battery compartment cover.

Fig. 254: Battery Removal



(1) Release tab

- Remove the battery cover.
- Pull the release tab to remove the battery.
- Check polarity and fit the new lithium battery.
- Push the release tab back into the battery holder.
- Re-fit the battery cover using the 4 x screws.



If a new battery is inserted incorrectly for more than a few seconds, do not use that battery for any application where a full battery life is required.

- Record the replacement date of the new battery. Affix a new label to the battery cover.
- Re-fit system covers.
- Re-set the ShockLog clock.
 - » At the start-up menu screen, select the download tab, followed by the 'Set Clock' button and follow the instructions.

- » The common print number (M6-020.840.01) from the Troubleshooting Guide is divided into separate Component print numbers.

Version 02

- Chapter Magnet System
 - Post-quench procedure reused from SMT document
 - Repairing leaks reused from SMT document
 - Alarm box reused from SMT document
 - Strategy information updated.
- Chapter MSUP
 - Strategy and procedure reused from SMT document
- Chapter Refrigerator
 - Strategy and procedure reused from SMT document
- Chapter Magnet Power Supply
 - Strategy information updated.

Version 03

- Procedure
 - Service platform added.
 - MSUP 2657 added.
 - MSUP download instruction updated.

Version 04

- Strategy
 - Link to "Magnet Error messages" has been added.

Version 05

- Procedure
 - Post Quench procedure updated.
 - CSW71: Corrected static pressure to 16.0 - 16.5 bar.

Version 06

- Procedure
 - Update of the Service platform installation.
 - New section: De-icing the syphon port.

Version 07

- Procedure
 - Update of the MSUP download instruction.

Version 08

- Procedure
 - Update of "What to do after a Quench".
 - F70 download instruction

Version 09

- Strategy
 - Compressor re-start time added.
 - Fuse location added.
 - MSUP 2306 added
 - Alarmbox added
- Procedure
 - Update of "De-icing of the Syphon Port".
 - F70 Hours run clock display added.
 - MSUP 2306 added

Version 10

- Procedure
 - Fuse Location F70 added. (→ Fuse Location F70 / Page 196)
 - Static Compressor Pressure Test added.(→ Static Compressor Pressure Test / Page 197)

Version 11

- Strategy
 - Link correction.
- Procedure
 - Updated: What to do after a Quench (→ What to do after a Quench (Avanto, Avanto Dot Upgrade, Espree with MSUP K2256) / Page 89) MR Charm_00453494.
 - Updated: What to do after a Quench (→ What to do after a Quench (Avanto Dot Ex-factory, Espree with MSUP K2306) / Page 124) MR Charm_00453494.

Version 12

- Procedure
 - Refrigerator (→ Refrigeration Failures with Magnet at Field / Page 194) MR Charm_00472208, added.

Version 13

- Strategy
 - Information added about ramp cable shorting links. (ID 209125)
- Procedure
 - Download and setup procedures for Mobile 298 ShockLog™ and Mobile RD 298 ShockLog added. (ID 209594)
 - Update of "What to do after a quench" (ID 212345)
 - Update of "Magnet / What to do after quench"
 - Update of "Magnet / Venting leak check"
 - Update of "Magnet / Repairing leaks on the turret"

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